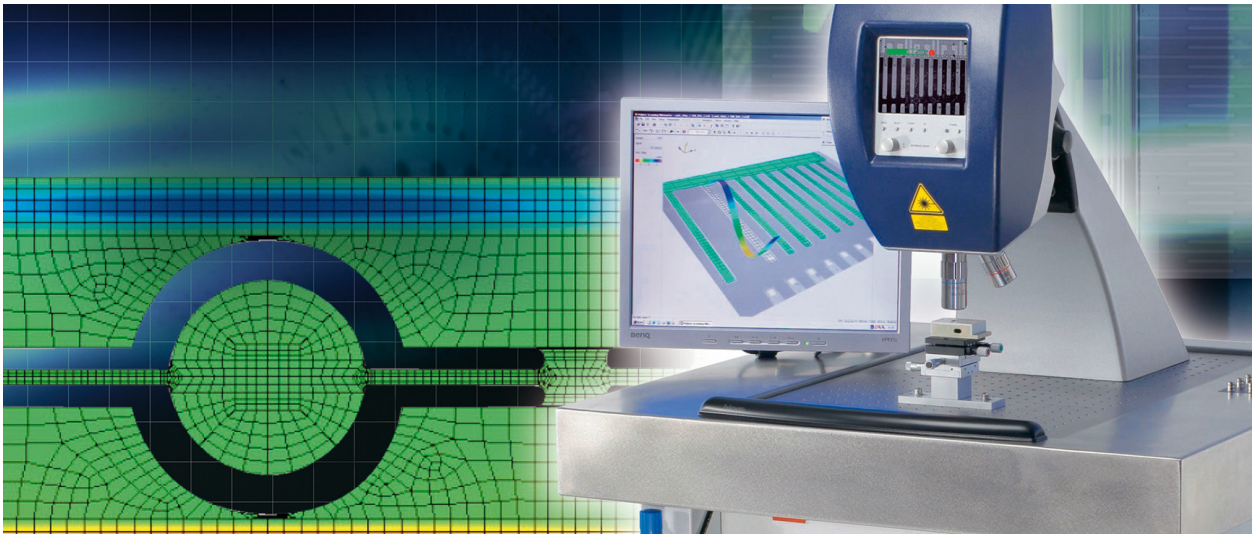
Polytec

Software Manual

Polytec Scanning Vibrometer

MSA Acquisition



MSA-650

MSA-600

MSA-500

Version 10.0

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Appendix A: Key Combinations in the Software

Index

1 Quick-Start

This chapter will familiarize you with important functions of the measurement system based on an FFT measurement. To do so, carry out a scan and evaluate it using some of the software functions. Use an object with simple geometry for the scan. If available, use a function generator to excite the object. Use this signal as the reference signal for the measurement. If you do not have a function generator, you can also excite the object with your own means.

ⓘ You can also carry out the measurement described here without a reference signal. In this case, you cannot animate the vibrations during evaluation.

Preparing

1. Connect up and start the measurement system as described in the operating instructions.
2. If available, connect the reference signal to the REF 1 jack on the front of the front-end/junction box.
3. Start the software in Acquisition mode by double-clicking the 

icon on the desktop.

   The application window of the software appears.
4. If you use automated axes from Polytec, you will be prompted to carry out a reference run of the axes. Select the desired option.

   The software makes a reference run.
5. You will be prompted to set the limits for the moving range of the axes. Enter the limits and click OK.

   You will see the video window at the top and an analyzer at the bottom. On the left you will see the project browser and on the right, the scanning head control.
6. In the menu bar, select Setup > Preferences.

   The Preferences dialog appears.
7. Display the Devices page.

   MSA-600:The hardware which has been identified by the software is displayed in the Junction Box list (e.g. MSA-F-600-M front-end).

   MSA-500: From the Junction Box list, select the hardware that is connected to the Data Management System.
8. If you are using a function generator, then set the Internal connection for Generator.
9. If you are using automated axes from Polytec, set the XY or XYZ connection for Axes.

10. Display the Scanning Head page.

    The sensor head which has been identified by the software is shown in the Scanning Head list.

11. Click OK.

12. Only MSA-500 with VDD option: Carry out test mode as described in SECTION 8.6.

You will find more information on setting up the software in CHAPTER 3.

Setting the Optics

You set the optics using the elements in the scanning head control.

1. Align the object under the objective as described in the operating instructions of the measurement system.
2. Focus the video image.

   This also focuses the laser beam.
3. Set alignment points, to align the positions on the live video image with the position of the laser beam on the measurement surface. To switch to the 2D alignment, select Setup > 2D Alignment in the menu bar (toolbar: ).

   The software will maximize the video window.
4. Move the laser beam to an appropriate alignment point on the object using the icons in the Position field in the scanning head control.

ⓘ The alignment points must be evenly distributed over the measurement surface. To scan a rectangular surface, select four alignment points near the corners.

5. In the video image, click the point which the laser beam is currently at.

   The target is displayed. The alignment point is set.

ⓘ If the laser beam is not precisely in the middle of the target, click the live video image with more precision.

6. Repeat steps 4 and 5 for at least three more alignment points.
7. Once you have aligned all points, exit the 2D alignment. To exit the 2D alignment, select Setup > 2D Alignment (toolbar: ).

You will find further information on 2D alignment in SECTION 5.1.

Defining Scan Points

1. You define scan points in Scan point Definition mode. To switch to the scan point definition, select Setup > Define Scan Points in the menu bar (toolbar: ).

   The software maximizes the video window and displays a graphics toolbar (the  icon is activated). The Object Properties dialog appears.
2. To delete all existing scan points, select Edit > Select All.
3. Select Edit > Delete.

   All scan points are deleted.
4. In the graphics toolbar, click  .

   You are in drawing mode.
5. Use the mouse to draw the largest possible rectangle on the object.

   You will see the rectangle with a red edge and white squares which mark the edge. You will see the scan points as blue squares inside the rectangle.
6. If the  icon in the Rectangle Properties dialog is active, click  to display the scan points.
7. Point the mouse at the grid.

   The mouse pointer becomes a cross.
8. To move the grid with the scan points, click the grid with the mouse and drag it to the required position.
9. To exit the scan point definition, select Setup > Define Scan Points in the menu bar (toolbar: ).
10. Point with the mouse at the live video image and click a scan point.

    The laser beam moves to this point on the object.
11. Click additional scan points to check and see if the laser beam is positioned precisely on the scan points.

You will find detailed information on scan point definition in CHAPTER 6.

Setting the Parameters

1. To set the parameters for a measurement, select Acquisition > Settings in the menu bar (toolbar: ).

   The Acquisition Settings dialog appears.
2. Display the pages in the dialog from left to right and set the parameters there. The suitable settings depend on the object.
3. Use the values in the following tables (MSA-600, MSA-500) if you do not know any suitable settings.

Table 1.1: Parameters and settings for an example measurement using the MSA-600

Page	Setting
General	Measurement Mode: FFT Averaging: Off Remeasure: Active
Channels[1]	Vibrometer: Active, Direction +Z Reference 1: Active, Direction +Z, Range 10 V, Coupling DC, Quantity Voltage, Factor 1
Filter	Vibrometer Channel: Filter Type No Filter, Int/Diff Quantity Velocity (0) Reference 1 Channel: Filter Type No Filter
Frequency	Bandwidth: 10 kHz From: 0 kHz To: 10 kHz FFT Lines: 400
Window	Vibrometer Channel: Function Rectangle Reference 1 Channel: Function Rectangle
Trigger	Source: Off
SE	Vibrometer Channel: Active Reference Channel 1: Not active Slider: Fast
Vibrometer	Velocity: Lowest available measurement range Tracking Filter: Slow
Generator	Active: Active

[1] The other parameters for the Vibrometer channel are fixed.

Table 1.2: Parameters and settings for an example measurement using the MSA-500

Page	Setting
General	Measurement Mode: FFT Averaging: Off Remeasure: Active
Channels[1]	Vibrometer: Active, Direction +Z, Range 10 V, Coupling DC Reference 1: Active, Direction +Z, Range 10 V, Coupling DC, Quantity Voltage, Factor 1
Filter	Vibrometer Channel: Filter Type No Filter, Int/Diff Quantity Velocity (0) Reference 1 Channel: Filter Type No Filter
Frequency	Bandwidth: 10 kHz From: 0 kHz To: 10 kHz FFT Lines: 400
Window	Vibrometer Channel: Function Rectangle Reference 1 Channel: Function Rectangle
Trigger	Source: Off
SE	Vibrometer Channel: Active Reference Channel 1: Not active Speckle Tracking: Active Slider: Fast
Vibrometer	Velocity: Lowest available measurement range Tracking Filter: Slow Low Pass Filter: 1.5 MHz (Off) High Pass Filter: Off
Generator	Active: Active

[1] The other parameters for the Vibrometer channel are fixed.

4. Click OK.

You will find detailed information on the parameters for data acquisition in CHAPTER 7.

Single Point Measurement

1. When using a function generator, start signal output by clicking Acquisition > Generator in the menu bar (toolbar: ).
2. To start a continuous measurement, select Acquisition > Continuous Measurement (toolbar: ).

   The analyzer shows the measurement signal on the vibrometer channel.
3. To automatically scale the y-axis, select Analyzer > Auto Scale (toolbar of the analyzer: ).

   The measurement signal is shown as large as possible in the diagram.

ⓘ To automatically modulate individual channels, click  for the corresponding channel in the scanning head control.

4. Click a scan point in the video image.

   The software positions the laser beam on the object at this scan point.

   If the OVERRANGE message is displayed in the status bar, the measurement range for the velocity is exceeded at this scan point.
5. Click several more scan points in the video image.
6. If the OVERRANGE message appears at several scan points, increase the measurement range for the velocity.
7. To change the measurement range, select Acquisition > Settings (toolbar: ).

   The Acquisition Settings dialog appears.
8. Display the Vibrometer page.
9. Select the next higher measurement range from the Velocity list.
10. Click OK.
11. Repeat steps 5 to 10 until the measurement range for the velocity is no longer exceeded.
12. To stop the continuous measurement, select Acquisition > Stop (toolbar: ).

You will find detailed information on making measurements in CHAPTER 8.

Scanning

1. To start a scan, select Acquisition > Start in the menu bar (toolbar: ).

   The Save As dialog appears.
2. Navigate to the saving location and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

3. Click Save.

   The scan starts. The software immediately saves every scan point measured.
4. In the video window you can follow the progress of the scan.

You will find detailed information on scanning in SECTION 8.2 and SECTION 8.3. You will find information on evaluating a scan in the Presentation software manual.

Displaying Data

The data is displayed in the analyzer.

Select data sets

1. To display a spectrum, in the menu bar select Analyzer > Domain > FFT (toolbar of the analyzer:  > FFT).
2. To automatically scale the y-axis, select Analyzer > Auto Scale (toolbar of the analyzer: ).
3. To display other signals apart from the measurement signal, select, for example, Analyzer > Signal > Acceleration (toolbar of the analyzer:  > Acceleration).

   While the  symbol is activated, the selected signal will be automatically scaled directly.

Zoom in

4. To zoom a range on the x-axis, point the mouse at the required lower frequency on the x-axis.

   The mouse pointer becomes a magnifying glass icon  .
5. Hold the mouse button pressed and drag to the right to the required upper frequency.
6. To undo zooming, select Analyzer > Zoom Out (toolbar of the analyzer: ).

Set a cursor

7. To activate a cursor, select Analyzer > Cursor in the menu bar (toolbar of the analyzer: ).
8. Click in the diagram to set the cursor in the analyzer.

   A vertical line will appear at the point in the diagram where you have clicked.

Read data

9. Read the coordinates at which the cursor is positioned in the readout to the right of the diagram. If the readout is not visible on the right in the analyzer, select Analyzer > Readout to display it.
10. Move the cursor to the right or left using the mouse.

    When moving, the readout is updated automatically.

You will find detailed information on displaying data in CHAPTER 9 or the Presentation software manual.

2 First Steps

The Micro System Analyzer (MSA) measurment system is used for non-contact and non-destructive measurement of three-dimensional movements in microsystems. You can use the PSV software to carry out the following tasks in the measurement system:

- Controlling the hardware
- Defining scan points
- Setting the parameters for data acquisition
- Generating excitation signals (as an option)
- Acquiring digital data
- Saving data
- Displaying data and evaluating it

First read the following sections to familiarize yourself with the basics of how to work with the PSV software.

Check whether your system meets the required conditions. See SECTION 2.1 on this.

Start up the data management system and configure it. See SECTION 2.2 on this.

Integrate the available hardware. See SECTION 2.3 on this.

Start the software. See SECTION 2.4 on this.

In the software you work in two different modes: Acquisition and Presentation. You can start an instance of the software in each of the Acquisition and Presentation modes and use them together. See SECTION 2.5 on this.

In the software, you work with different windows in Acquisition. See SECTION 2.6 on this.

The software offers you help on operation and information about the software. See SECTION 2.7 on this.

2.1 System Requirements

ⓘ For MSA-600 Data Acquisition Systems, only Windows 10 (64 Bit) is available as the operating system.

So that you can work optimally with the software, make sure that the following conditions for data acquisition systems are met:

- Operating system: Windows 10 (64 bit)
- Microsoft DirectX 11
- The respective drivers for your measurement system.
  You will find detailed information on this in the Release Notes.

Make sure that the following desktop system requirements are met so that you can work optimally with the software:

- Operating system: Windows 10 (64 bit)
- Microsoft DirectX 11
- Monitor with 32 bit color resolution, 1280 x 1024 pixel

  You will find detailed information on this in the Release Notes.



INFORMATION

First of all, carry out the installations as described in the Release Notes. See SECTION 2.2 on this as well.

2.2 Starting Data Management System and Installing the Software

Polytec DMS: License code

If you are using a Polytec Data Management System with Windows 7, you will need to enter the license code the first time you start it. You will find the label with license code on a Polytec Data Management System as follows:

- Industrial PC: Behind the front flap under the DVD burner
- Desktop PC: Outside on the right side of the housing
- Notebook: When you unfold the notebook, near the keyboard

Start Polytec DMS

To start the Polytec data management system for the first time, proceed as follows:

1. Switch on the Data Management System.

   The Windows setup is started.
2. Select the display language.

   The default language for the operating system is set.
3. Accept the license conditions displayed.
4. The Windows operating system has already been activated. Retain the default setting for activation.
5. Set up a new user.

   Windows 10 (as of 1809): On the Sign in with Microsoft page, click Domain join instead to create a local account or to use a domain account.
6. Choose a suitable password.
7. Allocate a computer name.
8. Follow any other instructions.

   You have now logged into the system and can start using the measurement system.

Network settings

You have to configure the network adapter for the following measurement systems:

- MSA-600 (all models)
- MSA-500 with Windows 10 and camera with network connection (GigE)

For all other measurement systems read on with Adapt Windows 10 settings or Adapt Windows 7 settings depending on the operating system.

Configure network adapter

The network settings are set by Polytec ex works. If these settings are lost, you must restore the names of the network adapters. To do so, proceed as follows:

1. In the task bar, click  .

   The start menu appears.

2. Select under All Programs the Polytec Configuration Tool folder and start the Polytec Configuration Tool.

   The application window for the configuration software appears.

3. Select your connected system from the Measurement System list.
4. Activate the Network check box and deactivate the Axis Controller check box.
5. Click Configure.

   The network adapter have suitable names assigned to them.

6. Close the application window.

Adapt Windows 10 settings

The settings for the way the Windows operating system behaves have been preset by Polytec. If you reinstall the operating system, you need to adjust the settings as follows:

1. Press the Windows and X key combination. Click Advanced System Settings. In the System Properties dialog, select the Hardware tab. Click Device Installation Settings and select No.

   Automatic installation of updates is suppressed.

2. Press the Windows and X key combination. Click Advanced System Settings. In the System Properties dialog, select the Advanced tab. Click Settings in the Startup and Recovery field. Deactivate the Automatically restart check box under System failure.

   Automatic restart after a Blue Screen is suppressed.

3. Press the Windows button. Select Power & sleep in the Settings. Under Related settings, click Additional power settings. In the Power Options dialog, select High performance.

   The Data Management System is not automatically switched off by power saving mode.

Adapt Windows 7 settings

The settings for the way the Windows operating system behaves have been preset by Polytec. If you reinstall the operating system, you need to adjust the settings as follows:

1. Select Start > Control Panel > System and Security > Windows Update > Change settings and select Download updates but let me choose whether to install them.

   Automatic installation of updates is suppressed.

2. Select Start > Control Panel > System and Security > System, click Advanced system settings and click Settings in the Startup and Recovery field. Deactivate the Automatically restart check box.

   Automatic restart after a Blue Screen is suppressed.

3. Select Start > Control Panel > Hardware and Sound > Power Options, click Show additional plans and click High performance.

   The Data Management System is not automatically switched off by power saving mode.

Install software

The software and all the associated system components are pre-installed by Polytec on delivery. If you have the reinstall the software, perform the following steps. If any problems occur during installation, contact your local Polytec representative.

1. Install the associated hardware as described in the operating instructions of the measurement system and in the manual of the data acquisition board.
2. Check whether your software protection (hardlock (dongle) or license file) is available and has been correctly installed.
3. Switch on the Data Management System and log in with administrator rights. See your operating system manual on this.
4. Run Polytec License Update that you have received from Polytec on a USB stick or as a link in an e-mail.

   Your license is activated and you can use Polytec Update to download Polytec software.

5. If Polytec Update is already installed on your Data Management System, proceed to step 8.
6. To download Polytec Update, copy the following link to your web browser:

   http://swdownload.polytec.com/polyupdate/Setup32.exe

7. Install Polytec Update by following the instructions.
8. Start Polytec Update.
9. Download the Polytec software that is available for your Data Management System and install it. You will find further information on this in the Polytec Update software manual.

ⓘ You will find information on updating Data Management Systems that are not connected to the Internet in the Polytec Update software manual.

Alternatively, install the required software as described in the Release Notes on your Polytec USB stick.

Storage of setting files

The files with settings for the PSV software are saved in different directories. The following table shows where the various setting files are saved.

ⓘ If the directories are not displayed, open the Folder and search options and activate the Show hidden files, folders, and drives check box on the View page.

Setting/File	Directory path
Example files	C:\ProgramData\Polytec\PSV\'Version'\Examples
Window settings - AnalysisSettings.set - BrowserSettings.set - WindowSettings.set	C:\Users\'Account name'\AppData\Roaming\Polytec\PSV\ 'Version'
Acquisition settings - <FrontEndName>_Ch<NumberOfChannels>.set (e.g. MSA-F-600_Ch4.set)	C:\Users\'Account name'\AppData\Local\Polytec\PSV\ 'Version'
Settings for camera and alignment - AcquisitionSettings.set	C:\Users\'Account name'\AppData\Local\Polytec\PSV\ 'Version'
Objective calibration[1] - LensCalibrations.set	C:\ProgramData\Polytec
License files	C:\ProgramData\Polytec\Licenses

[1] This file must never be deleted.

2.3 Integrating the Hardware

The software allows you to control a maximum of two Polytec vibrometers simultaneously. See SECTION 8.8 on this as well. Connect up the devices as described in the operating instructions of the measurement system, and switch the devices on.

Ethernet

All devices which are controlled by Ethernet are recognized automatically by the software and are set up.

USB

All devices which are controlled by USB are recognized automatically by the software and are set up.

RS-232

If you control vibrometers via RS-232, then the software recognizes the controller connected to the lower COM port as Vibrometer Controller 1.

GPIB

If you control some devices via GPIB, we recommend the following settings:

- Vibrometer Controller 1: GPIB address 5
- Vibrometer Controller 2: GPIB address 4

2.4 Starting the Software

To start the software, proceed as follows:

1. Check whether your software protection (hardlock (dongle) or license file) is available and has been correctly installed.
2. Connect up and start the measurement system as described in the operating instructions. You do not need administrator rights to start the software.
3. Start the software in Acquisition mode by double-clicking the 

icon on the desktop.

   The application window of the software appears. You may receive an error message and information on it. See SECTION 2.5 on this as well.

2.5 Modes in the Software: Acquisition and Presentation

In the software you work in two different modes: Acquisition and Presentation. You can start an instance of the software in each of the Presentation and Acquisition modes and use them together. You can switch between the windows of the software instances. Some functions are available for both instances.

ⓘ In the desktop version of the PSV software, the acquisition is not available.

Acquisition

When starting acquisition mode, the software searches the interfaces of the data management system for external devices and initializes them.

ⓘ You may receive an error message and information on it. You can switch off error messages as described in SECTION 3.8.

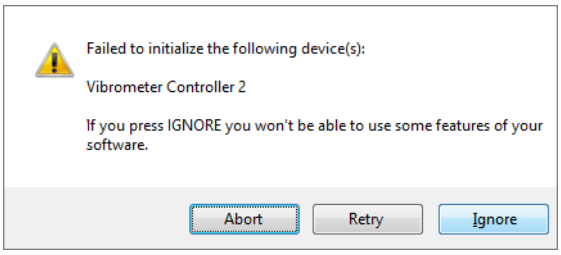


Figure 2.1: Error message when changing into acquisition mode

To check the connection between the devices and to correct it if necessary, select Setup > Preferences in the menu bar and display the Devices page. The connections to external devices are displayed.

ⓘ Some Polytec instruments are automatically initialized in acquisition mode and can not be changed. When ending Acquisition, all the hardware used is enabled. For this reason, you can use PSV software in parallel with other Polytec software products.

To change from Presentation to Acquisition, you have the following options:

- In the menu bar, select View > Acquisition.
- In the toolbar, click  .
- In the Windows task bar, click on the software instance for acquisition.
- Press the Alt + Tab key combination and select the software instance for Acquisition.

Presentation

To change from Acquisition to Presentation, you have the following options:

- In the menu bar, select View > Presentation.
- In the toolbar, click  .
- In the Windows task bar, click on the software instance for presentation.
- Press the Alt + Tab key combination and select the software instance for Presentation.

2.6 Windows in the Software

Read the following sections to familiarize yourself with the various windows in the software.

Application window

In the application window of the software, you work with windows to display and evaluate data. For this purpose, a video window and any number of analyzers are available to you. See SECTION 2.6.1 on this.

Video window

In the video window you can see the live video image. You can set different views and control the XY positioning stage. See SECTION 2.6.2 on this.

Analyzer

In the analyzer, measurement data are shown in xy diagrams. You can follow current measurements in the analyzer or use them to display measurement data from a file. You can open as many analyzers as you like. See SECTION 2.6.3 on this.

Project browser

In the project browser, you can display folders with the subdirectories and files they contain in a tree structure. See SECTION 2.6.4 on this.

Signal processor

You can use the signal processor to recalculate measurement data for further analysis and then compare the results with the original data. See SECTION 2.6.5 on this.

Message window

Critical or warning messages which appear during data acquisition are shown to you in a separate message window. See SECTION 2.6.6 on this.

Arrange windows

You can arrange all windows with the commands in the Window menu. See SECTION 2.6.7 on this.

Link windows You can link several analyzers with each other to ensure you are showing data with the same settings. See SECTION 2.6.8 on this.

Zoom in windows You can zoom in on image sections in the video window. See SECTION 2.6.9 on this.

Print windows You can print out analyzers and signal processors. See SECTION 2.6.10 on this.

General icons Several icons from the toolbars of the windows are similar to those in other software applications.

Table 2.1: General icons in the software

Icon	Key combination	Menu navigation	Function
	-	File > New Signal Processor	Create new signal processor document
	Ctrl + O	File > Open	Open existing document
	Ctrl + I	File > Info	Display file information
	Ctrl + P	File > Print	Print active document
	Ctrl + X	Edit > Cut	Cut selection and copy it to the Clipboard
	Ctrl + C	Edit > Copy	Copy selection onto the clipboard
	Ctrl + V	Edit > Paste	Insert selection from the clipboard
	Ctrl + Z	Edit > Undo	Undo last action
	Ctrl + Y	Edit > Redo	Redo previously undone action

You will find a description of the other icons in the following sections and chapters.

2.6.1 Application Window

After starting the software, the application window appears. At the top edge you will see the title bar, menu bar and toolbar, at the bottom the status bar. In the View menu, you can show/hide the toolbar and the status bar.

If you start PSV Acquisition for the first time, the application window will show the video window, an opened analyzer, the project browser, and the message window.

ⓘ Some Polytec instruments are automatically initialized in acquisition mode and can not be changed. But when ending acquisition, all the hardware used is enabled. This makes it easier for you to use the PSV software in parallel to PMA or TMS software for example with MSA systems.

The windows are described in the following sections.

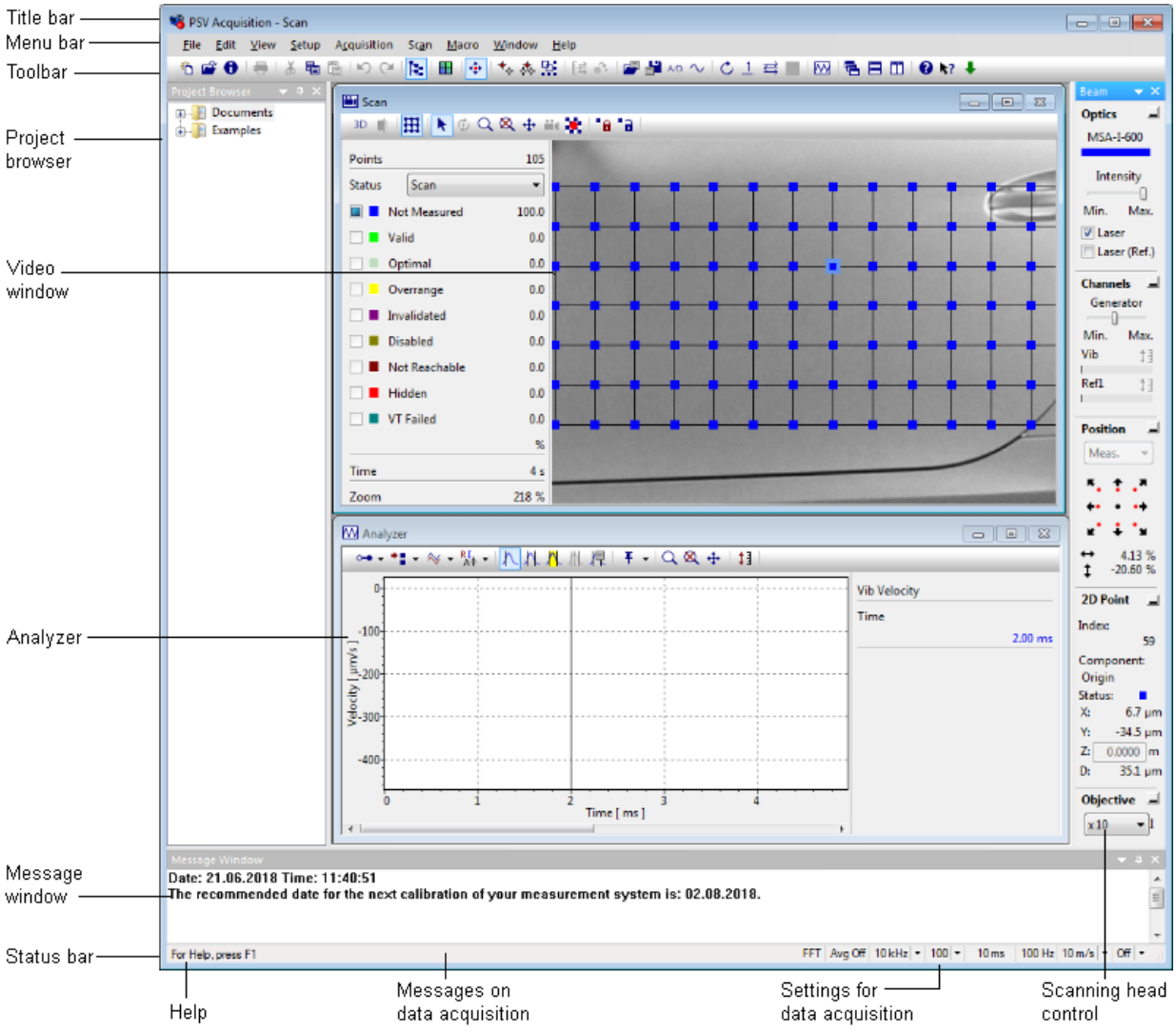


Figure 2.2: Example of the application window in acquisition mode (MSA-600)

In the application window you can display and evaluate data. For this purpose, a video window and any number of analyzers are available to you.

When you restart the software, then the layout and signal selection of the windows opened most recently will be restored. Read SECTION 2.6.7 on how to arrange windows.

Title bar In the title bar, you will see the name of the software. If you have already saved the measurement or have opened a measurement, you will see the file name of the measurement in the title bar as well.

Menu bar In the menu bar, every menu contains a group of commands for working with the software. Some menus are only visible if an analyzer or a presentation window is open.

Toolbar By clicking the icons, you can execute some commands quickly which you would otherwise have to select in the menus. The appropriate icons are available to you for every window that you have opened.

Software-specific icons Apart from the general icons, you will find more software-specific icons in the application window.

Table 2.2: Software-specific icons in the application window

Icon	Key combination	Menu navigation	Function
	-	View > Project Browser	Show/hide project browser window
	Ctrl + D	View > Presentation	Start Presentation (software instance)
	Ctrl + B	View > Scanning Head	Show/hide scanning head control
	-	File > Load Settings	Load settings from a file
	Ctrl + S	File > Save Current Settings As	Save the current settings
	F5	Acquisition > Settings	Define data acquisition settings
	F6	Acquisition > Generator	Switch on/off the generator
	F8	Acquisition > Continuous	Start a continuous measurement
	Shift + F8	Acquisition > Single Shot	Start a single shot measurement
	F9	Acquisition > Scan	Start scan
	F10	Acquisition > Stop	Stop measurement
	Ctrl + N	Window > New Analyzer	Open a new analyzer window
	-	Window > Cascade	Arrange windows so they overlap
	-	Window > Tile Horizontally	Arrange windows horizontally

Icon	Key combination	Menu navigation	Function
	-	Window > Tile Vertically	Arrange windows vertically
	F1	Help > Contents	Display the table of contents for the online help
	-	Help > Context Help	Display help for icons, menus and windows
	-	Help > Check for Updates	Search for Polytec Software updates
	-	Help > Check for Updates	Install Polytec Software updates

Icons for setting up the software

In addition to the software-specific icons you will find further icons to set up the software in the application window.

Table 2.3: Icons to set up the software in the application window

Icon	Menu navigation	Function
	Setup > 2D Alignment	Carry out 2D alignment
	Setup > 3D Alignment	Carry out 3D alignment
	Setup > Define Scan Points	Activate/deactivate definition of scan points

Icons from the Scan menu

In addition you will find further icons from the Scan menu in the application window.

Table 2.4: Icons from the Scan menu in the application window

Icon	Menu navigation	Function
	Scan > Geometry Scan	Perform geometry scan

Status bar

On the right, the status bar shows several settings for data acquisition. On the left you will see a short help text in certain situations. For example, point the mouse at one of the settings on the right and leave the mouse pointer there for a moment. Then on the left you will see which parameter it is for. You will find more information on how to get help in the software in SECTION 2.7.

When making a measurement, messages on data acquisition can appear in the middle of the status bar. See SECTION 8.4 on this.

Scanning head control

You adjust the optics using the scanning head control. See SECTION 4.1 on this.

Table 2.5: Icons for the scanning head control

Icon	Menu navigation (key combination)	Function
	View > Scanning Head (Ctrl + B)	Show/hide scanning head control
	Floating (double-click)	Display scanning head control in a separate window.
	Docking (double-click)	Fix scanning head control in the application window.
	Hide (Ctrl + B)	Hide scanning head control
	- (Ctrl + B)	Close scanning head control

You can adapt the display of the scanning head control, i.e. reduce its size and position it in a different place on the screen. If you drag the scanning head control with the mouse, then positioning aids appear at the edges of the software window. Drag the scanning head control to a suitable position and release the mouse button. The scanning head control is automatically placed at the corresponding window edge.

2.6.2 Video Window

In the video window, you can see the live video image.

Open

The video window is displayed if you change to acquisition mode or start PSV Acquisition directly.

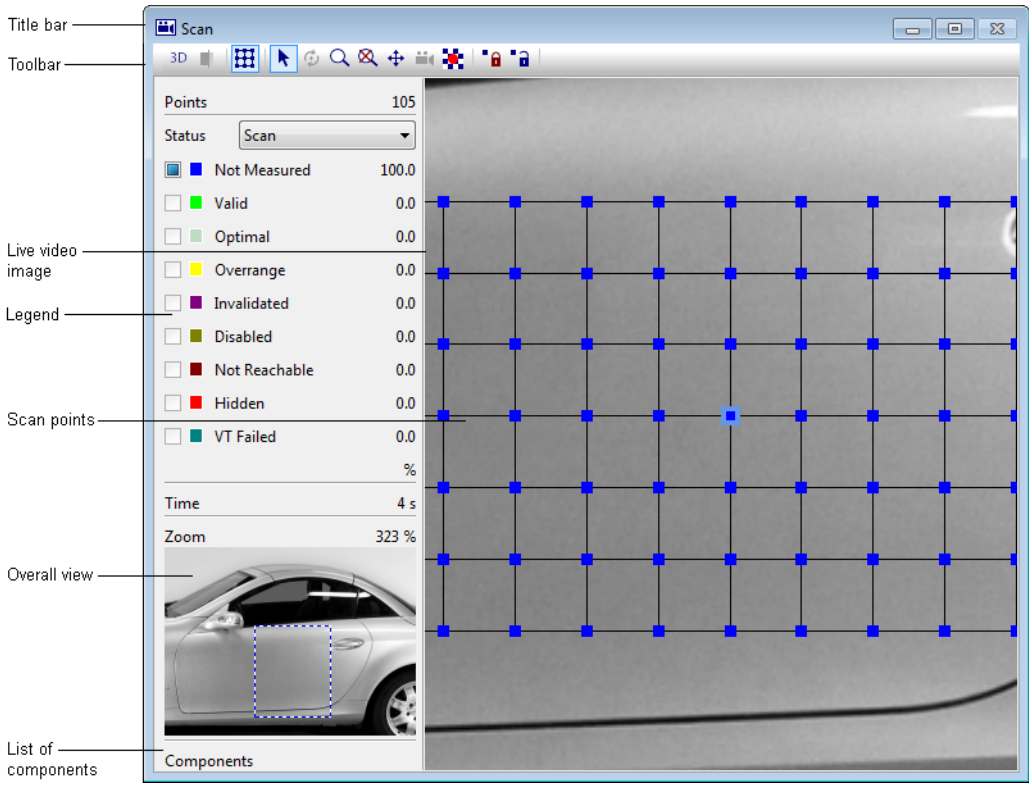


Figure 2.3: Example of the video window (MSA-500)

Toolbar

By clicking the icons, you can execute some commands quickly which you would otherwise have to select in the menus.

Icons in the video window

The appropriate icons are available to you for every view that you have displayed.

Table 2.6: Icons in the video window

Icon	Menu navigation (key combination)	Function
	Scan > 3D View	Toggle between 2D and 3D view
	Scan > Define Clipping Planes	Define clipping planes for 3D view
	Scan > Scan Points	Show/hide scan points in the live video image
	Scan > Select	Select scan points with the mouse
	Scan > Rotate	Rotate object in 3D view

Icon	Menu navigation (key combination)	Function
	Scan > Zoom In	Zoom the image section
	Scan > Zoom Out	Undo zooming the image section
	Scan > Pan	Pan the zoomed image section
	Scan > Position	Move XY-axes
	Scan > Camera Sight (Shift + Y)	Sight of the camera in 3D view
	Right mouse button > Disable (Ctrl + Space)	Disable selected scan points
	Right mouse button > Enable (Shift + Space)	Enable selected scan points

Views in the video window

You can display the live video image in 2D or 3D view and change the view of the different statuses of scan points (Scan Status, Geometry Status, etc.). To do so, proceed as follows:

1. To change between the 2D view and 3D view, click  .

   If you change from the 2D view to the 3D view, a snapshot of the live video image is shown as a fixed-image.
2. To set the line of sight directly in the video window with the mouse, click  .

   The mouse pointer become a rotation icon  .
3. Click the video image and drag the mouse in the direction required.
4. In the Status list, select the view of the scan points. Please read SECTION 8.3 on this as well.

Clipping planes

In 3D view, you can define clipping planes to control complex geometries. See SECTION 6.4.1 on this.

Scan points

To show/hide defined scan points in the video window, select Scan > Scan Points in the menu bar (toolbar: ). You see the scan points as markers in different colors. These colors show the status of the scan points.

You can directly disable or enable scan points in the live video image. The scan points are then disabled for measurements, geometry scans and for the Assign Focus Automatically function. See SECTION 6.5.3.1 on this as well.

ⓘ You will find a description of the different status displays in SECTION 8.3.

Zoom functions

In the video window you can zoom in on image sections and can pan the zoomed sections. You will find the necessary tools in the toolbar of the video window. See SECTION 2.6.9 on this as well.

Control the XY positioning stage

If you are using an XY positioning stage, you can control the axes with the software. To do so, proceed as follows:

1. In the toolbar of the video window, click  .
2. Then click the video image with the left mouse button and drag in the direction you require. If you hold the Shift key pressed while dragging, the XY positioning stage only moves in the corresponding main direction.
3. Zoom the video window to be able to traverse the XY positioning stage more precisely.

Full screen display on 2nd monitor

The MSA-600 sensor head is equipped with two cameras. This allows you to display the live video image in full screen on a second monitor.

ⓘ If you only connect one monitor, then full screen display is over the software application window. In this case, click any key or click with the mouse to deactivate the full screen display.

With MSA-500, you can display the live video image in full screen on the measurement microscope monitor.

To deactivate or respectively activate the display on the second monitor, click  .

Live video image

In the live video image, you can execute the following functions:

- Check the position of the laser beams(s).
- Define scan points.
- Follow the progress of a scan.
- Adjust image settings (contrast, brightness, etc.) as described in SECTION 4.3.

To show the laser display in high-contrast, select Scan > High-Contrast Laser Display (toolbar: ). This means an image of the current live video image is captured and the high-contrast display of the laser spot is superimposed on it.

ⓘ While the high-contrast laser display is being active, the software follows the laser beams in the video image and marks them with red crosses. During a scan, this function is not available.

Using the high-contrast laser display, you have the following options:

- You will not see any interfering influences in the video image during the measurement.
- The video image is darkened. You can adjust the brightness as described in SECTION 4.3.
- The function is available in all modes (scan point definition, 2D alignment and 3D alignment).
- The function is automatically switched off when you make changes to the hardware (e.g. move automated axes of Polytec, zoom the camera, etc.). All manual changes to the camera are not taken into consideration.

Legend

On the left you see a legend for the status of the scan points. The legend also shows the number of defined scan points and the estimated time needed for the scan. The display of the status changes depending on which view (Scan, Geometry or Laser Focus) you selected in the list.

ⓘ In the legend, you can select the scan points on the basis of their status and display them in the video image. To do so, tick the corresponding check box of the status.

Overall view

Here you see a reduced view of the live video image. If you zoom in on the live video image, a blue frame shows the position of the zoomed section.

List of components

In acquisition and presentation mode you can define geometry components and assign scan points to the individual components. See SECTION 6.5.4 on this as well.

2.6.3 Analyzers

Analyzers show measured raw data and evaluated data in xy-diagrams. In analyzers, you can observe running measurements or display measurement data from a file. Read SECTION 9.2 on how to work in analyzers.

Open

An analyzer appears if you execute one of the following actions:

- You start a single shot or a continuous measurement.
- You select Window > New Analyzer in the menu bar.
- You click  in the toolbar.

If an analyzer is open, the menu bar of the application window is extended by menus for working with analyzers. You can open as many analyzers as you like.

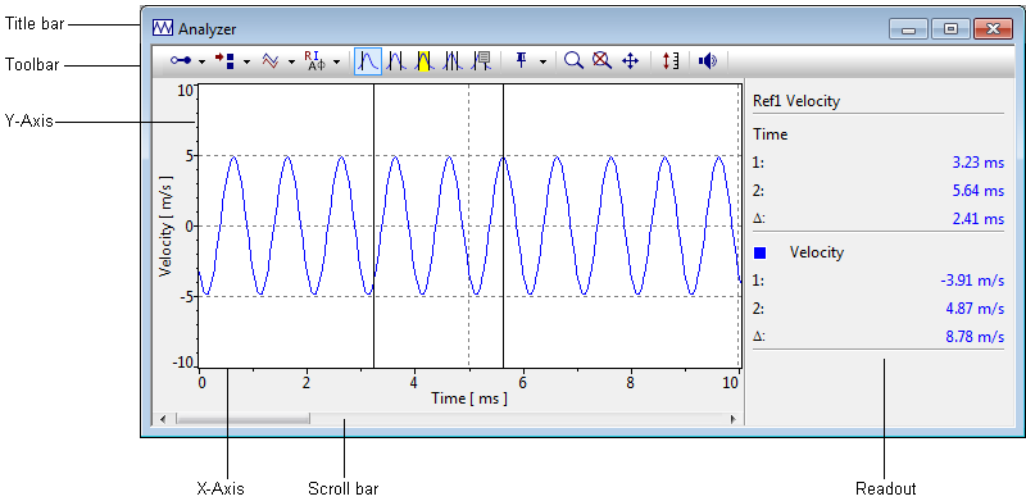


Figure 2.4: Example of an analyzer

Title bar If an analyzer has the function of a measurement window in acquisition mode, then in the title bar you will see the Analyzer name (if applicable with a number).

Toolbar You can show/hide the toolbar by selecting Analyzer > Toolbar. By clicking the icons you can quickly execute some commands from the Analyzer menu.

Icons in the analyzer In the following table, you will find the default icons of the analyzer summarized.

Table 2.7: Icons in the analyzer

Icon	Menu navigation (key combination)	Function
	Analyzer > Domain	Select a domain
	Analyzer > Channel	Select a channel or combination of channels
	Analyzer > Signal	Select a signal
	Analyzer > Display Type	Select a display type
	Analyzer > Cursor	Place a cursor in the graph
	Analyzer > Differential Cursor	Place two cursors in the graph
	Analyzer > Band Cursor	Display a band cursor in the graph
	Analyzer > Harmonic Cursor	Display a harmonic cursor in the graph
	Analyzer > Cursor Properties	Edit cursor properties
	Pin	Select analyzer settings to link the windows
	Analyzer > Zoom In	Zoom graph's x- and y-ranges simultaneously
	Analyzer > Zoom Out	Undo last zoom action; press Shift key to undo all actions
	Analyzer > Pan	Pan the zoomed section
	Analyzer > Auto Scale (Ctrl + K)	Scale the ordinate to min/max values of current data

Zoom functions In the analyzer you can zoom in on diagram sections and can pan the zoomed section. You will find the appropriate tools in the toolbar of the analyzer.

Diagram The data are shown as a graph in xy diagrams. You can set up the look of the diagrams and the graphs individually.

Readout On the right you can show/hide the readout for the diagram by selecting Analyzer > Readout in the menu bar.

Scroll bar You can use the scroll bar to scroll if you have zoomed the abscissa.

2.6.4 Project Browser

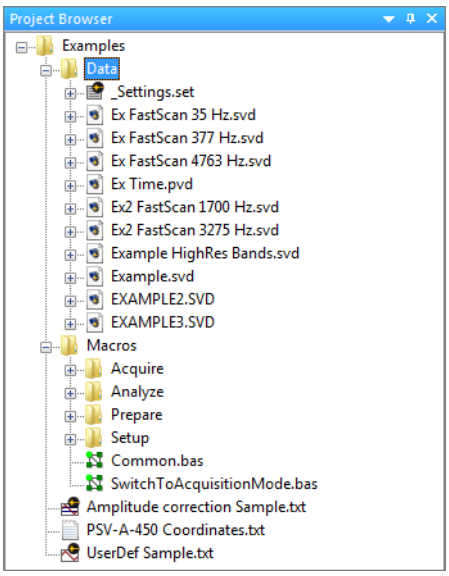


Figure 2.5: Example of the project browser

Description

In the project browser, you can display folders with the subdirectories and files they contain in a tree structure. To do so, open a program on file management, e.g. Windows Explorer and use the mouse to drag the required folder or folders into the project browser. You can now manage these folders from here.

ⓘ The folders in the project browser are not copies but direct links to the folders on your hard disk or network drives. All actions, such as deleting or renaming subdirectories and files, are directly executed in the respective folder.

Open/position

The contents of the project browser can be configured separately for every Windows user. The contents of the project browser and selected entries are saved when the program ends and are displayed again the next time the software is started. You can arrange the docking window of the project browser in different ways in the application window. The following overview shows the available functions.

Table 2.8: Icons for the project browser

Icon	Menu navigation or name	Function
	View > Project Browser	Show/hide project browser window
	Window Position	Select the window position for the docking window from the list: - Floating - Docking - Auto Hide - Hide
	Auto Hide	Hide the docking window automatically, that means the docking window is shown minimized at the edge. As soon as you point the mouse pointer there, the docking window is displayed completely.
	Docking	Fix the docking window at the edge
	Close	Close the docking window

You can adapt the display of the project browser, i.e. reduce its size and position it in a different place on the screen. If you drag the project browser with the mouse, then positioning aids appear at the edges and in the middle of the application window. Drag the project browser to a suitable position and release the mouse button. The project browser is automatically placed at the corresponding window edge.

Add folder

To add another project folder to the project browser, you have the following options:

- Click the project browser with the right mouse button, however not a file or a folder, and select Add Project Folder in the context menu.
- In the menu bar, select File > Add Project Folder.

The Select Folder dialog appears. Navigate to the required folder and click Select.

ⓘ You cannot transfer individual files to the project browser.

Remove folder

You can remove folders of the highest level. To do so, click the folder with the right mouse button and select Remove in the context menu.

ⓘ You can also delete individual files.

Copy and move

You can copy files or folders in the project browser into other directories. To do so, mark them and holding the left mouse button pressed, drag the marked objects onto the required directory.

To move files or folders in the project browser, proceed as described above. In addition, hold the Shift key pressed while dragging with the mouse.

Commands in the project browser

Click a file or a folder with the right mouse button to edit the contents in the project browser. The commands shown in the context menu differ depending on the level in the tree structure, the type and number of objects selected and the mode you are currently working in (acquisition or evaluation). The individual commands are explained in the following:

Explorer: Click here if you want to have the selected folder displayed in Windows Explorer as well. The command is only available for folders.

New Folder: Click here if you want to create a new subfolder in the selected folder. The subfolder appears in the project browser and can immediately be renamed respectively.

Refresh: Click here to update the display of folders and files in the project browser.

Remove: Click here if you want to remove the selected objects from the project browser, but do not want to delete them. The command is only available for folders on the top level.

Rename: Click here if you want to give the selected object a different name.

Delete: Click here if you want to irrevocably delete the selected objects. Before permanently deleting it, you have to confirm asecurity popup. The command is only available for files.

Load: Click here and the setting file in Settings Files (*.set) format which you have selected in the project browser will be loaded.

Open: In principle, you can open files from all programs installed on your PC in the project browser. Click Open and the selected file will be opened using the corresponding program. If you open a file in Signal Processor Data (*.spd) format, then the signal processor opens (SigPro option).

ⓘ The PSV software is run in 64 bit Windows systems as a 64 bit process. All applications started from the PSV software which are available as 64 bit and 32 bit versions are then also automatically started as a 64 bit process. This can cause some applications to behave differently than if they are started from either the PSV software or from Windows Explorer.

New Window: Click here if you want to display the file you have already opened again in a second window.

Info: Click here if you want to display the file information of the file selected. See SECTION 10.1.1 on this as well.

Close: Click here if you want to close the file you have already opened.

Save As: Click here if you want to save the selected file under a different name or in another directory.

Paste: This command is only available if you are using the optional signal processor. See SECTION 9.4 on this as well. If you have copied data in the signal processor, you can insert this data into the selected file by using the Paste command.

Edit: This command is available for macros in the WinWrapBasic (*.bas, *.obm, *.cls) format and for project files in the (*.wbp) format. Click Edit and a macro window appears with the macro you have selected in the project browser.

ⓘ Macros which are edited or run via the project browser appear as a list of files opened most recently in the Macro menu (max. 4). The lists are managed separately for the acquisition and presentation mode.

Run: Click here and then the macro you selected in the project browser is run.

Additional commands

Import: In the project browser, you can import geometries.

The following conditions must be met:

- You are in Acquisition mode.
- You have a valid 3D alignment.
- You are in Scan Point Definition.
- You have selected the Define Point Objects object type.

You will find an overview of the formats that you can import in SECTION 6.3.7.

2.6.5 Signal Processor

You can use the signal processor to recalculate measurement data for further analysis and then compare the results with the original data. You can read SECTION 9.4 on how to work with the signal processor.

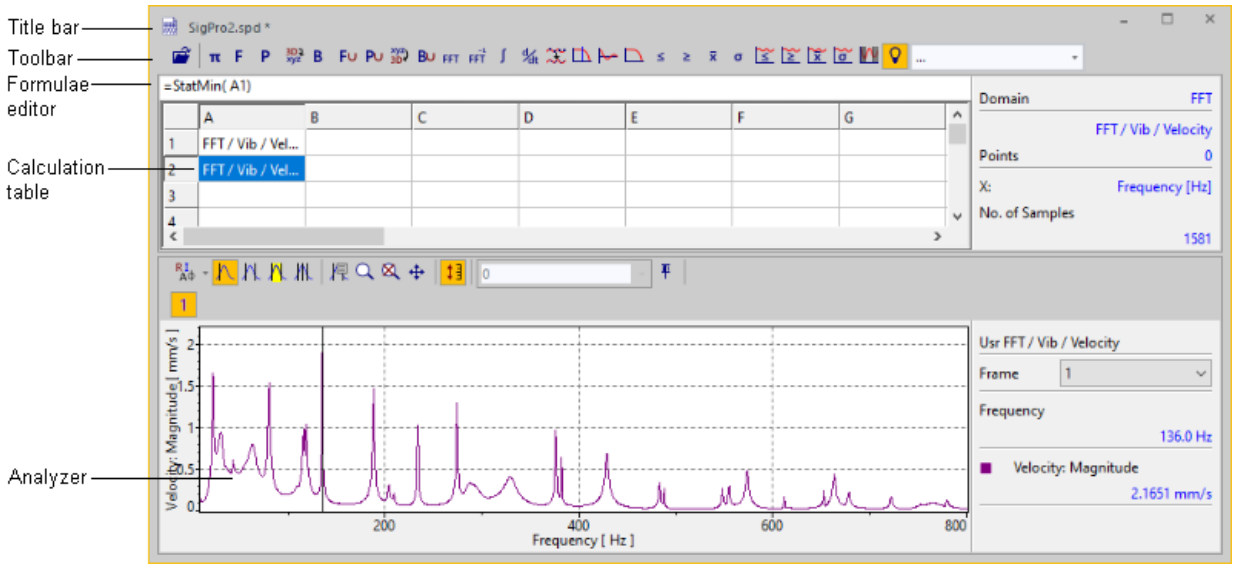


Figure 2.6: Example of recalculating measurement data in the signal processor

Toolbar

In the toolbar you will find the most important arithmetic operations you can execute with the signal processor. The icons are active as soon as the cursor is in the formulae editor.

You have the following options to enter the formulae in the formulae editor:

- Enter the formula using the keyboard.
- Click one of the icons.
- Select the formula in the list on the right of the toolbar.
- Click  and the desired icon to call up the function wizard. See SECTION 9.4.1 on this as well.

Formulae editor

The contents of the selected table cell is shown in the formulae editor. Here you enter text or formulae that should be in the selected cell.

Calculation table

You enter the measurement data and the formulae in the cells of the calculation table. By linking the individual cells via the formulae, the software can make various arithmetic operations using the measurement data and then display the results at any point in the computing chain. Apart from that, you can also enter comments into the cells.

ⓘ In the table cells, you can display either the formula or the result. To toggle, select View > Formulae in the menu bar (key combination: Ctrl + #).

For linking individual cells you establish references. A reference designates a cell or a cell range in the table and informs the signal processor where the values or data used in the formula are located. Using references, you can use data or values of a cell in different formulae.

Depending on the operation that you want to execute in the signal processor, you can either use relative or absolute cell links. Relative cell links are references to cells that are given in relative to the position of the formula, absolute references are cell links that always refer to cells at a certain position. If a dollar symbol is prefixed to the letter and/or the number, e.g. A1, then the column and/or cell link is absolute. Relative references are automatically adapted in case of copying, absolute references are not adapted.

Analyzer

The signal processing results are shown as a graph in the analyzer.

2.6.6 Message Window

Critical or warning messages which appear during data acquisition as well as general messages of the measurement system or of the software are shown to you in a separate message window. The message window is displayed automatically as soon as you start the software or the first message appears.

ⓘ If critical errors occur during a scan, then the software tries up to three times to repeat the measurement at a scan point before the scan is stopped. These messages are displayed in the message window in any case. This is not the case with measurement systems using Ethernet data acquisition. The scan will not be stopped.

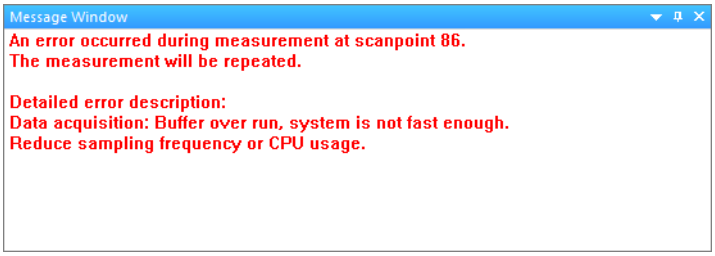


Figure 2.7: Example of the message window

Open/position The display can be configured separately for every Windows user.

Table 2.9: Icons of the message window

Icon	Menu navigation	Function
-	View > Message Window	Show/hide the message window
	-	Show message window minimized at the edge of the application window. As soon as you point the mouse pointer at this, then it is displayed completely (Auto Hide).
	-	Fix message window (Docking)
	-	Close message window

You can adapt the display in the message window, i.e. reduce its size and position it in a different place on the screen. If you drag the message window with the mouse, then positioning aids appear at the edges of the application window. Drag the message window to a suitable position and release the mouse button. The message window is automatically placed at the corresponding window edge.

Save The contents of the message window are automatically saved in the measurement file and can be displayed in the File Information dialog.

2.6.7 Arranging Windows

You can arrange all open windows in the application window using the commands in the Window menu.

New window

To open more analyzers, select Window > New Analyzer in the menu bar (toolbar: ).

Cascade

To arrange all windows/analyzers so that only the title bar and the ordinate name are visible respectively, select Window > Cascade in the menu bar (toolbar: ).

Tile horizontally and vertically

To display all windows/analyzers so that they are completely and maximum size, select Window > Tile Horizontally or Window > Tile Vertically in the menu bar (toolbar:  or ).

Close all

To close all windows/analyzers, select Window > Close All in the menu bar.

Window list

At the bottom of the Window menu, you will find a list of opened windows/ analyzers. To activate a window/analyzer – i.e. move it to the foreground – select it in the list.

Window settings

Window settings which you have made are automatically restored by the software when you newly open the file. For this purpose, the window settings are saved in a database under the complete path name of the file. This database is located on the local hard disk of your PC. Each user who is logged in Windows has their own database.

Change window settings

When starting the software, the window layout is restored by default to the way it was when the software was last closed. You can only load window settings from a setting file (*.set). See SECTION 10.1.3 on this.

2.6.8 Linking Windows

You can link several analyzers with each other to ensure you are showing data with the same settings.

While doing so, you can select the following properties:

- General
- Lines
- Abscissa
- Ordinate
- Cursor

Pin

To link several windows together, proceed as follows:

1. Open several windows.
2. Click  in the window in which you want to link properties together.

   The icon is permanently highlighted.
3. Click the arrow of the  icon and select the properties in the pull-down menu which are to be linked.

2.6.9 Zooming in Windows

Image section

You can zoom in on image sections in the video window. You will find the necessary tools in the toolbar of the video window.

Zooming in on the object helps you to define the points and carry out the alignment more accurately, thus enabling you to present the data more precisely. This is particularly advantageous if your object has complicated geometry or your scan points are very close together.

ⓘ If you zoom in the video window, on the left in the overall view you will see the presentation of the object with a blue frame marking the zoomed section. This means you can always ascertain which part of the object under investigation you are currently viewing.

Zoom in

If you would like to take a closer look at a certain image section, you can zoom in on it. To do so, proceed as follows:

1. Click  .

   The mouse pointer becomes a magnifying glass icon  .
2. Holding the left mouse button pressed, draw a rectangle around the area of the image which you want to zoom in on. You can repeat this step several times.

ⓘ You can also zoom by scrolling with the mouse wheel. If you point with the mouse pointer at the area you want to zoom, and then scroll with the mouse wheel, exactly this area is zoomed.

Pan image section

You can pan the zoomed image section. To do so, you have the following options:

- Click in the zoomed view using the middle mouse button (mouse wheel). Hold the mouse button pressed and pan the image section. If you have given the middle button a special function (e.g. double-click), then you will have to proceed as described below.
- Click  . The mouse pointer becomes an arrow cross icon  . Hold the left mouse button pressed and pan the image section.

Zoom out

You can zoom out again. To do so, scroll up with the mouse wheel. To undo all previous zoom actions at once, click  .

End zooming

To leave the zoom function again, click  . The last image section selected remains.

2.6.10 Printing Windows

You can print out analyzers and signal processors. You can not print out video windows.

ⓘ The title of the window as well as the date and time are automatically printed as a header.

Set up printer

To set the paper size, feed and orientation, select File > Print Setup in the menu bar. The Print Setup dialog appears. Here you can select the printer and set up the page.

Preview

To see a print preview, select File > Print Preview in the menu bar.

Start printing

To print out the active window, proceed as follows:

1. In the menu bar, select File > Print (toolbar: ).

   If you can see two views in the active window (e.g. in the signal processor), then the Select View dialog will appear. Otherwise the Print dialog appears. Then proceed with step 4.
2. Select whether you want to print out the top, the bottom, or both graphics.
3. Click OK.

   The Print dialog appears.
4. Select the printer and set additional print properties.
5. To start printing, click OK in the Print dialog.

   The corresponding window is printed.

2.7 Help Texts and Software Information

You can get help in the software as online help, as context-sensitive help and through brief explanations of commands. You can also display the theory manual and the Basic Engine manual as well as the version of the software and the software options which are installed.

Online help

To open the online help, you have the following options:

- Press the F1 key on the keyboard.
- In the toolbar, click  .
- In the menu bar, select Help > Contents.
- In the menu bar, select Help > Search.
- In the menu bar, select Help > Index.

Context-sensitive help

For dialogs and windows you can get context-sensitive online help. To do so, proceed as follows:

1. Activate the window or dialog you need help with. To do so, click him.
2. Press the F1 key or click in the dialog Help.

   The corresponding page of the online help appears.

ⓘ You can also get context-sensitive help if you click  and then the corresponding window or element.

Command description

You can also get explanations for icons and commands without opening the online help. To do so, point the mouse at the command or the icon and leave the mouse pointer there for a moment. The explanation appears on the left in the status bar. For icons, next to the mouse pointer you will see a brief description.

Theory and Basic Engine

For background information about various features in the software, you can display the theory manual of the PSV software. To do so, select Help > Theory Manual in the menu bar.

For interesting facts from Basic Engine programming, you can display the Basic Engine manual of the PSV software. To do so, select Help > Basic Engine Manual in the menu bar.

Software version and options

To display the version of the software, select Help > About PSV. The About Polytec Scanning Vibrometer dialog appears.



Figure 2.8: About Polytec Scanning Vibrometer dialog

To display a system summary, click System Info. The System Information dialog appears. Here you will find all information on the operating system (version, serial number, etc.) as well as the resources (memory, available hard disk space, etc.) of the data management system.

To display the software options installed, click Options.

3 Setting Up the Software

Before making a measurement, set up the software for data acquisition first.

ⓘ In almost all views and modes, you can use the right mouse button to call up via the context menu several commands and functions that are often needed at this point.

Options

To set up the software, proceed as follows:

1. In the menu bar, select Setup > Preferences.

   The Preferences dialog appears.
2. The dialog has several pages, each with a group of parameters. To display a certain page, click the name of the group.
3. Set the parameters on all pages.
4. When you have set all parameters, click OK.

You will find information on the pages in the dialog and the associated parameters in the following sections:

Devices

On the Devices page you set which measurement system and which external devices you want to control with the software. See SECTION 3.1 on this.

Channels

On the Channels page, you set which channels you want to use to control Polytec vibrometers. See SECTION 3.2 on this.

Scanning Head

You can set up the optics of the measurement system on the Scanning Head page. See SECTION 3.3 on this.

Geometry

On the Geometry page you can have hidden points calculated by the software (GeoWorks option). Apart from that, you can use the 3D coordinates to allow calculating the focus values automatically. See SECTION 3.4 on this.

Objective calibration

Enter the name or the type of the objective used and its resolution on the Objective Calibration page. See SECTION 3.5 on this.

dB Reference

On the dB Reference page you can define dB = 0. This allows you to follow an online measurement with a certain dB reference directly. See SECTION 3.6 on this.

Display

On the Display page, you can change the scaling of the toolbar. See SECTION 3.7 on this.

Messages

On the Messages page, you select which system messages you would like to have displayed. See SECTION 3.8 on this.

MSA-500 with VDD option

If you have installed the VDD option for the MSA-500 (with VDD decoder), then run test mode. See SECTION 8.6 on this.

3.1 Setup: Devices

Settings

On the Devices page you set which measurement system and which external devices you want to control with the software.

Junction Box: This list is only active for certain hardware configurations. Junction boxes connected via the Ethernet or USB interface are recognized and displayed automatically.

Device: The connected or implemented devices are displayed.

Type: Select the type of camera you are using.

Connection: Select which interfaces are to be searched for devices. For the vibrometer controllers you have to state where the devices are connected (RS-232 or GPIB).

ⓘ If the vibrometer controller is connected via an Ethernet or USB interface, then it will be automatically identified. You do not have to make any more settings for this controller.

If you are using a generator, then you can also see what type of device it is:

- Integrated in the PC (Internal)
- Connected via Ethernet (Ethernet)
- Connected via Visa (Visa)
- Not connected (None)

Address: If you have given a fixed connection, then select the address here or carry out an automatic search for it (Auto).

Warning: Mark the devices which you want a warning message issued for if the software can not find them.

ⓘ These warnings are not shown if under Setup > Preferences on the Messages page, you have selected Errors or Show None respectively. See SECTION 3.8 on this as well.

Features

Depending on the measurement system and equipment, the page may look different.

- Setting up devices for MSA-600/MSA-650 - See SECTION 3.1.1 on this.
- Setting up devices for MSA-500 - See SECTION 3.1.2 on this.

3.1.1 Setting up Devices for MSA-600/MSA-650

The only difference between the pages for the measurement systems is the display of the Junction Box.

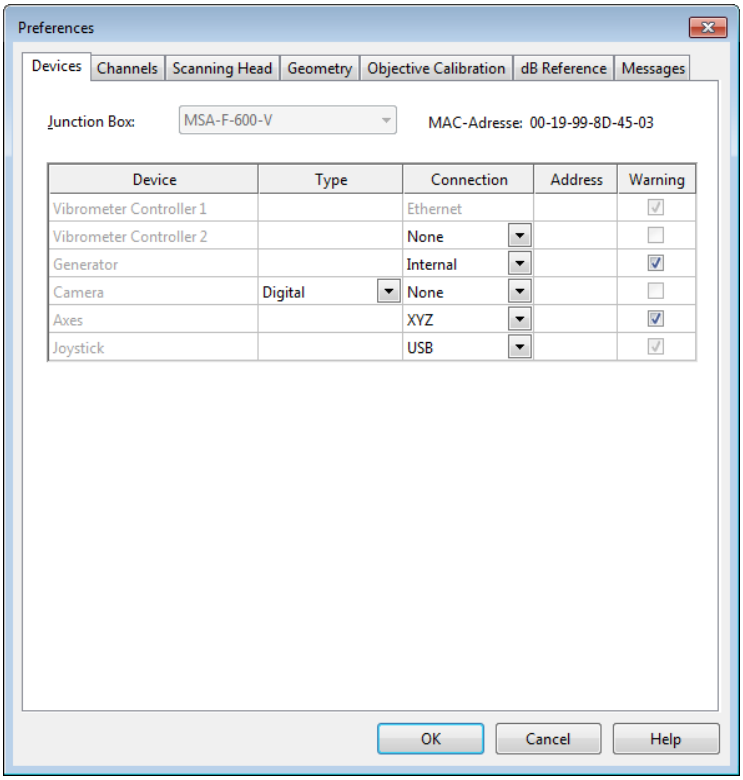


Figure 3.1: Devices page (for MSA-600)

ⓘ You will find a general description of the dialog fields in SECTION 3.1.

Features

Device: If you are using the measurement system with automated Axes from Polytec, then this additional device is also shown. You can control all axes using the software. If you want to control the automated axes from Polytec with the aid of the optional Joystick, you need to set up the connection.

Type: Select the Digital camera.

Connection: If you are using the measurement system with a Generator, you need to set up the connection to the generator. To do so, select Internal.

If you are using the measurement system with automated Axes from Polytec, you need to set up the connection to the axes. To do so, depending on the equipment, select XYZ, XY or Z.

In order to use the optional Joystick, you need to set up the connection to the joystick. To do so, depending on the type, select Internal or USB. If you do not want to use a joystick, select None.

Warning: The software searches the PC interfaces for devices and initializes them when you start PSV Acquisition or have changed settings.

3.1.2 Setting up Devices for MSA-500

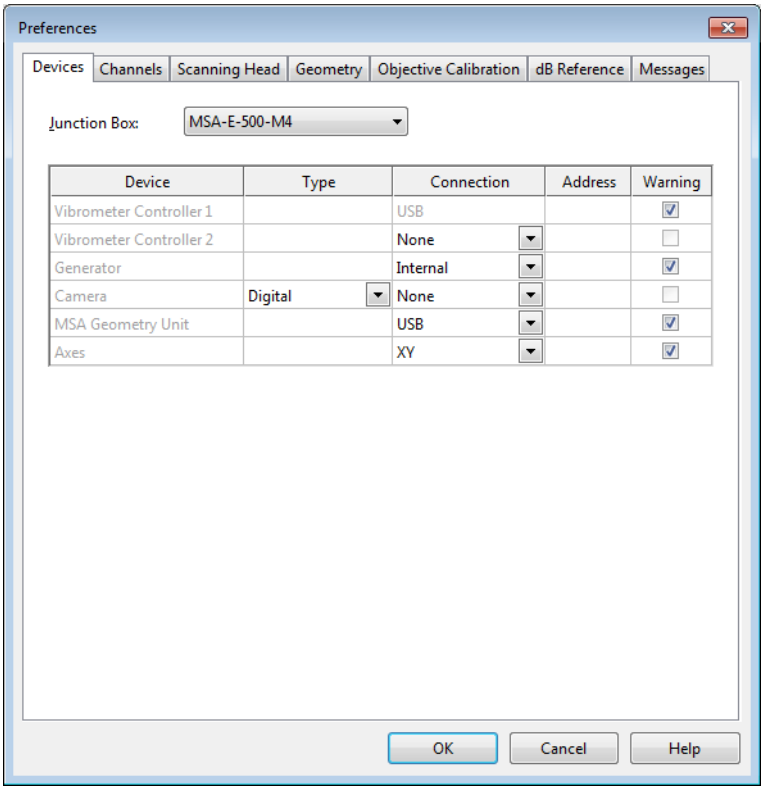


Figure 3.2: Devices page for MSA-500

ⓘ You will find a general description of the dialog fields in SECTION 3.1.

Features

Junction Box: To use the VDD option, you need to select the MSA-E-500 (VDD) junction box.

ⓘ Via USB (e.g. OFV-5000): To be able to run test mode automatically, the controller needs to be remotely controlled via USB and needs to be connected on the Channels page with VDD Vib Controller 1. See SECTION 3.2 on this as well.

Device: If you are using the MSA-500 with the MSAGeo option, then this additional device is also shown (MSA Geometry Unit). With the MSAGeo option in the MSA-500 you can focus the laser beam, carry out a 3D alignment and define 3D geometries.

With the GeoWorks option, you can carry out a 3D alignment and import 3D geometries.

If you are using the measurement system with automated Axes from Polytec, then this additional device is also shown. You can control all axes using the software. If you want to control the automated axes from Polytec with the aid of the optional Joystick, you need to set up the connection.

Type: Select the Digital camera (for VDD option as well).

Connection: If you are using the measurement system with a Generator, you need to set up the connection to the generator. To do so, select Internal.

If you are using the MSA-500 with the MSA Geometry Unit (MSAGeo option), then you need to set up the connection. To do so, select USB.

If you are using the measurement system with automated Axes from Polytec, you need to set up the connection to the axes. To do so, depending on the equipment, select XYZ, XY or Z.

In order to use the optional Joystick, you need to set up the connection to the joystick. To do so, depending on the type, select Internal or USB. If you do not want to use a joystick, select None.

3.2 Setup: Channels

Settings

On the Channels page, you set which channels you want to use to control Polytec vibrometers.

Channel: The number of channels displayed depends on which acquisition mode you are using and which junction box has been selected on the Devices page. See SECTION 3.1 on this as well.

ⓘ However, the number of channels is only updated if you close the Preferences dialog by clicking OK and then reopen it again.

Name: A meaningful name is displayed for every channel.

Abbr.: An abbreviation is displayed for every channel.

Vibrometer: Select which vibrometer provides the signal for this channel. You can use the software to control the connected vibrometers as described in SECTION 8.8. For certain hardware configurations and measurement modes, channels are already permanently connected.

ⓘ In this column, a channel is only active if a Polytec vibrometer is connected, the vibrometer is switched on and the correct connection is set on the Devices page.

Features

Depending on the measurement system and equipment, the page may look different.

- Setting up channels for MSA-600/MSA-650 - Refer to SECTION 3.2.1.
- Setting up channels for MSA-500 - See SECTION 3.2.2 on this.

3.2.1 Setting up Channels for MSA-600/MSA-650

If you want to make a measurement with the MSA-600 or the MSA-650, then the front-end needs to be displayed as the Junction Box on the Devices page. See SECTION 3.1 on this as well. You can only adjust settings on the Channels page if you have connected up an additional reference vibrometer.

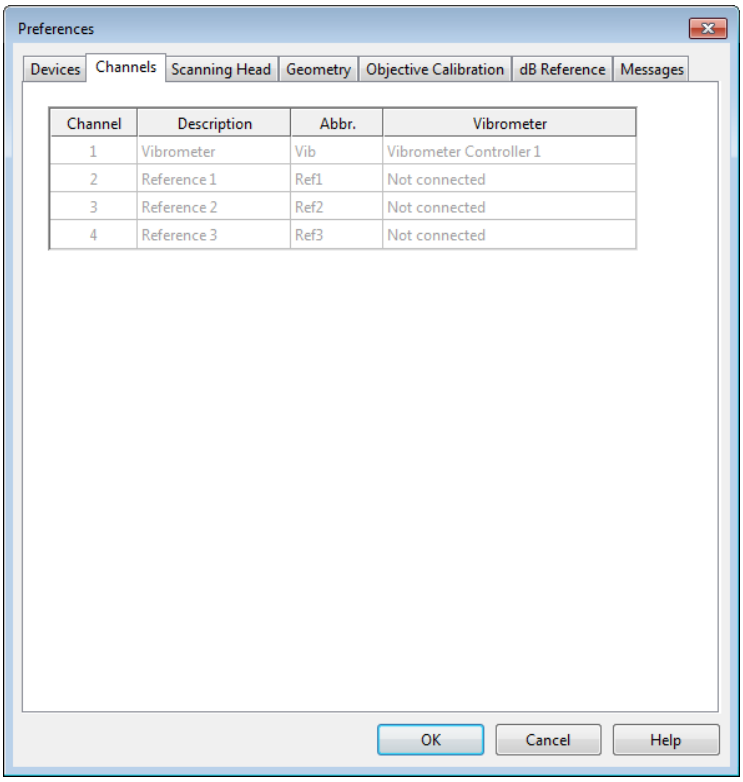


Figure 3.3: Channels page (for MSA-600)

ⓘ You will find a general description of the dialog fields in SECTION 3.2.

Allocation

In the following table you will find the allocation of the channels to the different connections of the front-end.

Software		Front-End	
Channel	**Abbr.**	**MSA-F-6x0-M (M system)**	**MSA-F-6x0-V (V system)**
1	Vib	-	-
2	Ref1	REF 1	REF 1
3	Ref2	REF 2	REF 2
4	Ref3	REF 3	REF 3

3.2.2 Setting up Channels for MSA-500

To make a measurement with the MSA-500, set up the corresponding junction box on the Devices page. If you want to make a measurement using the VDD option, then you have to connect the channel to VDD Vib Controller 1 and set the appropriate junction box on the Devices page. See SECTION 3.1 on this as well.

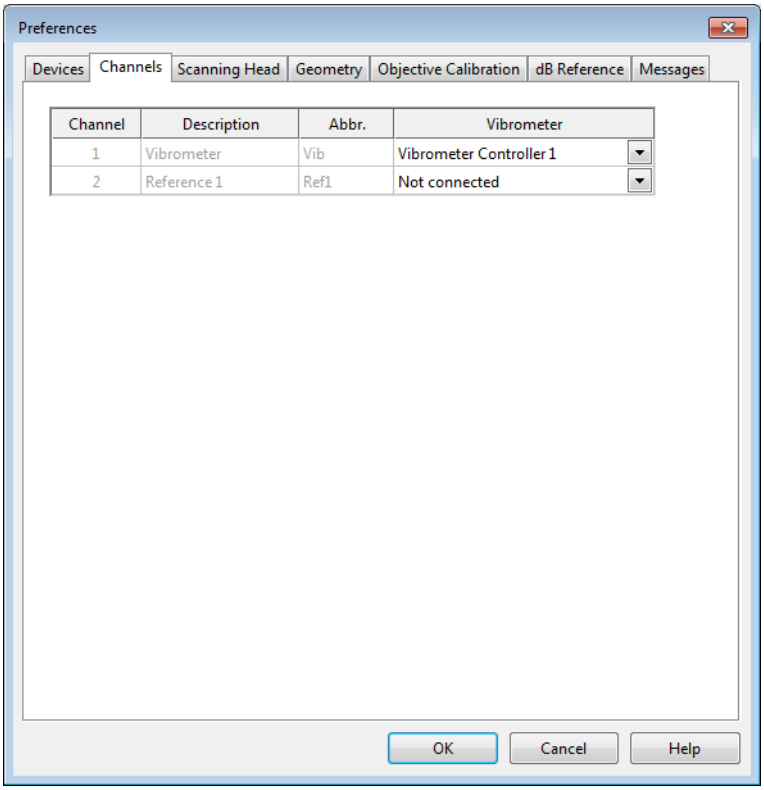


Figure 3.4: Channels page for the MSA-500

ⓘ You will find a general description of the dialog fields in SECTION 3.2.

Allocation

In the following tables you will find the allocation of digital channels to the different connections of the individual junction boxes.

Software		Junction Box	
Channel	**Abbr.**	**MSA-E-500**	**MSA-E-500 (VDD)**
1	Vib	VELO IN	Vibrometer signal[1]
2	Ref1	REF IN	REF IN

[1] The vibrometer signal is not fed in through the junction box; it is only available directly in the software (only with VDD option).

3.3 Setup: Scanning Head

You can set up the optics of the measurement system on the Scanning Head page.

Features

Depending on the measurement system and equipment, the page may look different.

- Setting up scanning head for MSA-600/MSA-650 - Refer to SECTION 3.3.1.
- Setting up scanning head for MSA-500 - See SECTION 3.3.2 on this.

3.3.1 Setting up Scanning Head for MSA-600/MSA-650

The only difference between the pages for the measurement systems is the display of the Scanning Head.

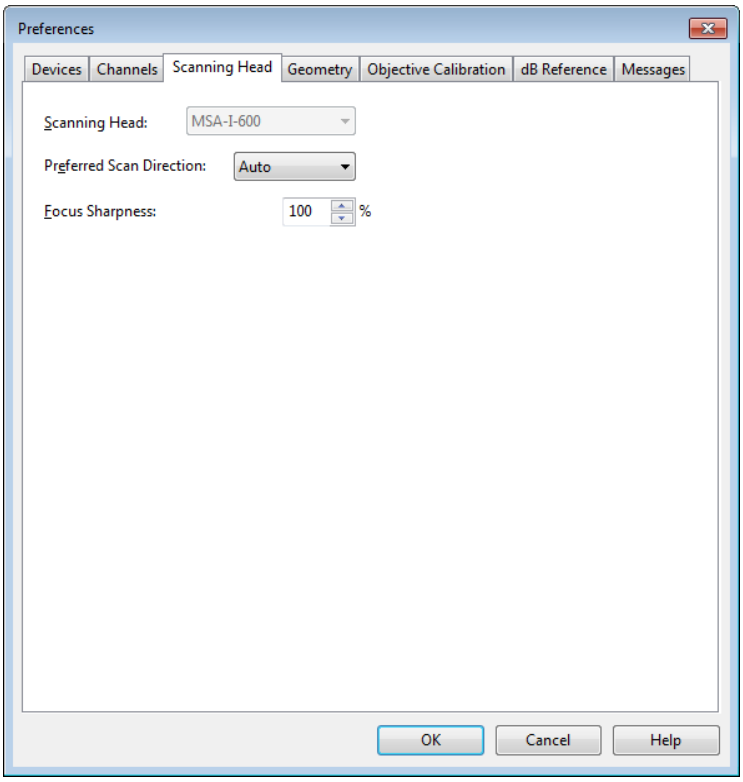


Figure 3.5: Scanning Head page (for MSA-600)

Settings

Scanning Head: The scanning head of the measurement system is displayed.

Preferred Scan Direction: Select how the laser beam is to be moved when scanning.

Auto:	The software independently optimizes the order.
Horizontal:	The scan points are to be hit row by row.
Vertical:	The scan points are to be hit column by column. In this case the scanning time is shorter because the scanner mirror for the Vertical (y-direction) is smaller and thus has a shorter settling time.
Index:	The scan points are to be hit in ascending order according to the index. You can individually set the order by changing the indices of the scan points. See SECTION 6.4.2 on this as well.

Focus Sharpness: Set the range for the depth of field. The smaller the value you select, the less often the focus is corrected during a measurement. The default value is 100%. For very fast measurements you should select a value of 200%.

ⓘ In order to reach the previously determined focus values during the measurement, select the Focusing during Scan > Assigned Values command in the Scan menu.

Indicator Position: Select in the list whether or where on the second monitor you want the displays (laser beam activity (on/off) and signal level) to be shown.

Show Crosshair on Display: Tick the check box if a crosshair is to be shown on the second monitor.

3.3.2 Setting up Scanning Head for MSA-500

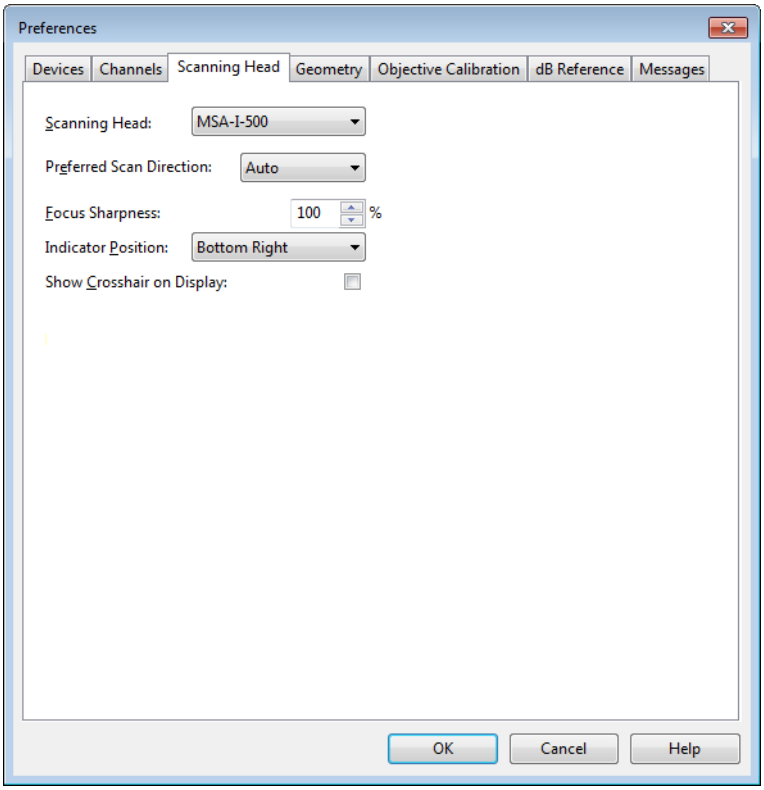


Figure 3.6: Scanning Head page for the MSA-500

Settings

Scanning Head: The scanning head of the measurement system is displayed.

Preferred Scan Direction: Select how the laser beam is to be moved when scanning.

Auto:	The software optimizes the order. Among other things, the software takes the focus valuesof scan points into account.
Horizontal:	The scan points are to be hit row by row.
Vertical:	The scan points are to be hit column by column.
Index:	The scan points are to be hit in ascending order according to the index. You can individually set the order by changing the indices of the scan points. See SECTION 6.4.2 on this as well.

Focus Sharpness: Set the range for the depth of field. The smaller the value you select, the less often the focus is corrected during a measurement. The default value is 100%. For very fast measurements you should select a value of 50%. For best focusing you should select a value of 200%.

ⓘ In order to reach the previously determined focus values during the measurement, select the Focusing during Scan > Assigned Values command in the Scan menu.

Indicator Position: Select whether or where on the MSA monitor you want the displays (laser beam activity (on/off) and signal level) to be shown.

Show Crosshair on Display: Tick the check box if a crosshair is to be shown on the MSA monitor. With the help of this crosshair, you can find the exact center of the screen and launch the laser beam precisely.

3.4 Setup: Geometry

On the Geometry page you can have hidden points calculated by the software (GeoWorks option). Apart from that, you can use the 3D coordinates to allow calculating the focus values automatically.

ⓘ The functions on this page will only become active when a valid 3D alignment is available.

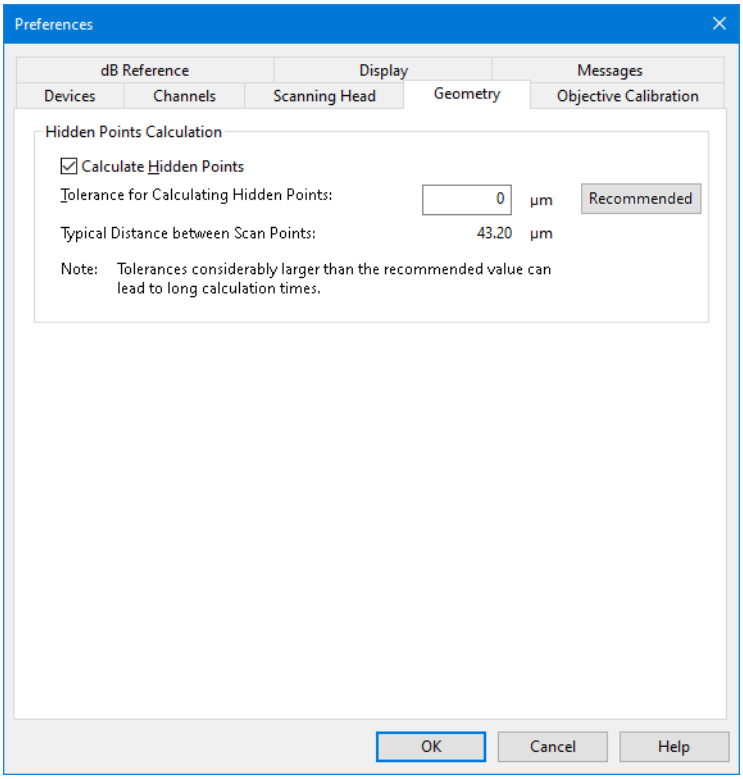


Figure 3.7: Geometry page (for MSA-650)

Hidden Points Calculation

Calculate Hidden Points: Tick this check box if you want the software to calculate which points are hidden by other parts of the geometry. For this to work, the scan points need to have 3D-coordinates.

ⓘ If you are working with a large number of scan points, this calculation can take some time. In this case, you can deactivate the calculation here.

Tolerance for Calculating Hidden Points: Enter a tolerance value for recognizing hidden points. The tolerance value is used to compensate for the deviations between the defined geometry and the actual shape of the object under investigation Scan points which can only be reached by the laser beam if they pass the geometry at a distance smaller than the tolerance value, are marked as Hidden.

Recommended: Click here and, depending on the typical distance between scan points (approx. half), the software will give you a recommendation for the tolerance value.

Typical Distance between Scan Points: Here the software shows the typical value for the distance between two scan points.

Features for MSA-500

For the MSA-500, you can have the focus values calculated automatically. Prerequisite for this is a valid 3D alignment.

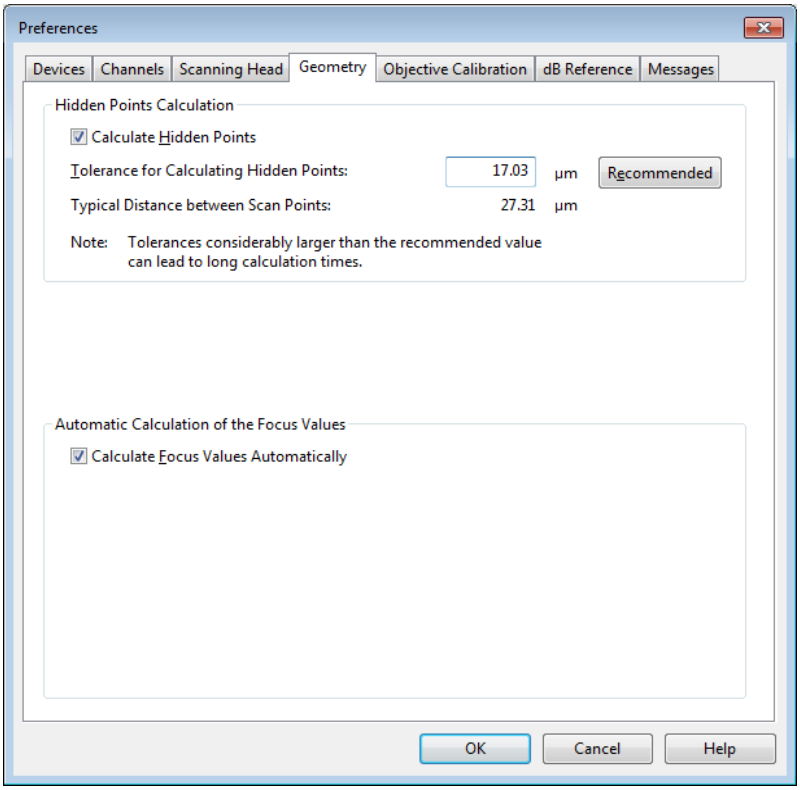


Figure 3.8: Geometry page (for MSA-500)

Calculate Focus Values Automatically: Tick the check box if you want all focus values automatically being calculated based on the 3D coordinates.

3.5 Setup: Objective Calibration

Objective settings

Enter the name or the type of the objective used and its resolution on the Objective Calibration page. The dialog already contains some objectives with the respective values by default. You can overwrite these entries and add any number of other objectives.

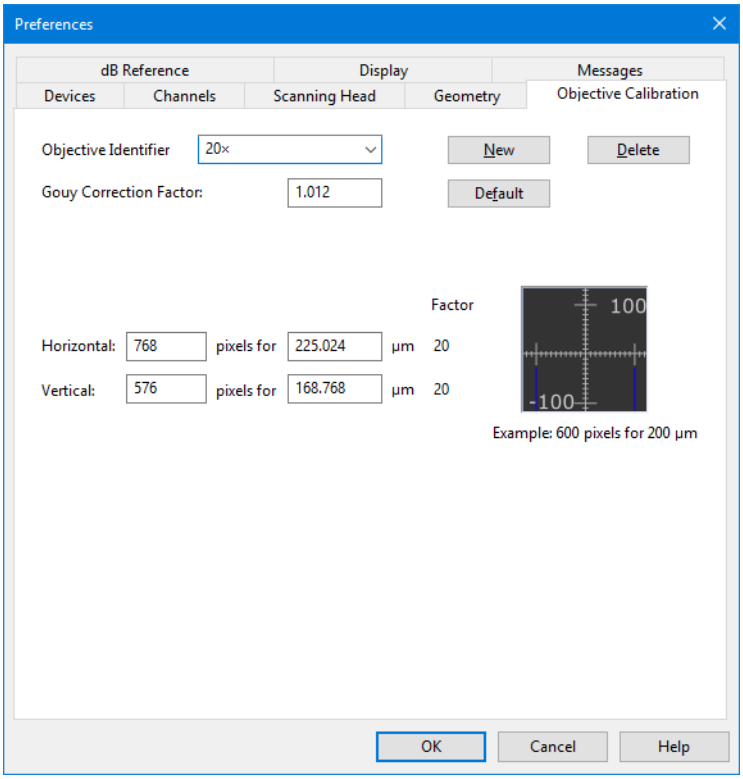


Figure 3.9: Objective Calibration page (for MSA-600)

Objective Identifier: Enter a name for the objective or the zoom factor of the objective. This identifier is also saved in the scan.

Gouy Correction Factor: The Gouy correction factor is used to adjust the objective calibration at objectives with high magnification. The value is specified by Polytec and can be adjusted if required. In this case, contact your local Polytec representative.

ⓘ Adjust the Gouy correction factor when exchanging the objective.

Default: Click here if you want to reset the Gouy Correction Factor to the default value set by Polytec.

Horizontal … pixels for … µm: Enter the number of pixels in a horizontal direction which corresponds to a certain section length your microscope image in µm.

Vertical … pixels for … µm: Enter the number of pixels in a vertical direction which corresponds to a certain section length your microscope image in µm.

Factor: The zoom factor of the objective is shown on the basis of your input. It approximately matches the value printed on the objective.

Features for MSA-650

For the MSA-650, an individual value is required due to the necessary optical path length alignment for each objective.

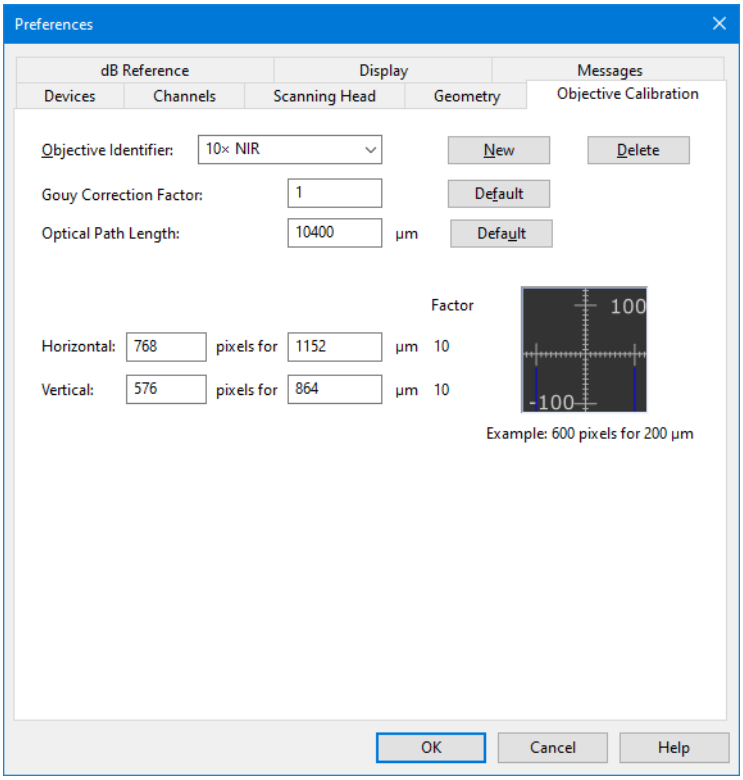


Figure 3.10: Objective Calibration page (MSA-650)

Optical Path Length: The optical path length is the starting value for the path length alignment. Each objective has its own optical path length in glass which influences the length of the measurement path accordingly. For this reason, enter an approximate optical path length here. The values for the specified objectives are factory-set. You can determine the optical path length yourself by carrying out the path length alignment on a silver mirror (not a dielectric mirror). Refer to SECTION 5.4 for the path length alignment.

Default: Click here if you want to reset the Optical Path Length to the default value set by Polytec.

New objectives

New: Click here to add a new objective to the Objective Identifier list. The new objective is given the same name as the last one selected, supplemented by a consecutive number, e.g. x20 (2). The number of pixels and the distance (horizontal and vertical) of the last objective selected are also applied. Overwrite the entries with the appropriate values.

Delete: To delete a objective, select it in the Objective Identifier list and click Delete. Then, if necessary, select another objective and click OK.

Determine pixels/distance

To determine the pixels and the distance for the objective used, proceed as follows:

1. Place the stage micrometer supplied under the microscope objective.
2. MSA-600: Switch on the laser beam in the software.

   MSA-500: Use the keys on the MSA-E-500 Junction Box to switch on the illumination.

   In the video window you will see the stage micrometer and the laser spot.
3. Set up the stage micrometer as precisely as possible to be parallel to the edges of the image. Adjust the image so that the zero point of the stage micrometer is at the bottom left at the edge of the image.
4. Focus on the stage micrometer.

   The video image is in sharp focus once the laser spot has reached the smallest possible diameter.
5. Set the contrast so that the bright grid lines do not appear overexposed (light gray).

   MSA-600: Use the automatic exposure control.

   MSA-500: Click the video image with the right mouse button and select Video Properties in the context menu. Move the Brightness and Contrast sliders correspondingly and click OK.

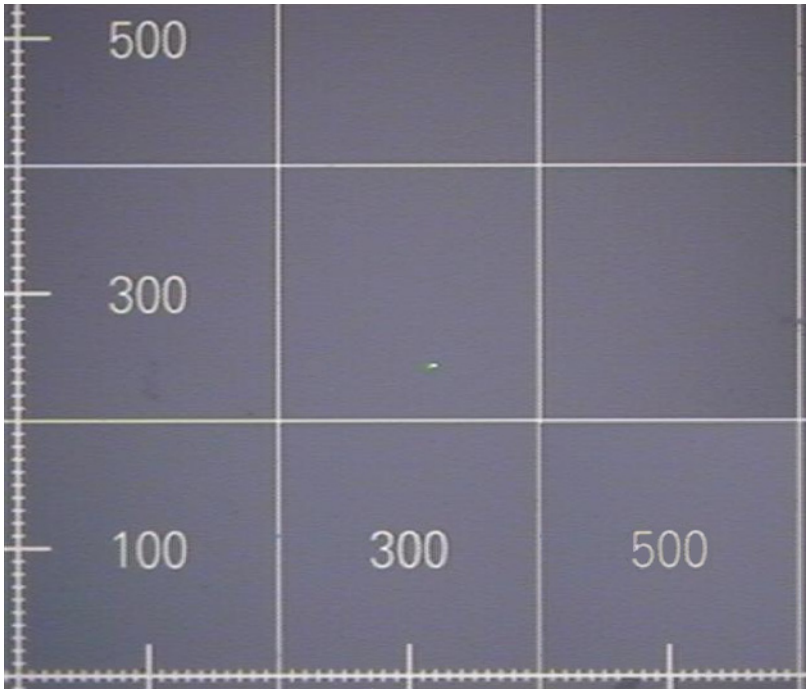


Figure 3.11: Video image of the stage micrometer

Set scan points

6. In the menu bar, select Setup > Define Scan Points (toolbar: ).

   You are in Scan Point Definition. The software maximizes the video window and displays a graphics toolbar.

7. Activate the point objects by clicking  in the graphics toolbar.
8. Delete the existing scan points by selecting all scan points with the mouse and pressing the Del key.
9. In the graphics toolbar, click  to activate the function to define new scan points.
10. Place one scan point respectively on a graduation line of the stage micrometer as far as possible to the right in a horizontal direction and as high as possible in a vertical direction.

ⓘ To increase the accuracy, zoom in on the video image.

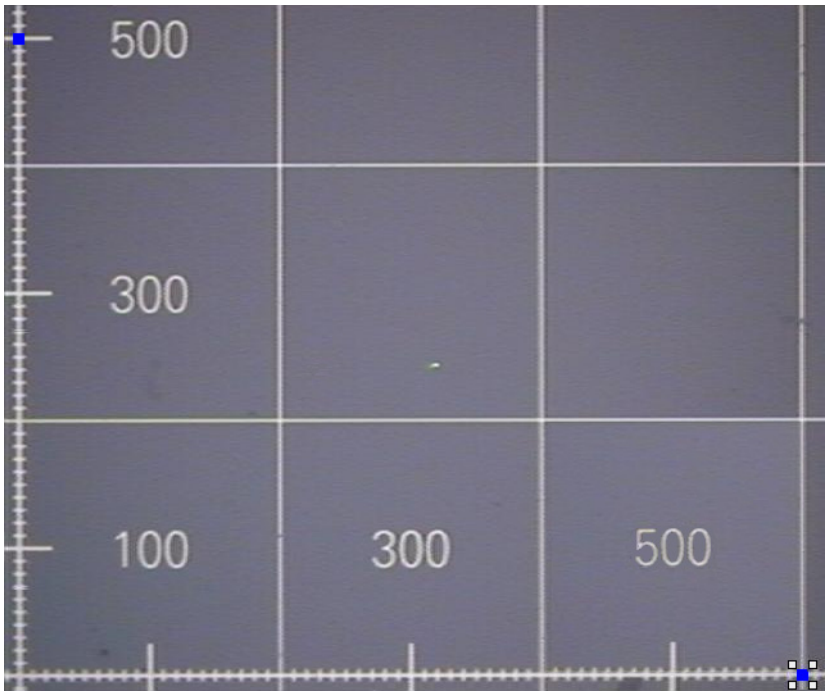


Figure 3.12: Stage micrometer with defined scan points

Run macro

11. In the project browser, select the Examples > Macros > Setup directory.
12. Using the right mouse button, click CalibrateObjective.bas and select Run in the context menu.

    The Calibrate Objective dialog appears with a description of the macro.

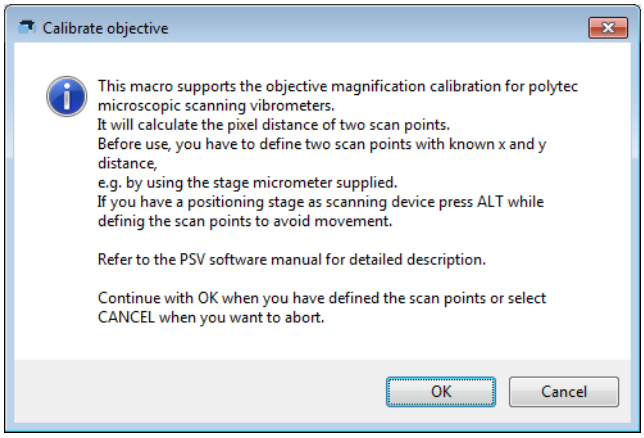


Figure 3.13: Calibrate Objective dialog

13. Click OK.

    The ObjectiveCalibrationResult.txt file appears in the text editor and shows the results.

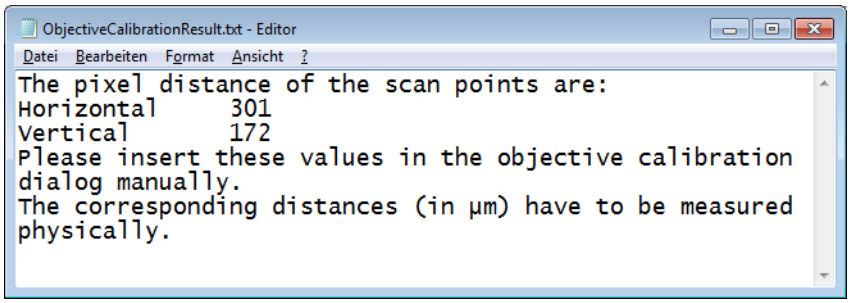


Figure 3.14: Results in the text editor

14. Make a note of the values for the horizontal and vertical pixels.
15. Make a note of the distances between the scan points with the help of the partition on the stage micrometer in horizontal and vertical direction. The distance between two graduations corresponds to 10 µm.

Enter values

16. Open the Preferences dialog by selecting Setup > Preferences in the menu bar.

    The Preferences dialog appears.
17. Display the Objective Calibration page.
18. Enter the width in pixels and in µm under Horizontal.
19. Enter the height in pixels and in µm under Vertical.
20. Click OK.

ⓘ MSA-500: To make a measurement, switch off the illumination again.

3.6 Setup: dB Reference

On the dB Reference page you can define dB = 0. This allows you to follow an online measurement with a certain dB reference directly.

ⓘ Changes you make here only have an effect on the global 0 dB reference (applies to all Polytec products). If you want to change the local 0 dB reference for the active document, you have to open the page in presentation mode. See the Presentation manual as well.

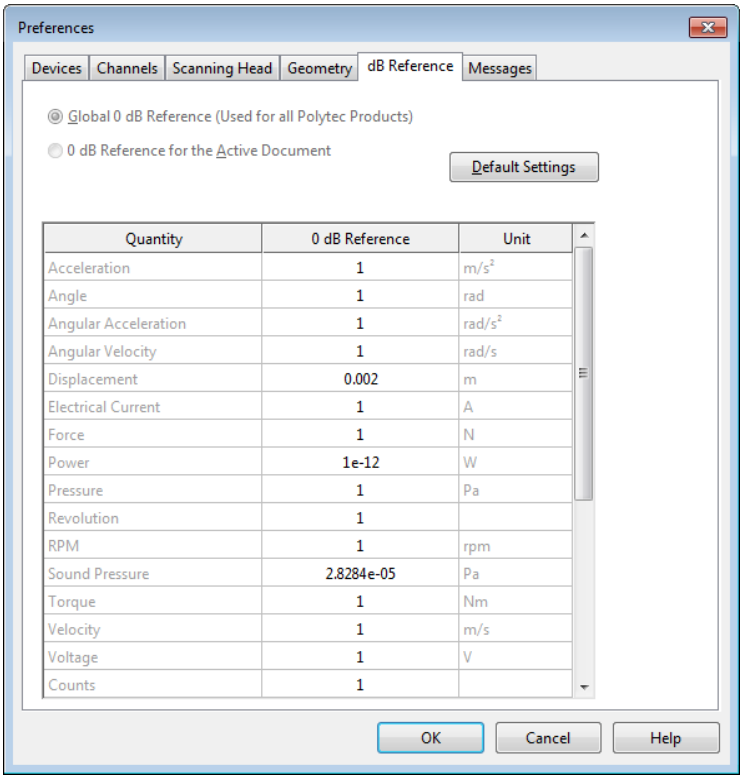


Figure 3.15: dB Reference page

Settings

Default settings: If you want to recover the original settings for the 0 dB Reference, click here.

Quantity: All available physical quantities are shown here.

dB Reference: Enter the factor for the respective unit here.

Unit: The unit for the respective physical quantity is shown here.

Example of 0 dB Reference

The 0 dB reference for displacement is 2 mm. Enter 0.002 or 2e-3 in the 0 dB Reference column.

Quantity	0 dB Reference	Unit
Displacement	0.002	m

Figure 3.16: Example of 0 dB Reference

3.7 Setup: Display

On the Display page, you can change the scaling of the toolbar.

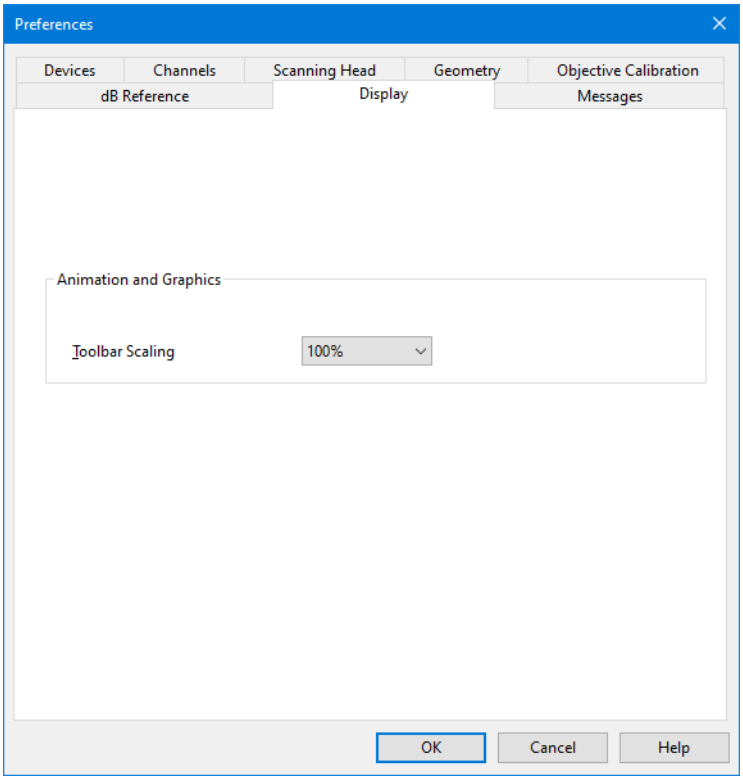


Figure 3.17: Display page

Animation and graphic

Toolbar Scaling: Select the required scaling from the list (default 100%).

3.8 Setup: Messages

On the Messages page, you select which system messages you would like to have displayed.

ⓘ Message windows which contain queries or which advise you of faulty input will appear in any case.

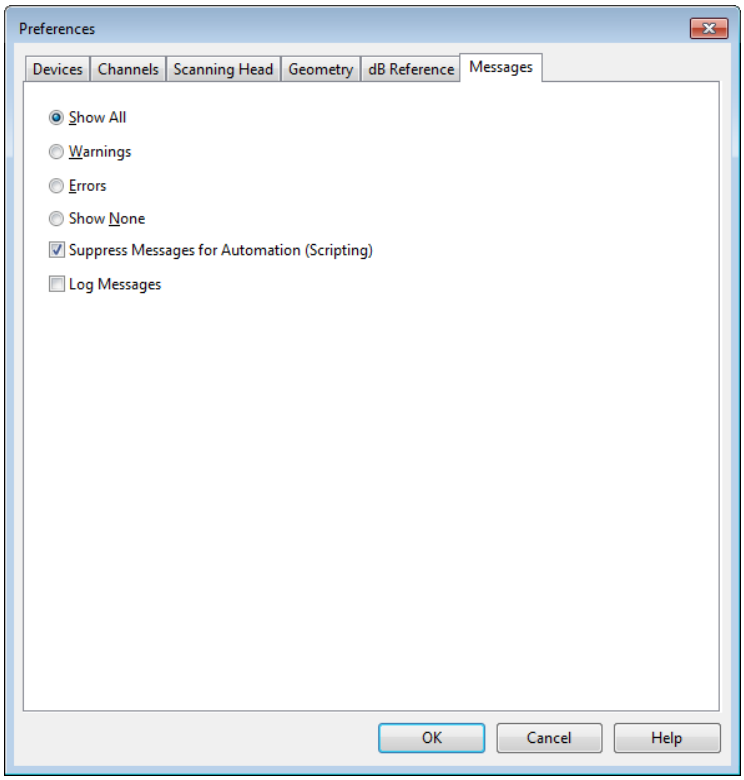


Figure 3.18: Messages page

Settings

Show All: All messages are displayed (notes, errors and warnings).

Warnings: Only warnings and errors are displayed.

Errors: Only error messages are displayed.

Show None: No system message is displayed.

Suppress Messages for Automation (Scripting): Tick the check box if you want to suppress messages during an automatic activity (Scripting). Otherwise it is possible that the activity of macros is disrupted by these messages.

Log Messages: Tick the check box if you want to log all messages. You will find the log file in the Event Viewer (refer to the following section).

Log file (Windows)

To open the log file under Windows 7, 8, or Windows 10, proceed as follows:

1. Press the Windows key on your keyboard.
2. Enter Event Viewer in the search field.
3. Click Event Viewer in the files shown.

   The Event Viewer window appears.
4. Open the Application and Services Logs folder.

   There you will find the logged messages in the Polytec directory.

To save the log file, proceed as follows:

5. Click the Polytec directory with the right mouse button and select Save All Events As in the context menu.

   The Save As dialog appears.
6. Navigate to the saving location and enter the file name.
7. Click Save.

4 Setting the Optics

Depending on which measurement system model you have, you can control certain functions of the optics:

Controlling the hardware - See SECTION 4.1 on this.

Automatically focusing the laser beam - See SECTION 4.2 on this.

Image settings for the live video image - Here you regulate contrast, brightness and color of the video image. See SECTION 4.3 on this.

4.1 Controlling the Hardware

You can control the optics in different ways:

- Controlling the optics with the software - See SECTION 4.1.1 on this.
- Fields in scanning head control - See SECTION 4.1.2 on this.

You can control certain functions of the hardware:

ⓘ You can remotely control either the vibrometer controller or respectively the front end you have connected. To do so, proceed as described in SECTION 7.2.8 or use the range changer in the status bar.

MSA-600

For the MSA-600, you can carry out the following functions with the software:

- Switch on or off the laser beam
- Position the laser beam
- Adjust the light intensity of the laser
- Switch on or off and position the reference laser beam for differential measurements (MSA-A-DIF option)
- Focus the laser beam (optional automated Z-axis from Polytec)
- Select objective for measurement (same setting on objective turret required)
- Control axes (optional automated axes from Polytec)

You set the other functions using the hardware. See the operating instructions of the measurement system on this.

MSA-650

For the MSA-650, you can carry out the following functions with the software:

- Switch on or off the laser beam
- Position the laser beam
- Adjust the light intensity of the laser
- Focus the laser beam (optional automated Z-axis from Polytec)
- Select objective for measurement (same setting on objective turret required)
- Aligning the Path Length
- Control axes (optional automated axes from Polytec)

You set the other functions using the hardware. See the operating instructions of the measurement system on this.

MSA-500

For the MSA-500, you can carry out the following functions with the software:

- Switch the laser beam on or off (beam shutter on the sensor head)
- Position the laser beam
- Adjust the light intensity of the laser (optional dimmer in the sensor head)
- Focus the laser beam and move the z-axis in the measurement microscope to zero position (MSAGeo option)
- Select objective for measurement (same setting on measurement microscope required)
- Control axes (optional automated axes from Polytec)

You set the other functions using the hardware. See the operating instructions of the measurement system on this.

4.1.1 Controlling the Optics with the Software

To control the optics with the software, proceed as follows:

1. If you do not see the scanning head control in the application window, select View > Scanning Head in the menu bar (toolbar: ).
2. In the Objective list, select the zoom factor which is set.

ⓘ If you can not see the Objective list in the scanning head control, then first of all set up the software for your measurement system. See CHAPTER 3 on this.

3. If necessary, maximize the video window. To do so, in the title bar of the video window on the right, click  .
4. If necessary, adjust the image settings of the live video image. See SECTION 4.3 on this.

With the fields in the scanning head control, you can now work as described in SECTION 4.1.2.

4.1.2 Fields in Scanning Head Control

Depending on the equipment your measurement system has, you will see the following fields in the scanning head control:

Optics In the Optics field, you can switch the laser beam on or off, adjust the light intensity of the laser, and monitor the signal level. See SECTION 4.1.2.1 on this.

Channels In the Channels field you can watch the amplitude level of the active channels, for example as soon as you have started a continuous measurement. See SECTION 4.1.2.2 on this.

Position In the Position field you can move the laser beam by clicking the icons. See SECTION 4.1.2.3 on this.

2D Point The 2D Point field is displayed if there is 2D geometry. There you will see information on selected scan points. See SECTION 4.1.2.4 on this.

3D Point The 3D Point field is displayed if there is a 3D geometry. There you will see information on selected scan points. See SECTION 4.1.2.5 on this.

Objective In the Objective field, you select the name of the objective that you have set on the objective turret/measurement microscope. See SECTION 4.1.2.6 on this.

Path length In the Path Lgth. field, you can carry out the path length alignment for the MSA-650. For a description of the field, see SECTION 4.1.2.7. For a description of the path length alignment, see SECTION 5.4.

Axes In the Axes field you can set up and control automatized axes from Polytec. See SECTION 4.1.2.8 on this.

4.1.2.1 Scanning Head Control: Optics

Control laser beam for MSA-600/ MSA-650

In the Optics field, you can switch the laser beam on or off, adjust the light intensity of the laser, and monitor the signal level.

ⓘ With the MSA-600-X/-U, the signal level is only updated while the measurement is running.

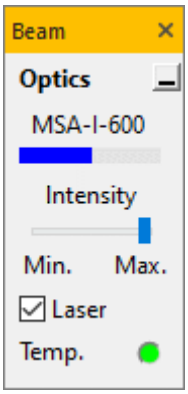


Figure 4.1: Optics field for MSA-600

At the top you can see the type of scanning head. Below that you can see the optical signal level as a bar. This makes manual focusing easier.

ⓘ Usually, not signal level is displayed for the MSA-650 because of the short coherence. The image sharpness is used to focus the laser beam.

Intensity: To reduce the light intensity of the laser, move the slider with the mouse in the Min. direction.

Laser: You can switch the laser beam on or off (key combination Ctrl + 1). Activate the check box to switch on the laser beam.

Features for MSA-600-X/-U:

- If the laser beam is switched off before you start a measurement, it will only be switched on at the start of the measurement for the duration of the actual data acquisition. You will see the laser spot flashing on the object. After the measurement, the laser beam is switched off.
- If the laser beam is switched on before you start a measurement, it remains switched on at the start of the measurement for the duration of the data acquisition and after the measurement.

Temp.: The temperature in the sensor head is monitored. The statuses are displayed using the status display.

The following statuses are possible:

- Status display lights up green: The temperature in the sensor head is within the specified range.
- Status display lights up red: The temperature in the sensor head has exceeded 55 °C. In this case it is important to switch off the measurement system and allow the sensor head to cool down.

ⓘ After exceeding the temperature the status display remains lighting red, even though the temperature in the sensor head is fallen below 55 °C in the meantime. Only switching the measurement system off and on again resets the status of the status display.

Features for optics for MSA-600/ MSA-650

If you are using the MSA-600-M/-V or the MSA-650 with an automated Z-axis from Polytec, you can focus the laser beam via the software.

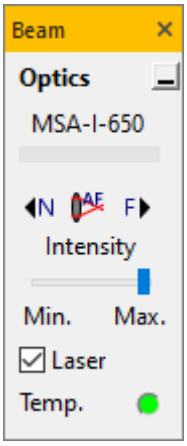


Figure 4.2: Optics field for MSA-650 with automatized Z-axis

You focus the laser beam using the icons as follows:



Near - This means that the Z-axis is moved downwards.



Far - This means that the Z-axis is moved upwards.



Automatically (key combination Ctrl + F) - This means that the Z-axis is automatically moved first upwards and then downward in a specified range, and is then stopped at the position with minimum noise level. You will find more information on this in SECTION 4.2.

ⓘ You can focus automatically more quickly by holding the Control key pressed and clicking the icon (Ctrl + E key combination). Thereby the autofocus is only performed in a limited range.

When using the MSA-600 with the MSA-A-DIF option, you can carry out differential measurements. In addition to the measurement laser beam you can additionally switch the reference laser beam on or resp. off via the software.

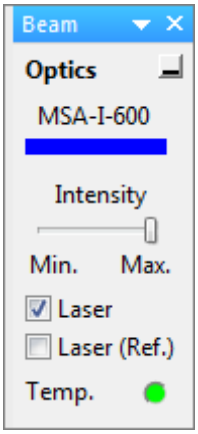


Figure 4.3: Optics field for MSA-600 with MSA-A-DIF option

Laser (Ref.): You can switch the reference laser beam on or off. Tick the check box to switch on the reference laser beam. In the Position field, you can switch between the measurement laser beam and the reference laser beam and you can position the laser beam. See SECTION 4.1.2.3 on this.

When you are using the MSA-600 with the MSA-A-DIF option and an automated Z-axis from Polytec, you can carry out differential measurements and you can focus the measurement laser beam via the software (see description above).

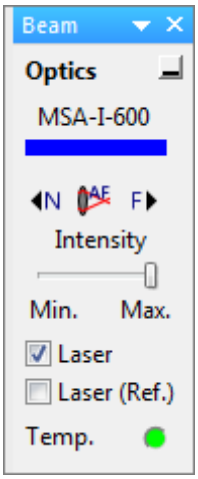


Figure 4.4: Optics field for MSA-600 with MSA-A-DIF option and automated Z-axis

Control laser beam for MSA-500

In the Optics field, you can switch the laser beam on or off, adjust the light intensity of the laser, and monitor the signal level.

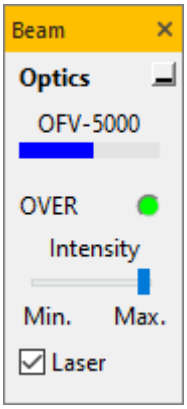


Figure 4.5: Optics field for MSA-500

At the top you can see the type of controller. Below that you can see the optical signal level as a bar. This makes manual focusing easier.

OVER: The status display shows when the set measurement range is exceeded. In this case it lights up red. If it is permanently lit up, select the next highest measurement range.

Intensity: To reduce the light intensity of the laser, move the slider with the mouse in the Min. direction. This function is available as an option for the OFV-551/-552 Sensor Head.

Laser: You can control the beam shutter and use it to switch the laser beam on or off. Deactivate the check box to switch off the laser beam. You can also switch the laser beam on or off directly by pressing the LASER key on the OFV-551/-552 Sensor Head. See your sensor head operating instructions on this.

Features for optics for MSA-500

If you are using the MSA-E-500 junction box with the MSA-I-500 scanning head and the MSAGeo option, you can focus the laser beam via the software.

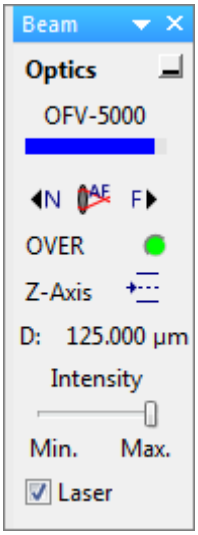


Figure 4.6: Focusing the laser beam for MSA-500

You focus the laser beam using the icons as follows:



To close-up (N) - This means that the z-axis in the measurement microscope is moved upward.



To infinity (F) - This means that the z-axis in the measurement microscope is moved downward.



Automatically (AF, key combination Ctrl + F) - This means that you automatically travel along the z-axis in the measurement microscope and stop at the position where the signal level is maximum.

ⓘ You can focus automatically more quickly by holding the Control key pressed and clicking the icon (Ctrl + E key combination). Thereby the autofocus is only performed in a limited range.

Z-Axis: You can position the z-axis in the measurement microscope directly at zero position. To do so, click  . The z-axis is then positioned in the middle of the range of travel. The position D of the z-axis is displayed under the icon (0…250 µm).

2nd vibrometer controller

If an additional vibrometer controller is connected and active, then another field is shown. As this vibrometer controller is not connected to the system via the junction box, only those functions are available here which are offered by the second vibrometer itself (refer to the following figure OFV-5000 with OFV-551/OFV-552).

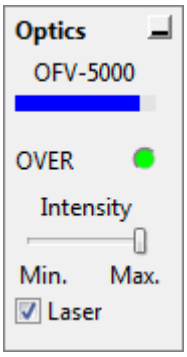


Figure 4.7: Optics field for 2nd vibrometer controller

4.1.2.2 Scanning Head Control: Channels

With the MSA-600-X/-U, you can not modulate the channels individually (Vib, Ref).

ⓘ You can control up to four channels depending on the device configuration.

Display channels

In the Channels field, you can watch the amplitude level of the active channels, for example as soon as you have started a continuous measurement.

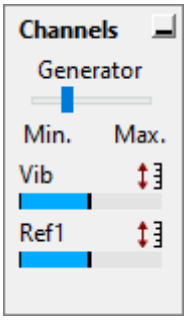


Figure 4.8: Channels field

Generator: If you are using a generator, you can use the slider to change the amplitude of the generator signal directly. The scaling of the slider is not linear. If you point the mouse at the slider, you will be shown the value of the amplitude. The minimum or maximum amplitude of the slider is determined by the value you entered for the generator signal. See SECTION 7.2.9 on this as well.

Vib, Ref: You can observe the amplitude level of the active channels. Apart from that, you can modulate individual channels automatically by clicking  for the corresponding channel. This optimizes the input range of the data acquisition board or the sensitivity range of the vibrometer front-end/controller respectively. You can modulate all channels together automatically by clicking  while holding the Shift key pressed.

4.1.2.3 Scanning Head Control: Position

Position laser beam for MSA-600/ MSA-650

In the Position field you can move the laser beam by clicking the icons. To do so, click the icon for the direction required.

When you click  , the laser beam will be exactly positioned so that it emits at the center of the scan area (position 0).

ⓘ You can also move the laser beam by pressing the Ctrl + Shift + arrow key (↑, ↓, →, ←) key combination.

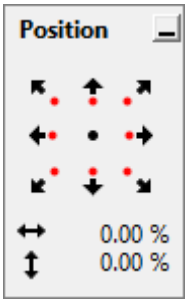


Figure 4.9: Position field for MSA-600

Under the icons the position of the laser beam is displayed as a % of the mirror deflection. At 100% the laser beam is at the far right/at the top.

When using the MSA-600 with the MSA-A-DIF option, you can carry out differential measurements. You can switch between the measurement laser beam (Meas.) and the reference laser beam (Ref.) and you can position the selected laser beam. Read SECTION 4.1.2.1 on how to switch the reference laser beam on or resp. off.

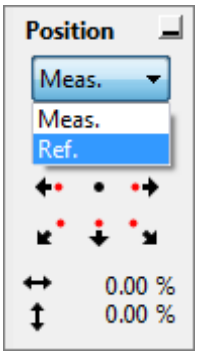


Figure 4.10: Position field for MSA-600 with MSA-A-DIF option

Position laser beam for MSA-500

In the Position field you can move the laser beam by clicking the icons. To do so, click the icon for the direction required.

When you click  , the laser beam will be exactly positioned so that it emits at the center of the scan area (position 0).

ⓘ You can also move the laser beam by pressing the Ctrl + Shift + arrow key (↑, ↓, →, ←) key combination.

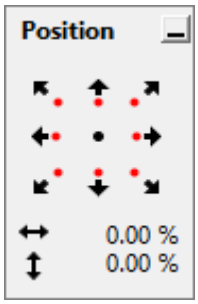


Figure 4.11: Position field for MSA-500

Under the icons the position of the laser beam is displayed as a % of the mirror deflection. At 100% the laser beam is at the far right/at the top.

4.1.2.4 Scanning Head Control: 2D Point

Axis coordinates

The 2D Point field is displayed if there is 2D geometry.

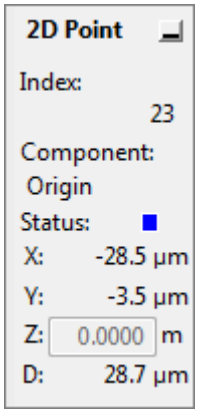


Figure 4.12: 2D Point field for MSA

Index: The index of the selected scan point is displayed.

Component: The name of the geometry component is displayed to which the selected scan point belongs. Read SECTION 6.5.4 on how to define geometry components.

Status: The status of the selected scan point is displayed. See SECTION 8.3 on this as well.

X, Y, Z: The coordinates of the scan point are displayed. The coordinate system used here is the one which you determined when setting the vibrometer channel. It depends on the direction of the vibrometer channel in the Acquisition Settings dialog, which of the three axes you can specify the distance for. See SECTION 7.2.2 on this as well. The figures here apply to the +Z direction.

ⓘ You can read the coordinates with great accuracy if you leave the mouse pointer on the corresponding coordinate.

D: The software displays the calculated distance, i.e. the distance of the selected scan point to the middle of the video image (origin of coordinate system).

4.1.2.5 Scanning Head Control: 3D Point

Axis coordinates

The 3D Point field is displayed if there is a 3D geometry.

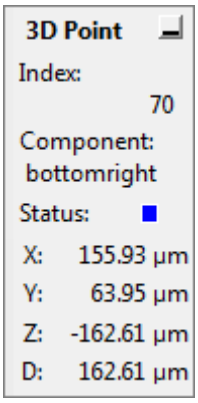


Figure 4.13: 3D Point field for MSA

Index: The index of the selected scan point is displayed.

Component: The name of the geometry component is displayed to which the selected scan point belongs. Read SECTION 6.5.4 on how to define geometry components.

Status: The status of the selected scan point is displayed. See SECTION 8.3 on this as well.

X, Y, Z: The coordinates (X, Y, Z) of the scan point are displayed. The coordinate system used here is the one which you determined with the 3D alignment. See SECTION 5.2 on this as well.

ⓘ You can read the coordinates with great accuracy if you leave the mouse pointer on the corresponding coordinate.

D: The software displays the position D of the z-axis (0…250 µm). This also corresponds to the distance Z on the z-axis in the measurement microscope relative to the zero point.

ⓘ The equivalence with Z is only applicable if Z has been set as the vibrational direction and with the 3D alignment, no offset has been entered.

4.1.2.6 Scanning Head Control: Objective

MSA-600/ MSA-650

The software detects automatically on which position the objective turret is located and displays this position in the Objective field in the list.

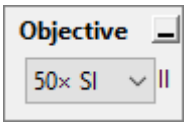


Figure 4.14: Objective field for MSA-650 (50x objective on position II)

You have to assign the name of the mounted objective to the position on the objective turret (I, II or III) set for measuring. To do so, turn the objective turret to the wanted position and select the suitable objective from the list.

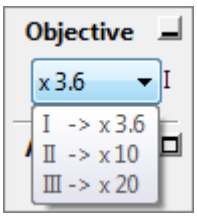


Figure 4.15: Objective assignment for the positions on the objective turret (for MSA-600)

Error at the objective assignment or resp. at the positioning of the objective turret are marked with an appropriate icon.

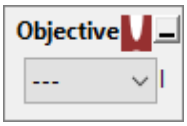


Figure 4.16: Error message at the objective assignment

The following error messages are possible:

- No objective was allocated for the position set on the objective turret: Select the suitable objective from the list.
- One of the following errors occurred:

  1. The objective turret is not engaged on a valid position: Turn the objective turret to position I, II or III.

  2. The objective turret is not mounted: Mount the objective turret, screw the suitable objectives in the objective turret and assign a position to the objectives.

MSA-500

In the Objective field, you select the name of the objective that you have set on the measurement microscope.

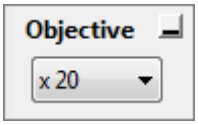


Figure 4.17: Objective field for MSA-500

Select the name of the objective that you have set on the measurement microscope. The software uses the respective zoom factor and the position of the scan points on the video image to calculate the coordinates of the scan points.

4.1.2.7 Scanning Head Control: Path Length

MSA-650

In the Path Length field, all required functions for the path length alignment are available. For a description of the path length alignment, see SECTION 5.4.

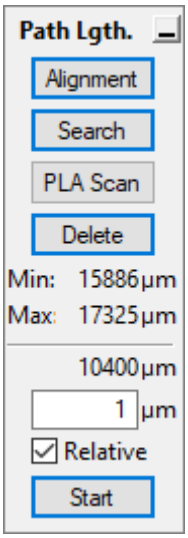


Figure 4.18: Path Lgth. field for MSA-650

Alignment: Click here to start the path length alignment. The measurement is performed in a range of ±2 mm around the path length stored in the Objective Calibration dialog for the selected objective.

Search: Click here to start a search measurement in the range defined by Min and Max. The travel axis stops at the position with maximum signal level.

Delete: Click here to delete the path length alignment.

Min, Max: The minimum and maximum value of the path length is displayed, in the range where the search for the maximum signal level runs. The current path length of the reference mirror/reference arm is displayed below. If you want to specify a path length, enter it in the field below.

Relative: Tick the check box if you want to move the current position of the reference mirror/reference arm by an offset

Start: Click here to move the reference mirror/reference arm absolutely or relatively. In the case of a relative displacement by an offset, you can reverse the direction of movement of the reference mirror/reference arm by simultaneously pressing the Shift key.

4.1.2.8 Scanning Head Control: Axes

Control axes

You can control automated axes by Polytec via the software.

ⓘ When starting the software or setting up the axes, the software automatically performs a reference run.

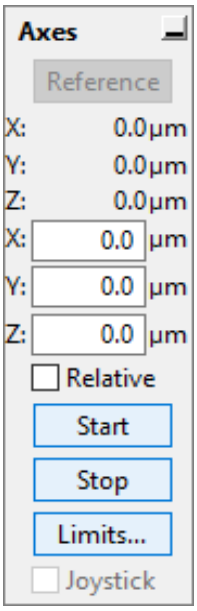


Figure 4.19: Axes field

Functions for axes

Reference: A reference run is carried out if one of the following conditions is met:

- An automated Z-axis by Polytec is detected by the software.
- Automated XY-axes by Polytec are used for scanning.

If the reference run was not successful or was aborted when the software was started, you have to start a reference run yourself. To do so, click Reference. A message box appears, warning you of possible collisions. You can select the following options:

- Perform reference run and restore last known position.

  The axes are moved along to the reference switches and then close to the last position. Below the button, the absolute position (X, Y, Z) is displayed.
- Perform only reference run

  The axes are moved to the reference switches.

ⓘ If the axes are not referenced, they can not be moved.

If you have manually moved one of the axes, you have to restart the software for performing a reference run.

Relative: Tick the check box if you want to move from the current position of the axes by an offset.

X, Y, Z: Enter the desired value for a movement of the axes. For a relative movement of the axes the following direction convention applies:

+X: The X-axis moves to the right.

–X: The X-axis moves to the left.

+Y: The Y-axis moves backwards.

–Y: The Y-axis moves forwards.

+Z: The Z-axis moves upwards.

–Z: The Z-axis moves downwards.

Start/Stop: To move the axes, enter the desired values into the X, Y or Z field and click Start. To stop moving the axes, click Stop.

Limits: If you want to set limits for the axes to prevent from collisions, click Limits. The Limits for Travel Range of Axes dialog appears.

ⓘ If you want to move the automated axes from Polytec for setting the limits (Current Position), you have to activate the Joystick check box in the scanning head control before opening the dialog.

You can activate the joystick at any time by pressing the 1 button on the joystick.

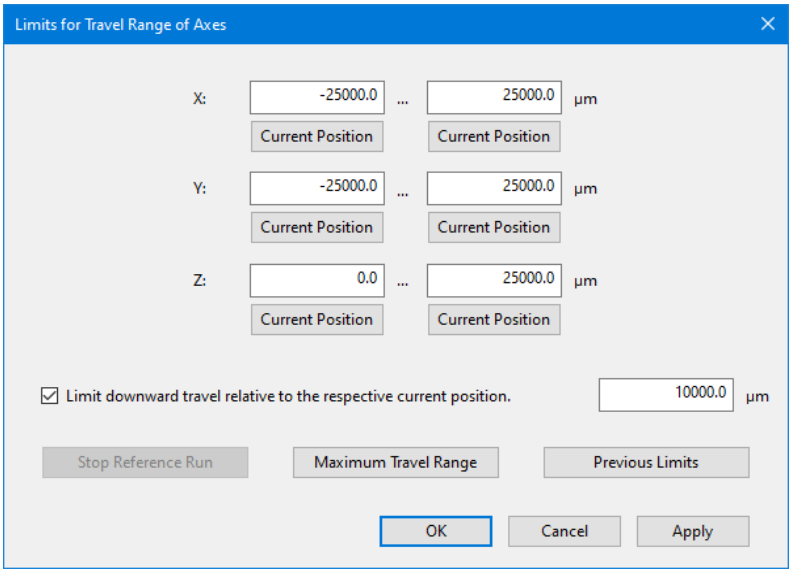


Figure 4.20: Limits for Travel Range of Axes dialog

ⓘ Each time you start the software the dialog is displayed while the axes will be referenced. You have to enter the limits before you can close the dialog.



CAUTION!

Pinching hazard or damage caused by careless operation!

During setup the axes are moved to determine the limits.

» Always be careful when working with the automated axes.

» Before setting up, make sure that the objective cannot collide with the object. Adjust the limits of the axes travel range (Limit downward travel relative to the respective current position function).

» Do not put any body parts into the range of travel of the axes.

» In principle, during traveling the axes, observe the distance between the objective and object to prevent the object from being damaged.

Enter lower and upper limit values for the axes or click Current Position respectively to apply the current position of the axis. Adapt the following settings to your measurement task:

- Limit downward travel relative to the respective current position: If you use an automated Z-axis from Polytec, you can limit the travel range downwards. To do so, tick the check box and enter a corresponding value. The software checks with all movements (apart from movements activated by the joystick) the limit and displays either a message or stops the movement at the limit.
- Stop Reference Run: Click here to cancel the current reference run. As soon as you have aborted a reference run, you cannot access the axes anymore (Functions of the axes like focusing automatically with the aid of the z-axis or scanning with the aid of the XY positioning stage are not possible anymore). However, you can enter limits in the dialog and close it. Restart the software to perform a reference run.
- Maximum Moving Range: Click here to restore the maximum available moving range of the axes.
- Previous Limits: Click here to restore the previous limits of the moving range.

Joystick: Tick the check box to control the axes with the aid of the optional USB joystick. Set up the joystick beforehand as described in SECTION 3.1.



CAUTION!

Pinching hazard or resp. damage caused by operating error!

If you cause a movement of the axes using the joystick, the software can not execute a limits check.

» Always cause critical movements using the software.

» If you use the joystick, check always the movements visually on the specimen.

The keys on the USB joystick are numbered from 1 to 6. The keys are programmed with the following functions:

- Key 1: Activate or deactivate the joystick.
- Key 2 in Define Scan Points mode: For point objects (GeoWorks option), you can define individual scan points.
- Key 3: The axes are traversed according to the amount of relative movement (Relative check box ticked, direction convention as described before under X, Y, Z).
- Key 4: Not applicable.
- Key 5: If you hold the key pressed and move the joystick, the axes traverse with maximum speed.
- Key 6: The axes are traversed according to the amount of relative movement (Relative check box ticked) into the opposite direction of key 3.

A-PST-A010 XY Positioning Stage

When using the A-PST-A010 XY Positioning Stage with the MSA-500, connect the joystick to the positioning stage directly.



CAUTION!

Pinching hazard caused by automatic start!

If you tick the check box without having connected a joystick, the axes are moving into an undefined position. This leads to a risk of pinching for the hands.

» Connect the joystick to the XY positioning stage before starting the PSV software.

Joystick: Tick the check box to control the XY positioning stage with the aid of the optional joystick. See your XY positioning stage operating instructions on this.

4.2 Automatically Focusing the Laser Beam

MSA-600-M/-V or MSA-650

To be able to automatically focus the laser beam with the MSA-600-M/-V or the MSA-650, the following conditions have to be fulfilled:

- Automated Z-axis from Polytec (e.g. A-STD-TST-01)
- A-CTR-110 or A-CTR-120 Axis Controller

To start the Autofocus function, you have the following options:

- Click  .
- Press the key combination Ctrl + F.

ⓘ MSA-600-M/-V: During autofocus travel, the vibrometer signals will be continuously acquired.

MSA-650: During autofocus travel, individual images are captured from the video image. The image sharpness of each shot is analyzed. The positions with maximum image sharpness are usually located on the different planes of the MEMS component.

The autofocus travel starts at the position the Z-axis is currently located. All limits defined for the travel range of the axes remain active. Read SECTION 4.1.2.8 on how to define limits.

Process (MSA-600-M/-V)

The process of an autofocus travel is described as follows:



CAUTION!

Pinching hazard or damage caused by automated process!

During the autofocus travel, the z-axis of the positioning device is moved to focus the laser beam.

» Always be careful when working with the positioning device.

» Make sure that the sensor head objective cannot collide with the object before the autofocus travel. Adjust the limits of the axes travel range (Limit downward travel relative to the respective current position function).

» Do not put any body parts into the range of travel of the axes.

» In principle, during traveling the axes, observe the distance between the objective and object to prevent the object from being damaged.

» Make sure that the specimen will survive a longer exposure with maximum laser light intensity undamaged. If not, renounce an autofocus travel.

1. The Z-axis travels up to 1 mm upward.
2. If the laser beam is switched off, it will be automatically switched on via the software (Laser check box activated).
3. If the laser is not set to maximum light intensity, it will be automatically set to maximum light intensity via the software (Intensity slider to Max. position).
4. A predefined vibrometer measurement range is set.
5. The Z-axis travels up to 2 mm downward. The noise level is measured on the vibrometer channel.

   If a minimum of the noise level is detected, the Z-axis travels back to the position with minimum noise level at the end of the autofocus travel.

ⓘ It may be that the software can only recognize a minimum of noise level as such at the end of the autofocus travel. Also then the Z-axis travels back to this position.

If no minimum noise level is recognized, the Z-axis stands still at the end of the autofocus travel (up to 1 mm lower than the start position).

6. The vibrometer measurement range is reset to initial state.
7. The laser light intensity is reset to initial state.
8. The laser beam is reset to initial state.

Process (MSA-650)



CAUTION!

Pinching hazard or damage caused by automated process!

During the autofocus travel, the z-axis of the positioning device is moved to focus the laser beam.

» Always be careful when working with the positioning device.

» Make sure that the sensor head objective cannot collide with the object before the autofocus travel. Adjust the limits of the axes travel range (Limit downward travel relative to the respective current position function).

» Do not put any body parts into the range of travel of the axes.

» In principle, during traveling the axes, observe the distance between the objective and object to prevent the object from being damaged.

» Make sure that the specimen will survive a longer exposure with maximum laser light intensity undamaged. If not, renounce an autofocus travel.

1. The Z-axis travels up to 1 mm upward.
2. The Z-axis travels up to 2 mm downward. Individual images are captured at the positions with good image sharpness.

   At the end of the autofocus travel, the Z-axis returns to the position with the highest image sharpness.

ⓘ The software may not be able to recognize the position with the highest image sharpness as such until the end of the autofocus travel. Also then the Z-axis travels back to this position.

If no sufficiently sharp image is detected, the Z-axis stops at the end of the autofocus travel (up to 1 mm lower than the start position).

Automatically focus quickly

If the laser beam is already roughly focused, you can focus automatically much more quickly. To do so, you have the following options:

- Hold the Control key pressed and click  .
- Press the key combination Ctrl + E.

The Z-axis only travels up to 0.2 mm and down to 0.4 mm to find the optimum focus position.

ⓘ MSA-650: After the end of the autofocus travel, in addition to the position with the highest image sharpness, you can move to all other positions with high image sharpness individually (i.e. the individual levels of the MEMS component). Press the Ctrl + arrow key (↑ oder ↓) key combination.

MSA-500

If you are using the MSA-E-500 junction box with the MSA-I-500 scanning head and the MSAGeo option, you can focus the laser beam via the software. See SECTION 4.1.2.1 on this as well.

You can focus the laser beam automatically. This means you automatically travel along the z-axis in the measurement microscope and stop at the position at which the signal level is highest.

To start the Autofocus function, you have the following options:

- Click  .
- Press the key combination Ctrl + F.

Automatically focus quickly

If the laser beam is already roughly focused, you can focus automatically much more quickly. To do so, you have the following options:

- Hold the Control key pressed and click  .
- Press the key combination Ctrl + E.

The z-axis in the measurement microscope travels only a few millimeters upward and downward to find the maximum noise level.

4.3 Image Settings of the Live Video Image (2D View)

You can adjust the image settings of the live video image in the Display Properties dialog.

General properties

On the General page you set up the view of the video window on the screen. See SECTION 4.3.1 on this.

Video properties

You adjust the camera settings on the Video page. See SECTION 4.3.2 on this.

Open dialog

To open the dialog, you have the following options:

- Double-click the live video image.
- Click the live video image with the right mouse button and select General Properties or Video Properties in the context menu.
- Activate the video window (click it, to do so) and select Scan > Properties. Display the appropriate page.

ⓘ Depending on the type of measurement and the geometry, not all settings options are shown in the dialog.

ⓘ The crosshair in the video image defines the center of the video image. This allows you to better adjust the laser to the middle of the video image.

4.3.1 Properties on the General Page

On the General page you set up the view of the video window on the screen.

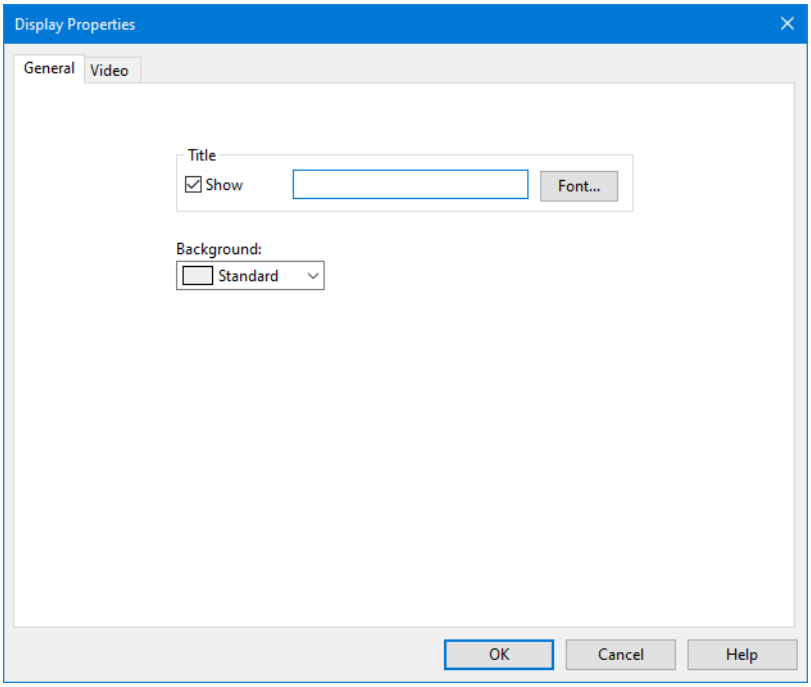


Figure 4.21: General page

Title

Show: If you would like to display a title above the video window, tick the check box. You enter the title on the right.

Font: Click here to format the title. The Font dialog appears. Set up the Font, Font style, Size, and if applicable Effects and click OK.

Background color

Background: Select the background color for the video window.

4.3.2 Properties on the Video Page

You adjust the camera settings on the Video page. The settings on this page take immediate effect.

MSA-600

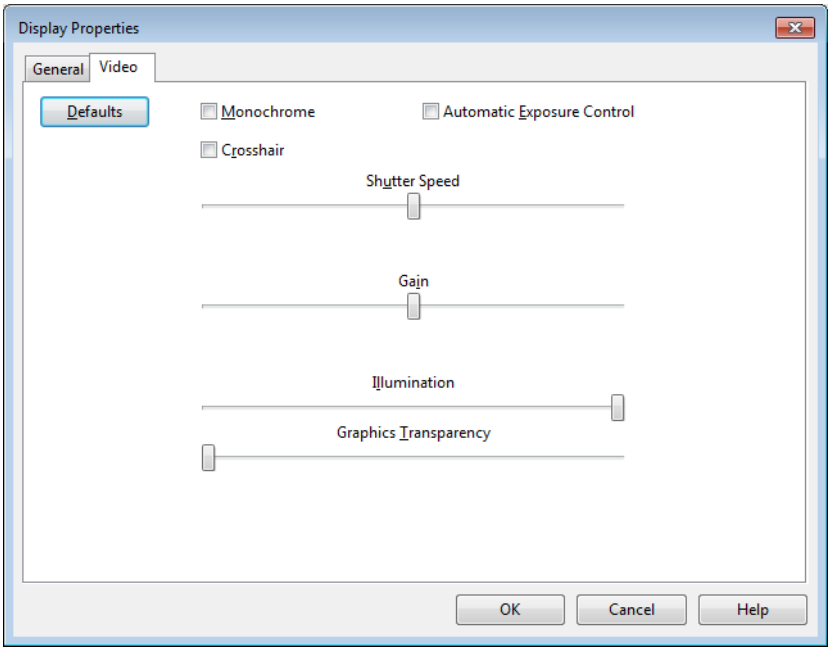


Figure 4.22: Video page for MSA-600

Settings

Defaults: Click here to load the factory settings for the live video image.

Crosshair: Tick the check box to show a crosshair on the live video image.

Automatic Exposure Control: Deactivate the check box if you want to switch off the automatic exposure control for the camera.

ⓘ If the automatic exposure control is active, then the camera autonomously controls the Shutter Speed and Gain parameters. For this reason you can only change these settings if you deactivate the automatic exposure control.

Shutter Speed: Use the slider to adjust the exposure time of the camera which affects the brightness of the live video image.

Gain: Use the slider to adjust the contrast of the live video image.

Illumination: Use the slider to adjust the brightness of the illumination.

Graphics Transparency: Use the slider to adjust the transparency of all inserted graphic elements.

MSA-650

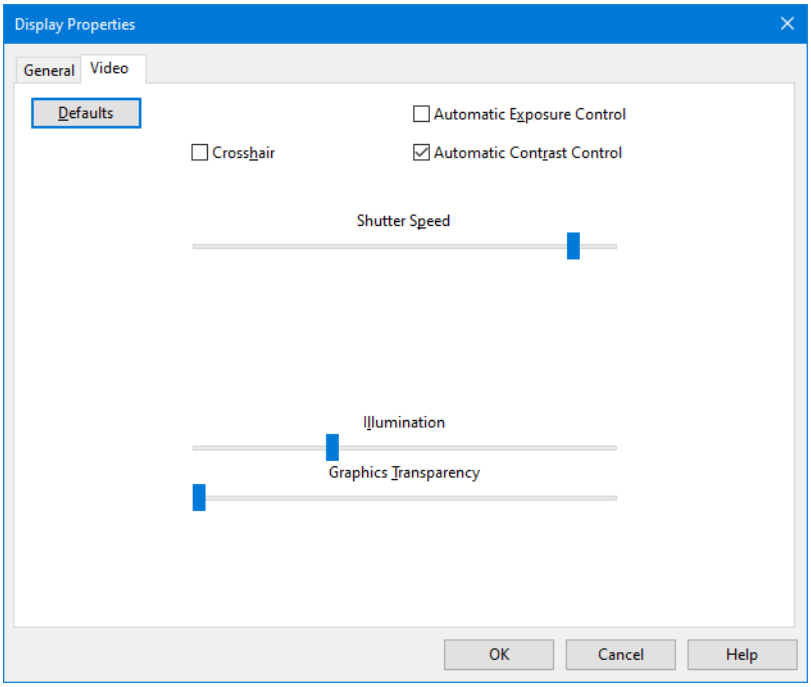


Figure 4.23: Video page for MSA-650

Defaults: Click here to load the factory settings for the live video image.

Crosshair: Tick the check box to show a crosshair on the live video image.

Automatic Exposure Control: Deactivate the check box if you want to switch off the automatic exposure control for the camera.

ⓘ If the automatic exposure control is active, then the camera controls the Shutter Speed parameter itself. For this reason, you can only change this setting if you deactivate the automatic exposure control.

Automatic Contrast Control: Tick the check box to improve the contrast of the live video image when measuring through silicon caps.

ⓘ If the video image appears black, it may be overexposed. Reduce the shutter speed or deactivate the Automatic Contrast Control check box.

Shutter Speed: Use the slider to adjust the exposure time of the camera which affects the brightness of the live video image.

Illumination: Use the slider to adjust the brightness of the illumination module.

Graphics Transparency: Use the slider to adjust the transparency of all inserted graphic elements.

MSA-500

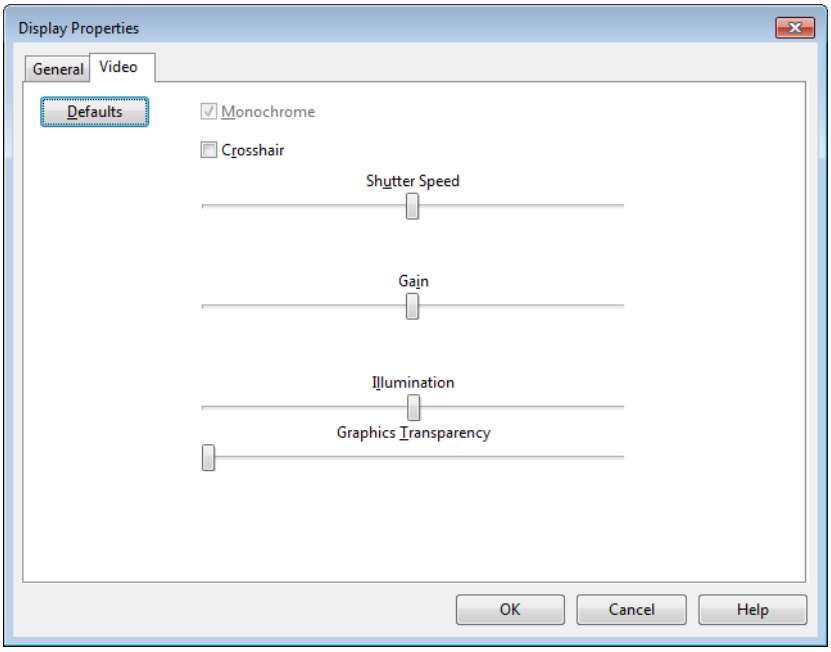


Figure 4.24: Video page for MSA-500

ⓘ The settings are identical to the MSA-600. The automatic exposure control is not available for the MSA-500.

5 Aligning Coordinates and Path Length

Depending on the design of the measurement system, you can align coordinates:

Carrying out 2D alignment - Here, the positions on the live video image have to be aligned with the position of the laser beam on the measurement plane. See SECTION 5.1 on this.

Carrying out 3D alignment - Here, the position and alignment of the scanning head are determined in the object's coordinate system. See SECTION 5.2 on this.

Further information on 3D alignment - See SECTION 5.3 on this.

For the MSA-650, you have to carry out a path length alignment before the measurement. See SECTION 5.4 on this.

Undo and redo You can undo up to 50 operations and redo them. In the menu bar, select Edit > Undo or Edit > Redo (toolbar:  or ).

5.1 Carrying out 2D Alignment

To define scan points on the live video image and to move the laser beam using the mouse, you have to align the positions on the live video image with the position of the measurement laser on the measurement plane. As you are aligning two-dimensional coordinate systems, you have to align at least two points with different horizontal and vertical coordinates.

ⓘ It is always necessary to carry out this 2D alignment.

How? You can carry out 2D alignment with the software as described in the following sections.

When? You have to realign if one of the following points is applicable:

- You measure with another stand-off distance.
- The 2D alignment was carried out before the end of the warm-up period.

Tips The following tips make it easier to align:

- The more points you align, the more precisely the software positions the laser beam.
- Align those points which you want to hit extremely precisely when scanning.
- Make sure that all areas that you want to define scan points in are covered by alignment points. If the object has areas at varying distances from the scanning head, then define alignment points at the edges of these areas.

The software divides the scan range into triangles using the alignment points as the vertices. If necessary, additional vertex points are added to avoid triangles with acute angles. Inside the triangles, the scanner angles which you have assigned to the alignment points are linearly interpolated. To improve the alignment, you can have the triangles you have generated displayed. To do so, click the video image with the right mouse button and select Triangulation Grid in the context menu.

- If after completing a 2D alignment, you would like to carry out a 3D alignment, define 2D alignment points near the 3D alignment points marked on the object. This then makes it easier for you to position the laser beam for 3D alignment.
- To increase precision, you can zoom in on image sections. Read SECTION 2.6.9 on this.

ⓘ If you want to import geometry data, you also have to perform a 2D alignment and a 3D alignment first. You will find detailed information on geometry import in SECTION 6.3.7.

Measurement system

Depending on the measurement system, there are different ways of carrying out the 2D alignment:

- Manual 2D alignment - See SECTION 5.1.1 on this.
- Automatic 2D alignment - See SECTION 5.1.2 on this.

Also, you can check the quality of the 2D alignment. See SECTION 5.1.3 on this.

5.1.1 Manual 2D Alignment

Start 2D alignment

To perform the 2D alignment, proceed as follows:

1. In the menu bar, select Setup > 2D Alignment (toolbar: ).

   The software maximizes the video window and displays the scanning head control.

2. To delete all existing alignment points, click the video image with the right mouse button (not an alignment point) and select Delete All in the context menu.

Position the laser beam

3. Move the laser beam to an alignment point on the object using the Position icons in the scanning head control. See SECTION 4.1 on this as well.

ⓘ You can also move the laser beam by pressing the Ctrl + Shift + arrow key (↑, ↓, →, ←) key combination.

With a bit of practice, you can position the laser beam more quickly for alignment by using the mouse to move it to approximately the right place first. If necessary, you can then use the scanning head control for fine-tuning as described above. Hold the middle mouse button (mouse wheel) pressed and drag across the video image with it. The laser beam follows the mouse pointer, although not very precisely, as 2D alignment has not yet been carried out. From the third alignment point onwards, the software calculates a preliminary alignment after every alignment point set. If the calculation is successful, then this preliminary alignment is used to position the laser beam if you click or drag with the middle mouse button.

ⓘ If you have allocated a special function to the middle mouse button (e.g. double-click), then this positioning method is not available.

Focus the laser beam

4. You need to focus the laser beam to increase the accuracy when positioning the alignment point.

   With the MSA-600/MSA-650, you can either manually focus the laser beam or via the software with the aid of an optional automatized Z-axis by Polytec.

   With the MSA-500, you can focus the laser beam via the software with the aid of the MSAGeo option. See SECTION 4.1.2.1 on this as well.

   Using the following key combinations, you can automatically focus the laser beam. See SECTION 4.2 on this as well.

Key combination	Description
Ctrl + F	Laser beam will be focused automatically.
Ctrl + E	Laser beam will be quickly focused automatically.

Define alignment points

5. In the live video image, point the mouse precisely at the laser beam and click.

   You can now see a target there.

6. If the laser beam is not precisely in the middle of the target, repeat step 5.
7. Repeat steps 3 to 6 for at least two more alignment points.

ⓘ You can display the triangulation grid which is used for 2D alignment. To do so, click the video image with the right mouse button and select Triangulation Grid in the context menu. Set more alignment points to improve the grid density.

8. Align all desired points.
9. To exit the 2D alignment, select Setup > 2D Alignment in the menu bar (toolbar: ).

Delete alignment points

You can delete alignment points. To do so, continue as follows:

10. To delete an alignment point, click it in the video image with the right mouse button and select Delete in the context menu.

    To delete all alignment points for one scanning head, click the video image with the right mouse button and select Delete All in the context menu.

Cancel 2D alignment

You can cancel the alignment at any time. To do so, select Setup > 2D Alignment in the menu bar (toolbar: ). You will be given a message that the alignment was not finished correctly and you can now decide whether you want to repeat it or not.

5.1.2 Automatic 2D Alignment

You can carry out an automatic 2D alignment. This allows you to define alignment points more quickly and easily.

Start 2D alignment

To perform the 2D alignment, proceed as follows:

1. In the menu bar, select Setup > 2D Alignment (toolbar: ).

   The software maximizes the video window and displays the scanning head control.

2. To delete all existing alignment points, click the video image with the right mouse button (not an alignment point) and select Delete All in the context menu.

Focus the laser beam

3. You need to focus the laser beam to increase the accuracy when positioning the alignment point.

   With the MSA-600/MSA-650, you can either manually focus the laser beam or via the software with the aid of an optional automatized Z-axis by Polytec.

   With the MSA-500, you can focus the laser beam via the software with the aid of the MSAGeo option. See SECTION 4.1.2.1 on this as well.

   Using the following key combinations, you can automatically focus the laser beam. See SECTION 4.2 on this as well.

Key combination	Description
Ctrl + F	Laser beam will be focused automatically.
Ctrl + E	Laser beam will be quickly focused automatically.

Align automatically

4. In the menu bar, select Scan > Automatic 2D Alignment (toolbar of the video window: ).

   The High-Contrast Laser Display (toolbar of the video window: ) is activated.

5. Use the left or middle mouse button to move the laser beam to the required position and release the mouse button.

   The software searches for the laser beam in the video image and defines an alignment point at this position.

6. Repeat this step until you get a valid 2D alignment (at least 3 alignment points).

   The software calculates a preliminary alignment after every alignment point set. If the calculation is successful, then this preliminary alignment is used to position the laser beam if you click or drag with the middle mouse button. After releasing the mouse button, the software searches for the laser beam and defines an alignment point at this position.

ⓘ You can switch off the High-Contrast Laser Display by clicking the icon again or pressing the Escape key. As long as you are working in automatic alignment, the high-contrast laser display is automatically reactivated when setting the next alignment point.

7. For additional alignment points, with the left mouse button, click the video image at the required positions.

   The software positions the laser beam at these positions and defines an alignment point there respectively.

From the third alignment point onwards, the software calculates a preliminary alignment after every alignment point set. If the calculation is successful, then this preliminary alignment is used to position the laser beam if you click or drag with the middle mouse button. After releasing the mouse button, the software searches for the laser beam and defines an alignment point at this position.

For the MSA-600/MSA-650 you can carry out an automatic 2D alignment scan as described in the following.

Perform 2D alignment scan

To perform a 2D alignment scan, proceed as follows:

1. Define at least 4 scan points as described in SECTION 6.2 or SECTION 6.3.

2. In the menu bar, select Setup > 2D Alignment (toolbar: ).

   The software maximizes the video window and displays the scanning head control.

3. In the menu bar, select Scan > 2D Alignment Scan (toolbar: ).

   The High-Contrast Laser Display (toolbar of the video window: ) is activated and the 2D Alignment Scan dialog appears.

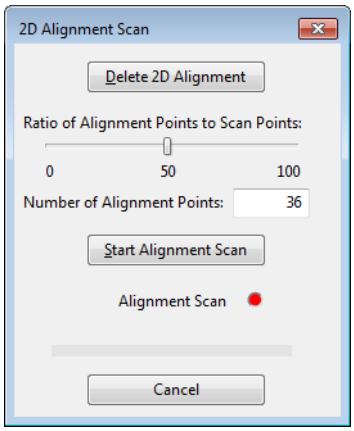


Figure 5.1: 2D Alignment Scan dialog before the alignment scan

4. To delete existing alignment points, click Delete 2D Alignment to delete them.

   There are no alignment points available.

5. Select the Ratio of Alignment Points to Scan Points or enter the desired Number of Alignment Points.
6. Click Start Alignment Scan.

   The software scans the defined scan points, respectively positions and focuses the laser beam and sets alignment points at these positions.
7. You can monitor the progress of the alignment scan in the dialog. To cancel the alignment scan click Cancel.

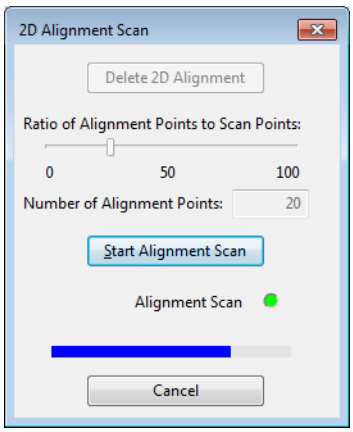


Figure 5.2: 2D Alignment Scan dialog during the alignment scan

   The dialog will be closed after the alignment scan has finished.
8. To exit the 2D alignment, select Setup > 2D Alignment in the menu bar (toolbar:).

5.1.3 Checking 2D Alignment

To check the 2D alignment, proceed as follows:

1. If the scan points are shown in the live video image, click to hide them.
2. Point at the live video image and click a point on the measurement plane.

   The laser beam moves to this point on the object.
3. Check if the position of the laser beam corresponds to that of the point clicked. If this is not the case, repeat the 2D alignment.

ⓘ The live video image is showing a two-dimensional image of the three-dimensional object. For this reason, the laser beam does not precisely hit points which are not on the measurement plane.

4. Repeat steps 2 and 3 for a few more alignment points.

5.2 Carrying out 3D Alignment

3D alignment is always necessary if you want to import or export 3D geometries or would like to generate combined 3D files. 3D alignment determines the position and alignment of the scanning head in the object's coordinate system.

For 3D alignment you need points on the measurement object for which the coordinate description is known in a Cartesian coordinate system. You have to hit these points with the laser. The software will then calculate the 3D alignment.

In the following section you will find information on:

- Overview of 3D alignment for the various measurement systems - See SECTION 5.2.1 on this.
- Information for a successful 3D alignment - See SECTION 5.3 on this.

5.2.1 3D Alignment for the Various Measurement Systems

ⓘ Before you carry out a 3D alignment, set the correct magnification in the Objective list in the scanning head control. The software uses the zoom factor set to calculate the position of the laser on the object under investigation.

MSA-600/ MSA-650

For an 3D alignment with the MSA-600/MSA-650 you will need one of the following options:

- GeoWorks option

or

- Automated Z-axis from Polytec connected and set up in the Options dialog on the Devices page.

MSA-600/MSA-650	
GeoWorks option	**Automated Z-axis from Polytec**
3D alignment for the scanning head system: During 3D alignment for the scanning head system, the coordinate system is automatically defined by the software depending on the vibrational direction of the vibrometer channel.	3D alignment for the object system: In the case of 3D alignment for the object system, the coordinate system is defined by you in the object coordinate system.
Read SECTION 5.2.2 for more information.	Read SECTION 5.2.3 for more information.

MSA-500

For an 3D alignment with the MSA-500 you will need one of the following options:

- GeoWorks option

or

- MSAGeo option

MSA-500	
GeoWorks or MSAGeo option	**MSAGeo option**
3D alignment for the scanning head system: During 3D alignment for the scanning head system, the coordinate system is automatically defined by the software depending on the vibrational direction of the vibrometer channel.	3D alignment for the object system: In the case of 3D alignment for the object system, the coordinate system is defined by you in the object coordinate system.
Read SECTION 5.2.2 for more information.	Read SECTION 5.2.3 for more information.

5.2.2 3D Alignment for the Scanning Head System

During 3D alignment for the scanning head system of the MSA, the coordinate system is automatically defined by the software in the scanning head system depending on the vibrational direction of the vibrometer channel.

ⓘ Before you carry out a 3D alignment, set the correct magnification in the Objective list in the scanning head control.

If you are using the MSA with the GeoWorks option, then you will have to import the 3D geometry after 3D alignment. See SECTION 6.3.7 on this as well.

If you are using the MSA-600/MSA-650 with an automated Z-axis from Polytec or the MSA-500 with the MSAGeo option, then you can carry out a geometry scan instead. See SECTION 6.6 on this as well.

For 3D alignment, proceed as follows:

ⓘ An automatic 3D alignment for the scanning head system is only possible if a valid 2D alignment is available.

2D alignment

1. If you have not carried out a 2D alignment yet, do it now as described in SECTION 5.1.

Open dialog

2. In the menu bar, select Setup > 3D Alignment (toolbar: ).

   The video window is maximized and the 3D Alignment dialog appears.

ⓘ You can show/hide the dialog as follows:

In the menu bar, select Setup > Dialog 3D Alignment.

In the toolbar of the video window, click  .

Use the right mouse button to click the video image and select Dialog 3D Alignment in the context menu.

3. If your MSA-600/MSA-650 is equipped with an automated Z-axis from Polytec or your MSA-500 is equipped with the MSAGeo option, then you can carry out the 3D alignment for the object system as well. See SECTION 5.2.3 on this.

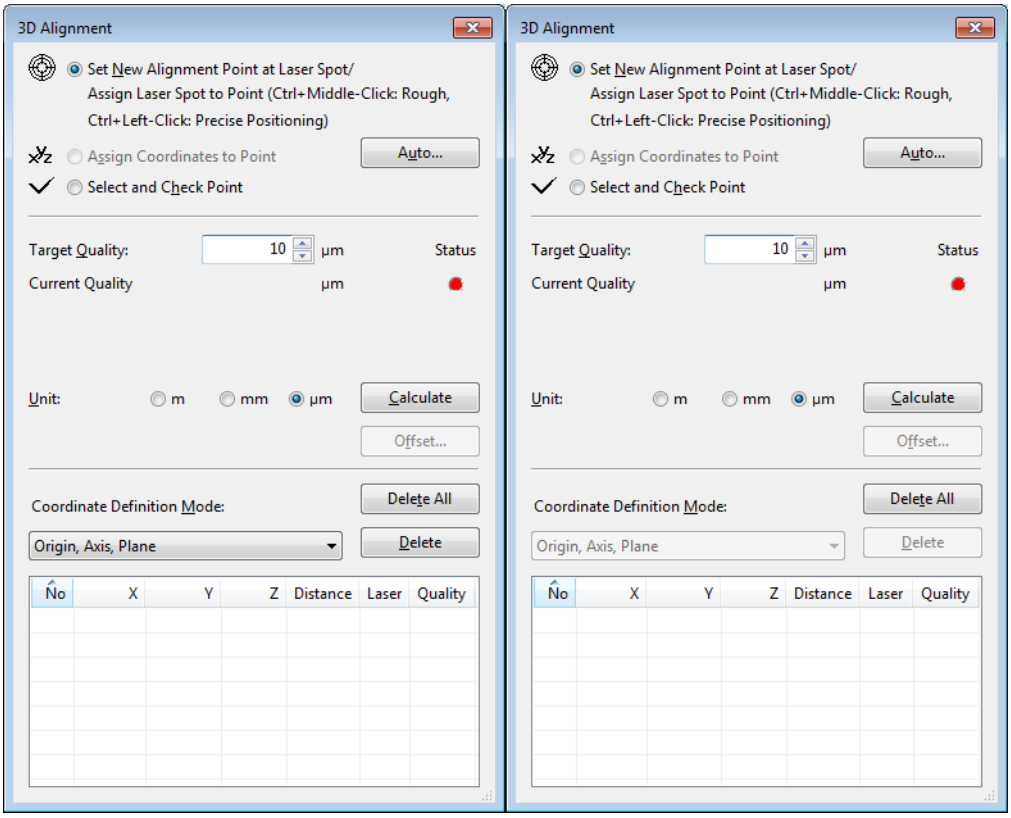


Figure 5.3: 3D Alignment dialog (scanning head system) for the MSA-600/MSA-650 or MSA-500 with MSAGeo option (on the left) and for the MSA with the GeoWorks option (on the right)

Select the unit

4. Select the unit for the coordinates (m, mm or µm).

Specify quality

5. The value in the Target Quality field is the maximum amount by which the alignment points may deviate from the given coordinates.

ⓘ During automatic 3D alignment, the value for the Current Quality is always optimized which is why an entry for the Target Quality does not have any impact.

Carry out and calculate alignment

6. Click Auto.

   The coordinate definition mode is changed to Origin, Axis, Plane.

ⓘ If you are using the MSA with the GeoWorks option, then the coordinate definition mode is already set to Origin, Axis, Plane and can not be changed.

7. The software will inform you that the vibrational direction of the vibrometer channel is used to align the scanning head system and the 3D alignment is calculated and carried out automatically. Confirm this message with Yes.

ⓘ If you still want to change the vibrational direction, click No and change the direction of the vibrometer channel in the Acquisition Settings dialog on the Channels page. See SECTION 7.2.2 on this as well.

The software defines the coordinate system in the scanning head system and automatically carries out the 3D alignment. The origin of the scanning head system can be found in the middle of the video image and at position D = 0 µm.

Determining the coordinates automatically is derived from the objective used and therefore does not have to be adjusted. The values along the vibrational direction selected above are equal to zero for all generated alignment points, as it is only possible to scan the object at right angles to this axis (refer to the following figure).

ⓘ You can sort the entries in the table in ascending or descending order in every column. To do so, click the head of the respective column.

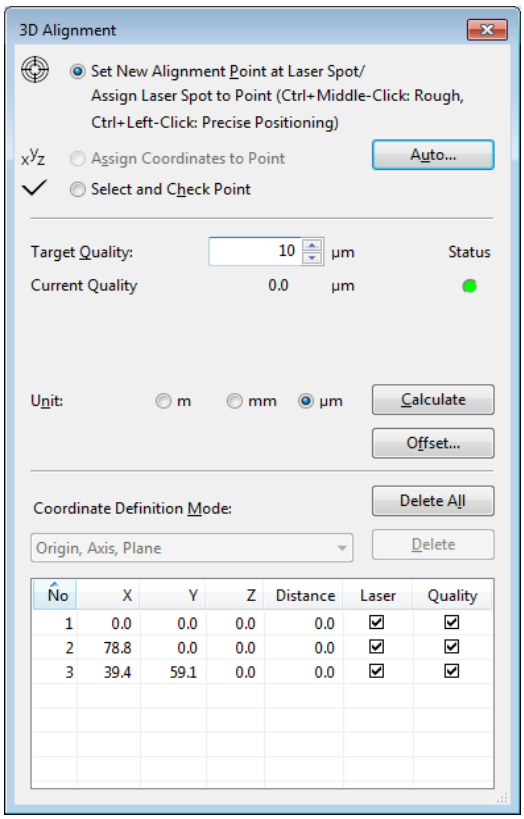


Figure 5.4: Automatic 3D alignment for the MSA (scanning head system)

Check alignment points

8. You can retrospectively check the assignment of the alignment points on the video image with the coordinates in the table. To do so, go to Select and Check Point.
9. Click an alignment point in the video image.

   The corresponding line in the table is marked.
10. Or click a line in the table.

    The corresponding alignment point is displayed blue in the video image.

ⓘ If a valid 3D alignment is available, the laser beam is moved to the position calculated from the X-, Y- and Z-coordinates. You can visually check the quality of the alignment at the respective alignment point. The more accurately the laser beam hits the points on the object, the higher the quality of the alignment.

Check the objective

11. If the position of the laser beam should clearly deviate from the alignment points, check the objective selected or calibrate the objective selected as described in SECTION 3.5.

Delete alignment points

12. To delete all alignment points, click the video image with the right mouse button and select Delete All in the context menu.

    You can also click Delete All in the 3D Alignment dialog.

Offset

You can add an offset if for example you move the object for further measurements or would like to make measurements on different height levels of an uneven object to merge individual measurements (Stitching).

13. As soon as you move the object or the focus level of the measurement beam and want to start making a new measurement, you will have to add an appropriate offset to the coordinates in the corresponding direction. When an offset is added, the software automatically switches to the Points with Free Coordinates mode. Continue as follows:

Add offset

14. Once you have calculated the 3D alignment, you can add an offset to the coordinates by clicking Offset.

    The Add Offset dialog appears.

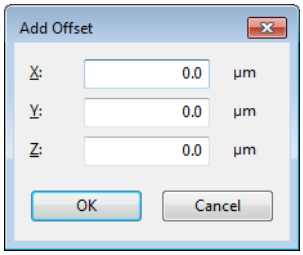


Figure 5.5: Add Offset dialog

15. Enter the desired offset.
16. Click OK.

    The offset is added to all coordinates and the alignment is calculated again.

End alignment

17. To exit the 3D Alignment dialog, select Setup > 3D Alignment in the menu bar (toolbar: ).

5.2.3 3D Alignment for the Object System

The 3D alignment for the object system is only available for the following systems:

- MSA-600/MSA-650 with automated Z-axis from Polytec
- MSA-500 with MSAGeo option

In the case of 3D alignment for the object system of the MSA, the coordinate system is defined by you in the object coordinate system.

ⓘ Before you carry out a 3D alignment, set the correct magnification in the Objective list in the scanning head control.

For 3D alignment, proceed as follows:

2D alignment

1. If you have not carried out a 2D alignment yet, do it now as described in SECTION 5.1.

ⓘ When performing the 2D alignment, already define alignment points near the 3D alignment points marked on the object. This then makes it easier for you to position the laser beam for 3D alignment.

Open dialog

2. In the menu bar, select Setup > 3D Alignment (toolbar: ).

   The video window is maximized and the 3D Alignment dialog appears.

ⓘ You can show/hide the dialog as follows:

In the menu bar, select Setup > Dialog 3D Alignment.

In the toolbar of the video window, click  .

Use the right mouse button to click the video image and select Dialog 3D Alignment in the context menu.

3. If your MSA-600/MSA-650 is not equipped with an automated Z-axis from Polytec or if your MSA-500 is not equipped with the MSAGeo option, see SECTION 5.2.2.

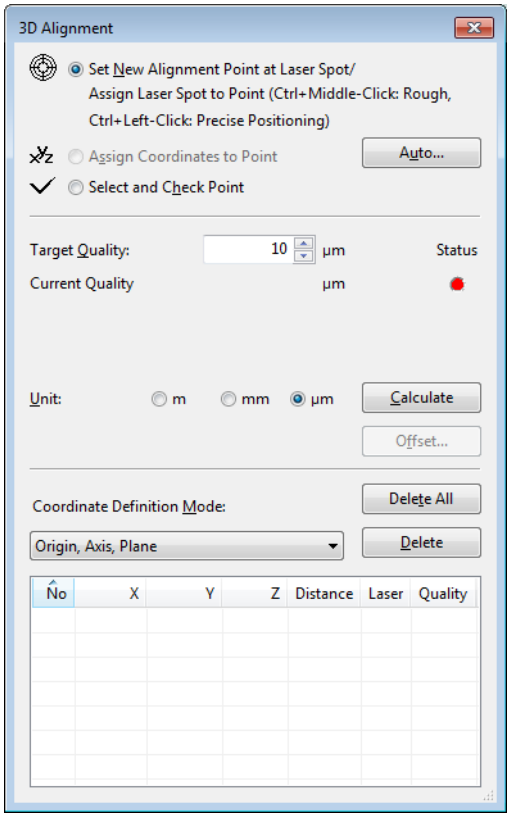


Figure 5.6: 3D Alignment dialog for the MSA-600/MSA-650 or the MSA-500 with MSAGeo option (object system)

Select the unit

4. Select the unit for the coordinates (m, mm or µm).

Position the laser beam

5. Move the laser beam to an alignment point on the object using the Position icons in the scanning head control. See SECTION 4.1 on this as well.

ⓘ You can also move the laser beam by pressing the Ctrl + Shift + arrow key (↑, ↓, →, ←) key combination.

Alternatively, you can also use the mouse directly. Using the middle mouse button (mouse wheel), click in the video image. This will make the laser beam jump to your mouse pointer. To position the beam, hold the middle mouse button pressed and drag with the mouse until the laser beam is positioned on the required point on the object.

ⓘ If you have allocated a special function to the middle mouse button (e.g. double-click), then this positioning method is not available.

6. To increase the accuracy when hitting the required point, zoom the video image. See SECTION 2.6.9 on this as well.

Focus the laser beam

7. You need to focus the laser beam to increase the accuracy when positioning the alignment point.

   With the MSA-600/MSA-650, you can either manually focus the laser beam or via the software with the aid of an optional automatized Z-axis by Polytec.

   With the MSA-500, you can focus the laser beam via the software with the aid of the MSAGeo option. See SECTION 4.1.2.1 on this as well.

   Using the following key combinations, you can automatically focus the laser beam. See SECTION 4.2 on this as well.

Key combination	Description
Ctrl + F	Laser beam will be focused automatically.
Ctrl + E	Laser beam will be quickly focused automatically.

Define alignment points

8. Once you have found the precise position of the required point on the object, select  Set New Alignment Point at Laser Spot/Assign Laser Spot to Point.
9. In the video image, click the point which the laser beam is currently at.

   An alignment point will be defined there. The alignment point is marked with a black circle  and a line is added to the table which is automatically assigned to the alignment point. In the Distance column, the distance measured to this alignment point is entered. The Laser column shows whether the point has been defined.

ⓘ When defining each alignment point the software automatically focuses the laser beam and from the position of the z-axis in the measurement microscope, determines the coordinates of the alignment point.

10. Keep the Control key pressed and first click the video image with the middle mouse button (mouse wheel) and then with the left mouse button at the position where you want to define the next alignment point.

    The software uses VideoTriangulation to position the laser beam at this point and defines an alignment point there respectively.

11. Define three to twenty alignment points in this way. You will find information on appropriate arrangement of the alignment points in SECTION 5.3.2.

Delete alignment points

You can delete alignment points. To do so, continue as follows:

12. To delete an alignment point, click it in the video image with the right mouse button and select Delete in the context menu.

    To delete all alignment points for one scanning head, click the video image with the right mouse button and select Delete All in the context menu.

13. You can also mark the corresponding line of the table in the 3D Alignment dialog and click Delete.

Define the object coordinate system

You have the following options to define the object coordinate system (object system):

Points with Free Coordinates: Mark at least three points and enter all the coordinates for these points.

Origin, Axis, Plane: Mark the origin, any point of your choice along the positive x-, y- or z-axis and any point in the positive half plane limited by the selected axis (e.g. 0, +x, x/+y).

Three Points on Axes: Mark any point of your choice on the positive x-, y- and z-axis respectively.

Define and assign coordinates

In the following, the Points with Free Coordinates option is described in detail. Both other alternatives are briefly explained to you further on under Alternative coordinate definition.

1. Select Points with Free Coordinates from the Coordinate Definition Mode list.
2. Click the alignment point with the right mouse button and select Point with free coordinates in the context menu.

   The Assign Coordinates dialog appears.

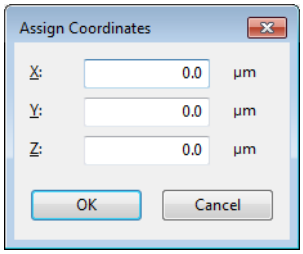


Figure 5.7: Assign Coordinates dialog

3. Enter the alignment point coordinates and click OK.

   XYZ will appear next to the point.
4. Repeat steps 2 and 3 until you have assigned coordinates to at least three alignment points.

ⓘ If you have defined more than three alignment points, then you do not have to assign coordinates to all the alignment points. However, only alignment points which have had coordinates assigned to them will be used to calculate the 3D alignment. For further information, please read SECTION 5.3.

Specify quality

5. In the Target Quality field, enter the maximum value by which the alignment points may deviate from the given coordinates (e.g. 10 µm).

Calculate the alignment

6. Once you have defined the alignment points and have assigned coordinates to them, you can calculate the alignment. To do so, click Calculate.

   The software will calculate the alignment. The Current Quality of this alignment is shown in the dialog. If the 3D alignment has been successful, the Status display will turn green. If the status display lights up red, then the alignment is not valid.

ⓘ If the value for the current quality exceeds the value for the target quality, then a message will appear. The Status display will turn yellow. You can use the 3D alignment anyway.

7. You can check the quality for the individual alignment points. The alignment points for which the quality value is lower than the value in the Target Quality field will have a tick in the Quality column.

ⓘ You can sort the entries in the table in ascending or descending order in every column. To do so, click the head of the respective column.

8. You have successfully carried out the 3D alignment.

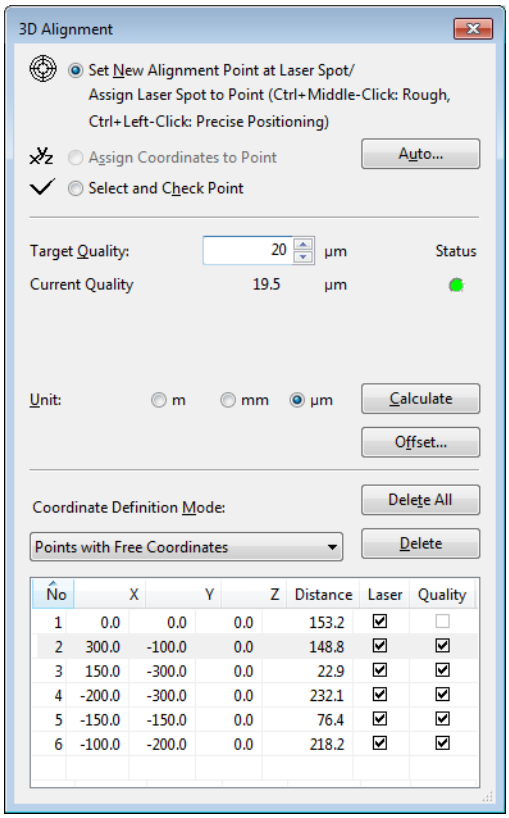


Figure 5.8: 3D alignment successfully carried out for the MSA (object system)

Reassign the laser position

9. You can also reassign new laser positions to existing alignment points to increase the quality of the measurement results. To do so, select  Set New Alignment Point at Laser Spot/Assign Laser Spot to Point.

10. Move the laser beam to the point on the object and left-click the alignment point in the video image.

The software focuses the laser beam again and from the position of the z-axis in the measurement microscope, calculates the coordinates of the alignment point. The new position and d coordinates are used to recalculate the alignment.

Check alignment points

11. You can retrospectively check the assignment of the alignment points on the video image with the coordinates in the table. To do so, go to Select and Check Point.
12. Click an alignment point in the video image.

    The corresponding line in the table is marked.
13. Or click a line in the table.

    The corresponding alignment point is displayed blue in the video image.

ⓘ If a valid 3D alignment is available, the laser beam is moved to the position calculated from the X-, Y- and Z-coordinates. You can visually check the quality of the alignment at the respective alignment point. The more accurately the laser beam hits the points on the object, the higher the quality of the alignment.

Offset

You can add an offset if for example you move the object for further measurements or would like to make measurements on different height levels of an uneven object to merge individual measurements (Stitching).

14. As soon as you move the object or the focus level of the measurement beam and want to start making a new measurement, you will have to add an appropriate offset to the coordinates in the corresponding direction. When an offset is added, the software automatically switches to the Points with Free Coordinates mode. Continue as follows:

Add offset

15. Once you have calculated the 3D alignment, you can add an offset to the coordinates by clicking Offset.

    The Add Offset dialog appears.

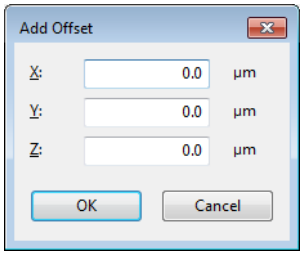


Figure 5.9: Add Offset dialog

16. Enter the desired offset.
17. Click OK.

    The offset is added to all coordinates and the alignment is calculated again.

End alignment

18. To exit the 3D Alignment dialog, select Setup > 3D Alignment in the menu bar (toolbar:).

Cancel alignment

You can cancel the alignment at any time. To do so, select Setup > 3D Alignment (toolbar: ). You will be given a message that the alignment was not finished correctly and you can now decide whether you want to repeat it or not.

Alternative coordinate definition

As an alternative to Points with Free Coordinates, the following coordinate definitions are available to you. They are both described in the following.

- Coordinate definition: Origin, Axis, Plane
- Coordinate definition: Three Points on Axes

ⓘ The Target Quality has to be entered also in these coordinate definition modes. The Current Quality determined is always 0. If you have defined more than three alignment points, the excess alignment points are not taken into consideration when calculating the 3D alignment.

Origin, Axis, Plane

To define the object coordinate system this way, replace steps 1 to 4 in the Define and assign coordinates section with the following:

1. Select Origin, Axis, Plane from the Coordinate Definition Mode list.
2. With the right mouse button, click the alignment point which you want to define as the origin and select Origin in the context menu.

   A 0 will appear next to the alignment point. The coordinates of the origin (0, 0, 0) are displayed in the table.
3. With the right mouse button, click the alignment point which you want to define on the x-axis for example and select Point on +x-axis in the context menu.

   A +x will appear next to the point which will have an arrow drawn through it. The distance and coordinates (+x, 0, 0) of the point are displayed in the table.
4. With the right mouse button, click the alignment point which you want to define in a half-plane and select for example Point on x/+y-plane in the context menu.

   An x/+y will appear next to the point. The distance and coordinates (x, +y, 0) of the point are displayed in the table.
5. Now proceed with the 3D alignment as described from step 5 onwards in the Use mirror section above.

3 Points on Axes

To define the object coordinate system this way, replace steps 1 to 4 in the Define and assign coordinates section with the following:

1. Select Three Points on Axes from the Coordinate Definition Mode list.
2. Use the right mouse button to click the alignment point which the x-axis of the coordinate system is to run through and select Point on +x-axis in the context menu.

   A +x will appear next to the point which will have an arrow drawn through it.
3. Repeat step 2 for one more alignment point each on the y and z-axis.

   A +y or +z respectively will appear next to the points and an arrow will run through them.
4. Now proceed with the 3D alignment as described from step 5 onwards in the Use mirror section above.

5.3 Further Information on Successful 3D Alignment

In the following sections you will receive information on how you can optimally carry out the 3D alignment and what you need to take into consideration.

- Edit the 3D coordinates - See SECTION 5.3.1 on this.
- Special features of 3D alignment - See SECTION 5.3.2 on this.

5.3.1 Editing 3D Alignment Points

If you have defined 3D alignment points as described in SECTION 5.2.3, you can edit them. While doing so, you can edit the following functions:

- Defining object system alignment points as alignment points - See SECTION 5.3.1.1 on this.
- Changing coordinate definition mode - See SECTION 5.3.1.2 on this.
- Correcting coordinates of alignment points in Points with Free Coordinates definition mode - See SECTION 5.3.1.3 on this.

5.3.1.1 Defining Object System Alignment Points as Alignment Points

To define an object system alignment point (an alignment point defined with the aid of a coordinate definition mode) as a normal alignment point, click it with the right mouse button and select Alignment Point in the context menu. The alignment point remains but it is no longer an object system alignment point. The special identification which identifies it as an object system alignment point disappears as well.

5.3.1.2 Changing Coordinate Definition Mode

Independently of which coordinate definition mode you have defined the alignment points in (Points with Free Coordinates; Origin, Axis, Plane; Three Points on Axes), you can still retrospectively change the definition mode. To do so, proceed as follows:

1. In the 3D Alignment dialog, select another definition mode.

   The object system alignment points become alignment points without identification. If a valid 3D alignment had already been carried out however, the coordinates of the alignment points are retained.

2. With the right mouse button, click the alignment point which you want to modify and select the corresponding command in the context menu.

5.3.1.3 Correcting Coordinates of Alignment Points with Free Coordinates

To modify free coordinates of alignment points which have already been set, proceed as follows:

1. Click an alignment point with the right mouse button and select Point with Free Coordinates in the context menu.

   The Assign Coordinates dialog appears.

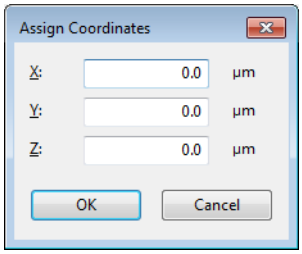


Figure 5.10: Assign Coordinates dialog

2. Enter the correct values and click OK.

5.3.2 Special Features of 3D Alignment

ⓘ In this section, the points on the object which are used for alignment are called alignment points.

Positioning Alignment Points Appropriately

If you are working with the MSA-600/MSA-650 and an automated Z-axis from Polytec, then the laser position and the position of the Z-axis are used to calculate the position and orientation of the object coordinate system.

If you are working with the MSA-500 and the MSAGeo option, then the laser position and the position of the z-axis in the measurement microscope are used to calculate the position and orientation of the object coordinate system.

At least three alignment points are necessary for 3D alignment. The alignment points have been optimally selected if the surface of the triangle formed by these points in space is as large as possible.

So it is particularly important that the alignment points in three-dimensional space are not along one line, as this would mean the surface of the triangle formed would be zero. Despite this, the software can successfully calculate 3D alignment. The calculated position of the object coordinate system however is then one of any number of solutions and thus undefined.

When using the Points with Free Coordinates coordinate definition mode, you can set up to twenty alignment points with free coordinates. By setting additional alignment points, errors in the coordinates when positioning the laser and the z-axis have less effect on the results of the 3D alignment. Setting additional alignment points is particularly recommended if the uncertainty of the coordinates is greater than 10 µm.

Merging Individual Measurements (Stitching)

With the aid of 3D alignment, areas of large measurement objects can be acquired one after the other and the individual results can be retrospectively combined. If you move the object between the individual measurements, then you need to carry out a new 3D alignment in the new position and/or add a corresponding offset.

5.4 Path Length Alignment (only MSA-650)

ⓘ Before the path length alignment, the laser must stabilize. For this reason, switch on the laser beam at least 20 seconds before starting a measurement (tick Laser check box in the scanning head control). Carry out the alignment afterwards.

For a path length alignment, proceed as follows:

Align path length

1. Position your MEMS component underneath the objective.
2. Adjust the exposure time (Shutter Speed) of the camera. If necessary, activate Automatic Exposure Control as well as Automatic Contrast Control in the Display Properties dialog. See SECTION 4.3.2 on this.
3. Focus on the required surface of your MEMS component. If necessary, adjust the settings for the objective.
4. Position the laser on the area of the MEMS component, where you want to measure later.
5. Click Alignment in the Path Lgth. field in the Scanning Head Control.

   The reference mirror/reference arm is moved in the range of ±2 mm by the path length stored for the objective in the Objective Calibration dialog. The noise power of the vibrometer channel is measured and recorded. The measured noise is inverted and normalized and displayed in a special analyzer.

ⓘ If you want to use the whole travel range, keep the Shift key pressed while clicking Alignment.

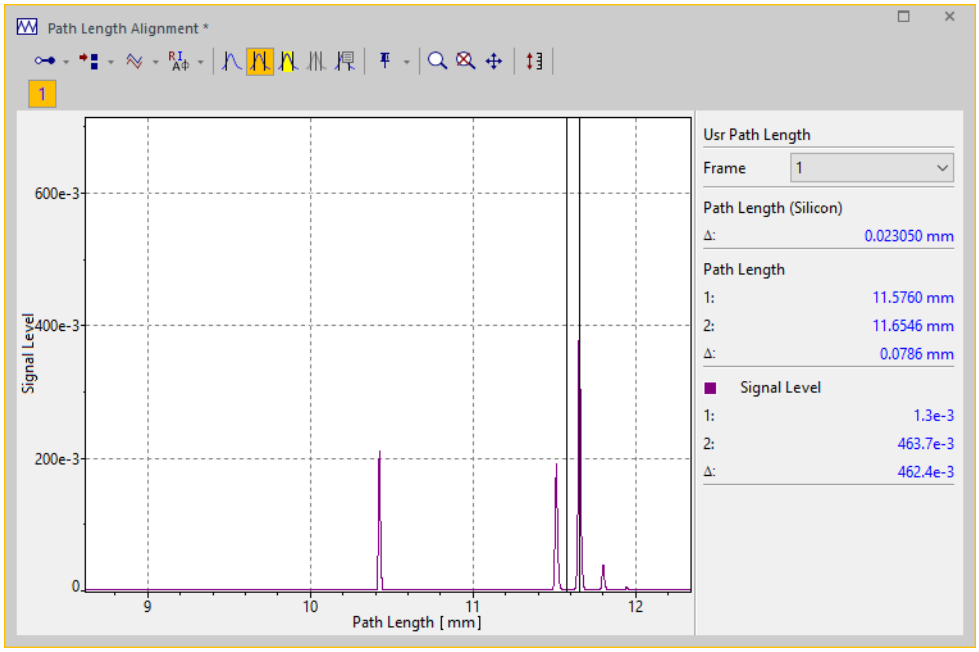


Figure 5.11: Analyzer for path length alignment

Single surfaces of the component appear as maxima of the signal level. However, maxima can also occur due to multiple reflections in the component.

ⓘ You will have to decide for yourself which of the maxima belongs to the focused surface based on your sample setup and the number of surfaces penetrated.

For objectives with large magnification (from about 20x), the focused surface provides the largest signal level, due to additional confocal suppression.

6. If necessary, you can measure the distances of your surfaces using the differential cursor.

Save

7. To save the measured path length alignment as a file in Path Length Data (*.pla) format, select File > Save As in the menu bar.

   The Save As dialog appears.

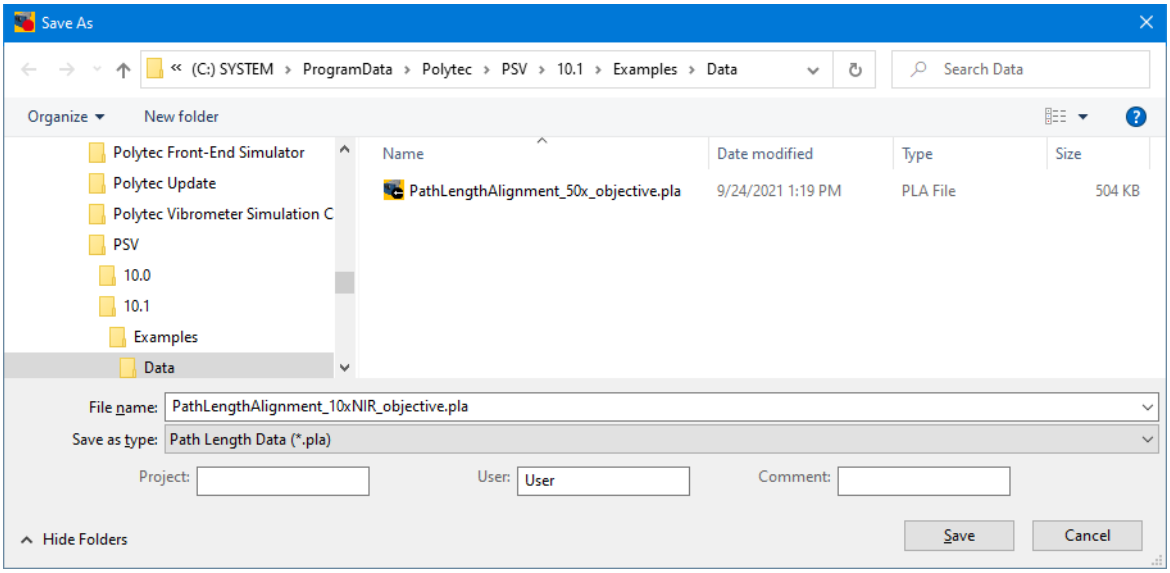


Figure 5.12: Save As dialog for path length alignment

8. Navigate to the saving location and enter the file name.
9. Click Save.

Apply values

10. Select Analyzer > Differential Cursor in the menu bar (toolbar of the analyzer: ).
11. Position the differential cursor in front of and behind the maximum belonging to your surface.

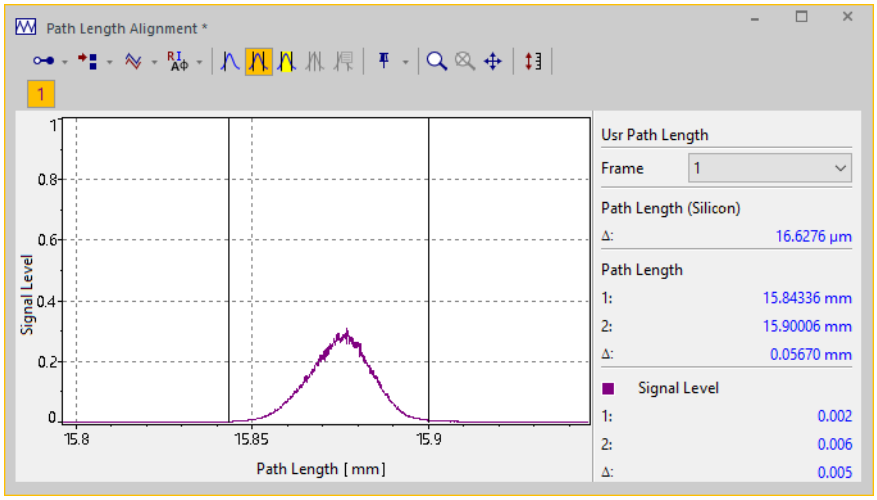


Figure 5.13: Differential cursor to limit the path length range

This defines the travel range for searching the exact path length.

12. Click the analyzer with the right mouse button and select Apply Path Length Range in the context menu.

    The values are applied to the Path Lgth. field in the scanning head control (Min, Max).

13. Click Search in the Path Lgth. field in the Scanning Head Control.

    The position of the signal level maximum is determined in the selected range. The signal level display shows full scale.

ⓘ If the signal level display does not show full scale, the path length alignment was not successful. In this case, repeat the alignment.

6 Defining and Editing Scan Points

Before a scan you start by defining the scan points. The software offers the following functions for this:

Procedure for scan point definition - You will get information on defining scan points (APS mode = Setup > Define Scan Points). See SECTION 6.1 on this.

Define scan points as standard objects - You draw geometric figures on the live video image as you would in graphics software. See SECTION 6.2 on this.

Define scan points as point objects - You do not draw figures but edit single scan points and their connections. See SECTION 6.3 on this.

Common functions - Some functions are available for different modes or object types (APS/not APS or standard objects/point objects). See SECTION 6.4 on this.

Edit scan points (not APS) - You can edit scan points if you have already finished scan point definition (APS mode). See SECTION 6.5 on this.

Make a distance measurement - If you are using the MSA-600/MSA-650 with an automated Z-axis from Polytec, you can still determine the focus point position if the object geometry is unknown. If you are using the MSA-500 with the MSAGeo option, you can determine the position of the focus point on the z-axis in the measurement microscope even if you do not know the geometry of the object. See SECTION 6.6 on this.

Teach-in focus values - You have the option of assigning focus values to the scan points. This makes it easier to focus if several objects are to be scanned which are at different distances from the scanning head or if the object has three-dimensional geometry. See SECTION 6.7 on this.

Undo and redo

You can undo up to 50 operations and redo them. In the menu bar, select Edit > Undo or Edit > Redo (toolbar:  or ).

Standard object

You define standard objects by drawing rectangles, ellipses, polygons and lines. You can set up the grid for every figure individually and it can be polar, hexagonal and irregular.

Point objects

You define point objects by defining and editing individual scan points and their connections rather than figures. You can convert standard objects to point objects and refine them.

Overview The following table contains an overview of the geometries which you can define using the MSA.

System	Standard objects	Standard objects with geometry scan	Point objects	Point objects with geometry scan	Point objects with geometry import
MSA-600/ MSA-650	2D	-	2D	-	3D
MSA-600/ MSA-650 (with automated Z-axis)	2D	3D	2D	3D	3D
MSA-500	2D	-	2D	-	3D
MSA-500 (with MSAGeo option)	2D	3D	2D	3D	3D

ⓘ With the MSA, 2D geometries are created for standard and point objects. With a valid 3D alignment you can also import 3D geometries.

You can teach-in scan points if you are using an automated Z-axis from Polytec (MSA-600/MSA-650) or if you are using the MSA geometry unit (MSA-500). To do so, proceed as follows:

1. Define scan points as standard or point objects as described in SECTION 6.1, SECTION 6.2, and SECTION 6.3.
2. Exit scan point definition and perform a geometry scan as described in SECTION 6.6.
3. Switch to editing point objects. In doing so, the scan points and the measured coordinates are retained.

6.1 Procedure for Scan Point Definition

Overview

To define scan points, change to the scan point definition (APS mode) as described in the following and select the respective object type:



Define standard object



Define point object
(Only with APS Professional, GeoWorks, or GeoPro option)



Convert standard objects to point objects
(Only with APS Professional, GeoWorks, or GeoPro option)

Define

To define scan points, proceed as follows:

1. If necessary, align the positions on the live video image with the position of the laser on the measurement plane or carry out a 3D alignment. See SECTION 5.1 or SECTION 5.2 on this.

2. In the menu bar, select Setup > Define Scan Points (toolbar: ).

   You are in APS mode. The software maximizes the video window and displays a graphics toolbar. Areas of the live video image can be crosshatched. These areas are outside the range you can scan.

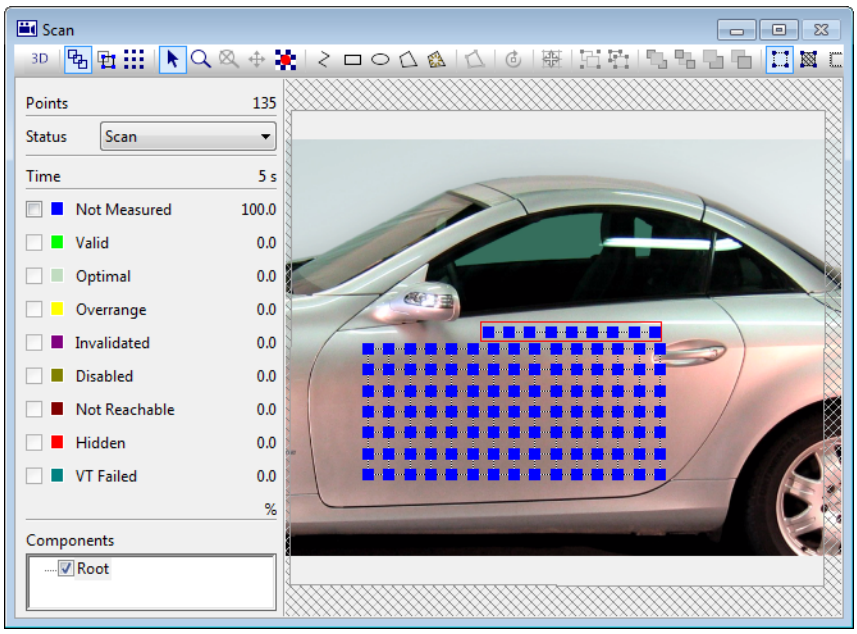


Figure 6.1: Available range for defining the scan points

ⓘ The available scan range depends on the scanning head and correct 2D or 3D alignment.

3. Select the required object type (standard object or point object).

ⓘ For point objects, you require the GeoWorks option (requires APS Professional).

4. Define scan points as described in SECTION 6.2 and SECTION 6.3.

ⓘ To check the signal quality of scan points in the analyzer, you can execute single or continuous measurements.

You can see how many scan points you have defined in the live video image legend. Below that you see the estimated time needed for the scan. The estimate is based on the time for data acquisition at all scan points and the settling time of the scanner mirrors.

ⓘ You can also define scan points in the crosshatched areas. Scan points, which cannot be reached, are not scanned.

5. In the menu bar, select Setup > Define Scan Points (toolbar: ).

   You exit scan point definition (APS mode).

Zoom in

To position the scan points exactly, you can zoom in on the video image. To define scan points outside the video image, you can zoom out. Read SECTION 2.6.9 on this.

Check

As the live video image shows a two-dimensional image of the three-dimensional object, it is possible that the laser beam is not hitting some of the scan points precisely. So check the scan points as follows:

1. Point with the mouse at the live video image and click a scan point.

   The laser beam moves to this point on the object.

2. Check if the position of the laser beam on the object corresponds to that on the live video image.

3. Repeat steps 1 and 2 for a few more scan points.

6.2 Defining Scan Points as Standard Objects

ⓘ If you are using an automated Z-axis from Polytec (MSA-600/MSA-650) or the MSA geometry unit (MSA-500), and define scan points as standard objects, 3D coordinates have already been assigned to these points. These coordinates were calculated from the previous 3D alignment. In other cases, import the scan points as point objects.

ⓘ You can also define standard objects outside the video image. These are then given the Not Reachable scan status.

Icons and functions

To define standard objects, draw geometric figures. The following figures and functions for editing and arranging figures are possible:

Icon	Menu navigation	Function
	Scan > Line	Draw lines - See SECTION 6.2.1 on this.
	Scan > Rectangle	Draw rectangles and squares - See SECTION 6.2.2 on this.
	Scan > Ellipse	Draw ellipses and circles - See SECTION 6.2.3 on this.
	Scan > Polygon	Draw polygons - See SECTION 6.2.4 on this.
	Scan > Automatic Polygon	Automatically generate polygons - See SECTION 6.2.5 on this.
	Scan > Edit Polygon	Change polygon - See SECTION 6.2.6 on this.
	Scan > Rotate	Rotate objects - See SECTION 6.2.7 on this.
	Scan > Move Grid	Move scan point grid - See SECTION 6.2.11 on this.

Apart from that you can use the following functions for figures:

- Arrange figures in different levels - See SECTION 6.2.8 on this.
- Duplicate already defined figures - See SECTION 6.2.9 on this.
- Show or hide scan points on figures - See SECTION 6.2.10 on this.
- Set up the scan point grid of a figure - See SECTION 6.2.11 on this.
- Define the number of vertex points of ellipses - See SECTION 6.2.12 on this.
- Define the density of scan points on lines - See SECTION 6.2.13 on this.

To define standard objects, click  in the graphics toolbar. The Object Properties dialog appears.

ⓘ The Object Properties dialog changes its name depending on which object you have drawn (e.g. Rectangle Properties, Polygon Properties, etc.).

 : Errors when entering numerical values in the dialog are displayed by the software. Move the mouse pointer onto the icon to get a detailed error message.

6.2.1 Drawing Lines

To draw a line, proceed as follows:

1. Click  .
2. Point at the live video image.

   The mouse pointer becomes a cross.
3. Click the corner points of the line.
4. Double-click the last corner point.

6.2.2 Drawing Rectangles and Squares

To draw a rectangle or a square, proceed as follows:

1. Click  .
2. Point at the live video image.

   The mouse pointer becomes a cross.

Rectangle

You can draw a rectangle in the following ways:

1. Point at a corner of the rectangle.
2. Press the mouse button and drag to the diagonally opposite corner.

or

1. Point at the center of the rectangle.
2. Press the Control key and the mouse button simultaneously and drag across to a corner point.

Square

You can draw a square in the following ways:

1. Point at a corner of the square.
2. Press the Shift key and the mouse button simultaneously and drag to the diagonally opposite corner.

or

1. Point at the center of the square.
2. Press the Control key, Shift key and the mouse button simultaneously and drag across to a corner point.

6.2.3 Drawing Ellipses and Circles

To draw an ellipse or a circle, proceed as follows:

1. Click  .
2. Point at the live video image.

   The mouse pointer becomes a cross.

Ellipse

You can draw an ellipse in the following ways:

1. Point at a corner of the rectangle enclosing the ellipse.
2. Press the mouse button and drag to the diagonally opposite corner.

or

1. Point at the center of the ellipse.
2. Press and hold the Control key and the mouse button and drag across to a corner point of the rectangle enclosing the ellipse.

Circle

You can draw a circle in the following ways:

1. Point at a corner of the square enclosing the circle.
2. Press the Shift key and the mouse button simultaneously and drag to the diagonally opposite corner.

or

1. Point at the center of the circle.
2. Press the Control key, Shift key and the mouse button simultaneously and drag across to a corner point of the square enclosing the circle.

6.2.4 Drawing Polygons

To draw a polygon, proceed as follows:

1. Click  .
2. Point at the live video image.

   The mouse pointer becomes a cross.
3. Click at least three corners of the polygon.
4. Double-click the last corner point.

   The software connects it to the first corner.

6.2.5 Automatically Drawing Polygons

You can automatically generate polygons. For this purpose, the software identifies the color of the point you have clicked and determines the area that contains this point and other points of the same color. The edge of the detected area is approximated with an APS polygon.

To automatically generate a polygon, proceed as follows:

1. Click  .

   The Automatic Polygons dialog appears. The mouse pointer becomes a cross.
2. If already known, select the values for the parameters.
3. Click the desired point of the object on the video image.

   The software generates the polygon.
4. Expand existing polygons or add new ones by holding the Control key pressed and clicking additional points on the video image.

   The new defined areas are added to the existing ones.
5. Reduce or delete existing polygons by holding the Alt key pressed and clicking additional points on the video image.

   The new defined areas are subtracted from the existing ones.
6. If necessary, change your settings.

   The changes are reflected immediately being visible in the video image.

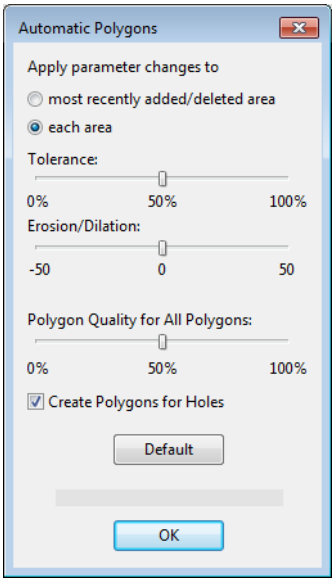


Figure 6.2: Automatic Polygons dialog

Apply parameter changes to: Select which area you want to adapt the settings for.

You can apply the settings to the following areas:

- Most recently added/deleted area
- Each area

Tolerance: Using the slider select the tolerance in % for determining similar colors.

Erosion/Dilation: Using the slider select the corresponding filter (Erosion: –50…0, Dilation: 0…50) to expand or reduce respectively the selected areas.

Polygon Quality for All Polygons: Using the slider select the quality in % and thus the number of vertex points for the generated polygons.

Create Polygons for Holes: Tick the check box if objects bordered by the polygons include areas of different color (e.g. holes in a plate) that should be left blank for generating polygons.

Defaults: Click here to reset the settings to default values.

6.2.6 Changing the Shape of a Figure

To retrospectively add corner points to a figure or to move individual corner points, proceed as follows:

1. Select the figures as described in SECTION 6.4.3.
2. Click  .

Insert corner point

To insert a corner point, proceed as follows:

1. Point at the edge of the figure.

   The mouse pointer becomes an insertion icon  .
2. Click the edge of the figure.

   The software inserts a corner point at this place.

Move a corner point

To move a corner point, proceed as follows:

1. Point at the corner.

   The mouse pointer becomes an arrow icon  .
2. Press the mouse button and drag in the direction required.

6.2.7 Rotating Figures

To rotate figures, proceed as follows:

1. Select the figures as described in SECTION 6.4.3.
2. Click  .
3. Point at one of the white markers which mark the edge of the figures.

   The mouse pointer become a rotation icon  .
4. Press the mouse button and drag in the direction you want to rotate in. The anchor point is the center of the figures.

6.2.8 Arranging Figures

The arrange figures in different levels, select the figure as described in SECTION 6.4.3 and click the respective icon (refer to the following table).

Icon	Menu navigation	Function
	Scan > Object to Foreground	The figure moves to the foreground.
	Scan > Object Before	The figure moves one level forwards.
	Scan > Object to Background	The figure moves to the background.
	Scan > Object Behind	The figure moves one level backwards.

6.2.9 Duplicating Figures

To duplicate previously defined figures, proceed as follows:

1. Select the figure as described in SECTION 6.4.3.
2. Press the D key on the keyboard or in the menu bar, select Edit > Duplicate.

   A copy of the figure is created and is displayed slightly staggered to the original.
3. To teach-in an offset to the original, position the copy as desired. To do so, drag the figure with the mouse.
4. If you want to define geometries for regular repeated structures, you can create additional copies at the desired distance. To do so, repeatedly press the D key or select Edit > Duplicate.

   Additional copies of the figure are created with the offset taught-in.

6.2.10 Showing Scan Points on Figures

ⓘ The Object Properties dialog changes its name depending on which object you have drawn (e.g. Rectangle Properties, Polygon Properties, etc.).

 : Errors when entering numerical values in the dialog are displayed by the software. Move the mouse pointer onto the icon to get a detailed error message.

With the icons in the Object Properties dialog, you can show or hide scan points on the figures. To do so, select the figures as described in SECTION 6.4.3 and click the corresponding icon (refer to the following list):

 You are showing scan points inside and on the edges of the figures (not active for lines).

 You are showing scan points only within the figures (not active for lines).

 You are only showing scan points on the edges of the figures.

 You are hiding the scan points inside and on the edges of the figures.

6.2.11 Setting up the Grid

ⓘ The Object Properties dialog changes its name depending on which object you have drawn (e.g. Rectangle Properties, Polygon Properties, etc.).

 : Errors when entering numerical values in the dialog are displayed by the software. Move the mouse pointer onto the icon to get a detailed error message.

When defining standard objects, every figure has got its own grid To set up the grid, select the figure as described in SECTION 6.4.3 and adapt the following settings in the Object Properties dialog:

- Select the type of grid
- Set the density of the grid points
- Rotate the grid

You can not set up a grid for lines and figures which only have scan points on the edge. See SECTION 6.2.10 on this as well.

Select the type of grid

To select the type of grid, click one of the following icons:

 You are selecting a rectangular grid.

 Your are selecting a circular grid (polar grid).

 You are selecting a hexagonal grid.

 You are selecting an irregular grid. Once you have shown all scan points (inside and on the edges of the figure), there will be an irregular distribution of scan points.

Set density of grid points

You enter the density of the grid points in the Object Properties dialog in the respective Density fields. You can also enter decimals.

ⓘ The density does not depend on the dimensions of the figure, it always refers to the height of the live video image.

For a figure with a right-angled grid, the Density X and Density Y fields appear in the dialog. The density of the grid points $d_{x,\,y}$ determines the distance between the grid points $a_{x,\,y}$ as follows:

$$a_{x,\,y} = \frac{h_y}{d_{x,\,y}}$$ Equation 6.1

h_y … Height of the live video image

For a figure with a polar grid, the Density Radial and Density Angle fields appear. The radial density d_{radial} determines the distance between the grid points on the beams a_{radial} as follows:

$$a_{radial} = \frac{h_y}{d_{radial}}$$ Equation 6.2

h_y… Height of the live video image

The tangential density d_{tang} is equal to the number of grid points on the circles.

For a figure with a hexagonal grid or an irregular grid, the Density field appears. The density of the grid points d determines the distance between neighboring grid points as follows:

$$a = \frac{h_y}{d}$$ Equation 6.3

h_y… Height of the live video image

ⓘ The software approximately calculates the average distance of the scan points on the object and shows this distance next to the Density field.

Rotate grid

To rotate the grid relatively to its figure, enter the rotation angle into the Rotation field of the Grid group.

Move grid

To move the grid within its figure, proceed as follows:

1. Select the figures as described in SECTION 6.4.3.
2. Click  .
3. Point at the grid of the figure.

   The mouse pointer becomes an arrow cross icon  .
4. Press the mouse button and drag in the direction you want to move the grid in.

6.2.12 Setting Vertex Points of Ellipses

ⓘ The Object Properties dialog changes its name depending on which object you have drawn (e.g. Rectangle Properties, Polygon Properties, etc.).

 : Errors when entering numerical values in the dialog are displayed by the software. Move the mouse pointer onto the icon to get a detailed error message.

The software approximates ellipses by using polygons (at least quadrangles). You set the number of vertex points, proceed as follows:

1. Select the ellipse as described in SECTION 6.4.3.

   In the Ellipse Properties dialog in the Line group, the Vertex Points field appears.

2. Enter the required number of vertex points in the field.

   The number of vertex points then also applies for all ellipses which you subsequently draw.

ⓘ If you define scan points on the edge of an ellipse, every vertex point becomes a scan point. See SECTION 6.2.10 on this as well.

6.2.13 Setting Properties of Lines

ⓘ The Object Properties dialog changes its name depending on which object you have drawn (e.g. Rectangle Properties, Polygon Properties, etc.).

 : Errors when entering numerical values in the dialog are displayed by the software. Move the mouse pointer onto the icon to get a detailed error message.

You can not set up a grid for lines and figures which only have scan points on the edge. However, you can set the density of the scan points on the lines.

To set the density of the scan points on the lines, proceed as follows:

Resolution

1. Select the figure as described in SECTION 6.4.3.

   In the Object Properties dialog in the Line group, the Resolution field appears. The resolution reflects the number of scan points on one line, the length of which corresponds to the height of the video image This also applies to rectangles, polygons and ellipses whose scan points are only on the edge.

2. Enter a value.

Width

For lines, the Width field also appears. The width determines the size of a shadow around the line. Within the shadow, the scan points that are behind the line are hidden. This means, for example, you can delete scan points on cables which obscure the view of the object.

3. Enter a value.

6.3 Defining Scan Points as Point Objects

In order to define point objects, you need the GeoWorks option (requires APS Professional option). With point objects, you do not draw figures but edit single scan points and their connections. While doing so, the following functions are available to you:

- You can define single scan points with the software. See SECTION 6.3.1 on this.
- You have several options to connect scan points. See SECTION 6.3.2 on this.
- You can delete connections again. See SECTION 6.3.3 on this.
- If you have defined single scan points and connections, you can refine the resulting grid. This means you increase the number of scan points in the grid and thus their density. For this you need the GeoPro option. See SECTION 6.3.4 on this.
- If you have defined single scan points and connections, you can coarsen the resulting grid. This means you reduce the number of scan points in the grid and thus their density. For this you need the GeoPro option. See SECTION 6.3.5 on this.
- If you have defined single scan points and connections, you can merge neighboring scan points. This means you reduce the number of scan points in the grid and thus their density. For this you need the GeoPro option. See SECTION 6.3.6 on this.
- You can import geometries (scan points). Prerequisite is a valid 3D alignment. See SECTION 6.3.7 on this.
- If the object you are making measurements on does not have a continuous surface on which you can define a grid of scan points (e.g. a sheet of glass which has only been covered with retro-reflective film in certain places), you can define individual scan points using the optional geometry scan unit. See SECTION 6.3.8 on this.
- If you have defined single scan points as point objects, you can select them and modify the selected scan points. See SECTION 6.3.9 on this.

Convert standard object

To convert standard objects to point objects and to refine them, proceed as follows:

1. Draw the standard objects as described in SECTION 6.2.
2. Mark the figure you want to convert with the mouse.
3. In the graphics toolbar, click  .

   The software converts the figure into single points with connections and assigns them to the point objects.

ⓘ If you do not select a figure and click the icon, all standard objects will be converted to point objects.

4. Activate the point objects by clicking  .

ⓘ You can undo all actions.

You can additionally define individual scan points as described in SECTION 6.3.1.

You can edit connections of scan points as described in SECTION 6.3.2 and SECTION 6.3.3.

You can edit scan points as described in SECTION 6.3.9.

Additional functions

If you have carried out a 3D alignment, you have the following options available for defining scan points:

ⓘ For a 3D alignment, see SECTION 5.2.

- Importing scan points. See SECTION 6.3.7 on this.

  If you know the geometry of the object and it is saved in an external program, you can import the required scan points as an ASCII, OBJ, PLY, or STL file. You can also import Universal File, Vibrant ME'Scope structural data and files in ASAM ODS ATFX format.
- Teaching-in scan points with the MSA geometry unit. See SECTION 6.3.8 on this.

ⓘ Standard objects remain standard objects as long as you do not convert them to point objects. When changing the object type, scan points which have already been defined as standard objects are retained.

6.3.1 Defining Single Scan Points with the Software

There are different ways to define a scan point:

1. Click  .
2. Point at the live video image and click.

   A scan point is defined at this place.

Or

1. Move the laser beam on the object to the point at which you want to define a scan point. See SECTION 4.1 on this as well.
2. Click  .

   A scan point is defined at the position the laser beam is at.

6.3.2 Defining Connections between Scan Points

There are several ways of connecting scan points.

- Connect two scan points and thus define a line.
- Connect three scan points and thus define a triangle.
- Connect selected scan points and use them to define triangles.
- Automatically connect selected scan points with each other.

ⓘ To display the point objects select the  representation. All areas that are enclosed by scan points are displayed. For displaying figures, see SECTION 6.4.3.

Connect two scan points

To connect two scan points, proceed as follows:

1. Click  .
2. Point at the first scan point.

   The mouse pointer becomes a cross icon  .
3. Click the scan point.
4. Point at the second scan point and double-click it.

   The software connects this scan point to the first one.

Connect three scan points

To connect three scan points, proceed as follows:

1. Click  .
2. Point at the first scan point.

   The mouse pointer becomes a cross icon  .
3. Click the scan point.
4. Point at the second scan point and click it.
5. Point at the third scan point and click it.

   The software connects this scan point to the first one.

Connect selected scan points

To connect selected scan points, proceed as follows:

1. Select the points you want to connect as described in SECTION 6.5.2.
2. Click  .

   The software connects all selected point so that triangles are generated; existing connections are retained. The software avoids generating small triangles at the edges of the figure. This means the outline of the figure generated can be slightly concave.

ⓘ If the selected scan points are positioned approximately on a line, the software generates line segments to connect the scan points.

Connect scan points automatically

To automatically connect scan points, proceed as follows:

1. Change to 3D view by clicking 3D.
2. In the graphics toolbar, click .

   The Connect Points dialog appears.

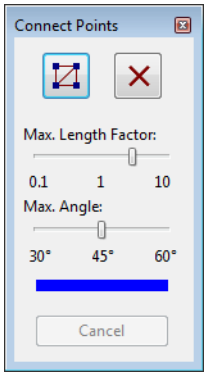


Figure 6.3: Connect Points dialog

3. Select the maximum length factor (Max. Length Factor). By moving this slider with the mouse you can set the length of connection still allowed (minimum 1/10 to maximum factor 10). The factor refers to the median of the length of the visible connecting lines that already exist.
4. Select the maximum permissible angle of the viewer (Max. Angle).

   The software only generate areas the normal vector of which includes a maximum of this angle with the line of sight of the viewer (minimum 30° to maximum 60°).
5. If applicable, select the scan points you want to connect as described in SECTION 6.5.2.

ⓘ If connections exist between the selected scan points, you have to delete these. Then you can connect the scan points automatically.

6. Click in the Connect Points dialog.

   The software will connect all scan points or the selected ones respectively, creating triangles. Existing connections are retained here.
7. If necessary, rotate the object further and repeat steps 5 and 7 until you have made all connections.
8. To delete connections which have been made automatically, select the scan points as described in SECTION 6.5.2 and click in the Connect Points dialog.

6.3.3 Deleting Connections (2D View)

To delete connections, proceed as follows:

1. Change to 2D view by clicking 3D.
2. Click .
3. Move the mouse pointer on a connection.

   The mouse pointer becomes a cross icon .
4. To select the connection, press the mouse button.

   To select further connections, click them while holding the shift key pressed. You can also draw a rectangle with the mouse and use it to select connections which are partially or completely in the rectangle.

   To remove a connection from the selection, click the connection while holding the shift key pressed.

   To select all connections, press the key combination Ctrl + A or select Edit > Select All.
5. To delete the selected connections, press the Delete key or select Edit > Delete.

ⓘ To delete a triangle, you only have to delete one of the connections. The software automatically deletes the other connections.

6.3.4 Refining Scan Point Grid

You require the GeoPro option. If you have defined single point objects and connections, you can refine the resulting grid. This means you increase the number of scan points in the grid and thus their density. The software calculates the distances between the selected scan points and sets an additional scan point in the middle on the connection. In the case of a quadratic polygon, an additional scan point is set in the middle.

ⓘ You can also fine-tune the scan point grid in 3D view.

To refine the grid, proceed as follows:

1. Select several connected scan points as described in SECTION 6.5.2.
2. In the menu bar, select Scan > Refine Grid (graphics toolbar:).

   The software automatically inserts one more scan point respectively between the selected scan points.

6.3.5 Coarsening Scan Point Grid

ⓘ If no connection lines are defined between the scan points, then the command and the icon for Coarsen Grid are inactive.

You require the GeoPro option. If you have defined point objects and connections, you can coarsen the resulting grid. This means you reduce the number of scan points in the grid and thus their density. To do this, the software initially searches for pairs of scan points which are connected to each other and which are near each other. The software then deletes one point in the pair and retains the other one unchanged (in contrast to merging scan points). You can choose if you want to delete the points or just disable them for the scan.

ⓘ You can also coarsen the scan point grid in 3D view.

To coarsen the grid, proceed as follows:

1. Select several connected scan points as described in SECTION 6.5.2.

ⓘ You can also not select a scan point to enlarge the entire grid.

2. In the menu bar, select Scan > Coarsen Grid (graphics toolbar: ).

   The Coarsen Grid dialog appears.

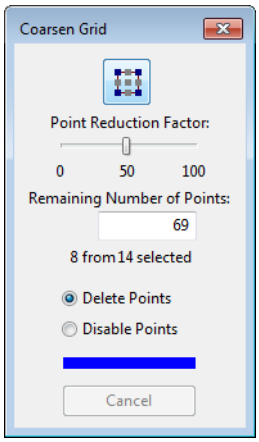


Figure 6.4: Coarsen Grid dialog

3. Use the slider to select the Point Reduction Factor as a percentage.

ⓘ You can also enter the Remaining Number of Points. The Point Reduction Factor is adjusted correspondingly.

The total number of remaining points (Remaining Number of Points) as well as the number of the remaining points within the selection (x from x selected) are shown below.

4. To delete undesired points or to disable them for the measurement, select Delete Points or Disable Points.

5. Click  in the Coarsen Grid dialog.

   The software automatically deletes or disables the calculated number of scan points.

6.3.6 Merging Scan Points in the Grid

You require the GeoPro option. If you have defined single point objects and connections, you can merge neighboring scan points. This means that you reduce the number of scan points in the grid and thus their density. To do this, the software first calculates the distances between the selected scan points. Then, starting from the minimum distance found, the software merges two points into one point if their distance is less than the minimum distance +50%. The positions of the scan points on the video image and, if available, the 3D coordinates of the scan points are calculated by averaging the values of the two points that have been merged.

ⓘ You can also merge scan points in 3D view.

To merge scan points, proceed as follows:

1. Select several connected scan points as described in SECTION 6.5.2.
2. In the menu bar, select Scan > Merge Points (graphics toolbar: ).

   The software will merge neighboring scan points as described above.

6.3.7 Importing Scan Points

The icon and the command for geometry import are only available for the following measurement systems:

- MSA-600/MSA-650 with automated Z-axis from Polytec and GeoWorks option
- MSA-500 with the GeoWorks option
- MSA-500 with the MSAGeo and GeoWorks options

A prerequisite for importing scan points is a valid 3D alignment. See SECTION 5.2 on this as well.

Formats

With the GeoWorks option, you can import the following formats:

- Point coordinates in ASCII Files (*.txt) format

  Here, only the point coordinates are imported, not the connections between the points.
- Geometries in STL Files (*.stl) format
- Geometries in the Universal Files (*.unv, *.uff, *.asc) format

ⓘ If you are importing geometry from a file in Universal File format in which geometry components are defined, then these are only taken into account for data set 82. See SECTION 6.5.4 and SECTION 10.2.9 on this as well.

- Geometries in ME'Scope5 Project Files (*.vtprj) format

If you import geometries from ME'Scope5, then ME'Scope5 is started automatically, the corresponding file is opened and then the data is imported. If you have defined substructures in ME'Scope5, these are displayed as geometry components in the software.

- With ExpME option: Geometries in ME'Scope Struktur Datei (*.str) format
- With ExpASM option: Geometries in ASAM ODS ATFX Files (*.atfx) format
- With GeoPro option: Geometries in OBJ Datei (*.obj) and PLY Files (*.ply) format

Import geometry initially

To initially import scan points of a geometry, proceed as follows:

1. In the menu bar, select Scan > Import Geometry (graphics toolbar: ).

   The Open dialog appears.

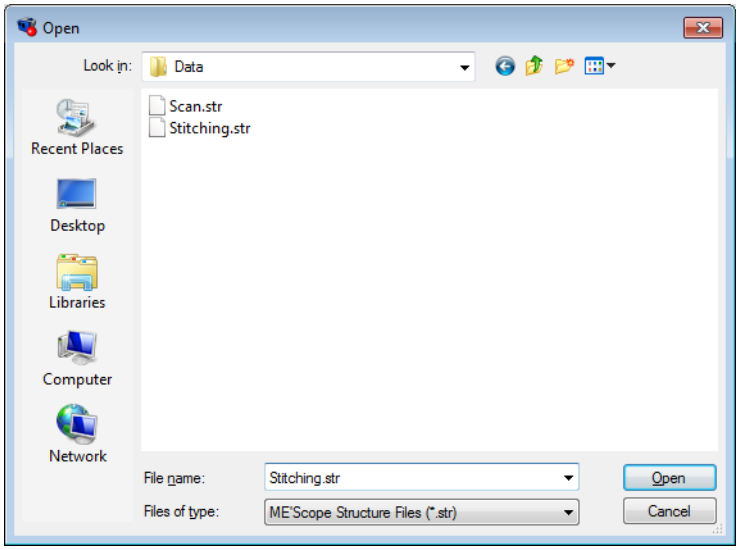


Figure 6.5: Open dialog when importing geometries

2. Select the required File Type.
3. Navigate to the saving location and select the required file.
4. Click Open.

   A PSV message window appears asking whether you want to exactly position the scan points.

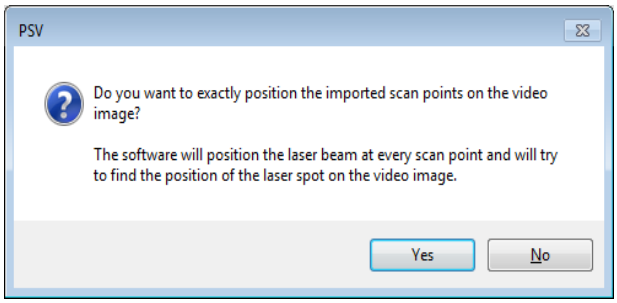


Figure 6.6: PSV message window

If you click Yes, the software will position the laser beam at every scan point to find the position of the laser spot on the video image. So the scan points are positioned exactly. The indices of the scan points are applied.

If you click No, the software imports all scan points from the selected file directly into the video image. The indices of the scan points are also applied.

ⓘ If you carry out a geometry import from a file (in ASCII, Universal, STL, OBJ, or PLY format) which does not contain a data set 164 with the units of the coordinates, then you will have to enter a scaling factor in the Scaling of Imported Coordinates dialog. Factor 1 stands for the unit 1 m, factor 0.01 for the unit 1 cm etc.

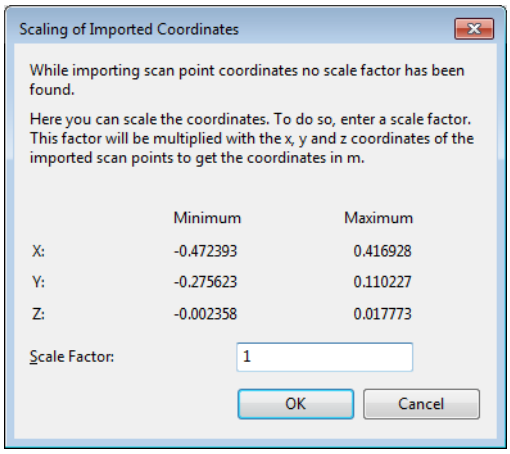


Figure 6.7: Scaling of Imported Coordinates dialog

ⓘ If the object is deep, the imported points are not precisely on the object in the on-screen display. Despite this, during scanning, the laser beams hit the scan points precisely. If you want to correct the display on the screen, you can move the single points in the video image to the right position. This, however, does not have any effect on the measurement results.

Complete geometry

If you import geometries, you can effectively handle conflicts to existing geometries. The software recognizes the existing geometry and offers the following options:

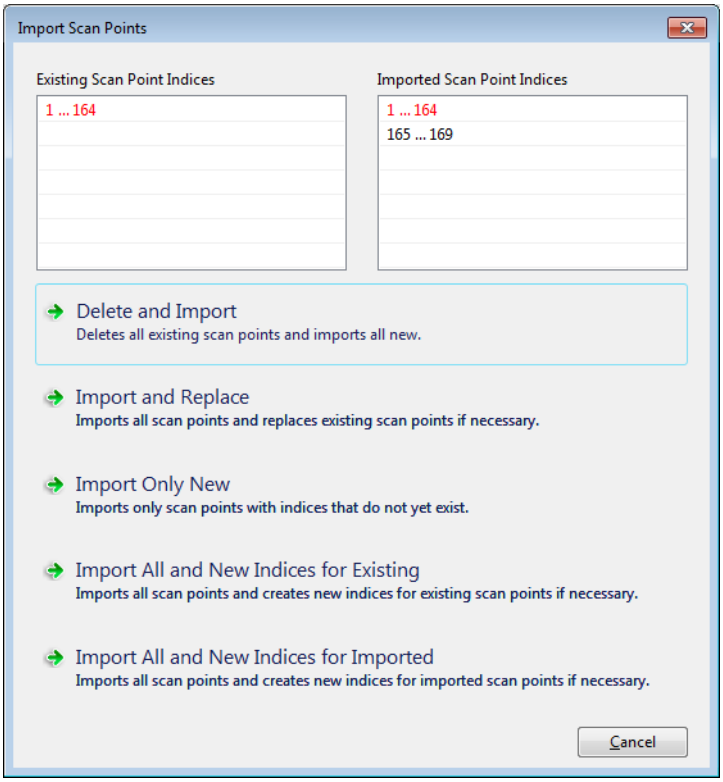


Figure 6.8: Import Scan Points dialog

Existing/Imported Scan Point Indices: The indices of the existing and imported scan points are displayed.

Delete and Import: All existing scan points will be deleted and all new scan points will be imported with their indices.

Import and Replace: All new scan points will be imported and the indices of existing scan points will be replaced if necessary.

Import Only New: Only the new scan points with indices that do not yet exist will be imported.

Import All and New Indices for Existing: All new scan points will be imported and all existing scan points will get new indices if necessary.

Import All and New Indices for Imported: All new scan points will be imported and will get new indices if necessary.

Work with large geometries

If the imported geometry is larger than the image section of the object in the video window, the video window is reduced correspondingly and the geometry is shown completely in both the 2D view as well as the 3D view.

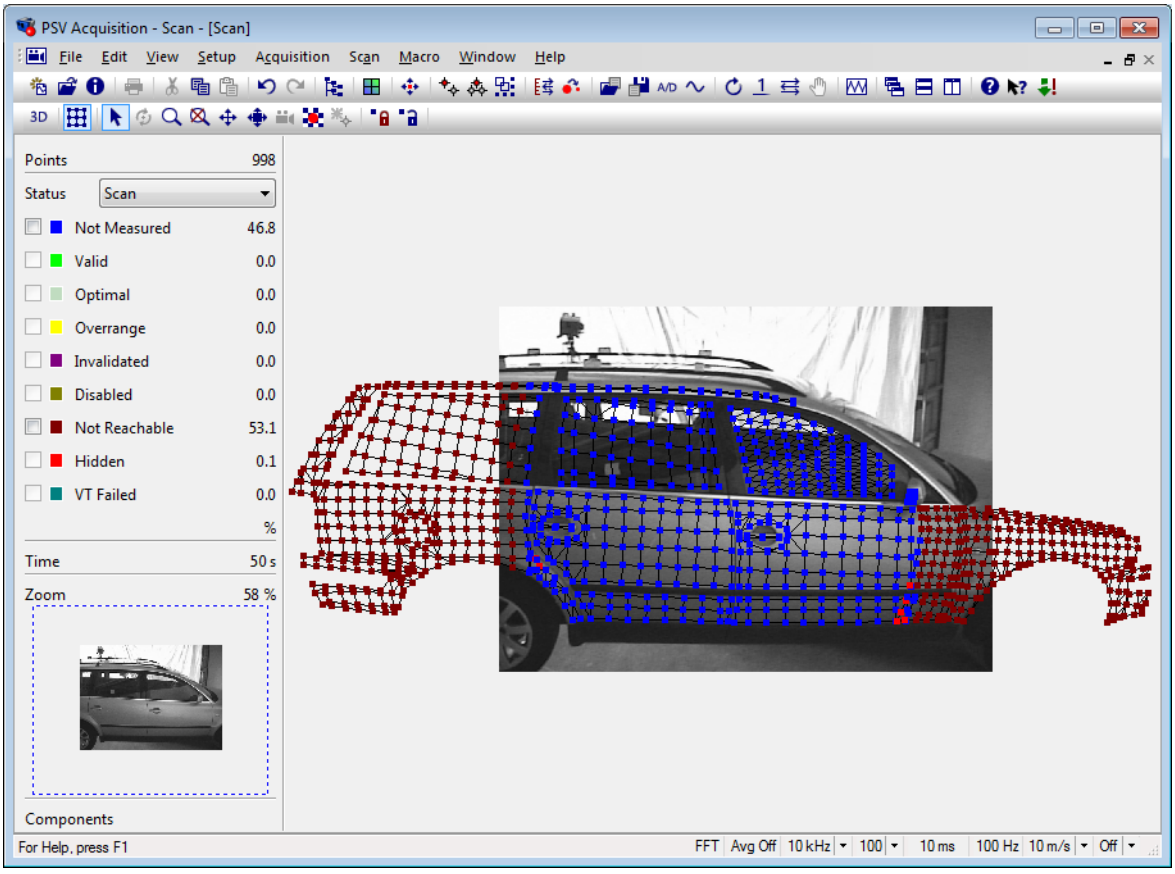


Figure 6.9: Example of large geometry with the PSV

Generally several scan points will then be given the Not Reachable or Hidden status. See SECTION 8.3 on this as well. Points that are not reachable will not be measured in a scan, hidden points can be measured after a warning message.

To acquire the entire geometry of the object, proceed as follows:

1. Select the scan points which are not to be measured with the current measurement position. See SECTION 6.5.2 on this as well.
2. Assign the Disabled status to these scan points. See SECTION 6.5.3 on this as well.
3. Start a scan as described in SECTION 8.2.
4. Change the position of the object/measurement setup so that you can scan another part of the object.

ⓘ If you are using the XY positioning stage, you can move this by a suitable distance and adjust the 3D alignment by adding a corresponding offset. See SECTION 5.3 on this. Do not recalculate the 3D coordinates of the scan points when you leave 3D alignment.

5. Carry out a new 3D alignment as described in SECTION 5.2.
6. Disable the scan points which should not be measured.
7. Start another scan.
8. Repeat steps 4 to 7 until you have acquired all of the geometry.

ⓘ If you are using 3D geometries, you can then merge the individual measurements to a combined file as described in the Presentation.

6.3.8 Teaching-In Single Scan Points

If your object does not have a continuous surface which you can define a grid or scan points on, you can define single scan points.

For this, you will need:

- MSA-600/MSA-650 with automated Z-axis from Polytec
- MSA-500 with MSA geometry unit (MSAGeo option)

Approach

To define single scan points, proceed as follows:

1. Position the laser beam at the point at which you want to define a scan point.
2. In the menu bar, select Scan > Point at Laser Position (graphics toolbar: ).

   The software makes a distance measurement. At the same time, both the coordinates of the point on the object and the position of the point on the video image are determined automatically.
3. Once you have taught-in all the scan points, you can link them together to become a surface as described in SECTION 6.3.2.

6.3.9 Editing Scan Points (APS)

If you have defined single scan points as point objects, you can select them and modify the selected scan points.

ⓘ Click next to the selected scan points with the right mouse button and select Modify Selected Points followed by the desired modification in the context menu.

Select scan points

You can select scan points as described in SECTION 6.5.2 and modify the selected points as described in the following. To do so, precisely click the selected scan points with the right mouse button and select the required modification in the context menu.

ⓘ If you have selected an individual scan point, you can change the point selection and thus move the laser beam by holding the Control key pressed and pressing the arrow key for the desired direction.

Select marginal points

In order to select the marginal points only, you have the following options:

- Click with the right mouse button next to the scan points and select Select All Marginal Points in the context menu.
- Select Edit > Select All Marginal Points in the menu bar.

All marginal points are selected.

If you have defined several figures and have selected a scan point of a figure, you have the following options:

- Click the scan point with the right mouse button and select Select Marginal Points in the context menu.
- Click next to the scan point with the right mouse button and select Modify Selected Points > Select Marginal Points in the context menu.

The marginal points of the figure are selected. If you have selected several figures and have selected scan points of several figures, the marginal points of these figures are selected.

Disable

You can disable selected scan points which are not to be measured. To do so, select Disable or press the key combination Ctrl + space bar. The scan points are disabled for measurements, geometry scans and for the Assign Focus Automatically function.

Enable

You can unlock selected scan points. To do so, select Enable or press the key combination Shift key + space bar.

Modify 3D coordinates

If 3D geometry data is available, you can modify the 3D coordinates of single scan points. See SECTION 6.5.3.2 on this.

With the MSA you can then only interpolate the 3D coordinates of single scan points if there is 3D geometry, i.e. after importing geometry or after a geometry scan. See SECTION 6.3.7 or SECTION 6.6 on this as well.

Smooth 3D coordinates

In order to smooth the 3D coordinates of the selected scan points, 3D geometry data are required. Select Smooth 3D Coordinates. The software calculates a new 3D coordinate for every selected scan point taking the coordinates of the neighboring scan points into account. The weighting of the neighboring points reduces with their spatial distance.

ⓘ The measurement object becomes smoother if you carry out the command several times.

Interpolate 3D coordinates

In order to interpolate the 3D coordinates of the selected scan points, 3D geometry data are required. Select Interpolate 3D Coordinates.

For the interpolation scan points are used that are defined as follows:

- The scan points are not selected.
- The scan points have 3D coordinates.
- The scan points are not connected directly or indirectly to the selected scan points.

The interpolation is carried out by taking the position of the scan points on the video image into account.

With the MSA you can then only interpolate the 3D coordinates of single scan points if there is 3D geometry, i.e. after importing geometry or after a geometry scan. See SECTION 6.3.7 or SECTION 6.6 on this as well.

Clear 3D coordinates

To clear the 3D coordinates of the selected scan points, select Clear 3D Coordinates. The software clears the 3D coordinates and uses the values of the 2D alignment to hit the scan points.

Geometry scan

To carry out a geometry scan for the selected scan points, select Geometry Scan. The software automatically switches over to the geometry laser. The geometry laser is positioned at every scan point, measures the distance from the scanning head, and determines every x-, y- and z-coordinate. See SECTION 6.6 on this as well.

6.4 Common Functions

Some functions are available for different modes or object types (APS/not APS or standard objects/point objects). They are described in the following sections:

- Define clipping planes in 3D view (all modes or object types) - See SECTION 6.4.1 on this.
- Change indices for point objects (APS mode) and outside scan point definition (not APS) - See SECTION 6.4.2 on this.
- Edit figures for standard and point objects (APS mode) - See SECTION 6.4.3 on this.

6.4.1 Defining Clipping Planes

In 3D view, you can define clipping planes to control complex geometries. Thereby in acquisition mode, it does not matter whether you are in the APS mode or not. The 3D view is only available if you have defined scan points.

ⓘ You can only define clipping planes using DirectX technology.

Depending on your measurement system, you can define a front and/or rear clipping plane respectively perpendicular to the coordinate axes and the sight of the camera. All parts of the geometry which lie outside the cutting area are removed from displaying. This makes it easier to analyze complex geometries and deflection shapes.

Clipping planes

To define the clipping planes in APS mode, start with step 1. To define clipping planes outside APS mode, start with step 3. To define clipping planes, proceed as follows:

1. In the menu bar, select Setup > Define Scan Points (toolbar:).

   You are in Scan Point Definition (APS mode). The software maximizes the video window and displays a graphics toolbar.

2. Activate the point objects by clicking in the graphics toolbar.

3. Change to 3D view by clicking 3D.

   If you change from the 2D view to the 3D view, a snapshot of the live video image is shown as a fixed-image.

4. Click .

   The Clipping Planes dialog appears.

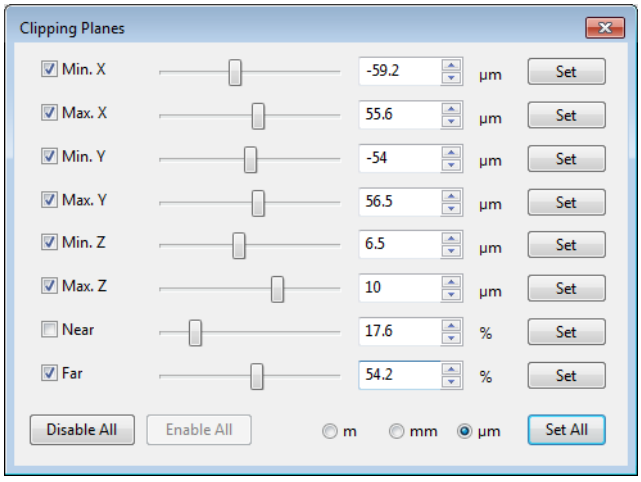


Figure 6.10: Clipping Planes dialog

5. Tick the check boxes of the clipping planes that you want to define. A dialog description can be found below.

ⓘ Only activated clipping planes can be configured.

6. To finish defining the clipping planes and to exit the dialog, click  .

   The current configuration of clipping planes is saved and restored when you reopen the dialog.

Dialog description

Enable All: All available clipping planes are enabled.

Min. X/Y/Z and Max. X/Y/Z: Define the position of a clipping plane either using the associated slider or entering the directional coordinate value in the associated input field.

m, mm, µm: Select the appropriate length unit for the X/Y/Z-values of the positions of the clipping planes. An own step size belongs to every length unit for adapting the position values using the sliders and the arrow keys. The step sizes are defined as follows:

Length unit	Increment
m	10 mm
mm	10 µm
µm	100 nm

Near, Far: Define the position of these clipping planes either using the sliders or entering percentage values into the input fields. The whole percentage range covers the expansion of the amount of all selected scan points from the camera's perspective.

Set, Set All: To position a clipping plane at the appropriate edge of the smallest rectangular solid that contains all selected scan points, click Set. To configure this for all clipping planes, click Set All.

Disable All: Click here to disable all available clipping planes.

6.4.2 Changing Indices

Display indices

You set up the display of the scan point indices in the video image in the Display Properties dialog in 3D view. To open the dialog, display the 3D view by clicking 3D.

In acquisition mode you open the dialog as folllows:

- Double-click the video image and display the Data page.
- Click the video image with the right mouse button and select Data Properties in the context menu.
- Activate the video window (to do so, click it) and select Scan > Properties in the menu bar. Display the Data page.

In presentation mode you open the dialog as folllows:

- Double-click the object in the presentation window and display the Data page.
- Click the presentation window with the right mouse button and select Data Properties in the context menu.
- Activate the presentation window (to do so, click it) and select Presentation > Properties in the menu bar. Display the Data page.

You will also find a detailed description of the dialog in SECTION 9.1.

Change indices

To change the indices of selected scan points, you have the following options:

- Click one or several scan points with the right mouse button ans select Change Indices in the context menu.
- Select one or several scan points and select Edit > Modify Selected Points > Change Indices in the menu bar.

The Change Indices dialog appears.

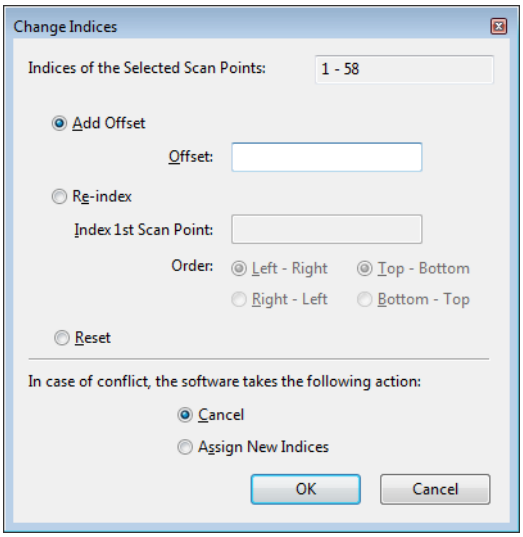


Figure 6.11: Change Indices dialog

Indices of the Selected Scan Points: The indices of the selected scan points are displayed.

Add Offset: Click here if you want to add a certain amount to the indices of the selected scan points or want to subtract it from them. Enter the required value in the Offset field. For a negative offset you have to enter additionally a minus sign (–).

Example:

The selected scan points have the indices 1 - 21.

Offset: 10

The new indices will be 11 - 31.

Re-index: Click here if you want to renumber the selected scan points. In the Index 1st Scan Point field, enter the value at which the new numbering is to start. In addition to that, select the direction of the numbering, starting with the first scan point.

Example:

The selected scan points have the indices 27 - 52.

Index 1st Scan Point: 30

Order: Left - Right, Top - Bottom

The new indices are 30, 31, 32, etc. They are assigned to the selected scan points from left to right and from top to bottom.

Reset: Click here if you want to undo the changes made to the indices. The indices are then reset to the scan point number. The scan point numbers are assigned by the software in ascending order and without gaps starting with one. After the measurement, the scan point number and the index are displayed in the readout of the presentation window.

In case of conflict, the software takes the following action: Here you select how the software is to react if there is a conflict after changing the indices, i.e. numbers were assigned twice.

- Cancel: The entire action is canceled, i.e. no changes are made.
- Assign New Indices: The selected scan points are assigned the required indices. Indices for other scan points which already exist are counted up, starting with the highest index +1.

Example:

You have defined 2500 scan points and want to newly assign the indices 1 - 200 to the scan points 200 - 400.

Index 1st Scan Point: 1

On conflict with these changes, take the following action: Assign New Indices

The selected scan points are given the indices 1 - 200. The scan points which originally had the indices 1 - 200, are given the indices 2501 - 2701.

ⓘ Independently of whether you abort assignment of new indices or carry it out, you will be notified whether a conflict has occurred.

6.4.3 Editing Figures

Display figure

Using the following icons, you can display the figures in different ways. To do so, select the figures as described below and click one of the following icons:



You can see the scan points and their connections.



You can see the scan points, their connections and cross-hatching of the areas covered by the scan points. As a general rule, these areas are different from the geometric figures you have drawn.

ⓘ In presentation mode, areas which are enclosed by scan points are shown as enclosed areas.



You see the connections between the scan points.



You see cross-hatching of the areas covered by scan points.

Select figure

To edit figures, you have to select them first. To do so, proceed as follows:

1. Select Scan > Select (toolbar: ).
2. Point at a figure.

   The mouse pointer becomes an arrow cross icon  .
3. Press the mouse button.

   The software marks the edge of the figure with white markers.
4. To select further figures, click them while holding the Control key pressed.

   The software marks every selected figure with white markers.

ⓘ To select all figures, select Edit > Select All in the menu bar (Ctrl + A key combination).

5. To remove a figure from the selection, click the figure while holding the Control key pressed.

   The white markers of the figure disappear.

Move figure

To move figures, select them as described above and then use the mouse to drag them in the direction required. You can also use the arrow keys.

Scan points which you have defined as point objects can be moved individually or in groups. You move single scan points, whether connected or freestanding, in exactly the same way as you move figures (see above). If you want to move several scan points at the same time, you have to select them. To do so, you have the following options:

- Hold the Control key pressed and click the single scan points. Then release the Control key.
- Mark the scan points you would like to move. To do so, use the mouse to draw a rectangle which encloses these scan points.

ⓘ If you want to add even more scan points to this selection, hold the shift key pressed and draw another rectangle.

Move the scan points with the mouse.

Scale figure

To scale figures, proceed as follows:

1. Select the figures as described above.

ⓘ To scale scan points which you have defined as point objects, select them and click  to group the selected scan points.

2. Point at one of the white markers which mark the edge of the figures.

   The mouse pointer becomes a double arrow.

3. Press the mouse button and drag in one of the directions the double arrow is pointing in. The anchor point is opposite the scan point you are dragging.

ⓘ Press the Control key and the mouse button and drag in one of the directions the double arrow is pointing in. The anchor point is the center of the figures.

Press the shift key and the mouse button and drag across to a corner point. The figures are resized in proportion horizontally and vertically.

Copy and paste figure

To copy and paste figures, proceed as follows:

1. Select the figures as described above.
2. In the menu bar, select Edit > Copy (toolbar: ).

   The figures are copied onto the clipboard.

3. In the menu bar, select Edit > Paste (toolbar: ).

   The figures a pasted from the clipboard.

Delete figure

To delete figures, proceed as follows:

1. Select the figures as described above.
2. In the menu bar, select Edit > Delete (press Delete key).

   The figures will be deleted.

Group figure

To group figures, proceed as follows:

1. Select the figures as described above.
2. Click  .

   The figures will be grouped.

ⓘ To undo grouping, click  .

Undo and redo

You can undo up to 50 operations and redo them. In the menu bar, select Edit > Undo or Edit > Redo (toolbar:  or ).

6.5 Editing Scan Points (not APS)

You can also edit scan points if you have already finished scan point definition. To do so, you have the following options:

- You can show/hide the scan points in the video image. See SECTION 6.5.1 on this.
- You have the following options to connect scan points. See SECTION 6.5.2 on this.
- You can retrospectively modify selected scan points. See SECTION 6.5.3 on this.
- You can define geometry components in the video window and assign scan points to the individual components. See SECTION 6.5.4 on this.

6.5.1 Showing or Hiding Scan Points

To show or hide scan points in the live video image, select Scan > Scan Points in the menu bar (toolbar of the video window: ). You will see the scan points as blue markers. You will see the connections between the scan points as lines.

6.5.2 Selecting Scan Points

Select with the mouse

To select scan points with the mouse, you have the following options:

- To select several scan points adjacent to each other, hold the Shift key and the left mouse button pressed and draw one or more rectangles across the required scan points.
- To select several scan points that are not adjacent to each other, hold the Control key pressed and click the required scan points.

Select marginal points

In order to select the marginal points only, you have the following options:

- Click with the right mouse button next to the scan points and select Select All Marginal Points in the context menu.
- Select Edit > Select All Marginal Points in the menu bar.

All marginal points are selected.

If you have defined several figures and have selected a scan point of a figure, you have the following options:

- Click the scan point with the right mouse button and select Select Marginal Points in the context menu.
- Click next to the scan point with the right mouse button and select Modify Selected Points > Select Marginal Points in the context menu.

The marginal points of the figure are selected. If you have selected several figures and have selected scan points of several figures, the marginal points of these figures are selected.

Select all

To select all visible scan points, you have the following options:

- Press the key combination Ctrl + A.
- Select Edit > Select All in the menu bar.
- Click the object with the right mouse button and select Select Points in the context menu.

  The Select Scan Points dialog appears.

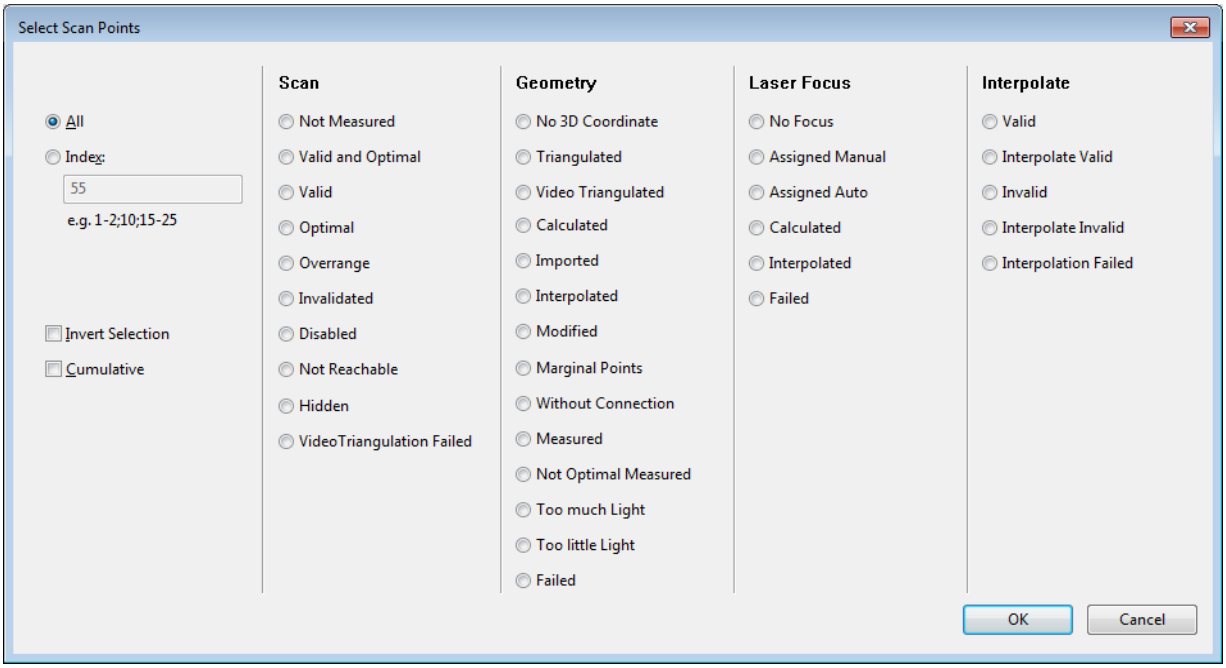


Figure 6.12: Select Scan Points dialog

Select All and click OK.

All scan points visible in the video image will be selected. Scan points in hidden components will not be selected.

Select with certain properties

To select scan points with certain properties, proceed as follows:

1. Select Edit > Select Points in the menu bar or click the video image with the right mouse button and select Select Points in the context menu.

   The Select Scan Points dialog appears (see above).

ⓘ You can select the scan points according to general criteria (left) or according to the corresponding status.

2. Select the criteria according to which you want to select points.

ⓘ If you want to select the scan points on the basis of their index, you can display the indices. See SECTION 9.1.2 on this.

3. If you want to select points with different properties, tick the Cumulative check box and repeat steps 1 and 2 as often as necessary.

ⓘ To invert the selection, also tick the Invert Selection box. To clear the selection, click  .

4. Once you have made your selection, click OK.

Select via legend for status

To display scan points of a certain status in the video window, tick the respective check box of the points status in the legend. The other way round, you can select scan points in the video image using the mouse, and in the legend the statuses of the selected scan points are displayed.

6.5.3 Modifying Scan Points

To retrospectively modify the selected scan points, you have the following options:

- In the menu bar, select Edit > Modify Selected Points and select the required command in the context menu.
- Click one of the selected scan points with the right mouse button and select the required command in the context menu.
- Click next to the selected scan points with the right mouse button, select Modify Selected Points and the required command in the context menu.

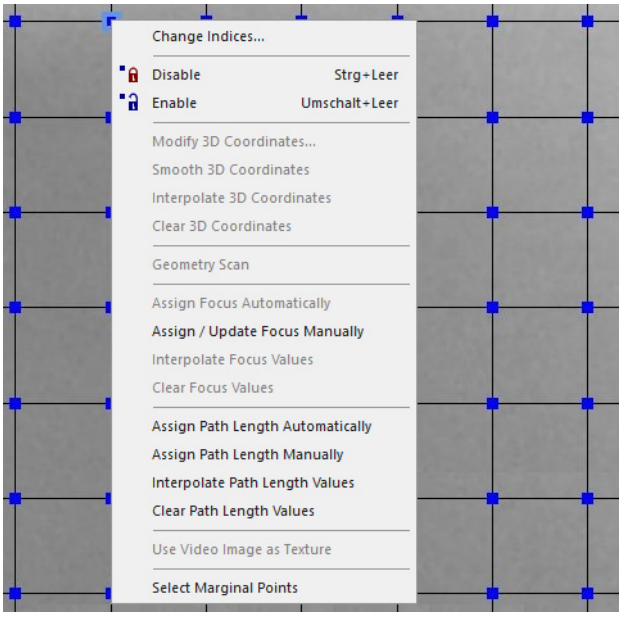


Figure 6.13: Command selection to modify scan points for the MSA-650

The following functions are available:

- You can change the indices of selected scan points. See SECTION 6.4.2 on this.
- You can disable selected scan points, that are not to be measured, and enable them again. See SECTION 6.5.3.1 on this.
- You can modify the 3D coordinates of single scan points. See SECTION 6.5.3.2 on this.
- If you want modify the 3D coordinates of selected scan points, choose Geometry Scan. The software will perform a geometry scan as described in SECTION 6.6.
- You can assign, interpolate or recalculate focus values. See SECTION 6.5.3.3 on this.
- You can assign, interpolate or recalculate path length values (only MSA-650). See SECTION 6.5.3.4 on this.

- Instead of the imported video image you can assign the current video image as texture to selected scan points. See SECTION 6.5.3.5 on this.
- If you want to select the marginal points on the figure of the marked scan point, choose Select Marginal Points.

6.5.3.1 Disabling/Enabling

Disable

To disable selected scan points that are not to be measured, you have the following options:

- Select Disable in the context menu.
- In the toolbar of the video window, click  .
- Press the key combination Ctrl + space bar.

The scan points are disabled for measurements, geometry scans and for the Assign Focus Automatically function.

Enable

You have the following options to disable the locking:

- Select Enable in the context menu.
- In the toolbar of the video window, click  .
- Press the key combination Shift key + space bar.

The scan points are enabled for measurements, geometry scans and for the Assign Focus Automatically function.

6.5.3.2 Editing 3D Coordinates

Modify 3D coordinates

To modify the 3D coordinates of individual scan points, select Modify 3D Coordinates for the selected scan point. The Modify 3D Coordinates dialog appears.

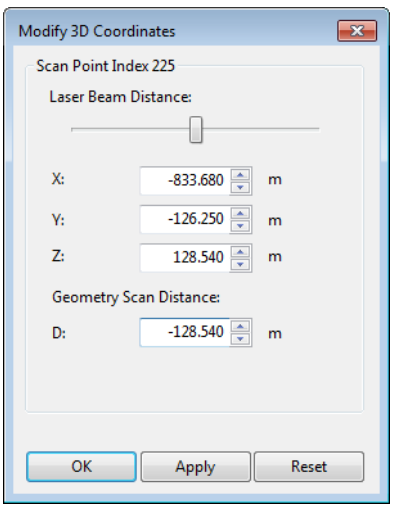


Figure 6.14: Modify 3D Coordinates dialog

You can modify the values of the X-, Y- and Z-coordinates and also the distance determined by the geometry scan (Geometry Scan Distance). If you move the laser beams with the Laser Beam Distance slider, the values in the fields are automatically updated. If however you modify the values in the fields, the slider remains in the set position.

- Apply: The changes are applied. The dialog remains open. You can select another scan point and modify its 3D coordinates.
- Reset: You can undo all changes you have made since you last clicked Apply.
- OK: The changes are applied. The dialog is closed.

You can modify the 3D coordinates of several scan points at the same time. To do so, proceed as follows:

1. Select several scan points as described in SECTION 6.5.2.
2. Click one of the scan points with the right mouse button and select Modify 3D Coordinates in the context menu.
3. Modify the 3D coordinates as described above and click OK.

   The change made to the 3D coordinates of that scan point is added to the 3D coordinates of all selected scan points. This allows you to move the position of several scan points in the same direction.

Smooth 3D coordinates

In order to smooth the 3D coordinates of the selected scan points, 3D geometry data are required. Select Smooth 3D Coordinates. The software calculates a new 3D coordinate for every selected scan point taking the coordinates of the neighboring scan points into account. The weighting of the neighboring points reduces with their spatial distance.

ⓘ The measurement object becomes smoother if you carry out the command several times.

Interpolate 3D coordinates

In order to interpolate the 3D coordinates of the selected scan points, 3D geometry data are required. Select Interpolate 3D Coordinates.

For the interpolation scan points are used that are defined as follows:

- The scan points are not selected.
- The scan points have 3D coordinates.
- The scan points are not connected directly or indirectly to the selected scan points.

The interpolation is carried out by taking the position of the scan points on the video image into account.

With the MSA you can then only interpolate the 3D coordinates of single scan points if there is 3D geometry, i.e. after importing geometry or after a geometry scan. See SECTION 6.3.7 or SECTION 6.6 on this as well.

Clear 3D coordinates

To clear the 3D coordinates of the selected scan points, select Clear 3D Coordinates. The software clears the 3D coordinates and uses the values of the 2D alignment to hit the scan points.

Geometry scan

You can carry out a geometry scan for the selected scan points. To do so, select Geometry Scan. The software focuses the laser beam on every scan point, determines the position of the z-axis in the measurement microscope and thus every X-, Y- and Z-coordinate. See SECTION 6.6 on this as well. The scan points are given the Assigned Auto laser focus status.

6.5.3.3 Editing Focus Values

A prerequisite for editing focus values is a valid 3D alignment.

ⓘ With a valid 3D alignment newly defined scan points will automatically get a focus value.

Assign focus manually

To manually assign a focus value to the selected scan points, select Assign/ Change Focus Manually. See SECTION 6.7.3 on this as well. The scan points are given the Assigned Manually laser focus status.

Interpolate focus values

You require the GeoPro option. To interpolate the focus values of the selected scan points, select Interpolate Focus Values.

Interpolation sequence:

- The software calculates the 3D coordinates for the selected scan points that have a focus value.
- The software interpolates the 3D coordinates for the selected scan points that have no focus value.
- From these temporary 3D coordinates, the software calculates the focus values and assigns them to the scan points without a focus value.
- The scan points are given the Interpolated laser focus status.

Clear focus values

As a prerequisite, deactivate the Calculate Focus Values Automatically function in the Preferences dialog. To clear the focus values of the selected scan points, select Clear Focus Values. The scan points are given the No Focus laser focus status.

Recalculate focus values

If, in the Preferences dialog on the Geometry page, you have ticked the Calculate Focus Values Automatically check box and the selected scan points have 3D coordinates, you can recalculate the focus values of the scan points. To do so, select Recalculate Focus Values. The scan points are given the Calculated laser focus status.

6.5.3.4 Editing Path Length Values (only MSA-650)

You can edit the path length values at individual scan points. If there is no valid path length value (scan point status Failed), no permissible signal level maximum was found within the specified limits (min/max). In this case, you have to remeasure or interpolate the points.

Assign path length automatically

All scan points are targeted with the laser. For each scan point, the maximum of the signal level is searched in the range defined by the min/max values. The detected values are stored for the scan point.

Assign path length manually

You apply the current position of the path length alignment for the selected points.

Interpolate path length values

Interpolation sequence:

- For the selected scan points that have no focus values, the software interpolates the maxima and assigns them to the scan points without path length value.
- The scan points are given the Interpolated status.

Delete path length values

To clear the path length values of the selected scan points, select Clear Path Length Values. The scan points are given the No Path Length laser focus status.

6.5.3.5 Using Video Image as Texture

Current video image as texture

For this you need the GeoPro option. The function is only available for the geometry import from suitable files (e.g. from OBJ files). To assign the current video image as texture to selected scan points instead of the imported video image, proceed as follows:

1. Select several scan points as described in SECTION 6.5.2.
2. Click one of the scan points with the right mouse button and select Use Video Image as Texture in the context menu.

   A message window appears.

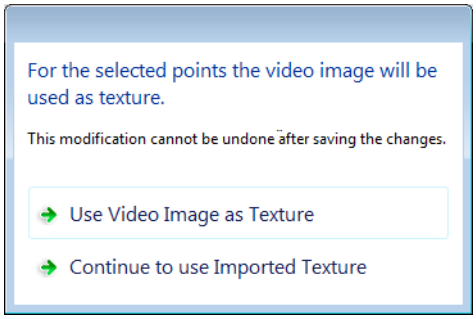


Figure 6.15: Message window for texture

3. To use the current video image, click Use Video Image as Texture.

ⓘ If you have selected all scan points, the imported video image is lost when you save. However, you can undo the step previously.

4. Alternative to step 3: To continue using the imported video image, click Continue to use Imported Texture.

ⓘ In 3D view, you can switch between the two kinds of displaying the video image. See SECTION 9.1.3 on this.

6.5.4 Defining Geometry Components

In the video window of acquisition mode and in the presentation window, you can define geometry components and assign scan points to the individual components. The components are displayed in a list on the left in the video window or presentation window respectively. By default, only one component initially exists, the root component.

ⓘ In acquisition and presentation mode, you can show (hide) the scan points allocated to a component. This does not affect the average spectrum of the points.

If you have selected scan points in the video window, then the name of the associated geometry component is shown in the scanning head control.

If you have selected scan points in the presentation window, then the name of the associated geometry component is shown in the readout.

Edit component To edit the root component or any other component, click the component with the right mouse button and select Edit in the context menu. The Edit Component dialog appears.

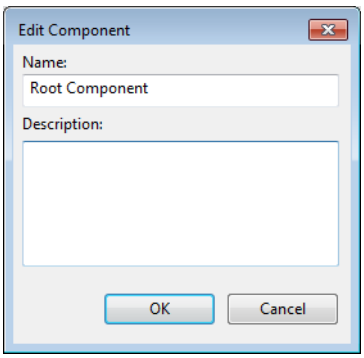


Figure 6.16: Edit Component dialog

Name: Enter the desired name for the component.

Description: Add a text to describe the component in greater detail.

ⓘ To change the name of the component, click the component with the left mouse button and press the F2 key. Enter the new name.

Add component

To add a component, with the right mouse button click the component below which you want to add a new component and select Add in the context menu. You can also click the components with the left mouse button and press Insert. The software generates a New Component at the required position. You can edit this component as described above.

ⓘ You can create any structure for the components.

Remove component

To remove a component, click the required component with the right mouse button and select Remove in the context menu. You can also click the components with the left mouse button and press the Del key. If you press the Shift + Del key combination, the selected component will be removed without a query.

ⓘ Scan points that were assigned to deleted components are automatically assigned to the root component.

Assign scan points

To assign scan points to the individual geometry components, proceed as follows:

1. Select the scan points you want to assign to a geometry component as described in SECTION 6.5.2.
2. Select the component which the scan points should belong to. To do so, click the component with the right mouse button and select Add Selected Points in the context menu.
3. Repeat these two steps until all scan points have been assigned to the required components.

ⓘ Scan points which you have not assigned to any other component are always automatically assigned to the root component.

Show and hide scan points

To show (hide) the scan points of individual components, activate or deactivate the corresponding check box in front of the component name.

Show number of scan points of component

To show the number of scan points in a component, click the name of the respective component and leave the mouse pointer there. The number of scan points is displayed and how many of these scan points have the Disabled scan status.

Select scan points of a component

To select the scan points of a component, click the component with the right mouse button and choose Select Points in the context menu. You can also double-click the component.

ⓘ To select the scan points of several components, double-click the required components while holding the Control key pressed.

Select all scan points of a component

To select all scan points of a component, i.e. the scan points of the secondary components as well, click the component with the right mouse button and choose Select All Points in the context menu.

6.6 Making a Distance Measurement

Requirements for MSA-600/ MSA-650

If you are using the MSA-600/MSA-650 with an automated Z-axis from Polytec, you can still determine the focus point position if the object geometry is unknown.

With an automated Z-axis from Polytec you can assign 3D coordinates to the scan points. To do this, start by defining the scan points as described in SECTION 6.1 and then determine the 3D coordinates of the scan points. For a geometry point measurement or a geometry scan, see SECTION 6.6.1.

Requirements for MSA-500

If you are using the MSA-500 with the MSAGeo option, you can determine the position of the focus point on the z-axis in the measurement microscope even if you do not know the geometry of the object.

You can use the MSAGeo option to assign 3D coordinates to scan points. To do this, start by defining the scan points as described in SECTION 6.1 and then determine the 3D coordinates of the scan points. For a geometry point measurement or a geometry scan, see SECTION 6.6.2.

6.6.1 Distance Measurement with MSA-600

Geometry point

To carry out a distance measurement between scanning head and object, proceed as follows:

1. Position the laser beam at any point on the object.
2. Select Scan > Geometry Point.

   The software will make a distance measurement to this point, will use the results of this measurement and the position of the automated Z-Axis to calculate the X, Y and Z-coordinates of this point on the object, and will display these coordinates.

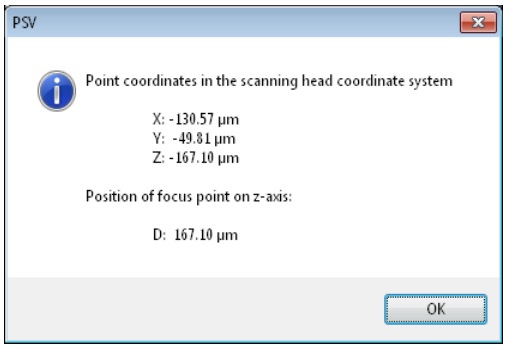


Figure 6.17: Display of coordinates for the MSA

If there is a valid 3D alignment, then the coordinates are given in the object coordinate system that you defined through the 3D alignment. Otherwise the coordinates are given in the scanning head system which you can define as described in SECTION 5.2.2.

Geometry scan

To scan an object that the software does not yet know the 3D geometry of, select Scan > Geometry Scan (toolbar: ). The software focuses the laser beam on every scan point, determines the position of the automated Z-axis and thus every X-, Y- and Z-coordinate as described above under Geometry point.

ⓘ A prerequisite for a geometry scan is a valid 3D alignment.

Once the software has determined the coordinates of the scan points, you can check the geometry by switching to the 3D view. You also can check individual coordinates of the scan points by selecting the corresponding scan point and reading the coordinates in the scanning head control. Afterwards you can start a scan.

ⓘ You can also select single scan points as described in SECTION 6.5.2 and only perform a geometry scan for these scan points. To do so, click one of the selected scan points with the right mouse button and select Geometry Scan in the context menu.

If the geometry scan has not been performed correctly or you have canceled it, you will get a status report.

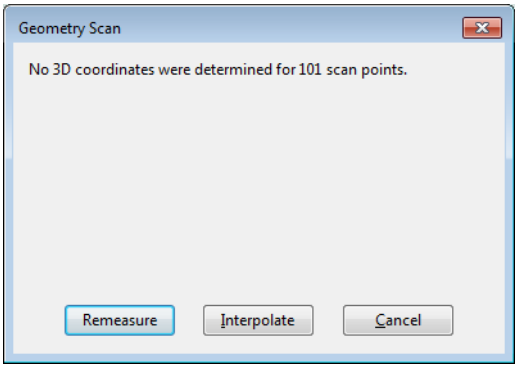


Figure 6.18: Status report if geometry scan is incomplete

You can decide whether to finally abort the geometry scan or whether to remeasure the remaining scan points or to have them interpolated. See SECTION 8.2 and your PSV theory manual on this. Independently of whether you have performed the geometry scan or have aborted it, you can subsequently check the status of the scan points and if necessary take further measures. See SECTION 6.5.2 and SECTION 8.3 on this.

The 3D coordinates of the geometry points are determined in the coordinate system of the alignment points (object system). The coordinates are displayed in the 3D Point element.

Convert 2D into 3D geometry

You can convert the 2D geometry of a three-dimensional object into 3D geometry. You need to have performed a 3D alignment beforehand. To convert 2D geometry into 3D geometry, start a geometry scan. This familiarizes the software with the three-dimensional geometry of the object.

6.6.2 Distance Measurement with MSA-500

Geometry point

To carry out a distance measurement between scanning head and object, proceed as follows:

1. Position the laser beam at any point on the object.
2. Select Scan > Geometry Point.

   From the results of this measurement, and the position of the z-axis in the measurement microscope, the software will calculate the X-, Y- and Z-coordinates of this point and will display these coordinates.

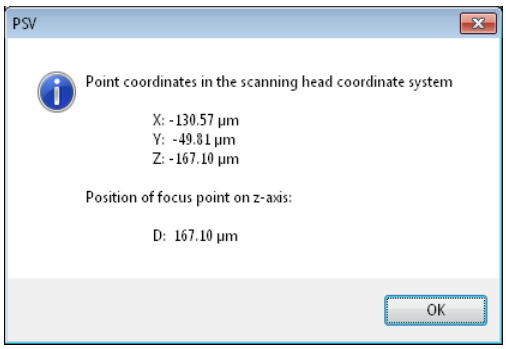


Figure 6.19: Display of coordinates for the MSA

If there is a valid 3D alignment, then the coordinates are given in the object coordinate system that you defined through the 3D alignment. Otherwise the coordinates are given in the scanning head system which you can define as described in SECTION 5.2.2.

Geometry scan

To scan an object that the software does not yet know the 3D geometry of, click  or select Scan > Geometry Scan. The software focuses the laser beam on every scan point, determines the position of the z-axis in the measurement microscope and thus every X-, Y- and Z-coordinate as described above under Geometry point.

ⓘ A prerequisite for a geometry scan is a valid 3D alignment.

If the software has determined the coordinates of all scan points, you can check the geometry by switching to the 3D view. You also can check individual coordinates of the scan points by selecting the corresponding scan point and reading the coordinates in the scanning head control. Afterwards you can start a scan.

ⓘ You can also select single scan points as described in SECTION 6.5.2 and only perform a geometry scan for these scan points. To do so, click one of the selected scan points with the right mouse button and select Geometry Scan in the context menu.

If the geometry scan has not been performed correctly or you have canceled it, you will get a status report.

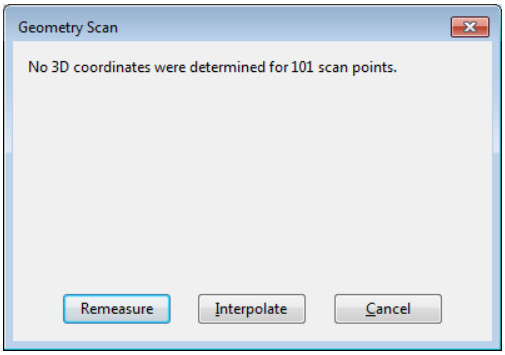


Figure 6.20: Status report if geometry scan is incomplete

You can decide whether to finally abort the geometry scan or whether to remeasure the remaining scan points or to have them interpolated. See SECTION 8.2 and your PSV theory manual on this. Independently of whether you have performed the geometry scan or have aborted it, you can subsequently check the status of the scan points and if necessary take further measures. See SECTION 6.5.2 and SECTION 8.3 on this.

The 3D coordinates of the geometry points are determined in the coordinate system of the alignment points (object system). The coordinates are displayed in the 3D Point element.

Convert 2D into 3D geometry

You can convert the 2D geometry of a three-dimensional object into 3D geometry. You need to have performed a 3D alignment beforehand. To convert 2D geometry into 3D geometry, start a geometry scan. This familiarizes the software with the three-dimensional geometry of the object.

6.7 Teaching-In Focus Values

You have the option of assigning focus values to the scan points. This makes it easier to focus if several objects are to be scanned which are at different distances from the scanning head or if the object has three-dimensional geometry. A focus value is determined when focusing automatically. You have the option of saving the focus values determined in this way with the object geometry. They can then be used for any number of measurements, providing the object geometry remains the same. Teaching-in the focus values is only available if you have the GeoWorks option (requires APS Professional option).

You have the following options to assign focus values to the scan points or to check them:

- You can automatically calculate focus values on the basis of different parameters. See SECTION 6.7.1 on this.
- Assign focus values automatically:

  MSA-600/MSA-650: With the GeoWorks option, you can automatically assign focus values to individual or to all scan points. See SECTION 6.7.2 on this.

  MSA-500: This function is carried out when you are performing a geometry scan. After a geometry scan the scan points are given the Assigned Auto laser focus status. For the description of the geometry scan, see SECTION 6.6. For the description of the laser focus status, see SECTION 8.3.

ⓘ With a valid 3D alignment newly defined scan points will automatically get a focus value.

- Assign focus values manually:

  MSA-600/MSA-650: With an automated Z-axis from Polytec you can manually assign focus values to single scan points. See SECTION 6.7.3 on this.

  MSA-500: You can manually assign focus values to single scan points. See SECTION 6.7.4 on this.
- You can display focus values of selected scan points and thus check the focus settings. See SECTION 6.7.5 on this.

6.7.1 Automatically Calculating Focus Values

In the Preferences dialog on the Geometry page, you can set whether you want to have the focus values automatically calculated on the basis of the 3D geometry information.

This function makes it easier for you to assign focus values after importing geometries.

For activating, proceed as follows:

1. In the menu bar, select Setup > Preferences.

   The Preferences dialog appears.

2. Display the Geometry page.

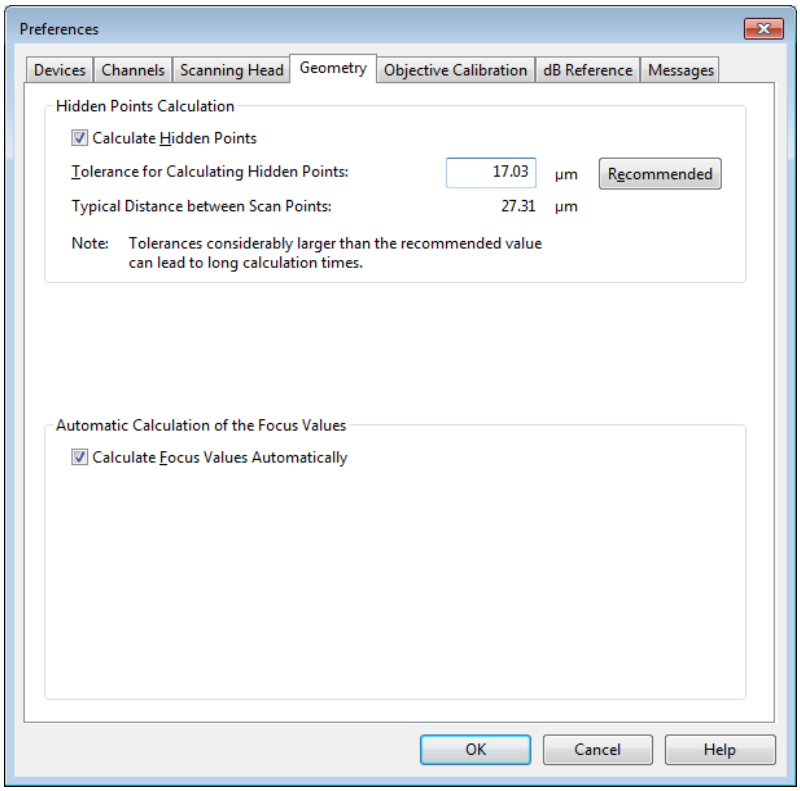


Figure 6.21: Geometry page for MSA-500

3. Tick the Calculate Focus Values Automatically check box.

   After a geometry import the scan points are given the Calculated laser focus status.

4. Click OK.

   The software uses the information from the 3D alignment or 3D coordinates respectively to calculate the focus values automatically.

6.7.2 Automatically Assigning Focus Values

You can automatically assign focus values to individual or to all scan points. To do so, you need at least the GeoWorks option.

Assign focus value to all scan points

To automatically assign the focus values to all scan points, proceed as follows:

1. In the menu bar, select Scan > Assign Focus Automatically (toolbar: ).

   A number of scan points determined by the software is automatically focused. The determined focus values are assigned to all scan points and saved with the object geometry.

   In the Scan menu, Focusing during Scan > Assigned Values is activated by the software. This has the effect that during a scan, the focus values are set for every scan point using the optics.

2. Check the focus settings as described in SECTION 6.7.5.
3. Start a scan.

ⓘ If you have not assigned focus values to all scan points, when the scan starts you will be given the message that not all scan points have a valid focus value. You can repeat Assign Focus Automatically or start the scan anyway.

Assign focus value to selected scan points

To automatically assign the focus values to the selected scan points, proceed as follows:

1. Select the scan points which you want to assign a focus value to as described in SECTION 6.5.2.
2. Click one of the scan points with the right mouse button and select Assign Focus Automatically in the context menu.

ⓘ If you click next to the selected scan points with the right mouse button, select Modify Selected Points and Assign Focus Automatically in the context menu.

A number of scan points determined by the software is automatically focused. The determined focus values are assigned to the selected scan points and saved with the object geometry.

In the Scan menu, Focusing during Scan > Assigned Values is activated by the software. This has the effect that during a scan, the focus values are set for every scan point using the optics.

3. Check the focus settings as described in SECTION 6.7.5.
4. Start a scan.

ⓘ If you have not assigned focus values to all scan points, when the scan starts you will be given the message that not all scan points have a valid focus value. You can repeat Assign Focus Automatically or start the scan anyway.

6.7.3 Manually Assigning Focus Values (with Z-Axis)

You can manually assign focus values to single scan points. This is particularly useful if the object is made up of one or more planes which are equidistant from the scanning head, i.e. the object is not at an angle to the scanning head.



CAUTION!

Pinching hazard or damage caused by automated process!

To focus the laser beam, the automated Z-axis is moved.

» Always be careful when working with the automated Z-axis.

» Before the measurement, make sure that the objective of the sensor head cannot collide with the object. Adjust the limits of the axes travel range (Limit downward travel relative to the respective current position function).

» Do not put any body parts into the range of travel of the axes.

» In principle, during traveling the axes, observe the distance between the objective and object to prevent the object from being damaged.

Assign a focus value

To manually assign a focus value, proceed as follows:

1. Use the middle mouse button to position the laser beam at the point at which you want to determine the focus value.

ⓘ If you have assigned a special function to the middle mouse button (e.g. double-click), position the laser beam with the XY positioning stage (Axes element in the scanning head control).

2. Move the Z-axis of the positioning device so that the laser beam is optimally focused.
3. Select the scan points which you want to assign a focus value to as described in SECTION 6.5.2.
4. Click one of the scan points with the right mouse button and select Assign/Update Focus Manually in the context menu.

ⓘ If you click next to the selected scan points with the right mouse button, select Modify Selected Points and Assign/Update Focus Manually in the context menu.

The focus value determined previously is assigned to all selected scan points.

5. Check the focus settings as described in SECTION 6.7.5.
6. Start a scan.

ⓘ If you have not assigned focus values to all scan points, when the scan starts you will be given the message that not all scan points have a valid focus value. You can repeat Assign/Update Focus Manually or start the scan anyway.

Assign an offset

You can assign an offset to existing focus values that is calculated from the current position of the Z-axis. This allows you to compensate long-term drifts of the Z-axis. To assign an offset, proceed as follows:

1. Move the Z-axis to correct the focus value at one scan point.

   At this position, the laser beam is optimally focused.

2. Click the scan point with the right mouse button and select Assign/Update Focus Manually in the context menu.

ⓘ If you click next to the selected scan points with the right mouse button, select Modify Selected Points and Assign/Update Focus Manually in the context menu.

A PSV query window appears.

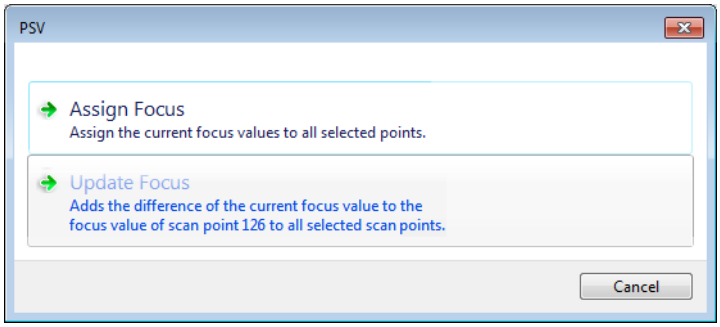


Figure 6.22: Query for Assign Offset

3. Click Assign Offset.

   The calculated offset is added to all selected scan points.

Interpolate focus values

To interpolate the focus values of the selected scan points, click the scan point with the right mouse button and select Interpolate Focus Values in the context menu.

Interpolation sequence:

- The software calculates the 3D coordinates for the selected scan points that have a focus value.
- The software interpolates the 3D coordinates for the selected scan points that have no focus value.
- From these temporary 3D coordinates, the software calculates the focus values and assigns them to the scan points without a focus value.
- These scan points are given the Interpolated laser focus status.

Clear focus values

To clear an assigned focus value, click the scan point with the right mouse button and select Clear Focus Values in the context menu. The scan points are given the No Focus laser focus status.

6.7.4 Manually Assigning Focus Values

You can manually assign focus values to single scan points. This is particularly useful if the object is made up of one or more planes which are equidistant from the scanning head, i.e. the object is not at an angle to the scanning head.

Assign a focus value

To manually assign a focus value, proceed as follows:

1. Use the middle mouse button to position the laser beam at the point at which you want to determine the focus value and then click  .

ⓘ If you have assigned a special function to the middle mouse button (e.g. double-click), position the laser beam using the Position icons in the scanning head control.

2. Select the scan points which you want to assign a focus value to as described in SECTION 6.5.2.
3. Click one of the scan points with the right mouse button and select Assign Focus Manually in the context menu.

ⓘ If you click next to the selected scan points with the right mouse button, select Modify Selected Points and Assign Focus Manually in the context menu.

The focus value determined previously is assigned to all selected scan points.

4. Check the focus settings as described in SECTION 6.7.5.
5. Start a scan.

ⓘ If you have not assigned focus values to all scan points, when the scan starts you will be given the message that not all scan points have a valid focus value. You can repeat Assign Focus Manually or start the scan anyway.

Interpolate focus values

To interpolate the focus values of the selected scan points, click the scan point with the right mouse button and select Interpolate Focus Values in the context menu.

Interpolation sequence:

- The software calculates the 3D coordinates for the selected scan points that have a focus value.
- The software interpolates the 3D coordinates for the selected scan points that have no focus value.
- From these temporary 3D coordinates, the software calculates the focus values and assigns them to the scan points without a focus value.
- The scan points are given the Interpolated laser focus status.

Clear focus values

As a prerequisite, deactivate the Calculate Focus Values Automatically function in the Preferences dialog. To clear the focus values of the selected scan points, click the scan point with the right mouse button and select Clear Focus Values in the context menu. The scan points are given the No Focus laser focus status.

Recalculate focus values

As a prerequisite, activate the Calculate Focus Values Automatically function in the Preferences dialog and 3D coordinates for the selected scan points are required. To recalculate the focus values of scan points, click the scan point with the right mouse button and select Recalculate Focus Values in the context menu. The scan points are given the Calculated laser focus status.

6.7.5 Checking Focus Settings

You have the following options to display the focus value of a scan point:

- Click the scan point and leave the mouse pointer on a button for focusing in the scanning head control.
- Display the laser focus status, click the scan point and leave the mouse pointer on the scan point.

The focus value of the scan point is displayed next to the mouse pointer.

MSA-600/ MSA-650

The current focus value is displayed as a percentage and as the position of the automated Z-axis.

- 0% (0 µm): The scanning head mounted on the Z-axis has been moved right to the end in the direction close-up (N).
- 100% (0 µm): The scanning head mounted on the Z-axis has been moved right to the end in the direction infinity (F).

MSA-500

The current focus value is displayed as a percentage and as the position of the z-axis in the measurement microscope.

- 0%: The optics in the measurement microscope have been moved right to the end in the direction close-up (N).
- 100%: The optics in the measurement microscope have been moved right to the end in the direction infinity (F).

At the same time, the optics are set to the focus value of this scan point.

Focus during scan

To ignore the focus values during the scan, select Scan > Focusing during Scan > Off. Focusing during a scan is disabled.

7 Parameters for Data Acquisition

There are several ways for you to set the parameters for data acquisition. See SECTION 7.1 on this.

You will find a description of all parameters in SECTION 7.2.

7.1 Setting Parameters

You have the following options when setting parameters for data acquisition:

- In the status bar
- In the Acquisition Settings dialog
- By loading the settings from a file

Status bar

You can set some parameters on the right in the status bar. The parameters which are shown there depend on the measurement mode set. To see which parameters they are, point at a setting and leave the mouse pointer there for a moment. The parameter is then shown in a tooltip next to the mouse pointer and also on the left in the status bar. To set a parameter, click the arrow on the right of it and select the required setting in the context menu.

Dialog

You can set all parameters in the Acquisition Settings dialog. To do so, proceed as follows:

1. In the menu bar, select Acquisition > Settings (toolbar: 

).

   The Acquisition Settings dialog appears.
2. The dialog consists of several pages each with a group of parameters respectively. To display a certain page, click the name of the group.
3. Display for example the General page and set the parameters. You will find information on this in SECTION 7.2.1.

   The pages in the dialog can change with the measurement mode selected.
4. Set the parameters on all other pages of the dialog. You will find information on this in SECTION 7.2.
5. When you have set all parameters, click OK.

File

You can load settings from setting files and measurements. See SECTION 10.1.3 on this. You can save the current settings as described in SECTION 10.2.3.

7.2 Description of Parameters

The individual measurement parameters are described in detail in the following sections:

General On the General page, you set the measurement mode, averaging and automatic remeasuring. See SECTION 7.2.1 on this.

Channels Activate the required measurement channels on the Channels page and set their parameters for digital data acquisition. For certain hardware configurations and measurement modes, parameters on the page are not available or are set permanently. See SECTION 7.2.2 on this.

Filter You set the digital filters for the input signals on the Filters page. See SECTION 7.2.3 on this.

Frequency You set the parameters for calculating frequency spectra on the Frequency page. See SECTION 7.2.4 on this.

Window You set windowing of the input signals on the Window page. With suitable windowing you can prevent leakage. See SECTION 7.2.5 on this.

Trigger You set the parameters for triggering on the Trigger page. See SECTION 7.2.6 on this.

SE Set the parameters for Signal Enhancement (SE) on the SE page. When scanning with Signal Enhancement, you get an approximately even noise level for all scan points. See SECTION 7.2.7 on this.

Vibrometer You set the parameters of the front-end or controller for data acquisition on the Vibrometer page. See SECTION 7.2.8 on this.

Generator You can control certain function generators with the software. You set the parameters for the signal output on the Generator page. See SECTION 7.2.9 on this.

FastScan You can scan individual frequencies quickly in FastScan measurement mode. You set the parameters for this on the FastScan page. See SECTION 7.2.10 on this.

Time In Time measurement mode, you can save time signals instead of spectra for scans. To do this, you need the TDD option (Time Domain Data). You set the parameters for time measurements on the Time page. See SECTION 7.2.11 on this.

7.2.1 General Settings

On the General page, you set the measurement mode, averaging and automatic remeasuring. The FFT measurement mode is always available as standard.

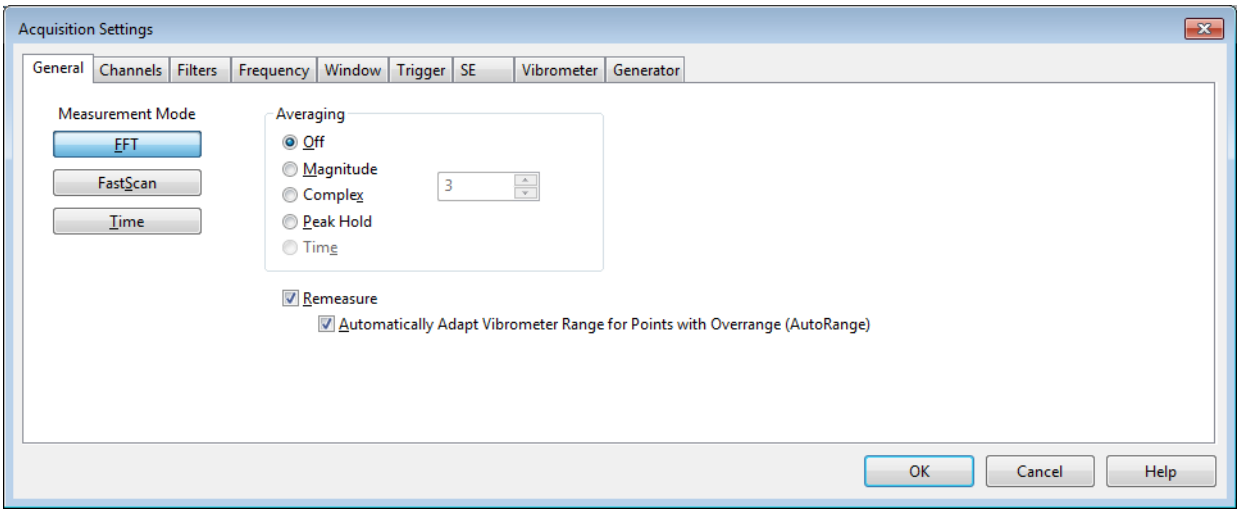


Figure 7.1: General page in FFT measurement mode

Measurement mode

On the left you select the measurement mode. You will find the availability of the various measurement modes in the following table.

Measurement mode	Measurement system		
	MSA-600-M/-V MSA-650	MSA-600-X/-U	MSA-500
FFT	Defaults	Defaults	Defaults
FastScan	Defaults	-	Defaults
Time	Optional	Optional	Optional
I/Q[1]	-	-	Optional

[1] Only with VDD option and only for MSA-500-M2 systems with a PCI-6110 data acquisition board

The pages in the dialog can change with the measurement mode selected. If, for example, you select FastScan measurement mode, then a page appears with parameters for this measurement mode. Other pages disappear.

ⓘ In FastScan measurement mode, Peak Hold and Time averaging are not available.

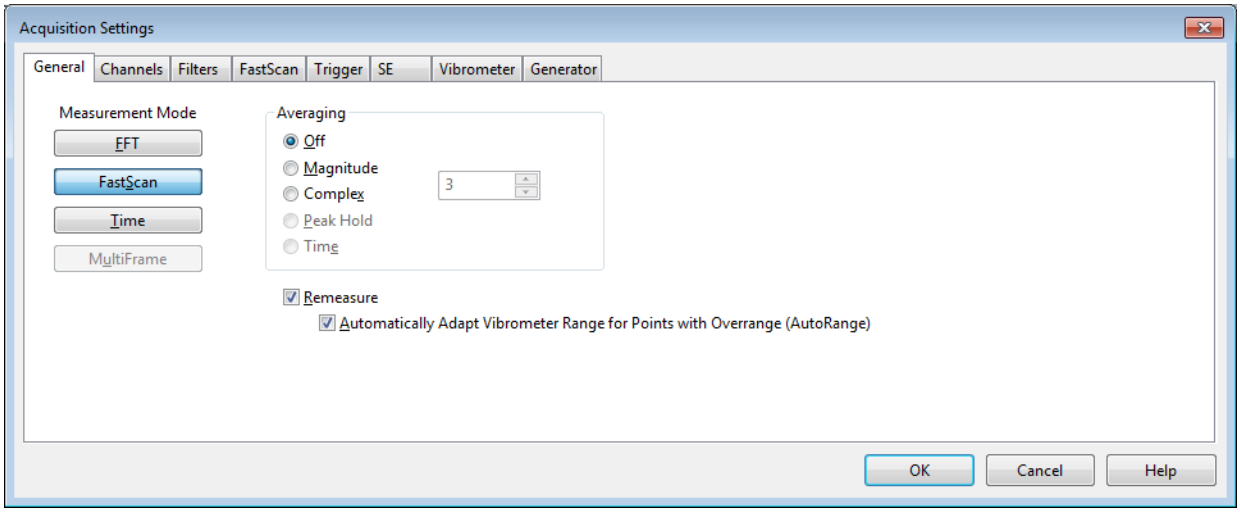


Figure 7.2: General page in FastScan measurement mode

The pages in the dialog can change with the measurement mode selected. If, for example, you select Time measurement mode, then a page appears with parameters for this measurement mode. Other pages disappear.

ⓘ In Time measurement mode, only Time averaging is available.

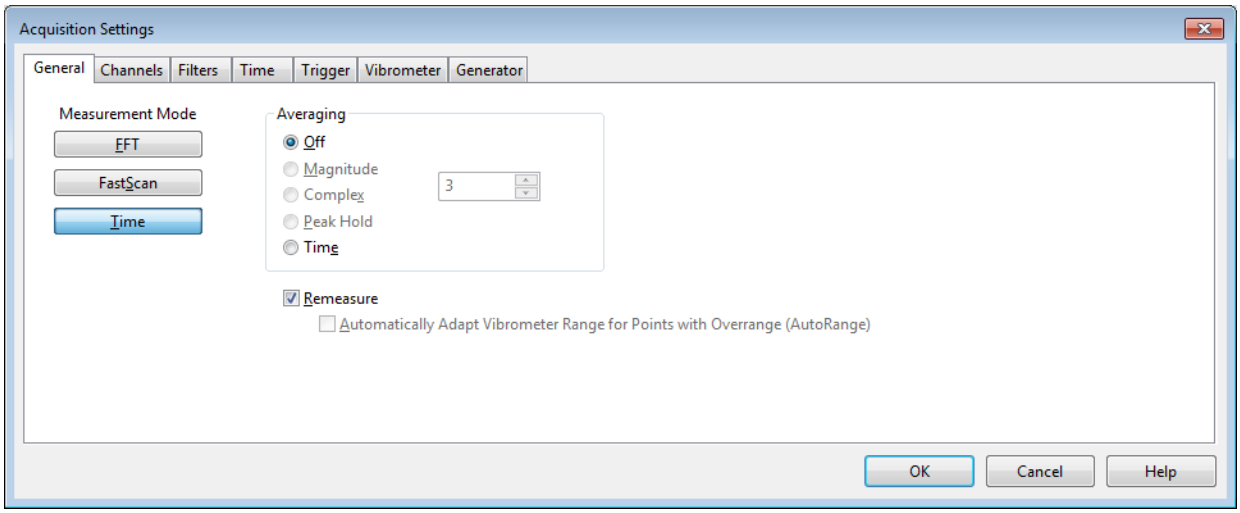


Figure 7.3: General page in Time measurement mode

The pages in the dialog can change with the measurement mode selected. If you select the I/Q measurement mode for the MSA-500, then other pages disappear.

ⓘ You cannot average in I/Q measurement mode (only with VDD option).

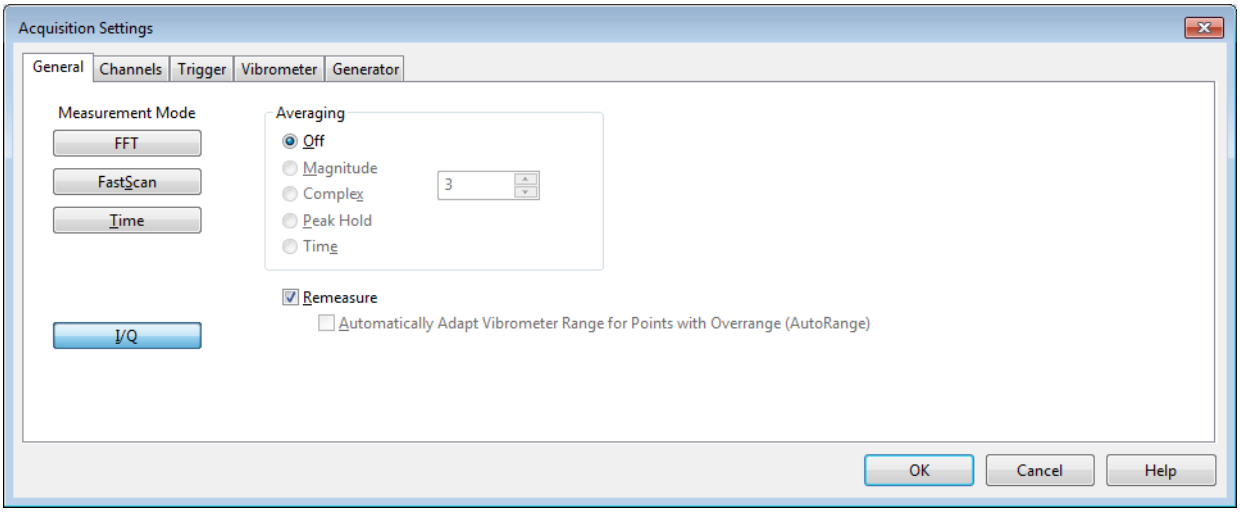


Figure 7.4: General page in I/Q measurement mode

In the following you will find a description of the individual parameters on the General page.

Averaging

On the right you set the averaging. By determining the mean value you improve the signal-to-noise ratio of spectra. You will find detailed information on averaging in your theory manual.

None: You do not use averaging.

For the following parameters, enter the number of averages next to it. The time needed for a measurement will increase by this factor.

Magnitude: Select magnitude averaging under the following conditions:

- You are neither measuring a reference signal nor triggering the measurement.
- You are using stochastic excitation – for example, in a wind tunnel.

Complex: Select complex averaging under the following conditions:

- You are measuring a reference signal or triggering the measurement.

  or
- You are using deterministic excitation.

Peak Hold: Select Peak Hold for averaging if you want to calculate and display the respective maximum of the spectra over the set number of averages.

You will find more information on averaging with Peak Hold in your PSV theory manual. You can also have the settings for Peak Hold calculated. Use the macro PeakHoldSettings.bas for this. You will find the macro in the Examples directory.

Time: In Time measurement mode, you can only select Time averaging. A prerequisite for this is that you trigger the measurements.

Features for averaging

SE	If you select Peak Hold or Time averaging, you can not make any further quality adjustment for Signal Enhancement (SE) (Fast…Excellent). See SECTION 7.2.7 on this as well.
Overrange	Spectra and time signals with the Overrange scan status are not taken into consideration when averaging. But if all spectra previously acquired have the Overrange status, then first of all these spectra will be averaged and displayed. Only once the first spectrum is captured without Overrange, the average counter is reset and the spectra measured previously with the Overrange status are rejected. Any other spectra with the Overrange status are then also rejected.

Remeasure automatically

Remeasure: Tick the check box to activate automatic remeasuring.

After a scan, all scan points are remeasured at a slightly different position in the following order:

- Not Measured status
- Invalidated status
- Overrange status
- Valid status (only when Signal Enhancement (SE) is active for at least one channel and you are in FFT measurement mode). See SECTION 7.2.7 on this as well.

ⓘ If the AutoRange setting is active, points with the Overrange status will be remeasured with less sensitive measurement ranges (if available).

Automatically Adapt Vibrometer Range for Points with Overrange: Tick the check box (AutoRange) if the software is to automatically adjust the vibrometer measurement range.

This function is available with the following limitations:

- OFV-5000 with firmware 2.0 or higher
- Not in Time measurement mode

Apart from that, a suitable, higher measurement range needs to be available.

ⓘ You can also start remeasuring manually. See SECTION 8.2 on this. After changing from acquisition to presentation mode you can also identify single scan points to be remeasured. See the presentation mode on this.

7.2.2 Settings for Channels

Work in the table

To accelerate adjustment of the settings, you can use the icons  (Copy) and  (Insert) above the table. To do so, proceed as follows:

1. Tick the required cell(s), line(s) or column(s) as described in the following.

- To mark a cell, click it. - To mark a line, click the first cell of this line. - To mark a column, click the head of this column.
- To mark several consecutive cells, lines or columns, hold the Shift key pressed and mark the first and last cell/beginning of line/column head with the mouse. - To mark several cells, lines or columns which are not consecutive, hold the Control key pressed and mark the respective cells/beginning of lines/column heads with the mouse. - If you want to mark the whole table, click the Channel cell.

2. Click  .
3. Mark the cell(s), line(s) or column(s) which you want to transfer the settings to and click  .

ⓘ You can also click with the right mouse button and select Copy or Paste in the context menu.

In the same way you can copy settings into and out of other applications such as Microsoft Excel.

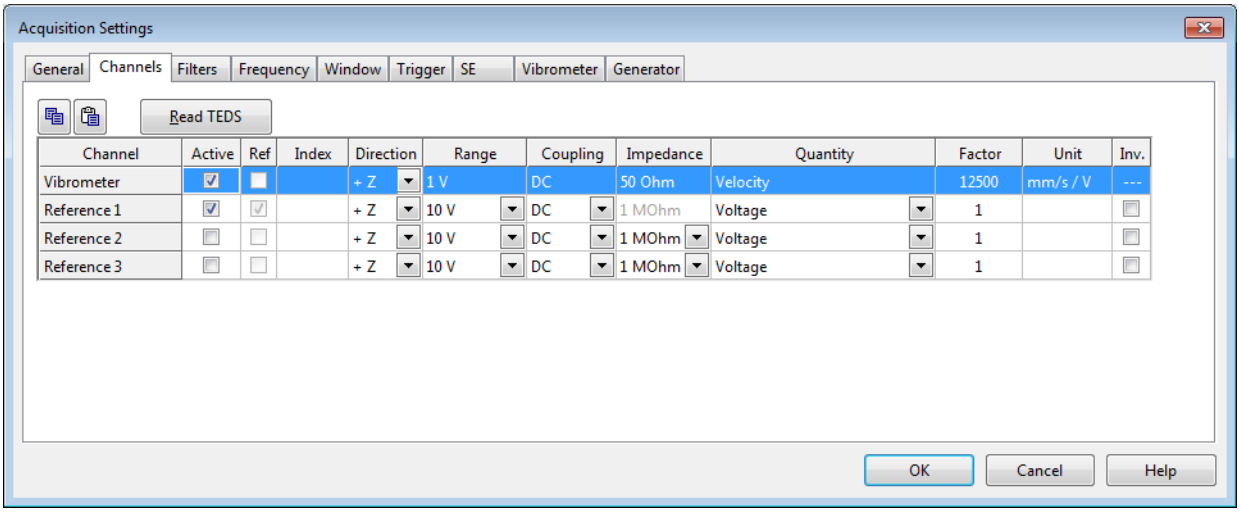


Figure 7.5: Channels page in FFT measurement mode

ⓘ You can control up to four channels depending on the device configuration.

Settings

Activate the required measurement channels on the Channels page and set their parameters for digital data acquisition. For certain hardware configurations and measurement modes, parameters on the page are not available or are set permanently.

Channel: The channels available for your system are displayed here.

Active: Mark the channels you want to make measurements on.

Ref: This column only appears if there are more than two input channels available in the software. A channel is either a response channel or a reference channel. The Vibrometer channel is always a response channel.

For systems with two channels, the Reference 1 channel is always a reference channel. For systems with more than two channels, you can select which channel is the reference channel (apart from Vibrometer). To do so, mark it in the Ref column. If the Ref column is not marked for one channel, it is used as the response channel, even if it is called Reference.

The first reference channel is used as the phase reference for the other channels if you average amplitudes or do not trigger. See SECTION 7.2.1 or SECTION 7.2.6 on this as well.

When calculating the frequency response function, all possible combinations of type response channel/reference channel are formed.

Example:

Vibrometer & Reference 1: Response channels

Reference 2 & Reference 3: Reference channels

The following signals are available as frequency response function:

- Vibrometer/Reference 2
- Vibrometer/Reference 3
- Reference 1/Reference 2
- Reference 1/Reference 3

When using more than one reference channel, frequency response functions are respectively formed as a total response signal/reference signal. This is correct with an excitation signal and several reference signals (e.g. self-excited system, vibrometer and microphone as reference).

ⓘ The measured frequency responses for the Experimental Modal Analysis can only be fed to a multi-reference procedure in the modal analysis software package by an additional determination of the transmission function. This calculation requires the total response signal from every measurement point being divided up into the proportions of the individual references. This is only possible if the reference signals are uncorrelated. See SECTION 7.2.9 on this as well.

Index: Enter the index of the point at which you are measuring the reference. The default setting is 0.

You can read the index under 2D Point or 3D Point in the scanning head control if you click the scan point with the mouse. Click the respective line in the Index column and enter the value displayed there.

ⓘ A reference point index of 0 is ignored when calculating the reference. In principle, the number of scan points is the highest possible value which you can enter in the Index field. However, the software also allows you to enter higher values which means you can make several measurements with the same index and then reunite them in an external program.

Direction: Enter the direction of the vibration. The default setting is +Z.

ⓘ The values from the Index and Direction columns are used for a geometry export in Universal File, ME'Scope and ASAM ODS AFTX. See SECTION 10.2.9, SECTION 10.2.10 or SECTION 10.2.11 on this as well. You can no longer change the vibrational direction during export.

You can also select another direction (±X/Y/Z) for all channels. With the direction of the vibrometer channel you set the orientation of the scanning head system. See SECTION 5.2.2 and your PSV theory manual on this.

ⓘ In presentation mode, you can also change the vibrational direction retrospectively.

Range: Select the input range of the data acquisition board for every channel.

Coupling: Select the input coupling for every channel. We recommend DC coupling.

Impedance: The column is only shown for the MSA-500 if the MI.3025 or M2i.3027 (M2-20 system) Spectrum data acquisition board is installed. In this case, set the input resistance of the card (50 Ω or 1 MΩ).

ⓘ You can not change the input resistance if the software automatically recognizes it and has set it up.

Quantity: Select the physical quantity which you want to measure on the respective channel.

ⓘ Using a digital filter, the software can integrate or differentiate the time signal of some quantities. See SECTION 7.2.3 on this. If the Displacement, Velocity or Acceleration quantity is selected, then the spectra of both the others are calculated by integration or differentiation. This is also applicable for the Angle, Angular Velocity, and Angular Acceleration quantities as well as for the Volume, Volume Velocity, and Volume Acceleration quantities.

Factor: Enter the calibration factor for every channel, i.e. the measurement value which corresponds to an analog input signal of one volt. If a vibrometer is connected, the software enters the calibration factor for this channel in accordance with the measurement range set on the Vibrometer page. See SECTION 3.2 and SECTION 7.2.8 on this as well.

ⓘ You can enter a calibration factor for reference channels with Voltage as the physical quantity, for example to take an amplifier into consideration.

You can also enter negative calibration factors.

Unit: Here, the software enters the SI unit of the set quantity.

Inv.: Tick the check box if you would like to invert the Unit and the Factor (form the reciprocal).

7.2.3 Settings for Filters

Description

You set the digital filters for the input signals on the Filters page:

- to limit the bandwidth
- for differentiation or integration

The Filters page is divided into two parts. In the first part you select the filter type to limit the bandwidth – high pass, low pass etc. – and select the integration or differentiation filter. In the second part you set the filter parameters to limit the bandwidth, i.e. cutoff frequency and quality. You can also see the frequency response there.

Work in the table

To accelerate adjustment of the settings, you can use the icons  (Copy) and  (Insert) above the table. To do so, proceed as follows:

1. Tick the required cell(s), line(s) or column(s) as described in the following.

- To mark a cell, click it. - To mark a line, click the first cell of this line. - To mark a column, click the head of this column.
- To mark several consecutive cells, lines or columns, hold the Shift key pressed and mark the first and last cell/beginning of line/column head with the mouse. - To mark several cells, lines or columns which are not consecutive, hold the Control key pressed and mark the respective cells/beginning of lines/column heads with the mouse. - If you want to mark the whole table, click the Channel cell.

2. Click  .
3. Mark the cell(s), line(s) or column(s) which you want to transfer the settings to and click  .

ⓘ You can also click with the right mouse button and select Copy or Paste in the context menu.

In the same way you can copy settings into and out of other applications such as Microsoft Excel.

Select filter

In the first part of the Filters page, select the filter types you want for each channel.

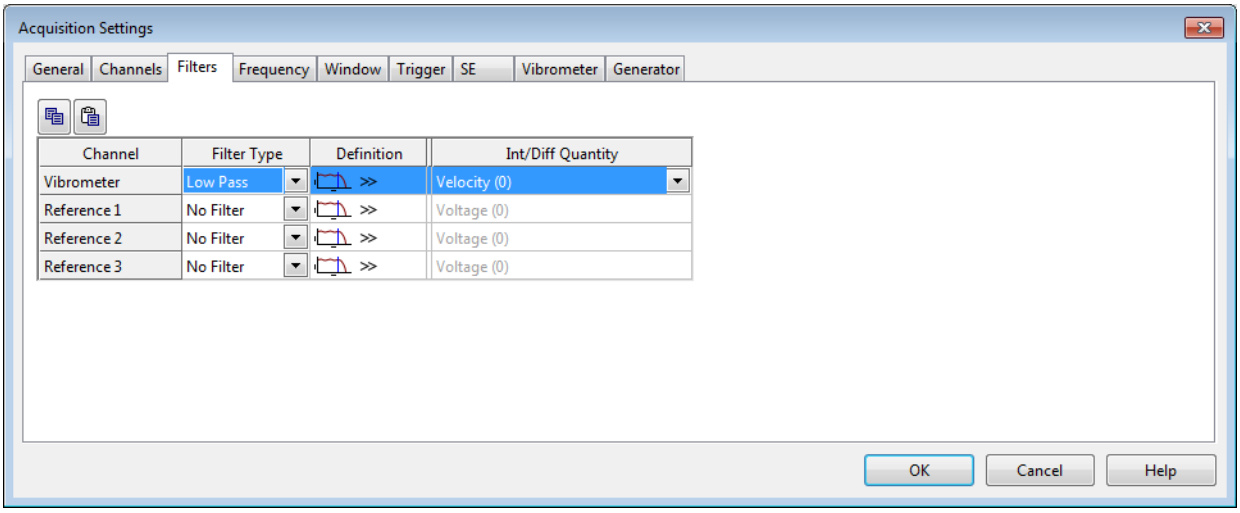


Figure 7.6: Filters page, part 1 to select the filter

Selection parameters

Channel: You can see the channels available for your system.

Filter Type: Select the type of filter to limit the bandwidth.

Definition: Click here to set the filter parameters to limit the bandwidth. The second part of the Filters page appears (see below).

Int/Diff Quantity: Using digital filters, the software can integrate or differentiate the time signal of some measured quantities. Select the physical quantity the software is to calculate. The number on the right next to the quantity tells you which arithmetic operation is being carried out. See the following table on this as well.

Number	Arithmetic operation
1	Integrate once
0	No filter
– 1	Differentiate once
– 2	Differentiate twice

ⓘ During integration and differentiation a time delay of half the sample interval occurs (sample interval = 1/sample frequency). This leads to a frequency-dependent phase shift. In case of a differentiation done twice, no time delay occurs.

Set the parameters

In the second part of the Filters page, you set the filter parameters to limit the bandwidth. There you can define the filters for several channels one after the other.

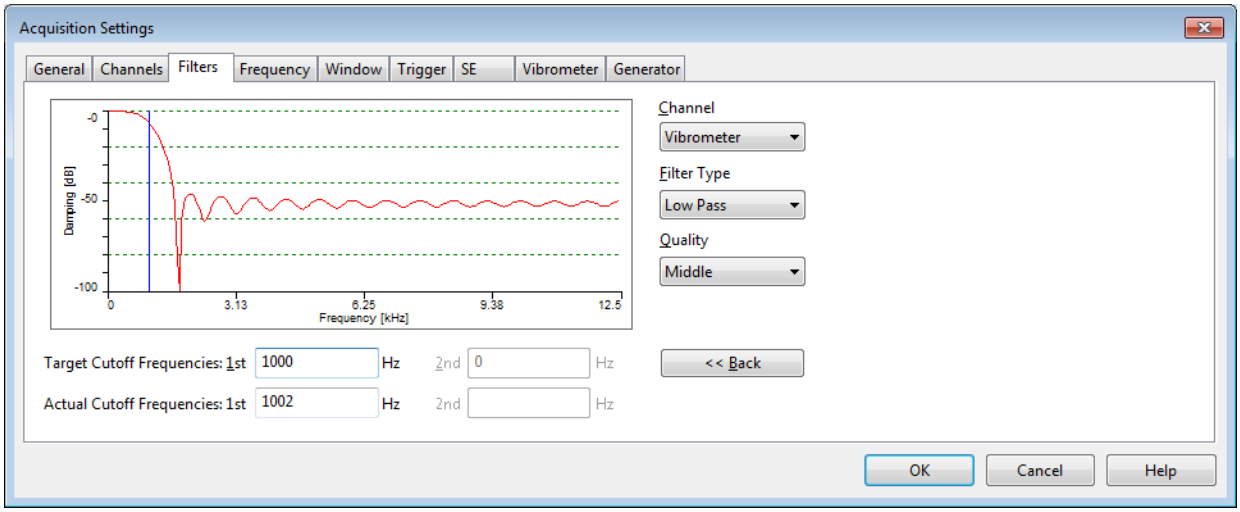


Figure 7.7: Filters page, part 2 to set the parameters

Settings

Channel: Select the channel you want to set the filter for.

Filter Type: Select the type of the filter.

The following filter types are possible:

- No filter
- Low Pass is a filter that passes signals with a frequency lower than a certain cutoff frequency and attenuates signals with frequencies higher than the cutoff frequency.
- High Pass is a filter that passes signals with a frequency higher than a certain cutoff frequency and attenuates signals with frequencies lower than the cutoff frequency.
- Band Pass is a filter that passes only signals of a frequency band. The frequency ranges below and above the pass band are thereby blocked or considerably attenuated.
- Notch is a filter that blocks or considerably attenuates signals of a frequency band. The frequency ranges below and above the stop band are thereby passed.

Quality: Select the quality of the filter.

The following qualities are possible:

- Very Low
- Low
- Middle
- High
- Very High
- Very High XXX: You can use digital filters with higher quality. The number represents the number of coefficients that are used for the digital filters.

Target Cutoff Frequencies: Enter the cutoff frequency/frequencies of the filter in Hertz.

Actual Cutoff Frequencies: The software shows the actually achievable cutoff frequency/frequencies of the filter in Hertz. If the actual value differs more than 5% from the target value, a warning message is issued.

The software uses the parameters set to calculate the frequency response and displays it.

To apply all settings, at the bottom right, click Back. The software returns to the first part of the Filters page (see above).

7.2.4 Settings for Frequency

You set the parameters for calculating frequency spectra in FFT measurement mode on the Frequency and Window pages. You will find the description of the Window page in SECTION 7.2.5.

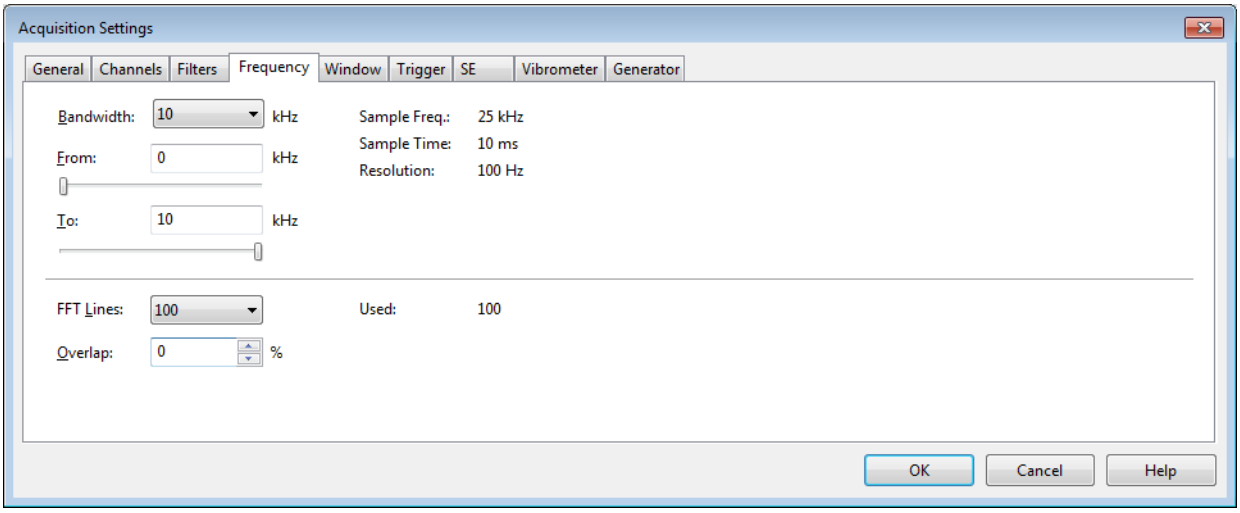


Figure 7.8: Frequency page in FFT measurement mode (MSA-600-M and MSA-650)

Frequency range

At the top left you set the frequency range to be measured.

Bandwidth: Select the bandwidth.

From, To: You can limit the frequency range which is displayed and evaluated. To do so, use the slider or enter the values. Invalid values are identified with a warning icon. Move the mouse pointer onto the warning icon to get the corresponding error message. If you control a function generator with the software, then with the Periodic Chirp, Pseudo Random, Burst Random, and White Noise signal shapes, only this frequency range is excited.

FFT

At the bottom you set the parameters for the Fast Fourier Transformation (FFT).

FFT Lines: Select the number of FFT lines that you want to analyze (up to 6400 or optionally up to 12800 or 819200 respectively).

ⓘ As an option to the 12800 FFT lines, you can still select up to 819200 FFT lines. You require the FFTExt option for this (Extended FFT Line Number). Note that if you are using this high number of FFT lines together with selecting many channels, your system can use its full capacity. Depending on the configuration of your system it is then not possible to use all settings.

Used: The number of FFT lines shown here corresponds to the frequency range between From and To (see above).

Overlap: This parameter only appears for certain hardware configurations. The software only takes the parameter into account for measurements with averaging and without trigger. With overlap you can significantly reduce the time for a measurement.

You will find more information on overlap in your PSV theory manual.

Bandwidth Extension: This parameter is only available for the MSA-600-U as an option. Select one of the extension options for the bandwidth. The maximum values of the BWX1 and BWX2 bandwidth extensions can only be reached if you do not select the amplitude values too high. The bandwidth extension makes sense for small amplitudes or low speeds.

The following settings are possible:

- Off: The default bandwidth of 0…600 MHz is available.
- BWX1 (only with PSV-S-Bw1G2 or PSV-S-Bw2G4 option): The bandwidth of 0…1.2 GHz is available. For technical reasons, the frequency range for the bandwidth of 2000 MHz is restricted (0…1200 MHz).
- BWX2 (only with PSV-S-Bw2G4 option): The bandwidth of 1.2…2.4 GHz is available. For technical reasons, the frequency range for the bandwidth of 4000 MHz is restricted (1235…2400 MHz).

Sample frequency, sample time and resolution

The software calculates the following parameters and shows them at the top right.

Sample Freq.: For the MSA-600/MSA-650, the software calculates the sample frequency f_{Sample} as follows:

$$f_{Sample} = 2.5 \cdot BW$$ Equation 7.1

BW… Bandwidth

For the MSA-500, the software calculates the sample frequency f_{Sample} as follows:

$$f_{Sample} = 2.56 \cdot BW$$ Equation 7.2

BW… Bandwidth

Sample Time: The software calculates the sample time t_{Sample} as follows:

$$t_{Sample} = \frac{n_{FFT}}{BW}$$ Equation 7.3

BW… Bandwidth

n_{FFT}… Number of FFT lines

Resolution: The software calculates the resolution Δf as follows:

$$\Delta f = \frac{1}{t_{Sample}}$$ Equation 7.4

t_{Sample}… Sample Time

Features for VDD

In I/Q measurement mode, an additional parameter appears on the Frequency page for VDD measurements. This parameter only appears for the MSA-500 with the VDD option (Digital Demodulation).

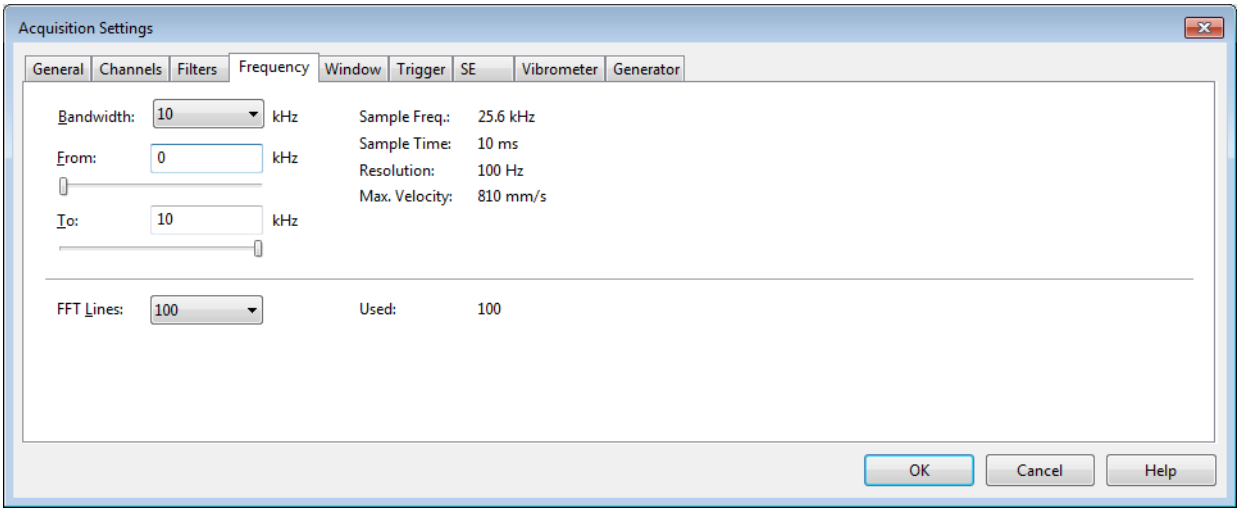


Figure 7.9: Frequency page in I/Q measurement mode (only MSA-500 with VDD option)

Max. Velocity: Because of the processing power of the PC, only signals up to this maximum vibrational velocity can be acquired with the VDD option (Digital Demodulation). The value depends on the bandwidth and the number of FFT lines (see above).

7.2.5 Settings for Window

You set the parameters for calculating frequency spectra on the Frequency and Window pages. You will find the description of the Frequency page in SECTION 7.2.4.

You set windowing of the input signals on the Window page. With suitable windowing you can prevent leakage. You will find information on this in your PSV theory manual.

Work in the table

To accelerate adjustment of the settings, you can use the icons  (Copy) and  (Insert) above the table. To do so, proceed as follows:

1. Tick the required cell(s), line(s) or column(s) as described in the following.

- To mark a cell, click it. - To mark a line, click the first cell of this line. - To mark a column, click the head of this column.
- To mark several consecutive cells, lines or columns, hold the Shift key pressed and mark the first and last cell/beginning of line/column head with the mouse. - To mark several cells, lines or columns which are not consecutive, hold the Control key pressed and mark the respective cells/beginning of lines/column heads with the mouse. - If you want to mark the whole table, click the Channel cell.

2. Click  .
3. Mark the cell(s), line(s) or column(s) which you want to transfer the settings to and click  .

ⓘ You can also click with the right mouse button and select Copy or Paste in the context menu.

In the same way you can copy settings into and out of other applications such as Microsoft Excel.

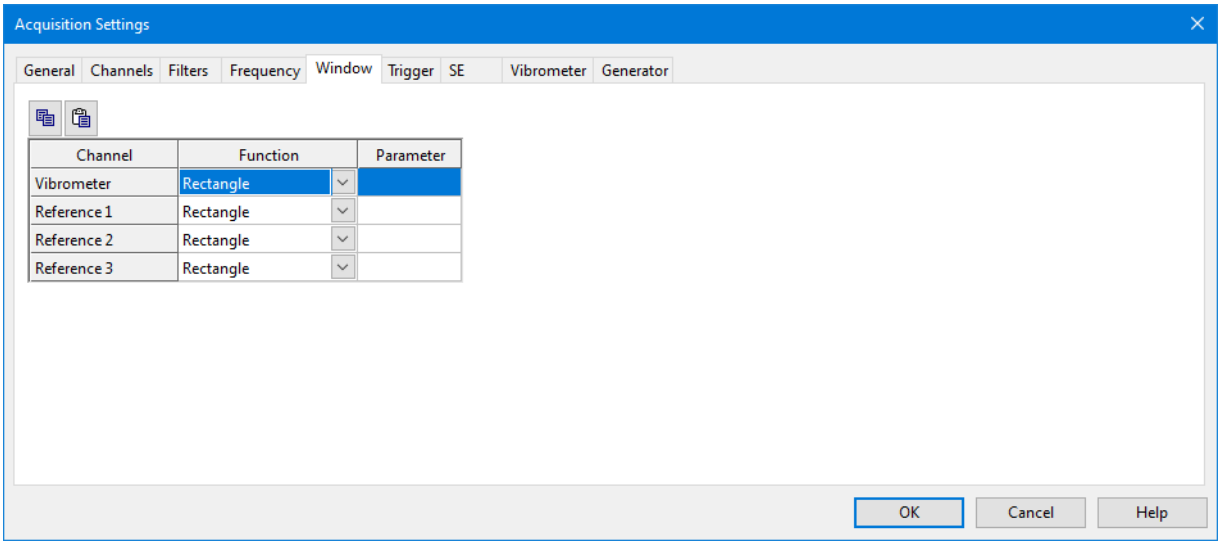


Figure 7.10: Window page

Channel: The channels available for your system are displayed.

Function: Select a window function which is suitable for your application. See your PSV theory manual on this as well.

Parameter: The window functions Tapered Hanning, Force, and Exponential can be fine-tuned with a parameter. The parameters are described in the following table.

Window function	Description of the parameter	Defaults
Tapered Hanning	Rise and decay time in percent of the sample time	10
Force	Length of the window in percent of the sample time	5
Exponential	Quotient from window length and decay time	4

7.2.6 Settings for Trigger

You set the parameters for triggering on the Trigger page. You will find information on the influence of the trigger in your PSV theory manual.

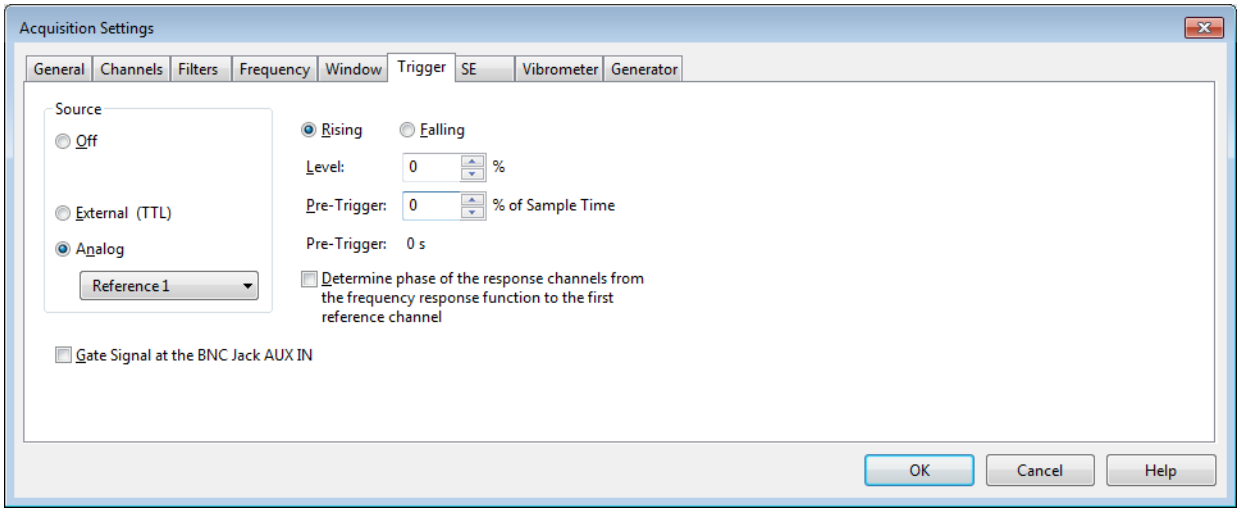


Figure 7.11: Trigger page in FFT measurement mode (MSA-600-M/-V and MSA-650)

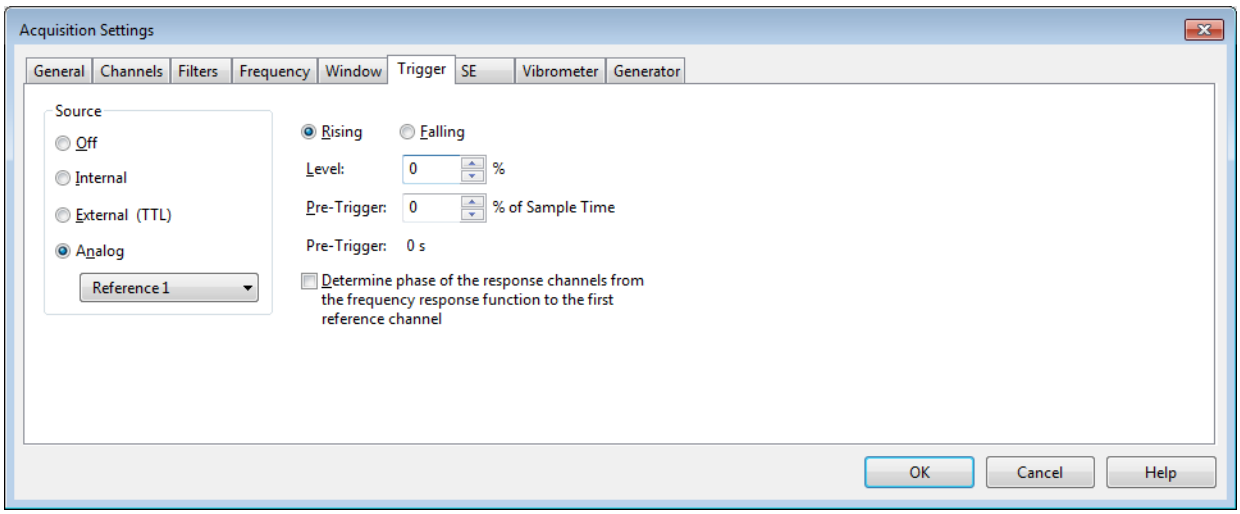


Figure 7.12: Trigger page in FFT measurement mode (MSA-500)

You can not trigger in FastScan measurement mode.

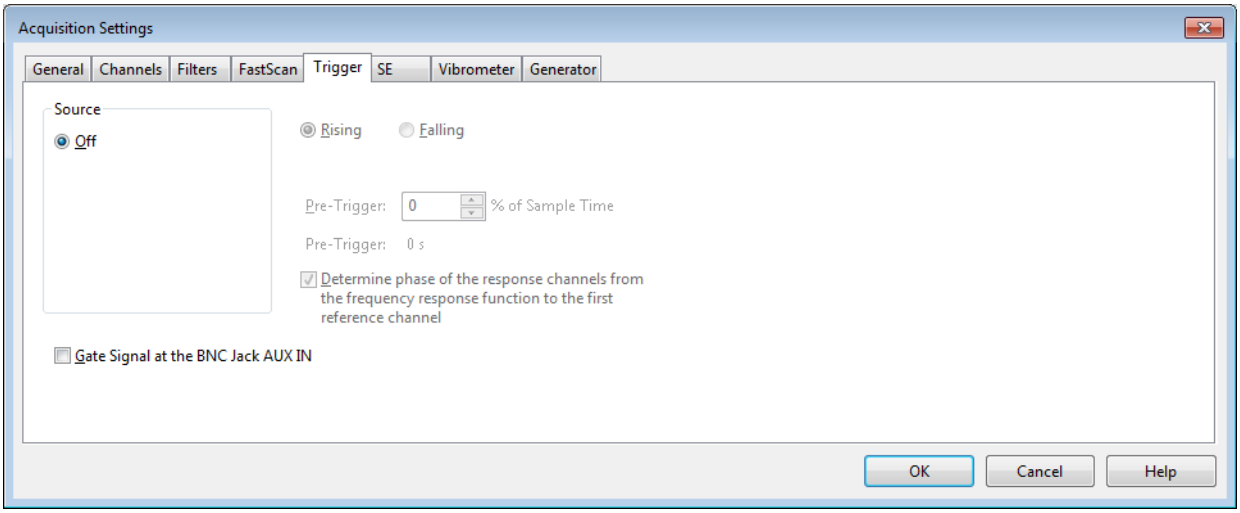


Figure 7.13: Trigger page in FastScan measurement mode (MSA-600-M/-V and MSA-650)

Source

Set the trigger source on the left.

Off: You are measuring without a trigger.

ⓘ To measure phases you need a trigger or a reference signal.

Internal: Trigger on the synchronization pulse of the internal generator. To do this, you do not need a cable connection to the junction box.

ⓘ Using the internal trigger signal is not available for all data acquisition systems: You will find limitations for the internal trigger signal below.

External (TTL): You trigger on an external trigger signal.



ATTENTION!

Damage to the data acquisition board caused by the voltage being too high!

» Only connect voltages in the range of 0 to 5 V to the input for the external trigger!

ⓘ Using the external trigger signal is not available for all bandwidths or sample frequencies depending on the data acquisition board. You will find limitations for the external trigger signal below.

Analog: You trigger on one of the analog input signals. Select the channel the trigger signal is connected to below.

ⓘ You will find limitations for the analog trigger signal in the following tables.

Conditions

On the right you set the trigger conditions.

Rising, Falling: Here you select which signal edge is used to trigger.

Level: If you trigger using one of the analog input signals, you set the trigger level here. 100% level corresponds to the input range of the channel. See SECTION 7.2.2 on this as well.

ⓘ You will find limitations for the analog trigger signal in the following tables.

Pre-Trigger: Here you enter the trigger time in % of the sample time. If the data acquisition is to start before the trigger time, enter a positive value. If data acquisition is to start after the trigger time (Post-Trigger), enter a negative value. The time for pre-trigger or post-trigger is displayed below.

ⓘ Decimal places may be entered.

Determine phase of the response channels: If you want to measure phases you need a trigger or a reference signal. You can then determine the phases differently. Tick the check box and the phase of the response channels is determined from the frequency response function to the first reference channel. Deactivate the check box and the phase is determined from the time relative to the trigger.

ⓘ If you measure without a trigger (Off), the phase of the response channel is determined as standard from the frequency response function to the first reference channel.

If you activate this function or if it is activated automatically, then all the phase information is no longer available for the inverse FFT (IFFT). This then means that time signals can no longer be reconstructed with the inverse FFT.

Limitations

If you use the analog and internal trigger, there are basically no limitations; for exceptions refer to the following table.

ⓘ If you use the analog trigger, then during scanning you do not trigger on the vibrometer signal.

Table 7.1: Limitations for using the internal trigger

System	Data acquisition board	Limitations
MSA-600-M	M2i.4961	no internal trigger
MSA-600-V	M2i.4961	no internal trigger
MSA-600-X	ADQ7	no internal trigger
MSA-600-U	ADQ7	no internal trigger
MSA-650-M	M2p.5942	no internal trigger
MSA-650-V	M2p.5942	no internal trigger
MSA-500-M2-20	M2i.3027	no internal trigger
MSA-500-M2-20	MI.3025	no internal trigger

You will find the limitations for using the external trigger in the following table. If you do not keep to the limitations, the external trigger will be switched off. For the MSA-600-X/-U, there are no limitations.

Table 7.2: Limitations for using the external trigger

System	Data acquisition board	Limitations
MSA-600-M	M2i.4961	> 1.25 MHz bandwidth: no limitation ≤ 1.25 MHz bandwidth: no pre-trigger
MSA-600-V	M2i.4961	> 1.25 MHz bandwidth: no limitation ≤ 1.25 MHz bandwidth: no pre-trigger
MSA-650-M	M2p.5942	> 1.25 MHz bandwidth: no limitation ≤ 1.25 MHz bandwidth: no pre-trigger
MSA-650-V	M2p.5942	> 1.25 MHz bandwidth: no limitation ≤ 1.25 MHz bandwidth: no pre-trigger
MSA-500-M2	PCI-6111	no pre-trigger
MSA-500-M4[1]	PCI-6110	no pre-trigger
MSA-500-M2-20	M2i.3027	> 2 MHz bandwidth: no limitation ≤ 2 MHz bandwidth: no pre-trigger
MSA-500-M2-20	MI.3025	> 2 MHz bandwidth: no limitation ≤ 2 MHz bandwidth: no pre-trigger

[1] Only with VDD option

Gate signal

You can control data acquisition with an external gate signal. To do this, you need the Gate Input option. You will find information on this in the operating instructions of the measurement system. You do not have to set special parameters for this.

Gate Signal at the BNC Jack AUX IN: This parameter is only available for the MSA-600-M/-V and the MSA-650. Tick the check box if you want to create a gate signal at the AUX IN BNC jack on the MSA-F-6x0 Front-End.

7.2.7 Settings for Signal Enhancement (SE)

You set the parameters for Signal Enhancement (SE) and Speckle Tracking on the SE page. When scanning with Signal Enhancement and Speckle Tracking, the noise level for all scan points is roughly the same. You will find information on measuring with Signal Enhancement in SECTION 8.9.1 and in your PSV theory manual.

ⓘ The Time measurement mode is an exception. The SE page only appears if you have set Time for averaging on the General page.

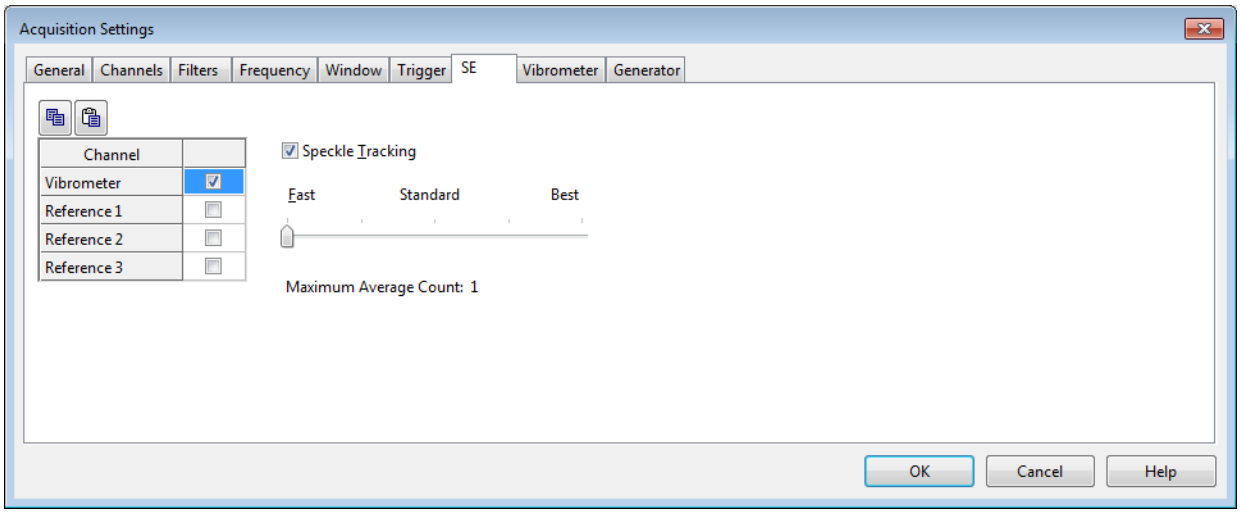


Figure 7.14: SE page (MSA-500)

ⓘ The Speckle Tracking parameter is only available for the MSA-500.

If you want to mark the whole table, click the Channel cell.

Activate On the left you can activate Signal Enhancement.

ⓘ Time measurement mode with Time averaging: With active SE, a median algorithm is used for averaging. This makes higher demands on your PC with regards to memory and calculation speed.

Channel: You can see the channels available for your system. Mark the channels which Signal Enhancement is to be active for.

ⓘ Activate Signal Enhancement for all channels which may have dropouts (e.g. vibrometer channel). For reference channels, the following applies: If a vibrometer is connected, measure on treated surfaces if possible and switch off Signal Enhancement. To treat surfaces, stick retro-reflective film on them or apply metallic paint. If you are measuring with a vibrometer, but not on treated surfaces, activate Signal Enhancement.

Speckle Tracking

Speckle Tracking: Tick the check box to activate Speckle Tracking. As a prerequisite, Signal Enhancement must be active for the vibrometer channel. Without Speckle Tracking, every scan point is measured at the same position when averaging. In doing so, you might always measure a dark speckle. With Speckle Tracking, the position of every scan point is slightly changed on averaging – at one meter stand-off distance by approx. 50 µm. It is very likely that you now measure some bright speckles as well.

Averaging

On the right you set how often additional averaging is carried out with Signal Enhancement.

ⓘ Spectra and time signals with the Overrange scan status are not taken into consideration when averaging. But if all spectra previously acquired have the Overrange status, then first of all these spectra will be averaged and displayed. Only once the first spectrum is captured without Overrange, the average counter is reset and the spectra measured previously with the Overrange status are rejected. Any other spectra with the Overrange status are then also rejected.

Fast: Averaging is only carried out as often as set on the General page. See SECTION 7.2.1 on this as well.

Defaults: Depending on the optical signal level, averaging is carried out up to three times as often as set.

Best: Averaging is carried out up to five times as often as set.

Maximum Average Count: The software shows the maximum number of averages depending on the setting for Signal Enhancement (Fast, Standard, Best).

ⓘ In the Time measurement mode, the number of averages always corresponds to the specification given on the General page.

7.2.8 Settings for Vibrometer

On the Vibrometer page, you set the parameters of the controller or front-end for data acquisition. You will find information on suitable settings in the associated operating instructions. If you are controlling a second vibrometer, you set the parameters for both controllers on the Vibrometer page. You will find information on controlling a second vibrometer in SECTION 8.8.

MSA-600/ MSA-650

For MSA-600-M/-V, the MSA-F-600 front-end is displayed. For MSA-650, the MSA-F-650 front-end is displayed. For the MSA-600-X/-U (MSA-F-620 front-end), the Vibrometer is displayed, but you can not set any parameters.

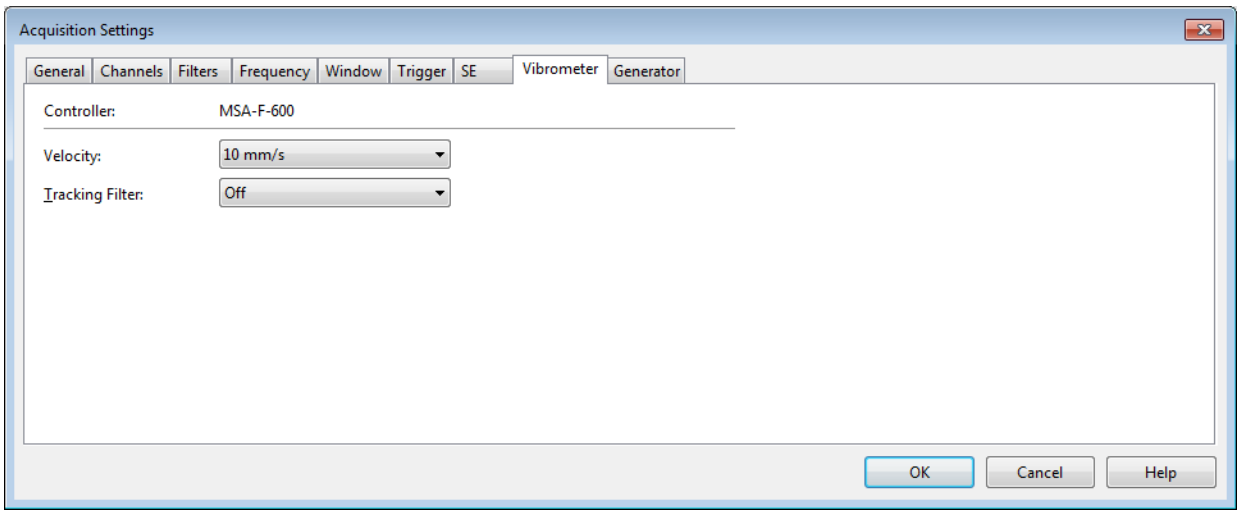


Figure 7.15: Vibrometer page for the MSA-F-600 front-end (MSA-600-M/-V)

Settings

Controller: The connected front-end is displayed.

Velocity: Select the measurement range for the velocity. You can only select the measurement ranges which are available for the selected bandwidth on the Frequency page. See SECTION 7.2.4 on this as well.

ⓘ The software queries the delay time of the measurement range set at the vibrometer front-end. The delay time is used to correct the phase of the spectra and the signals in the time domain. With time signals the delay time can only be corrected exactly to 1 sample.

Tracking Filter: Adjust the tracking filter. You will find information on suitable settings in the operating instructions of the measurement system.

MSA-500

For the MSA-500, the OFV-5000 controller is displayed.

It depends on the connected controller which parameters on the Vibrometer page appear. See SECTION 3.2 on this as well. The parameters for the OFV-5000 controller are described below. For information on other controllers, refer to the operating instructions of the vibrometer. You will also find information on suitable settings there.

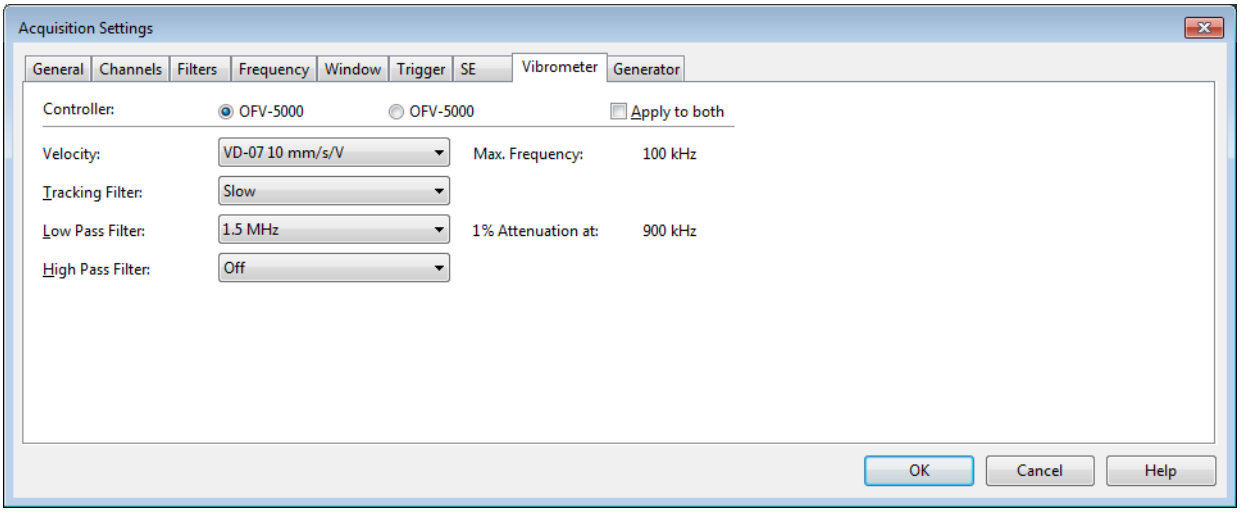


Figure 7.16: Vibrometer page for the OFV-5000 controller (MSA-500)

Settings

Controller: Select the controller or respectively the front-end which the settings are to apply to. If there are two controllers installed, you can switch between the two of them and define the settings for each of them individually.

Apply to both: If both controllers are identical, i.e. the same type and equipped with identical decoders, then the settings for one can be transferred directly to the other. To do so, tick the box.

Quantity: If your controller is equipped with the appropriate decoders, select the desired quantity from the list.

Velocity: Select the measurement range for the velocity. For the controllers or the front-end, the decoder names are displayed in full.

ⓘ OFV-5000 from firmware version 2.0: The software queries the delay time of the measurement range set on the vibrometer controller. The delay time is used to correct the phase of the spectra and the signals in the time domain. With time signals the delay time can only be corrected exactly to 1 sample. The high and low pass filter should be left switched off as their delay times are not being corrected. The correction can only be made approximately. The remaining phase error in the acoustic area is up to 3° and increases for frequencies above 20 kHz with increasing frequency up to 35°.

Displacement: This line only appears if the front-end or the controller is equipped with an optional displacement decoder. Select the measurement range for displacement.

ⓘ Even if you are only measuring and evaluating the displacement signal, the measurement range for the velocity must not be exceeded (see above).

ⓘ For the DD-900 and DD-500 displacement decoder, the delay time is not available on the vibrometer controller. Therefore, correcting the phase of the spectra and the signals in the time domain is not possible.

Tracking Filter: Adjust the tracking filter. You will find information on suitable settings in the operating instructions of the measurement system.

Low Pass Filter: Adjust the low pass filter.

ⓘ The settings for the low pass filter and the high pass filter only have an effect on the VELOCITY OUTPUT.

High Pass Filter: Adjust the high pass filter.

ⓘ If you are using the vibrometer to scan, switch the high pass filter off.

Max. Frequency: For the selected measurement range, the respective maximum frequency is shown.

1% Attenuation at: For the selected low pass filter or high pass filter, the frequency is shown for 1% attenuation of the measurement signal.

7.2.9 Settings for Generator

You can control certain function generators with the software. You will find information on this in SECTION 8.7. You set the parameters for the signal output on the Generator page.

ⓘ In FastScan measurement mode, a sinusoidal signal with the frequency of the FastScan is always used for excitation.

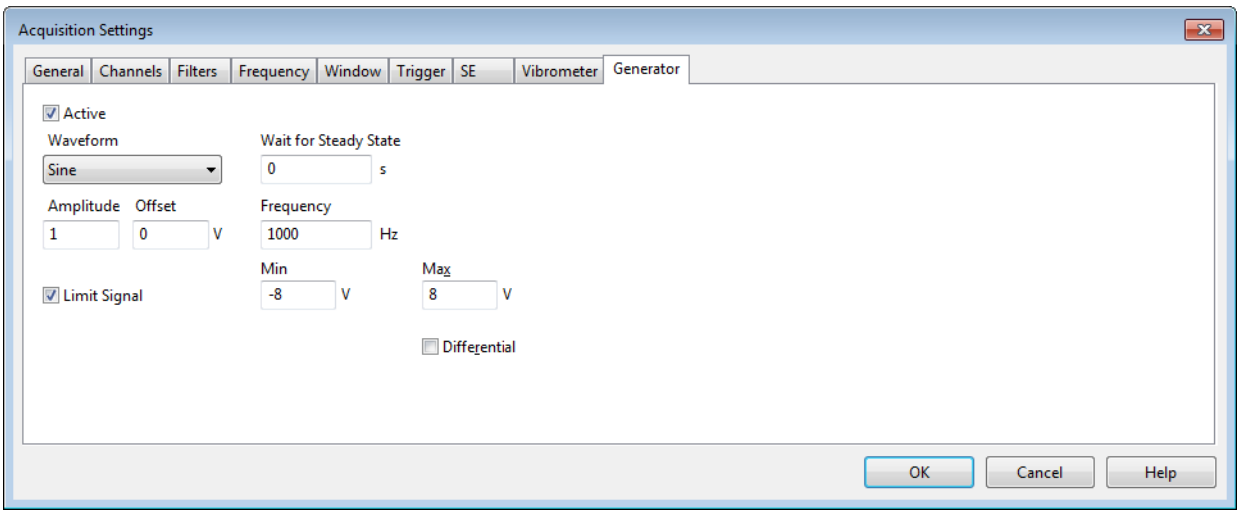


Figure 7.17: Generator page (MSA-600-M/-V and MSA-650)

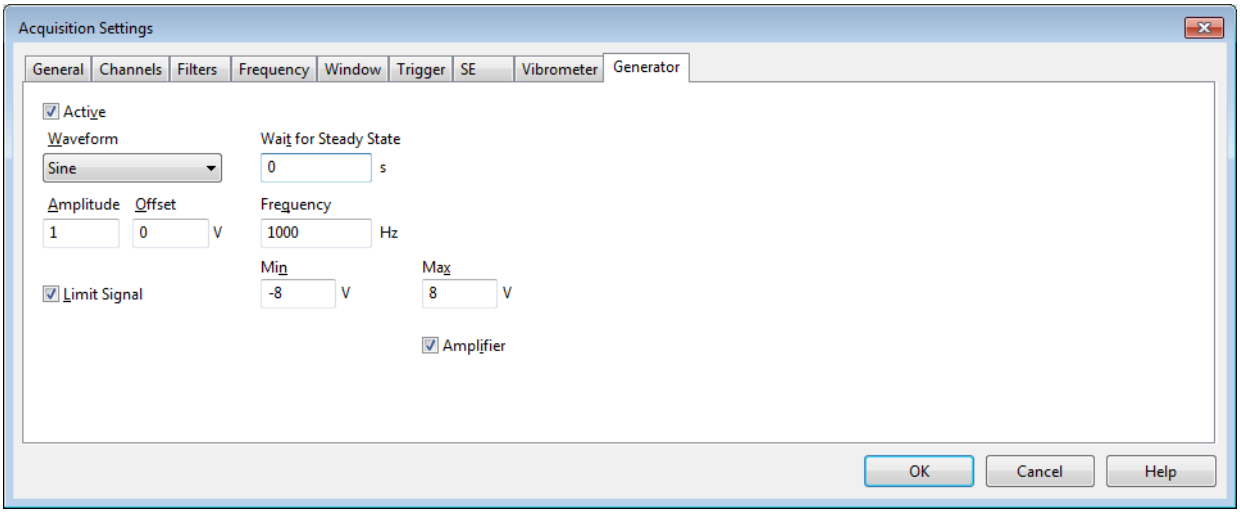


Figure 7.18: Generator page for Spectrum MI.6030 or M2i.6030 generator board respectively, without integrated amplifier (MSA-500)

Activate generator

Active: Tick the check box if you want to activate all generator channels being available for the measurement system. If you are not always working with the function generator, you can switch it off here for the software. Otherwise signal output starts automatically when scanning. Apart from that, you will suppress corresponding warnings and error messages.

Waveform: Select the waveform you want to emit. Some parameters on the Generator page change with the waveform selected. You will find an explanation of these parameters and information on some of the waveforms in the following.

Sine, Triangle, Square, Ramp

Frequency: For the waveforms Sine, Triangle, Square, and Ramp, enter the repeat frequency of the signal in Hertz.

Sweep

Start Frequency, End Frequency, Sweep Time: For a Sweep, enter the start and end frequency in hertz here. The start frequency can be higher than the end frequency. You also enter the duration of the sweep in seconds.

Periodic Chirp, Pseudo Random

For the waveforms Periodic Chirp and Pseudo Random, sinusoidal signals are emitted to all FFT lines at the same time, but only in the frequency range between From and To on the Frequency page. See SECTION 7.2.4 on this as well.

With Periodic Chirp, the phases of the sinusoidal signals are adapted so that the energy of the resulting signal is maximized. With Pseudo Random, the phase of the sinusoidal signals is random.

Amplitude Correction File: As standard, with Periodic Chirp and Pseudo Random all sinusoidal signals have the same amplitude. If you need other amplitudes or frequencies, you can generate a correction file and list it here. To do this, tick the check box and click Browse. The Open dialog appears. Navigate to your correction file and click Open.

You will find a short description of the format required in the file AmplitudeCorrectionSample.txt in the Examples directory (shortcut on the desktop). You can also use this file as a template for your own correction files.

Burst Chirp, Burst Random

The waveforms Burst Chirp and Burst Random are only available for certain hardware configurations. The number of samples for Burst Random is limited to 524 288 (1/2 Megasample).

ⓘ For the Burst Random waveform, the signals are only emitted in the frequency range between From and To on the Frequency page. See SECTION 7.2.4 on this as well.

Start Frequency, End Frequency: For a Burst Chirp, enter the start and end frequency in Hertz. The start frequency can be higher than the end frequency.

Burst Start, Burst Length: Enter when the burst is to start and how long it is to last (input as a percentage of the sample time).

White Noise

For the waveform White Noise, the signals are only emitted in the frequency range between From and To on the Frequency page. See SECTION 7.2.4 on this as well.

User Defined

Frequency: For the User Defined waveform you enter the repeat frequency of the signal in Hertz.

Browse: If you need other waveforms, you can generate them yourself. You will find a short description of the format required in the file UserDefSample.txt in the Examples directory (shortcut on the desktop). You can also use this file as a template for waveforms you want to generate. To load a waveform you have generated, select the User Defined waveform and click Browse. The Open dialog appears. Navigate to the file you have generated and click Open.

General parameters

Wait for Steady State: Enter the time in seconds which the excited object needs to attain steady-state condition. Once you have either starting signal output or have closed the Acquisition Settings dialog with OK, the software delays data acquisition by this time. This setting is stored in the measurement file and displayed in the File Information dialog.

ⓘ The generator signal is generated during this waiting time.

Amplitude, Offset: Enter the peak amplitude and DC offset of the signal in volts for high-impedance termination of the function generator. The following condition must be met:

$$\text{Magnitude} + |\text{Offset}| \leq 10\,\text{V} \qquad \text{Equation 7.5}$$

If the condition is not met, then the software changes the offset. Also note the features described further down.

Many function generators have an output impedance of 50 Ω. If you terminate such a function generator with 50 Ω, the output voltages are only half the size set.

Generator	Output impedance
PCI-4461	22 Ω (min. load 600 Ω)
PCI-6111	50 Ω
PCI-6110	50 Ω
PCI-6711	< 1 Ω
M2p.6570	50 Ω
MI.6030	< 1 Ω
M2i.6030 (without integrated amplifier)	< 1 Ω
M2i.6030 (with integrated amplifier)	50 Ω
M4i.6630-x8	50 Ω

The preset termination of the Spectrum data acquisition board is 50 Ω (default setting). To change the setting to 1 MΩ, display the Channels page in the Acquisition Settings dialog and make the change. See SECTION 7.2.2 on this as well.

Limit Signal: Tick this check box if you want to limit the signal strength of the generator signal. This allows you to protect sensitive components. Enter the appropriate values for Min and Max. The software takes these limits and the offset into consideration for the generator amplitude slider. See SECTION 4.1.2 on this as well.

Features for MSA-600-M/-V

Differential: Tick the check box to generate a symmetrical output signal. Deactivate the check box to generate an asymmetrical output signal. You will information on symmetric and asymmetric output signals in the operating instructions of the measurement system.

Features for MSA-500

Amplifier: This check box only appears for MSV-500-M2-20 systems with the Spectrum generator board MI.6030 or M2i.6030 without an integrated amplifier. Tick the check box to increase the voltage of the generator output from ±3 V to ±10 V.

Features for MSA-600-U

If you use the Polytec A-ESG-002 Signal Generator, you can make settings on the Generator page.

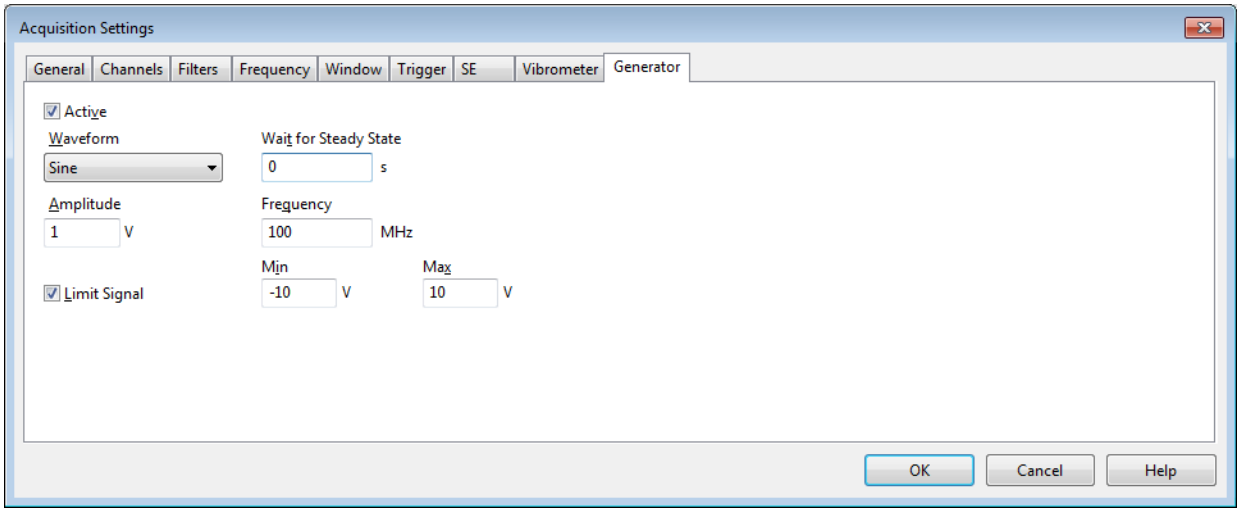


Figure 7.19: Generator page for the Sinus waveform (MSA-600-U)

Active: If you are not always working with the function generator, you can switch it off here for the software. This will suppress corresponding warnings and error messages.

Waveform: Select the waveform you want to emit. Some parameters on the Generator page change with the waveform selected.

Sinus	For the Sine waveform, enter the repeat frequency of the signal in Megahertz in the Frequency field (Time and FFT measurement mode).
MultiCarrierCW	For the MultiCarrierCW waveform, sinusoidal signals are emitted on all FFT lines at the same time, but only in the frequency range between Start Frequency and End Frequency (FFT measurement mode).

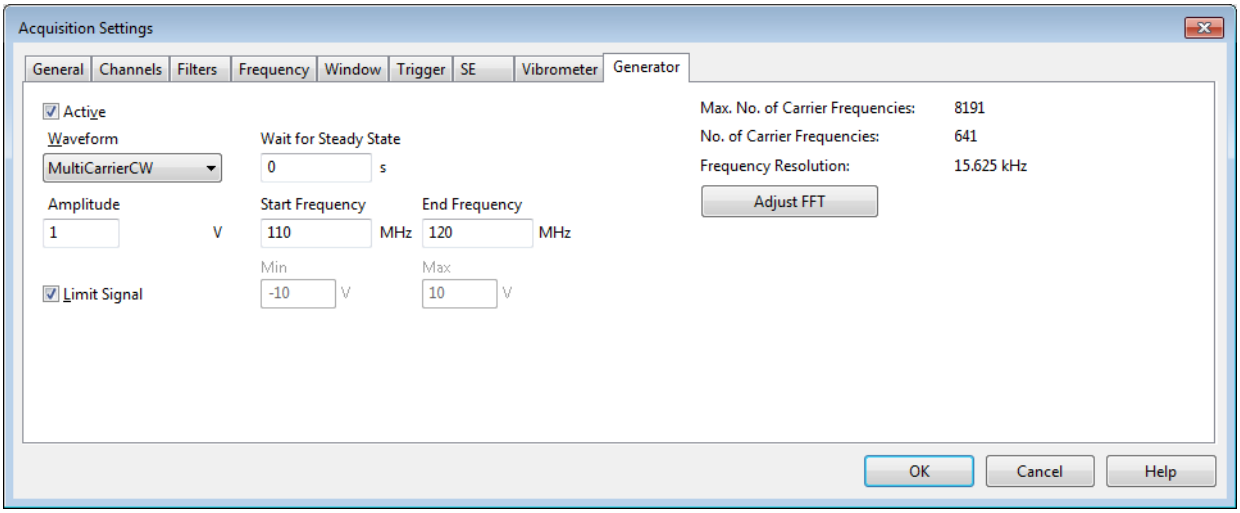


Figure 7.20: Generator page for the MultiCarrierCW waveform (MSA-600-U)

Start Frequency, End Frequency: For the MultiCarrierCW waveform, enter the start and end frequency in Megahertz.

Max. No. of Carrier Frequencies: The maximum number of carrier frequencies for the hardware is shown.

No. of Carrier Frequencies: The number of carrier frequencies is shown which result from the selected settings for the MultiCarrierCW waveform.

Frequency Resolution: The calculated resolution on the Frequency page is displayed. See SECTION 7.2.4 on this as well.

Adjust FFT: Click here if you want to adjust the values for Bandwidth and FFT Lines on the Frequency page on the basis of the settings for the generator.

7.2.10 Settings for FastScan

You can scan individual frequencies quickly in FastScan measurement mode. You will find information on this in SECTION 8.9.2. You set the parameters for this on the FastScan page.

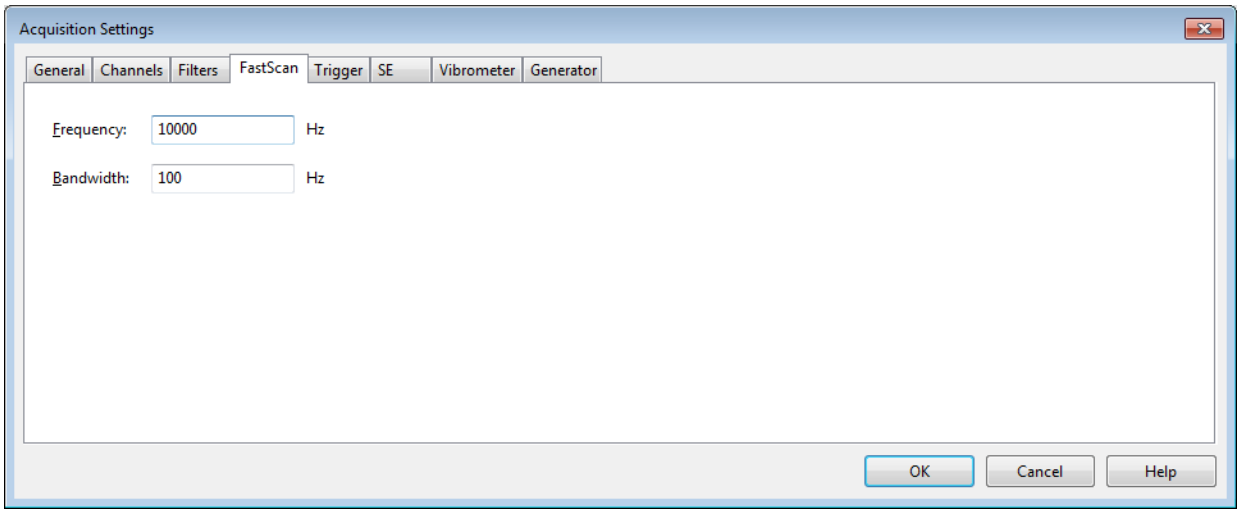


Figure 7.21: FastScan page (MSA-600-M/-V and MSA-650)

Settings

Frequency: Enter the frequency at which you want to scan fast.

ⓘ If you control a function generator using the software, then in the FastScan measurement mode, a sinusoidal signal at the FastScan frequency is always used for excitation.

Bandwidth: Enter the bandwidth.

ⓘ The smaller the bandwidth, the better the signal-to-noise ratio, but the slower the scan.

Features for VDD

In I/Q measurement mode, an additional parameter appears on the FastScan page for VDD measurements. This parameter only appears for the MSA-500 with the VDD option (Digital Demodulation).

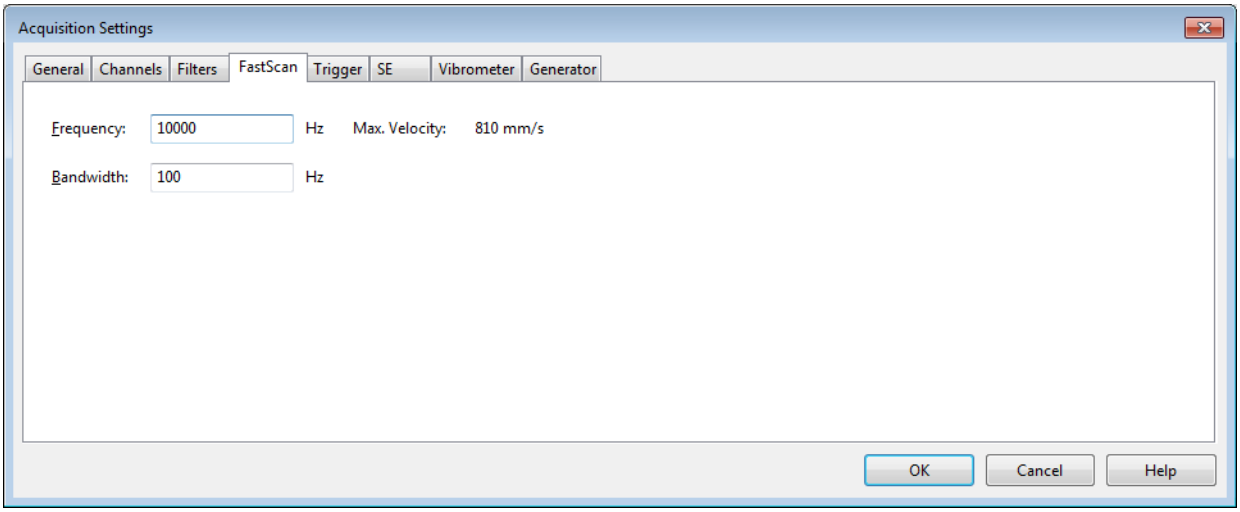


Figure 7.22: FastScan page in I/Q measurement mode (only MSA-500 with VDD option)

Max. Velocity: Because of the processing power of the PC, only signals up to this maximum vibrational velocity can be acquired with the VDD option (Digital Demodulation). The value depends on the frequency and the bandwidth (see above).

7.2.11 Settings for Time

In Time measurement mode, you can save time signals instead of spectra for scans. For this, you need the Time Domain Data option. You set the parameters for time measurements on the Time page.

With MSA-500-M2-20 systems, the maximum number of samples from a sample frequency of 10.24 MHz is limited by the data acquisition board memory.

Description

You can select up to 64 Megasamples (67 108 864 samples). However, please note that the amount of memory required increases with the number of samples. The minimum disk space a measurement requires can be approximated using the following formula:

File size in bytes > Active channels · Number of samples · 4

Depending on the number of samples and the hardware configuration, there are limitations on the Burst Chirp, Burst Random, and White Noise generator signals. In this case the software will give you corresponding messages. Either reduce the number of samples or select a different excitation signal. See SECTION 7.2.9 on this as well.

The higher the sample frequency you select, the more samples are possible. The software will try to use the main memory during data acquisition to buffer the acquired data.

While the contents of the buffer are being transferred to the hard disk, buffer overruns can occur which lead to the data acquisition being aborted. All data which had been acquired before this point in time will be saved. Make sure that your Data Management System has got a large enough main memory (RAM). In an ideal case this is 64 MB more than the file size calculated above. If the main memory used (RAM) is not big enough, the software will give you a warning message. If possible, close all other applications during data acquisition.

Data acquisition in Time measurement mode generates extremely large files with numerous samples. Once the file size has reached 2 GB, the software will give you a warning message.

Parameter Set the parameters for the Time measurement mode on the Time page.

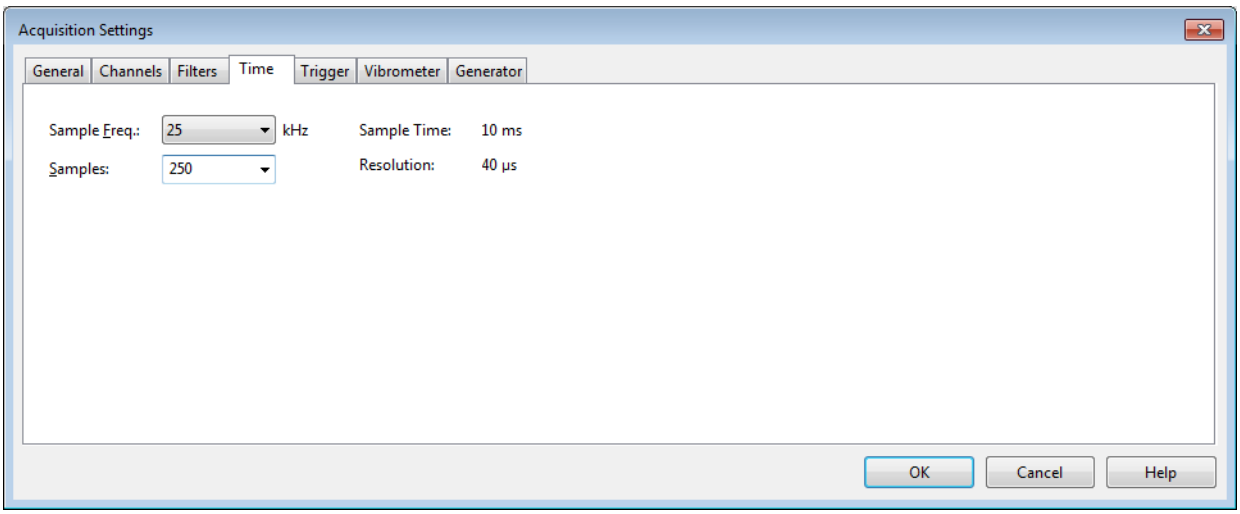


Figure 7.23: Time page (MSA-600/MSA-650)

Settings

Sample Freq.: Select one of the predefined sample frequencies from the list.

Samples: Select the number of samples. You can select in the list or enter any number.

Sample time and resolution

The software calculates the following parameters and shows them at the top right.

Sample Time: The software calculates the sample time t_{Sample} as follows:

$$t_{Sample} = \frac{n_{Sample}}{f_{Sample}}$$ Equation 7.6

n_{Sample} … Number of samples

f_{Sample} … Sample frequency

Resolution: The software calculates the resolution Δt

$$\Delta t = \frac{1}{f_{Sample}}$$ Equation 7.7

f_{Sample} … Sample frequency

Features for VDD

In I/Q measurement mode, an additional parameter appears on the Time page for VDD measurements. This parameter only appears for the MSA-500 with the VDD option (Digital Demodulation).

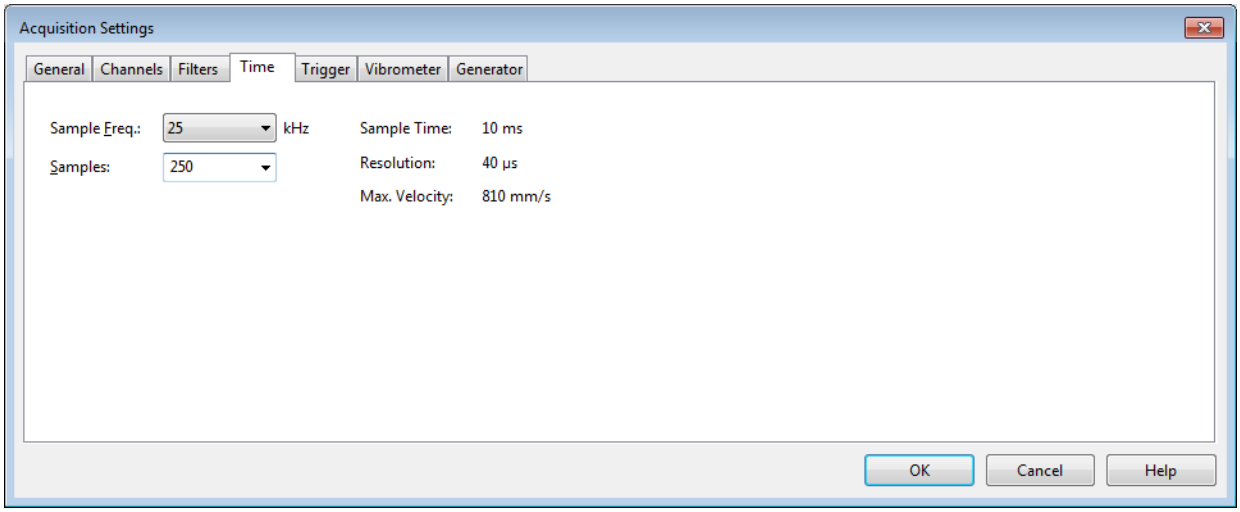


Figure 7.24: Time page in I/Q measurement mode (only MSA-500 with VDD option)

Max. Velocity: Because of the processing power of the PC, only signals up to this maximum vibrational velocity can be acquired with the VDD option (Digital Demodulation). The value depends on the sample frequency and the number of samples (see above).

8 Measuring

ⓘ Before a measurement or if the output power of the laser has changed, the laser needs to stabilize. For this reason, switch on the laser beam for at least 20 seconds (tick the Laser check box in the scanning head control). Carry out the measurement afterwards.

You can make single point measurements and scan using the software. See SECTION 8.1 and SECTION 8.2 on this.

In the video window, you can follow the progress of the scan. See SECTION 8.3 on this.

When a measurement is running, the status bar can display messages on it. See SECTION 8.4 on this.

In SECTION 8.5 you will find information on what to pay attention to if you want to stop a measurement.

Only MSA-500 with VDD option: You must run the test mode at regular intervals before making measurements. See SECTION 8.6 on this.

SECTION 8.7 describes how to control the function generator.

SECTION 8.8 describes how to control a second vibrometer from Polytec.

In SECTION 8.9 you will find a few application examples which give you detailed instructions on various measurement tasks.

8.1 Making a Single Point Measurement

You can make single point measurements as single shots or as a continuous measurement.

Single shot With a single shot, the software makes only one measurement and then ends data acquisition. You can also stop single shots manually.

Continuous measurement With a continuous measurement, the software repeats single shots until you manually stop data acquisition. You can only save the last single shot.

For a single point measurement, proceed as follows:

Prepare

1. If you are in presentation mode, select View > Acquisition in the menu bar (toolbar: ).

   You are in Acquisition mode.
2. Set the software up for data acquisition as described in CHAPTER 3.
3. Adjust the optics and position the laser beam to point at the place where you want to make a measurement. See CHAPTER 4 or CHAPTER 5 on this.
4. Focus the laser beam.
5. Set the parameters for data acquisition. See CHAPTER 7 on this.

Start

6. To start a single shot or continuous measurement, select Acquisition > Single Shot or Acquisition > Continuous (toolbar:  oder ).

   All lasers you have activated will be switched on.

Follow the measurement

7. During the measurement, you can display data in analyzers as described in SECTION 9.2 and SECTION 9.3.
8. During the measurement, the status bar and scanning head control can display messages on data acquisition. See SECTION 8.4 on this.

Stop

9. To stop the measurement, select Acquisition > Stop in the menu bar (toolbar: ). See SECTION 8.5 on this as well.

ⓘ Once the measurement is completed, you need to save it before you can process it any further.

8.2 Scanning

A scan is a sequence of single point measurements. The order in which the software approaches the scan points is determined by an internal algorithm.

For every scan point, the software carries out the following steps:

- Position the laser beam at the scan point.
- Set the optics of the scanning head to the focus value of the scan point, only with APS Professional option.
- Wait for settling time.
- Make a single shot.
- Assign scan point status.
- Save measurement data.

After a scan, the software can automatically remeasure certain scan points or you can start remeasuring manually. You can also mark single scan points in presentation mode to be remeasured and then remeasure this file in acquisition mode.

To scan, proceed as follows:

Prepare

1. If you are in Presentation mode, select View > Acquisition in the menu bar (toolbar: ).

   You are in Acquisition mode.
2. Set the software up for data acquisition as described in CHAPTER 3.
3. Adjust the optics. See CHAPTER 4 or CHAPTER 5 on this.
4. If applicable, define and edit scan points as described in CHAPTER 6.
5. Set the parameters for data acquisition. See CHAPTER 7 on this.

Start

6. To start the scan, select Acquisition > Scan in the menu bar (toolbar:).

   If the function generator is active, signal output starts. For the function generator, see SECTION 7.2.9. The software creates a file for the scan. The Save As dialog appears.

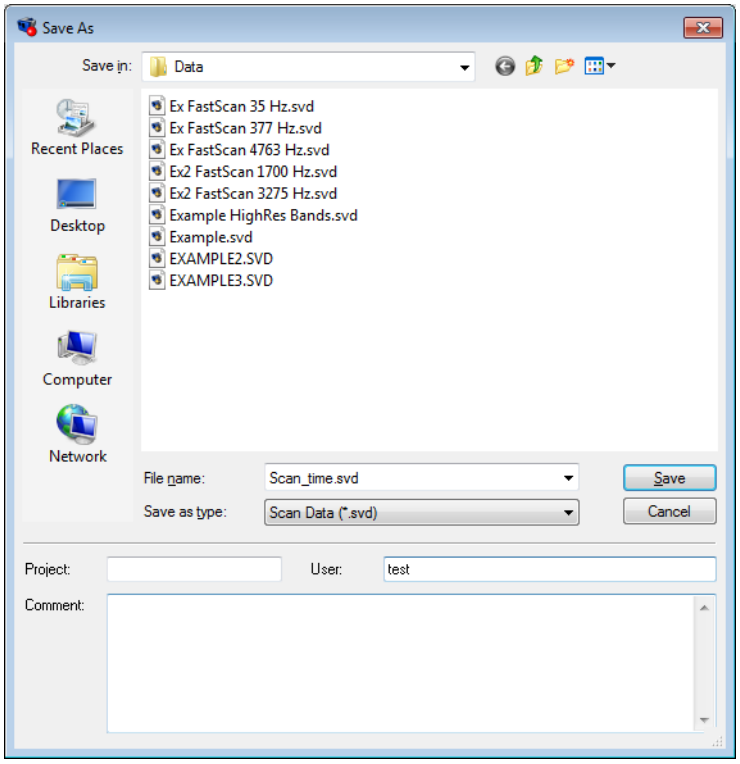


Figure 8.1: Save As dialog when starting a scan

Save

7. Navigate to the saving location and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

8. At the bottom in the dialog you can enter additional information to characterize the file more accurately.

ⓘ This information is part of the file properties which you can also view and edit in the Explorer. It is displayed when you open a file in presentation mode and also in the File Information dialog. See the presentation mode on this as well.

9. Click Save.

   The scan starts. The software immediately saves every scan point measured.

ⓘ If you have selected the 3D view in acquisition mode, the software will change over to the 2D view on starting the scan. This allows you to follow the movement of the laser beam on the object in the live video image.

Follow the scan

10. You can follow the progress of the scan in the video window as described in SECTION 8.3. During the scan, you can display data in analyzers as described in SECTION 9.2 and SECTION 9.3.

    During the scan, the status bar and scanning head control can display messages on data acquisition. See SECTION 8.4 on this.

Stop the scan and continue

11. Once the total sample time has expired, the software ends data acquisition and displays a message. You can also manually stop the scan as described in SECTION 8.5.
12. To continue a stopped scan, select Acquisition > Continue in the menu bar.

Remeasure

13. After the scan you can remeasure the scan points with the Overrange, Invalidated, and No Measured status by selecting Acquisition > Remeasure in the menu bar.

ⓘ If Signal Enhancement is active for at least one channel, then in FFT measurement mode, scan points with a Valid status are also remeasured. See SECTION 7.2.7 on this as well.

Remeasure file

If you have marked single scan points for remeasuring in presentation mode, you can remeasure them. To do so, proceed as follows:

1. In the menu bar, select Acquisition > Remeasure File.

   The Open dialog appears.
2. Navigate to the saving location and select the file.
3. Click Open.

ⓘ The software can also automatically remeasure the points. See SECTION 7.2.1 on this.

8.3 Following the Scan in the Video Window

In the video window you follow the progress of scans. While scanning, you can see what status the scan points get. You will find a description of the video window in SECTION 2.6.2.

ⓘ You can display the explanations for each status mentioned below in the software as well. To do so, leave the mouse pointer on the name of the status until the message text is displayed.

Display scan points

To show/hide scan points in the video window, select Scan > Scan Points in the menu bar (toolbar of the video window: ). To the left of the video image you can see a legend for the status.

You have different views to choose from here:

- Scan
- Geometry
- Laser Focus

Scan status

To change the views, select the required view in the Status list.

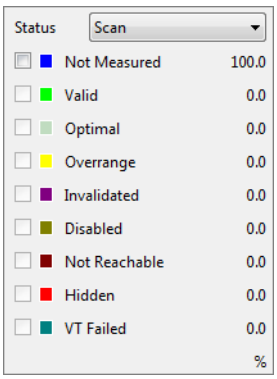


Figure 8.2: Display of the Scan status

Not Measured: Before a scan, all scan points have the Not Measured status.

Valid: When scanning, the software gives every scan point the Valid, Optimal or Overrange status.

Optimal: The Optimal status can only occur if Signal Enhancement is active for at least one channel. See SECTION 7.2.7 on this as well. At these scan points, the signal-to-noise ratio is relatively high.

ⓘ If Signal Enhancement is active, scan points can change their status between Valid and Optimal at the end of the scan.

Overrange: At scan points with Overrange status, the input range of the data acquisition board was exceeded on at least one channel.

ⓘ Note that the software may determine the Overrange scan status, even though the resulting time signal is displayed unaffected. This is due to the fact that, depending on the measurement settings, you are working with a higher sample rate than actually visible in the resulting time signal (Oversampling). If a brief voltage peak causes overloading, then the scan point is given the Overrange status, even though the decimated time signal is being displayed correctly.

Invalidated: After the scan, you can manually assign the scan points the Invalidated status. You will find details on this in presentation mode.

Disabled: If you do not want to scan single scan points, you can assign them the Disabled status. See SECTION 6.5.3 on this.

Not Reachable: If you define the scan points, then scan points which are outside the scan area are given the Not Reachable status.

Hidden: If you are working with 3D geometries and parts of the object hide scan points so that they cannot be reached by the laser beam, then these scan points are identified as Hidden. Prerequisite of this is that in the Preferences dialog on the Geometry page, you have activated Hidden Points Calculation. See SECTION 3.4 on this as well.

VT Failed: If VideoTriangulation fails, the geometry status remains unchanged, no vibration measurement data is recorded and the scan points are given the VT Failed scan status (option not available).

Geometry status

To change the views, select the required view in the Status list.

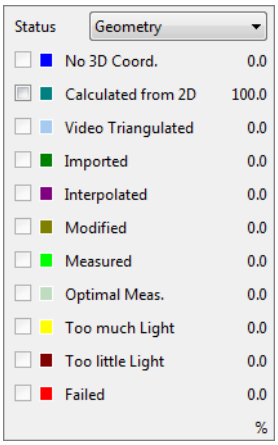


Figure 8.3: Display of the Geometry status

No 3D Coord.: If no 3D coordinates are available, the scan points are given the No 3D Coordinates status.

Calculated from 2D: If you are working with 3D geometry, you can see whether scan points were calculated from the 2D geometry.

Video Triangulated: If VideoTriangulation is successful, the scan points are given the Video Triangulated status (option not available).

Imported: If you import the geometry from ASCII, Universal File, ME'Scope or ASAM ODS ATFX files using the optional geometry import, the scan points are given the Imported status. See SECTION 6.3.7 on this as well.

Interpolated: If 3D coordinates that have not been calculated or have been calculated incorrectly are then interpolated from the neighboring scan points, they are given the Interpolated status.

Modified: If the coordinates of the scan points are retrospectively changed, they are given the Modified status. See SECTION 6.5.3 on this as well.

Measured, Optimal Meas.: If you are working with 3D geometry or distance measurement, you can see whether scan points have been measured with the MSA geometry unit (Measured or Optimally Measured status).

Too much Light, Too little Light: If the signal strength being too high or too low is the reason for the measurement failing, then you will see this from the Too much Light or Too little Light status (option not available).

ⓘ In the cases Too much Light, Too little Light and Failed, this means that these scan points do not have any valid 3D coordinates.

Failed: If the distance measurement has failed, the affected scan points are given the Failed status.

Laser Focus status

To change the views, select the required view in the Status list.

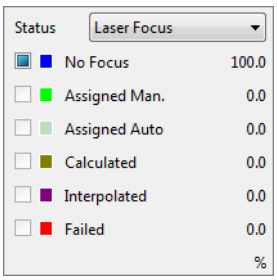


Figure 8.4: Display of the Laser Focus status

No Focus: If no focus value has been assigned, the scan points are given the No Focus status.

Assigned Man.: If you have allocated focus values with the aid of the Assign Focus Manually method, the scan points are given the Assigned Manually status. Read SECTION 6.7 on how to assign focus values.

Assigned Auto: If you have allocated focus values with the aid of the Assign Focus Automatically method, the scan points are given the Assigned Auto status. Read SECTION 6.7 on how to assign focus values.

Calculated: If, in the Preferences dialog on the Geometry page, you have ticked the Calculate Focus Values Automatically check box and the selected scan points have 3D coordinates, then the software will calculate the focus values automatically (e.g. with a geometry import). These scan points are given the Calculated status.

Interpolated: If the software interpolates the focus value of single scan points from those of the neighboring points, these scan points are given the Interpolated status.

Failed: Scan points for which no focus value could be determined using the method Assign Focus Automatically are labeled as Failed.

Path Length status (only MSA-650)

To change the views, select the required view in the Status list.

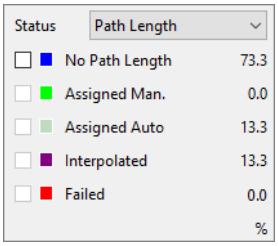


Figure 8.5: Display of the Path Length status

No Path Length: If no path length has been determined, the scan points are given the No Path Length status.

Assigned Man.: If you have assigned the path length values using the Assign Path Length Manually method, the scan points are given the Assigned Man. status. Read SECTION 6.5.3.4 on how to assign the path length values.

Assigned Auto: If you have assigned the path length values using the Assign Path Length Automatically method, the scan points are given the Assigned Auto status. Read SECTION 6.5.3.4 on how to assign the path length values.

Interpolated: If the software interpolates the path length value of single scan points from those of the neighboring points, these scan points are given the Interpolated status.

Failed: Scan points for which no path length value could be determined using the Assign Path Length Automatically method are labeled as Failed.

8.4 Messages when Making Measurements

During a measurement, the status bar and scanning head control can display messages on data acquisition. You will find a list of messages, what causes them and information on how to remedy the situation in the following table.

Table 8.1: Messages on data acquisition

Message	Cause	If the message appears permanently	See
Overrange Vib (status bar)	The signal on the vibrometer channel exceeds the input range set.	Click A/D and display the Channels page. If possible, set the input range to 10 V. Display the Vibrometer page and step by step, increase the measurement range for the velocity.	SECTION 7.2.2 and SECTION 7.2.8
Overrange Ref (status bar)	The signal on the reference channel exceeds the input range set.	Click A/D and display the Channels page. Set the next highest input range for this reference channel.	SECTION 7.2.2
No Trigger (status bar)	Triggering is set but the software can not detect a trigger signal.	Click A/D and display the Trigger page. Check the trigger settings and cabling.	SECTION 7.2.6 and operating instructions

ⓘ The Overrange Vib message can also appear briefly, caused by noise peaks due to dropouts. In this case, do not reset the input range.

8.5 Stopping the Measurement

With single shots and with scanning, the software ends data acquisition once the total sample time has expired. You can also stop measurements manually. Continuous measurements have to be stopped manually.

To stop the measurement, select Acquisition > Stop in the menu bar (toolbar: ■).

Single point measurement

If you stop a single point measurement, the following applies:

- If there are less than three seconds to the end of the sample time, the software carries out the measurement completely and then ends data acquisition.
- If there are more than three seconds to the end of the sample time, the software aborts the measurement. You can always display and save the time signal for aborted measurements in FFT measurement mode, but you may not be able to save the spectrum. For long sample times, single shot is recommended if you want to evaluate and save spectra.

Scan

If you stop a scan, the following applies:

All completely measured scan points are already saved. You can evaluate the scan the same as you can a completed one.

8.6 Test Mode

Mode of operation

In the test mode, the software calculates the errors which are caused by the analog components during signal processing. Prior to demodulation, the software corrects these errors using special algorithms.

ⓘ Test mode can be carried out automatically for the OFV-5000 controller if the controller is controlled remotely via the RS-232 interface. Test mode is then run every time the software is started.

Requirement

You need MSA-500 with VDD option (Digital Demodulation) for this.

ⓘ The VDD option is only available for MSA-500-M2 systems.

Run automatically during setup

Test mode is run automatically once you have installed the VDD option, and the hardware, and have also set up the software. To set up the software, proceed as follows:

1. Switch on the MSA-500 and wait for 30 minutes.

   The components are then in thermal equilibrium.
2. Start the software.
3. In the menu bar, select Setup > Preferences.

   The Preferences dialog appears.
4. Display the Devices page.
5. In the Junction Box list, select the MSA-E-500 (VDD) junction box. See SECTION 3.1 on this as well.
6. Click OK.

   Test mode is run automatically.
7. You can now start a measurement as described in SECTION 8.1 or SECTION 8.2.

ⓘ Test mode is now run every time the software is started.

Repeat several times

You need to regularly repeat test mode. The following applies:

- If you only want to measure to 10 nanometers accuracy or less, run the test mode again every three months.
- If you want to measure to 10 nanometers accuracy or more or if the ambient temperature has changed by ±5 °C or more, run the test mode every day.

Run manually

You can also run test mode manually. To do so, select Setup > VDD Test in the menu bar.

8.7 Controlling a Function Generator

With the software you can control the function generator for specific systems. To do this, you need the Signal Generator option.

ⓘ You need to set the connection to the generator in the Preferences dialog on the Devices page. You will find information on this in SECTION 3.1.

To control the function generator, proceed as follows:

Set the parameters

1. In the menu bar, select Acquisition > Settings (toolbar: ).
   The Acquisition Settings dialog appears.
2. Display the Generator page.
3. At the top left, tick the Active box to activate the function generator.
4. Set the parameters for the signal you want to emit. You will find information on this in SECTION 7.2.9.
5. Set the parameters on all other pages of the dialog. You will find information on this in SECTION 7.2.
6. Click OK.

Signal output

7. To start (stop) the signal output of the function generator, select Acquisition > Generator in the menu bar (toolbar: ).

You can now start a measurement as described in SECTION 8.1 or SECTION 8.2.

ⓘ When scanning, the signal output starts automatically.

8.8 Controlling a Second Vibrometer

You can control a second Polytec vibrometer together with the MSA.

You have the following options to control the second Polytec vibrometer:

- Via Ethernet - See SECTION 8.8.1 on this.
- Via USB, RS-232 or IEEE - See SECTION 8.8.2 on this.

Supported functions

The connection to the vibrometer supports the following settings:

- Controlling certain functions of the hardware. You will find a description in SECTION 4.1.
- Setting the parameters for data acquisition at the front-end or controller. You will find a description in SECTION 7.2.8.

8.8.1 Controlling a Second Vibrometer via Ethernet

To control a second vibrometer via the software, you need a free Ethernet interface in the Data Management System.

Supported vibrometers

You will find an overview of the supported vibrometers as well as information on the cable in the following table.

Vibrometer	Cable
VibroFlex Connect VFX-F-110	Gigabit Ethernet (M12/RJ45, X-coded)
VibroGo VGO-200	Gigabit Ethernet (M12/RJ45, D-coded)
VibroOne VIO-130	Gigabit Ethernet (M12/RJ45, X-coded)
IVS-500	Special Ethernet cable

Make a connection

To establish a connection, proceed as follows:

1. To set up the vibrometer and the Ethernet connection, refer to the operating instructions of the vibrometer. Follow the instructions given there regarding the IP addresses.
2. Switch on the vibrometer.
3. Start the software.
4. In the menu bar, select Setup > Preferences.

   The Preferences dialog appears.
5. Display the Devices page.
6. For the Vibrometer Controller, select Ethernet in the Connection list.

   Depending on the hardware configuration, you set further settings as described in SECTION 3.1.
7. Display the Channels page.
8. In the Vibrometer column, connect the Vibrometer Controller to the digital channel for which it provides the signal. You will find information on this in SECTION 3.2.
9. Click OK.

   The vibrometer is initialized.

You can now measure as described in SECTION 8.1.

8.8.2 Controlling a Second Vibrometer via USB, RS-232 or IEEE

To control a second vibrometer via the software, you need a free USB or RS-232 interface, or an IEEE-488/GPIB board in the Data Management System (depending on the vibrometer).

Supported vibrometers

You will find an overview of the supported vibrometers and interfaces as well as information on the cable in the following table.

Table 8.2: Supported vibrometers and interfaces

Vibrometer	USB	RS-232	IEEE-488/ GPIB	Cable
CLV-1000	-	x	-	RS-232 null modem cable (cross-wired)
CLV-2534-2/-3/-4	-	x	-	RS-232 null modem cable (cross-wired)
HSV-100	x	-	-	USB cable
IPV-100	x	-	-	USB cable
IVS-300/IVS-400	-	x	-	Special cable
IVS-500	-	x	-	Special cable
NLV-2500-2/-5	-	x	-	RS-232 null modem cable (cross-wired)
OFV-2500-2/-3	-	x	-	RS-232 null modem cable (cross-wired)
OFV-2520[1]	x	-	-	USB cable
OFV-2570	-	x	-	RS-232 null modem cable (cross-wired)
OFV-3001/OFV-3001 S	-	x	x	RS-232 cable (1:1)
OFV-3020/OFV-3020 S	-	x	x	RS-232 cable (1:1)
OFV-5000/OFV-5000-S[2]	-	x	-	RS-232 null modem cable (cross-wired)
OFV-5000 Xtra	-	x	-	RS-232 null modem cable (cross-wired)
RSV-150-B/-M	x	-	-	USB cable
VDD-E-600	x	-	-	USB cable

[1] Sensor head B is not supported.

[2] As of firmware version 1.22

Make a connection

To establish a connection, proceed as follows:

1. Connect the Data Management System to the vibrometer using the appropriate interface.

ⓘ If you are using an RS-232 interface, then set 115200 Baud on the controller. On the CLV-1000 and OFV-30xx controllers, set 9600 Baud.

2. Switch on the vibrometer.
3. Start the software.

4. In the menu bar, select Setup > Preferences.

   The Preferences dialog appears.
5. Display the Devices page.
6. Make the required settings. See SECTION 3.1 on this as well.
7. Display the Channels page.
8. In the Vibrometer column, connect the Vibrometer Controller to the digital channel for which it provides the signal. You will find information on this in SECTION 3.2.
9. Click OK.

   The vibrometer is initialized.

You can now measure as described in SECTION 8.1.

8.9 Example Measurements

Here you will find a few application examples which give you detailed instructions on various measurement tasks.

- Optimizing input signals with Signal Enhancement (SE) and Speckle Tracking - See SECTION 8.9.1 on this.
- Measurement with FastScan - See SECTION 8.9.2 on this.

8.9.1 Optimizing Input Signals with Signal Enhancement and Speckle Tracking

When scanning with Signal Enhancement and Speckle Tracking, the noise level for all scan points is roughly the same.

Speckles occur when laser light is scattered back from optically rough surfaces. The light scattered back from a certain point can cause constructive or destructive interference. Correspondingly, the detector sees a bright or a dark speckle. Dark speckles lead to dropouts in the vibrometer signal. Dropouts increase the noise level significantly.

Operating principle

Signal Enhancement works as follows:

- In FFT measurement mode, you suppress dropouts in the average spectrum with Signal Enhancement and Speckle Tracking. In addition, the software evaluates the noise level of the single scan points. You will find further information on this in your PSV theory manual.
- In Time measurement mode with Time averaging, the software uses a median algorithm for averaging with active Signal Enhancement.

Activate

To activate Signal Enhancement and Speckle Tracking, proceed as follows:

1. In the menu bar, select Acquisition > Settings (toolbar: 

).

   The Acquisition Settings dialog appears.
2. Display the SE page.
3. On the left, tick the channels which Signal Enhancement is to be active for. You will find information on this in SECTION 7.2.7.

4. If Signal Enhancement is active for the vibrometer channel, then you can activate Speckle Tracking for this channel on the right.
5. Set the parameters on all other pages of the dialog. You will find information on this in SECTION 7.2.
6. Click OK.

You can now start a measurement as described in SECTION 8.1 or SECTION 8.2.

8.9.2 Measurement with FastScan

You can fast scan at individual frequencies. To do this, you need the FastScan option. The principle of the FastScan is a regression in the time domain.

To perform a FastScan, proceed as follows:

Set the parameters

1. In the menu bar, select Acquisition > Settings (toolbar: ).

   The Acquisition Settings dialog appears.
2. Display the General page.
3. Select the FastScan measurement mode.

   In the dialog, a page with FastScan parameters appears. The Frequency, Window, and Trigger pages disappear.
4. Display the FastScan page.
5. Set the parameters. You will find information on this in SECTION 7.2.10.
6. Set the parameters on all other pages of the dialog. You will find information on this in SECTION 7.2.
7. Click OK.

Measure

8. You start and stop FastScans the same way as normal scans. See SECTION 8.2, SECTION 8.4 and SECTION 8.5 on this.

ⓘ Leakage does not occur with FastScans. Triggers are not active. If you are controlling a function generator with the software, then a sinusoidal signal with the frequency of the FastScan is used for excitation.

Display data

9. You can display measurement data for FastScans as described in CHAPTER 9 and in Presentation, but not spectra.

Evaluate

10. You can evaluate FastScans as described in Presentation, but only at the frequency of the FastScan. You can not define any other frequency bands.

9 Displaying Data

You can set up the properties of the data displayed. See SECTION 9.1 on this.

You work with the following windows in the software to display data:

- In acquisition mode: a video window, any number of analyzers and signal processors
- In presentation mode: any number of presentation windows, analyzers and signal processors

You will find a description of the analyzer in SECTION 2.6.3.

You can work in analyzers. Read SECTION 9.2 on this.

You can select the data set that is displayed in the analyzer. Read SECTION 9.3 on this.

Afterwards, you can use your measurement data as the basis for further calculations. Read SECTION 9.4 on this.

9.1 Setting Up Display Properties

In the Display Properties dialog, you can make the following settings:

- Set up title and background color for the object - See SECTION 9.1.1 on this.
- Set up video image, grid and indices for the view style of the measurement data - See SECTION 9.1.2 on this.
- Set up sight, illumination model and clipping planes for 3D view - See SECTION 9.1.3 on this.

In acquisition mode

To set up the view style of the measurement data and 3D view in acquisition mode, change to 3D view. To do so, select Scan > 3D View in the menu bar (toolbar: 

). You will no longer see a live video image but, as in presentation mode, a snapshot of the video image at the time at which you changed the view.

You have the following options to open the Display Properties dialog:

- Double-click the video window and display the desired page.
- Click the video window with the right mouse button and select General Properties, Data Properties or 3D View Properties in the context menu.
- Activate the video window (to do so, click it) and select Scan > Properties. Display the desired page.

ⓘ In acquisition mode, you can only switch to 3D view if you have defined scan points.

9.1.1 Setting up the Object

ⓘ The content of these pages differs depending on whether you are in acquisition or presentation mode. Depending on the type of measurement and the geometry, not all settings options are shown in the dialog.

If you are in acquisition mode (3D view), open the dialog as follows:

- Double-click the video window and display the General page.
- Click the video window with the right mouse button and select General Properties in the context menu.
- Activate the video window (to do so, click it) and select Scan > Properties. Display the General page.

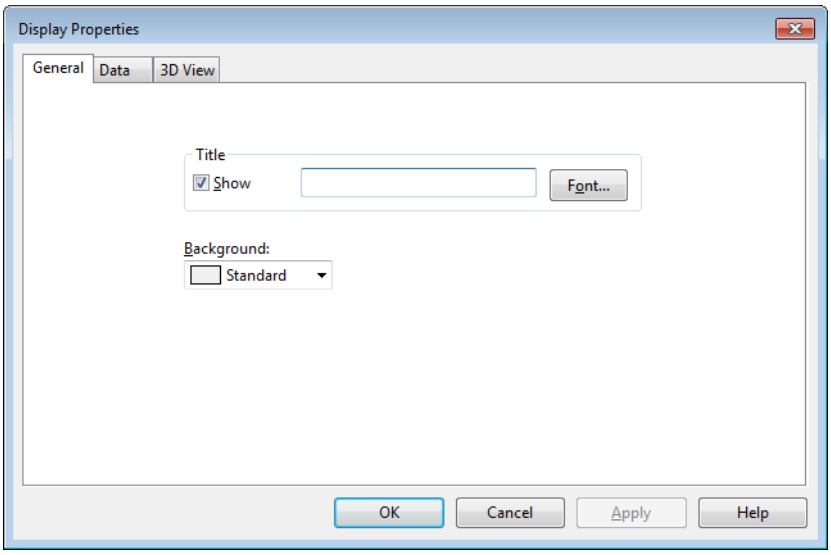


Figure 9.1: General page in acquisition mode

Title

Set up the object title.

Show: Tick the check box if you want to display a title above the object. You enter the title on the right.

Font: Click here to format the title. The Font dialog appears. Set up the Font, Font style, Size, and if applicable Effects and click OK.

Colors

Choose the background color of the object.

Background: Select the background color of the window.

9.1.2 Setting Up View Style of Measurement Data

ⓘ The content of these pages differs depending on whether you are in acquisition or presentation mode. Depending on the type of measurement and the geometry, not all settings options are shown in the dialog.

If you are in acquisition mode (3D view), open the dialog as follows:

- Double-click the video window and display the Data page.
- Click the video window with the right mouse button and select Data Properties in the context menu.
- Activate the video window (to do so, click it) and select Scan > Properties. Display the Data page.

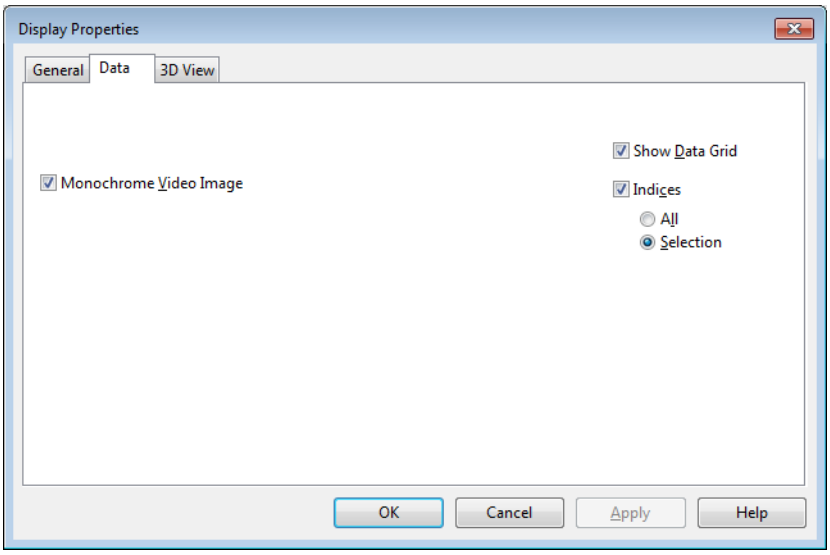


Figure 9.2: Data page in acquisition mode (3D view)

Monochrome Video Image: Tick the check box if you want to change the display of the video image from color to black&white.

Show Data Grid: Tick the check box if you want to see the grid that the scan points are on.

Indices: Tick the check box to show the index of scan points.

The following options are available:

- All: The indices of all scan points are shown.
- Selection: The indices of selected scan points are shown.

9.1.3 Setting up 3D View

ⓘ The content of these pages differs depending on whether you are in acquisition or presentation mode. Depending on the type of measurement and the geometry, not all settings options are shown in the dialog.

3D view

If you are in acquisition mode (3D view), open the dialog as follows:

- Double-click the video window and display the 3D View page.
- Click the video window with the right mouse button and select 3D View Properties in the context menu.
- Activate the video window (to do so, click it) and select Scan > Properties in the menu bar. Display the 3D View page.

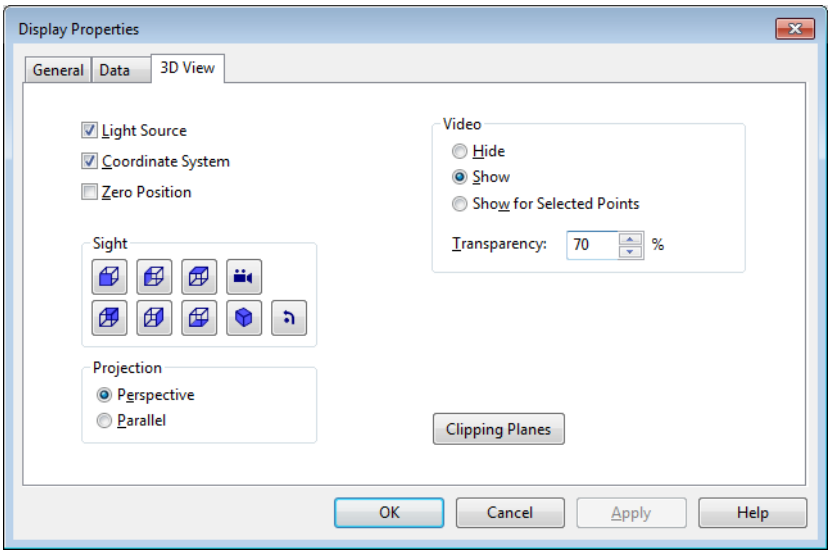


Figure 9.3: 3D View page in acquisition mode (GeoPro option)

Light Source: Tick the check box to switch on the illumination model. It simulates illuminating the object from the direction of the observer.

ⓘ Depending on the shape of the object, the display of light and shadow may have a detrimental effect on the presentation quality. In this case, deactivate the check box to switch off the illumination model.

Coordinate System: Tick the check box to display a coordinate system at the top left next to the object.

Zero Position: Tick the check box to show the object in its original position in addition to viewing the measurement results.

Use Imported Video Image as Texture: Tick the check box to show the imported video image of the object in its original view. The function is only available for the geometry import from suitable files (e.g. from OBJ files).

Sight

The icons in the Sight field show the line of sight of the object. By clicking these icons, you can move the object more precisely than by dragging with the mouse.

 View of the object from the front

 View of the object from behind

 View of the object from the left

 View of the object from the right

 View of the object from above

 View of the object from below

 View of the object at an angle from front, left and above

 Sight of the camera (only for 3D geometries; for this purpose the same limitations apply as with the display of the video image, see above under Video)

 Rotating the object around the line of sight

You can show/hide a coordinate system which shows you the resulting orientation of the object (see above, Coordinate System).

ⓘ You can also set the line of sight directly in the video window using the mouse. To select the Rotate mode, click  . Click on the object and drag it in the required direction with the mouse. Hold the Control key pressed and the object only turns around the line of sight.

Projection

Select the type of projection of the object.

Perspective: The object is shown in perspective, i.e. parts at the front of the picture are shown larger, parts towards the back of the picture are shown smaller.

Parallel: All parts of the object are shown the same size, regardless of their distance from the observer.

ⓘ Parallel projection makes it easier to distinguish between in-plane and out-of-plane vibrations. Particularly with out-of-plane vibrations, the view in perspective in top view can give the impression that the object is expanding and contracting again.

Video

For 3D geometries, the video image is only shown if the following is applicable:

- The 2D alignment is good quality, i.e. the laser beam hits the precise position on the video image.
- 3D coordinates are available for the scan points.
- You have defined at least four scan points which cover a larger area of the video image.

Select one of the following options to display the video image.

Hide: Hide the video image outside the defined scan points.

Show: To video image will be fully displayed.

Show for Selected Points: The video images will only be fully displayed for the video planes in which scan points have been selected (recommended for combined files).

ⓘ If you have not selected any scan points the video image will be hidden outside the defined scan points.

Transparency: Enter in % how transparently the video image is to be displayed.

Clipping planes

In 3D view, you can define clipping planes to check your geometry. Thereby, it does not matter whether you are in the APS mode or not. The 3D view is only available if you have defined scan points. To define clipping planes, click Clipping Planes. You will find a description of the dialog in SECTION 6.4.1.

9.2 Working in Analyzers

Analyzers show measurement data and evaluated data in x-y-diagrams. You will find a description of the analyzer in SECTION 2.6.3.

You select the data set to be displayed in the analyzer as described in SECTION 9.3.

Analyzers offer other user-friendly functions to present data quickly and sensibly:

- Set cursors and read single data points - See SECTION 9.2.1 on this.
- Link analyzer settings - See SECTION 2.6.8 on this.
- Zoom in analyzers - See SECTION 9.2.2 on this.
- Scale axes ranges - See SECTION 9.2.3 on this.

You will find more functions in the Analyzer Properties dialog.

- Set up view style of graphs - See SECTION 9.2.4 on this. There you can scale and set line style and color of the graph.
- Set up the diagram - See SECTION 9.2.5 on this. To do so, select Analyzer > Properties and display the corresponding page. Alternatively, click the analyzer with the right mouse button and select the required function in the context menu.

The analyzer simultaneously shows up to 0.5 Megasamples for time series as standard. If the measurement contains more data, then this is reloaded during scrolling.

ⓘ In acquisition mode, the analyzer always shows only a maximum of 0.5 Megasamples.

Features

You can not set a cursor in the I/Q measurement mode and in the Nyquist display type.

9.2.1 Setting Cursors and Reading Data

You can set cursors in the analyzer to read individual data points. Cursors are represented by thin vertical lines.

Single cursor

To set a single cursor, proceed as follows:

1. In the menu bar, select Analyzer > Cursor (toolbar: ).
2. Click in the diagram at the desired position.

   The cursor appears as a thin vertical line.
3. Once you move the mouse pointer onto the cursor, it becomes a double arrow ↔. In this way, you can move the cursor subsequently. To do so, click the cursor and drag it to the right or left.

   The data in the readout is simultaneously updated.

Read data

4. You can see the cursor's coordinates in the readout. If the readout is not visible on the right in the analyzer, select Analyzer > Readout in the menu bar to display it.

ⓘ If you have selected two data sets to be displayed, then the cursor is visible in both diagrams and the data from both diagrams is shown in the readout.

Cursor properties

In the cursor properties, you can set what is to be shown in the readout or in the diagram. To display the cursor properties, continue as follows:

5. In the menu bar, select Analyzer > Cursor Properties (toolbar: ).

   The Cursor Properties dialog appears.
6. Select the required additional properties and click OK.

   The data you selected in the cursor properties will appear in the readout or in the diagram.

Cursor Properties

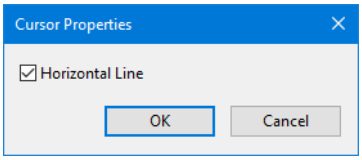


Figure 9.4: Cursor Properties dialog

Horizontal Line: Tick the check box if a horizontal line is to be shown respectively at the cursor position in the diagram.

Differential cursor

You can set two cursors and read the difference between the coordinates. To do so, proceed as follows:

1. In the menu bar, select Analyzer > Differential Cursor (toolbar: ).
2. In the diagram, point at the first desired point.
3. Press the mouse button.

   The mouse pointer becomes a double arrow ↔.
4. Drag the mouse to the left or right to the second point.

   The difference of the coordinates is also displayed in the readout.
5. Once you move the mouse pointer onto the cursor, it becomes a double arrow ↔. In this way, you can move the cursors subsequently. To do so, click the cursor and drag it to the right or left.

   The data in the readout is simultaneously updated.
6. To display the cursor properties, select Analyzer > Cursor Properties in the menu bar (toolbar: ).

   The Differential Cursor Properties dialog appears.
7. Select the required additional properties and click OK.

   The data you selected in the cursor properties will appear in the readout or in the diagram.

Differential Cursor Properties

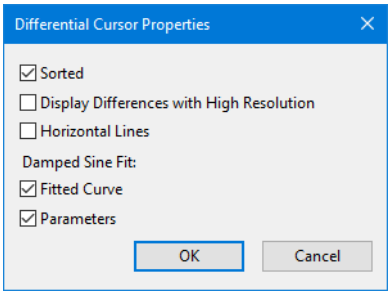


Figure 9.5: Differential Cursor Properties dialog

Sorted: Tick the check box if the cursor on the left is always to be named 1. The cursor on the right is then named 2. If you move a cursor so that the order changes, the numbers also change. If the check box is not marked, the cursors keep their numbers from the readout, independently of their position in the diagram.

Display Differences with High Resolution: Tick the check box to show the difference more clearly for values that are close to each other.

Horizontal Lines: Tick the check box if a horizontal line is to be shown respectively at the cursor positions in the diagram.

Damped Sine Fit: This property is only available for time signals. Tick the Fitted Curve check box to allow the adaptation of a damped sine fit to the signal being displayed between the cursor lines. Tick the Parameters check box to allow the adapted frequency f and the damping coefficient δ being displayed in the readout.

Band cursor

You can set a band cursor and can read various data about the marked area, e.g. the RMS value, the minimum and maximum, the mean value as well as the standard deviation. Apart from that, the software can fit the results to the frequency response of a one-dimensional oscillator and calculate the position of the peak as well as the attenuation. To do so, proceed as follows:

1. In the menu bar, select Analyzer > Band Cursor (toolbar: ).
2. In the diagram, point at the first desired point.
3. Press the mouse button.

   The mouse pointer becomes a double arrow ↔.
4. Drag the mouse to the left or right to the second point.

   The data you selected in the cursor properties will appear in the readout or in the diagram.
5. Once you move the mouse pointer onto the cursor, it becomes a double arrow ↔. In this way, you can move the cursors subsequently. To do so, click the cursor and drag it to the right or left.

   The data in the readout or in the diagram is simultaneously updated.
6. To display the cursor properties, select Analyzer > Cursor Properties in the menu bar (toolbar: ).

   The Band Cursor Properties dialog appears.
7. Select the required additional properties and click OK.

   The data you selected in the cursor properties will appear in the readout or in the diagram.

Band Cursor Properties

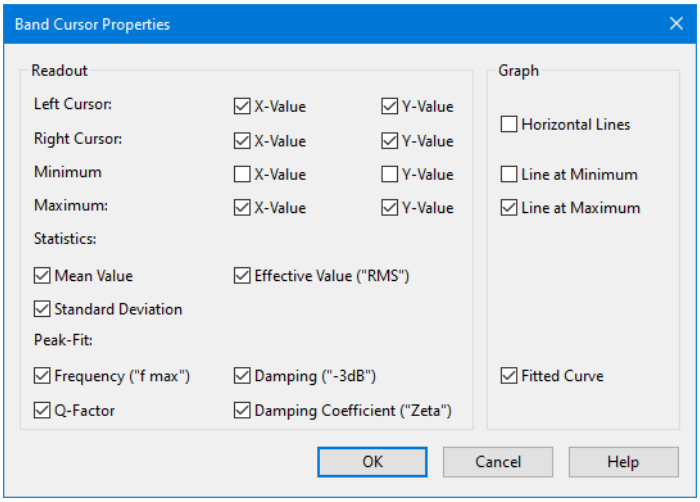


Figure 9.6: Band Cursor Properties dialog

Left Cursor, Right Cursor, Minimum, Maximum: Tick the required check boxes, so that the corresponding ordinate values are shown. In the Graph field on the right, you can select the corresponding line display in the diagram.

Statistics: Tick the desired check boxes for Mean Value, Effective Value ("RMS"), and Standard Deviation to see the corresponding ordinate values.

Peak-Fit: The Peak-Fit allows the frequency response function of a one-dimensional, harmonic oscillator being adapted to a real or complex spectrum. The function uses the least squares method for the adaptation. For a Peak-Fit, you can display the following values.

ⓘ Peak-Fit is only available for spectra.

Frequency ("fmax"): Tick the check box to display the frequency of the maximum of the fitted curve.

Damping ("–3 dB"): Tick the check box to show the damping. The damping corresponds to the full width of the fitted curve at a height of 3 dB below the maximum amplitude.

Q-Factor: Tick the check box to show the Q-Factor. The Q-factor is defined as follows:

$$\text{Q-factor} = f\text{max}/\text{Damping} \qquad \text{Equation 9.1}$$

Damping Coefficient ("Zeta"): Tick the check box to show the damping coefficient. The damping coefficient is defined as follows:

$$\text{Zeta} = \text{Damping}/(2 \cdot f\text{max}) \qquad \text{Equation 9.2}$$

ⓘ Calculating the Damping Coefficient ("Zeta") is only valid for small dampings.

Fitted Curve: Tick the check box to display the fitted curve for the selected band cursor in the diagram.

Harmonic cursor

You can set a harmonic cursor which allows you to display up to 12 cursor lines of the 2nd, 3rd, 4th etc. order of the fundamental frequency. To do so, proceed as follows:

1. In the menu bar, select Analyzer > Harmonic Cursor (toolbar: ).
2. Click the required frequency in the diagram.

   The displays you selected in the cursor properties will appear in the diagram. The data at the cursor position you selected in the analyzer will be shown in the readout. Additionally for simple spectra, the amplitude ratio (THD = Total Harmonic Distortion) will be displayed in [%] or [dB] respectively. THD is calculated as follows:

$$THD_{\%audio} = \frac{\sqrt{A_2^2 + A_3^2 + A_4^2 + \ldots + A_n^2}}{A_1}$$

ⓘ No THD is displayed for power spectra, PSD, and spectra from multiple channels.

3. Once you move the mouse pointer onto the cursor, it becomes a double arrow ↔. In this way, you can move the cursors subsequently. To do so, click the cursor and drag it to the right or left.

   The data in the readout or in the diagram is simultaneously updated.

4. To display the cursor properties, select Analyzer > Cursor Properties in the menu bar (toolbar: ).

   The Harmonic Cursor Properties dialog appears.

5. Select the required additional properties and click OK.

   The data you selected in the cursor properties will appear in the readout or in the diagram.

Harmonic Cursor Properties

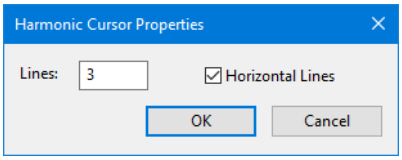


Figure 9.7: Harmonic Cursor Properties dialog

Lines: Enter the required number of lines that should be displayed.

Horizontal Lines: Tick the check box if a horizontal line is to be shown respectively at the cursor positions in the diagram.

Move cursor

You can move the cursor to the right and left. To do so, proceed as follows:

1. Point at the cursor.

   The mouse pointer becomes a double arrow ↔.
2. Press the mouse button and drag to the left or right to the required point.

   The data in the readout or in the diagram is simultaneously updated.

ⓘ You can also move a single cursor with the → and ← keys.

Delete cursor

To delete cursors which has been set, click another cursor icon ( ,  ,  or ).

Copy cursor values

You can use the clipboard to copy the currently set cursor values into a text file. To do so, proceed as follows:

1. Set the required single cursor, differential cursor, band cursor or harmonic cursor.
2. Click the diagram with the right mouse button and select Copy Cursor Values in the context menu.
3. You can now open a text editor, for example, and paste the cursor values from the clipboard there.

9.2.2 Zooming in Analyzers

You can zoom in on sections of the diagrams (zooming). You will find the appropriate tools in the toolbar of the analyzer.

Zoom the x- or y-axis

To zoom in on the x- or y-axis individually, proceed as follows:

1. Point at an edge point of the required section on the axis.

   The mouse pointer becomes a magnifying glass icon  .
2. Press the mouse button and drag horizontally or vertically to the other edge point.

Scroll the x-axis

The scroll bar below the diagram shows the position of the zoomed section on the whole x-axis range. You can scroll along the x-axis with the set zoom factor. To do so, move the bar in the scroll bar to the right or to the left. You can also click on the arrows to the left and right of the scroll bar.

Zoom the x- and y-axis

To zoom in on the x- and y-axis simultaneously, proceed as follows:

1. Click  or select Analyzer > Zoom In.

   The mouse pointer becomes a magnifying glass icon  .
2. In the diagram, point at an edge point of the required section.
3. Press the mouse button and drag diagonally to the other corner point.

Pan the section

You can pan the zoomed diagram section. To do so, you have the following options:

- Click in the zoomed diagram with the middle mouse button (mouse wheel), hold the mouse button pressed and pan the diagram section. If you have assigned a special function to the middle button (e.g. double-click), then you need to proceed as described below.
- Click  or select Analyzer > Pan. You can now pan the diagram section while pressing the left mouse button.

Zoom out

To undo the last zoom or autoscale, click  or select Analyzer > Zoom Out.

To undo all zooming, hold the Shift key pressed and click  . You can also hold the Shift key pressed and select Analyzer > Zoom Out.

9.2.3 Scaling Axes Ranges

Auto scale ordinate

If you have started a measurement or are displaying a different data set, then the ordinate range is often unsuitable. You can then autoscale, i.e. the software selects the ordinate displayed according to the y-range of data. To switch on autoscale, select Analyzer > Autoscale in the menu bar (toolbar: ). To switch off autoscale again, repeat this action.

Scale manually

In the Analyzer Properties dialog, you can scale manually and define additional scaling properties.

You have the following options to open the dialog:

- Point with the mouse pointer on an axis (the mouse pointer becomes a magnifying glass ) and double-click the axis.
- Click the diagram with the right mouse button and select Range Properties in the context menu.
- Activate the analyzer (to do so, click it), select Analyzer > Properties in the menu bar and display the Ranges page.

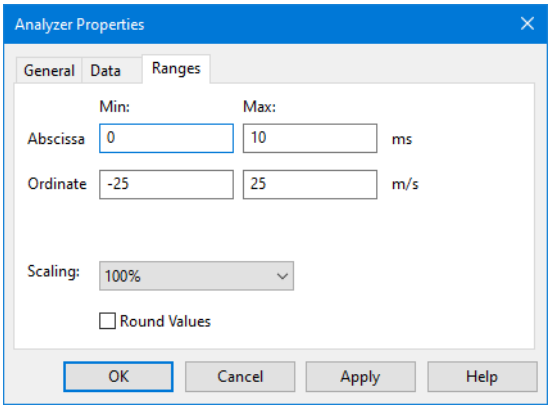


Figure 9.8: Ranges page

Abscissa Min, Max: Enter the range for the abscissa that you want to display.

Ordinate Min, Max: Enter the range for the ordinate that you want to display. If you want to display two diagrams (e.g. Mag.&Phase), you can respectively define the ordinate.

ⓘ The settings for abscissa and ordinate are only displayed temporarily. Any zooming or auto scaling will overwrite the values entered.

Scale: This box only appears if the active analyzer is showing a spectrum. Tick the check box if you want to scale Logarithmically. Deactivate the check box for linear scaling of the abscissa.

Round Values: Tick the check box if the values for abscissa and ordinate are to be displayed as round numbers. The software will then round the values in Min and Max appropriately. Round Values is also active when zooming. The zoomed section is then usually slightly larger than selected.

9.2.4 Setting up Presentation of Graphs

You set up the presentation of graphs in the Analyzer Properties dialog.

You have the following options to open the dialog:

- Double-click the diagram area.
- Click the diagram with the right mouse button and select Data Properties in the context menu.
- Activate the analyzer (to do so, click it), select Analyzer > Properties in the menu bar and display the Data page.

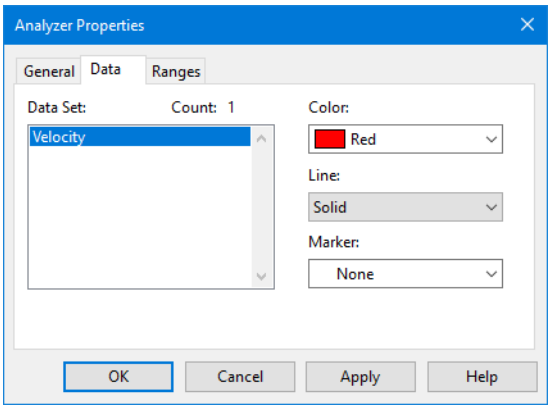


Figure 9.9: Data page

Data Set: Mark the data set which you want to set up. If the analyzer is showing several diagrams – e.g. magnitude and phase – select the data set which you want to set the layout for. To change the properties of multiple data sets at the same time, select the data sets before using the Control key.

ⓘ If you want to display the spectrum of different frames, then you can select the respective frame here as well (e.g. 1, 2, 3 etc.) and can set up the way it is displayed.

Color: Select the color of the graph.

ⓘ Changing the color affects the data set displayed in all display types.

Line, Marker: Select the view style of the graph and, if applicable, the marker. In the Marker, Histogram, Thin Bars and Thick Bars view styles, only measured y-values are shown. In the other view styles, the software interpolates between the measured values.

9.2.5 Setting up Diagrams

Set up the look of the diagram in the Analyzer Properties dialog. The current settings in the dialog also apply for all analyzers which you open afterwards.

You have the following options to open the dialog:

- Double-click an axis name.
- Click the diagram with the right mouse button and select General Properties in the context menu.
- Activate the analyzer (to do so, click it), select Analyzer > Properties in the menu bar and display the General page.

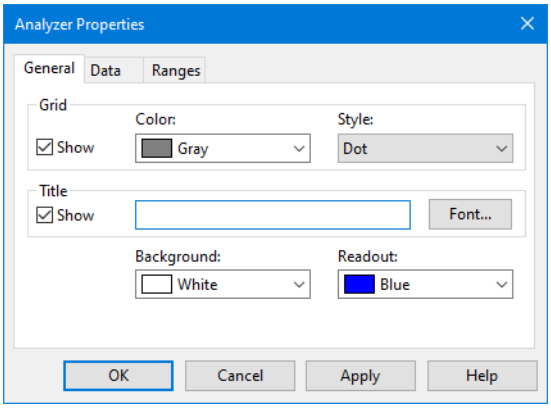


Figure 9.10: General page

Grid

Show: Tick the check box to display a grid in the diagram.

Color, Style: Select the color and the line style for the grid.

Title

Show: Mark the check box if you want to display a title above the diagram. You enter the title on the right.

Font: Click here to format the title. The Font dialog appears. Set up the Font, Font style, Size, and if applicable Effects and click OK.

Colors

Background: Select the background color of the diagram.

Readout: Select which color is to be used to display the readout beside the diagram.

9.3 Selecting the Data Set Displayed

You can already display the required data set in the analyzer while making the measurement.

Selection

To display a certain data set, select in the following order:

1. Time signal or type of spectrum (domain)
2. Channel or combination of channels
3. Signal or function
4. Display type used to display the signal or the function

Requirement

The selection depends on the following settings:

- Measurement mode
- Averaging, trigger and reference signal - See PSV theory manual on this as well.
- Which displays have already been selected - e.g. you can display a frequency response only for frequency spectra and a combination of channels.

To display a certain data set, proceed as follows:

Domain

1. In the menu bar, select Analyzer > Domain (toolbar: ).
2. Select the required setting (e.g. time signal, spectrum) in the pull-down menu or in the context menu.

Channel

3. In the menu bar, select Analyzer > Channel (toolbar: ).
4. Select a channel or a combination of channels in the pull-down menu or in the context menu. If you want to use user defined data sets, select the Usr, Usr X, Usr Y, Usr Z or Usr 3D channel. You will find information on this in the Polytec File Access documentation.

Signal

5. In the menu bar, select Analyzer > Signal (toolbar: ).
6. Select a signal or a function in the pull-down menu or in the context menu. The abbreviations used are explained in the following table.

Table 9.1: Abbreviations for functions

Abbreviation	Function
AP	Autopower
CP	Crosspower
FRF	Frequency Response Function
H1	Frequency Response
H2	Frequency Response
PSD[1]	Power Spectral Density
ESD[1]	Energy Spectral Density

[1] The PSD and ESD signals are only available in the analyzer.

ⓘ If you have more than 24 signals to choose from (Usr channel), select Analyzer > Signal > More Signals in the menu bar. The Signals dialog appears. Select the desired signal and click OK.

Display type

7. In the menu bar, select Analyzer > Display Type (toolbar: ).
8. Select the way that the vibration is to be presented, in the pull-down menu or in the context menu.

ⓘ In the Magnitude [dB(A)] display type, the measured amplitude is frequency-weighed according to the standard EN 60651 (IEC 60651) (A-weighting). The A-weighting approximately describes the human hearing range at low amplitudes. The following figure shows the frequency-weighting curve. Outside the range shown, the software evaluates as follows: under 10.0 Hz with –70.4 dB and over 20 000 Hz with –9,3 dB.

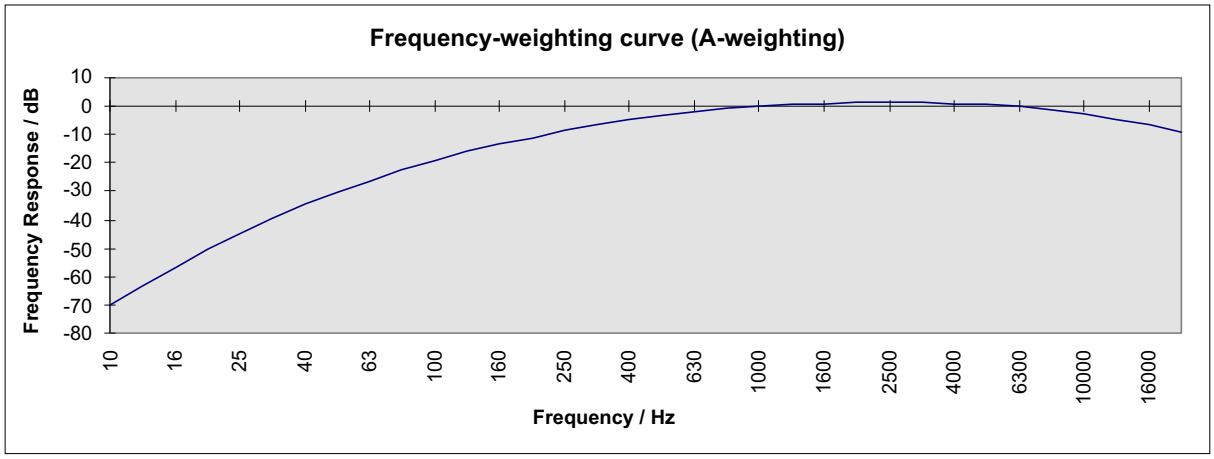


Figure 9.11: Frequency-weighting curve for the Magnitude [dB (A)] display type

ⓘ In the Nyquist display type, the real part of the amplitude is plotted along the x-axis and the imaginary part along the y-axis. You can limit the frequency range of the Nyquist display type by initially zooming the x-axis in a different display type so that the required frequency range is shown and then change to the Nyquist display type.

Features

Domain FFT measurement mode: To calculate a frequency spectrum and to display it, the software needs to have acquired the entire measurement time. Therefore, during a continuous measurement, the spectrum of the last complete measurement is displayed at the same time as the time signal of the current measurement.

In the following table you will find a summary of how to select Domain, Channel, Signal and Display Type.

Table 9.2: Selection of the data set displayed

Click	Or select Analyzer > or Presentation >	Display of	Examples
	Domain	Time signal, spectrum or RMS signal	FFT, 1/3 Octave RMS Time
	Channel	Channel or combination of channels	Vib, Vib 3D Ref 1, Ref 2, Ref 3 Vib & Ref 1, Vib & Ref 2, Vib & Ref 3 Vib 3D & Ref 1 Ref 1 & Ref 2, Ref 1 & Ref 3, etc.
	Signal	Signal or function	Displacement, Velocity, Voltage, Acceleration FRF Displacement / Voltage AP Displacement^2, AP Velocity^2, AP Voltage^2, AP Acceleration^2 CP Voltage * Velocity H1 Displacement / Voltage, H1 Velocity / Voltage, H1 Acceleration / Voltage H2 Displacement / Voltage, H2 Velocity / Voltage, H2 Acceleration / Voltage Coherence PSD Displacement^2/Frequency PSD Velocity^2/Frequency PSD Acceleration^2/Frequency ESD Displacement^2 * Time / Frequency ESD Velocity^2 * Time / Frequency ESD Acceleration^2 * Time / Frequency
	Display Type	Way in which the signal or function is shown	Magnitude Magnitude [dB], Magnitude [dB(A)] Mag. & Phase, Mag. dB] & Phase Phase Real Imaginary Real & Imag. Inst. Value Nyquist

9.4 Measurement Data in the Signal Processor

ⓘ When a signal processor document is active, in the menu bar the Analyzer menu is displayed.

In acquisition mode

You can display data in the signal processor in the acquisition mode. To do so, you can apply data from the analyzer during the measurement or open a scan or multi point file and apply the data from there into the signal processor. This allows you to check whether certain measurement conditions have not yet been met during acquisition.

In the menu bar, select File > New Signal Processor to open the signal processor in acquisition mode (toolbar: ).

You use the signal processor in acquisition mode in principle as described below for presentation.

In presentation mode

If you want to continue to analyze measurement data which has already been acquired, then you can recalculate it and show the results in direct comparison with the original data. For this purpose a signal processor is available which you can transfer the data to and can then process it further. The data calculated this way is shown in the analyzer of the signal processor. You can also save this data with the scan and display it again at any time. To do so, read the following sections:

- Copying and calculating measurement data - See SECTION 9.4.1 on this.
- Available arithmetic operations - See SECTION 9.4.2 on this.
- Calculation examples for arithmetic operations - See SECTION 9.4.3 on this.
- Setting up signal description - See SECTION 9.4.4 on this.
- Saving calculated measurement data (user defined data sets) - See SECTION 9.4.5 on this.
- Displaying measurement results - See SECTION 9.4.6 on this.
- Saving formulae - See SECTION 9.4.7 on this.
- Finding and replacing in formulae - See SECTION 9.4.8 on this.
- You will find application examples for processing measurement data in presentation mode.

9.4.1 Copying and Calculating Measurement Data

To process the measurement data in the signal processor, you need to open the respective measurement, apply the measurement data to the signal processor and carry out the required arithmetic operations there.

Open signal processor

To open the signal processor, proceed as follows:

1. Select File > New > Signal Processor in the menu bar (toolbar: > Signal Processor).

   The signal processor appears.

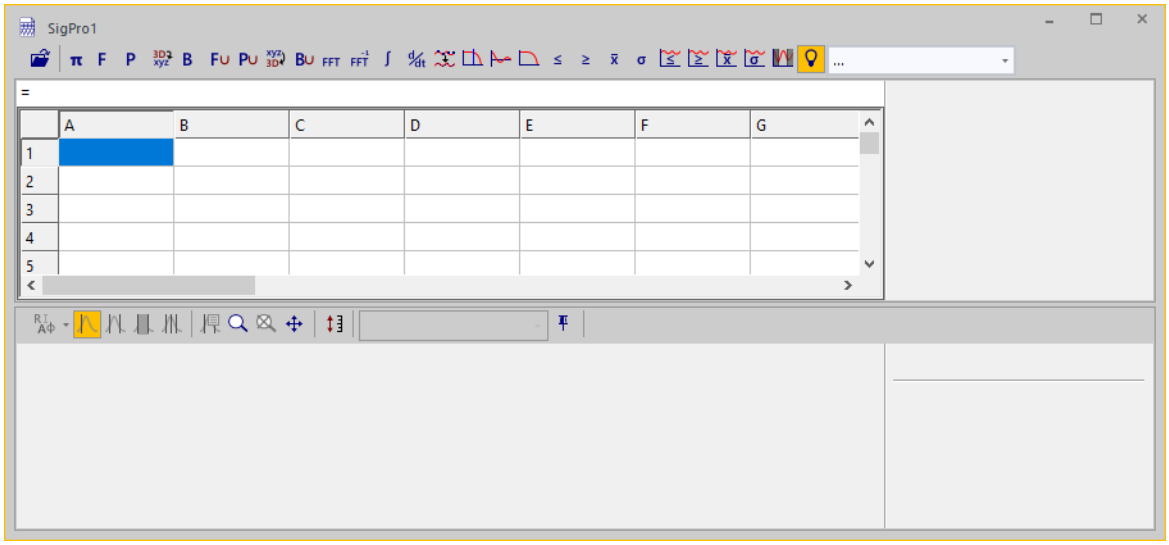


Figure 9.12: New signal processor

ⓘ You will find a description of the signal processor in SECTION 2.6.5.

Open scan

Open the scan in presentation mode. To do so, proceed as follows:

2. In the menu bar, select File > Open (toolbar:).

   The Open dialog appears.

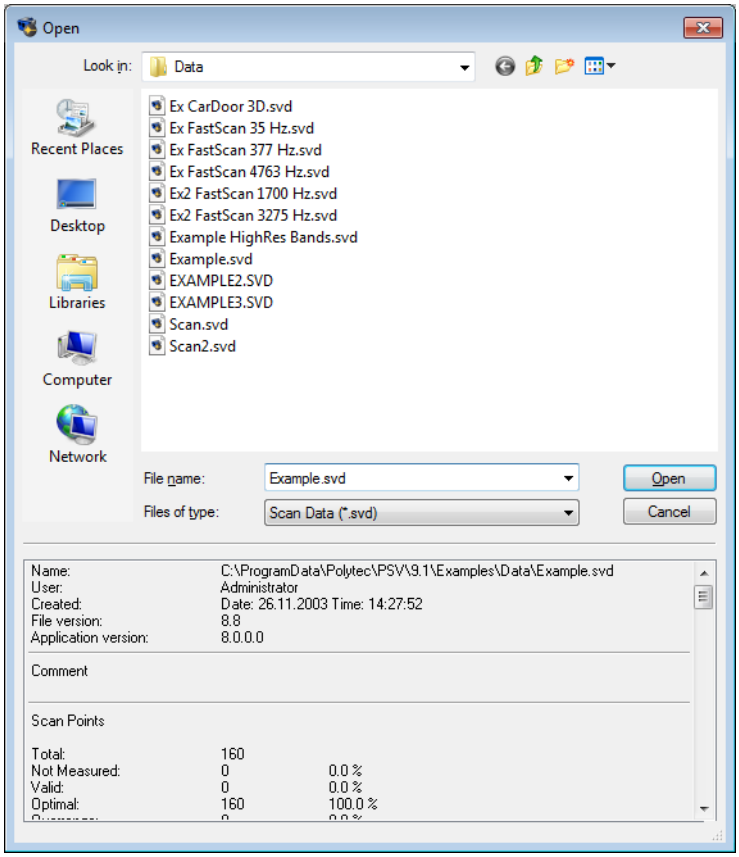


Figure 9.13: Open dialog for a scan

3. Navigate to the saving location and select Files of type and File name.

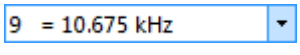
   At the bottom in the dialog you can see meta-information to be able to identify the files more easily. For example, the data acquisition board and the settings which were used to measure the data.

4. Click Open.

   If you open a scan, a presentation window appears. If you open a single point measurement, an analyzer appears.

Copy measurement data

You can apply point data or band data in the signal processor and create user defined data sets. To do this, copy the measurement data into the signal processor. To do so, proceed as follows:

5. To copy Point Data or Band Data, click  and select the desired in the pull-down menu. If you want to copy band data, then also select the required frequency band in the frequency band list  .

6. In the presentation window, use the mouse to select one or more scan points, click with the right mouse button and select Copy Selected Points in the context menu. If you have selected a frequency band, select Copy Selected Band.

   The data from the selected scan point or frequency band is copied into the signal processor.

ⓘ If you select Copy All Points or Copy All Bands in the context menu, then the data of all scan points or frequency bands is copied into the signal processor.

7. If you have selected the Average Spectrum view, click the analyzer with the right mouse button and select Copy in the context menu.

   The Copy dialog appears.

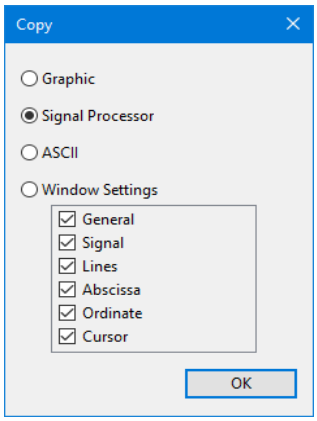


Figure 9.14: Copy dialog for signal processor

8. Select Signal Processor and click OK.

   The data is copied.

ⓘ You can also press the Ctrl + C key combination. The Copy dialog appears as long as you hold the Control key pressed. Select which data you want to copy and release the Control key. The data is copied.

Paste measurement data

9. Change to the signal processor.
10. Click a cell (e.g. A1) with the right mouse button and select Paste or Paste Link in the context menu.

    The copied data is inserted in the signal processor or inserted and linked. When inserting data from surface view, the signals of all channels are merged into one 3D signal for the defined scan points. In addition, a coordinate transformation into the coordinate system of the scan points will be carried out. When entering raw data, no transformation is carried out.

ⓘ Inserted data contains a reference to the original file and remains in the corresponding cell, even if you select other data in the sequence or close the scan or the signal processor. If you are using the RelativePath function, this ensures that the link still works if you copy or move the signal processor file with the original file. But while doing so, you need to retain the relative path names in the file system.

Linked data in contrast is updated in the formula as soon as you select other scan points and is lost as soon as you close the scan or the signal processor.

11. If you have selected and inserted several scan points or frequency bands, and would like to display the data from a single scan point or frequency band, then select the index or the frequency band in the list. See the following figure on this as well.

12. If you have applied the 3D data, you can select one of the vibrational directions for the display (X, Y, Z).
13. If you would like to display specific frames of the selected scan point or frequency band, click the number of the respective frame (1, 2, 3, 4, etc.). You can display an individual graph for each frame. See the following figure and SECTION 9.4.3 on this as well.

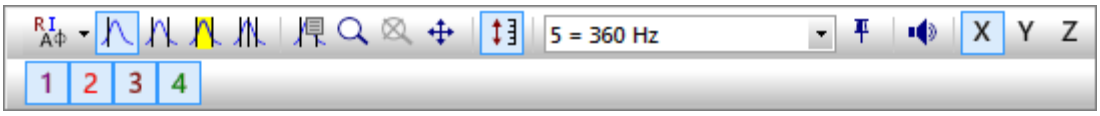


Figure 9.15: Selecting index/frequency band, vibrational direction and frames in the analyzer of the signal processor

Edit measurement data

14. Click another cell (e.g. A2) and then activate the input line by clicking it.

ⓘ You can insert a reference to another cell into the formula by holding the Alt key pressed and then clicking the required cell.

15. Enter the required command and confirm it with Enter.

ⓘ You can also enter any comments you like as text in the individual cells. In this case you just do not set an equal sign at the start of the respective text. Invalid formulae are shown in red.

Toolbar

You can also use the icons to enter some commands in the formulae editor directly or you can take over the formula in the formulae editor by clicking it in the list next to the icon for the function wizard.



Figure 9.16: Icons for calculations in the signal processor

You can also use the function wizard. To activate the function wizard, click  . Then when you click one of the icons, the Function Wizard dialog appears which supports you entering the parameters. The signal processor carries out this calculation and shows the result in the analyzer.

Example

You will find an example for the function wizard in SECTION 9.4.9.

ⓘ You will find an overview of the arithmetic operations stored in the signal processor in SECTION 9.4.2. You will find calculation examples for arithmetic operations in SECTION 9.4.3. Read SECTION 9.4.4 on how to set up the signal description.

9.4.2 Available Arithmetic Operations

In the following table you will find an overview of all arithmetic operations that you can carry out.

Table 9.3: Overview of the arithmetic operations of the signal processor

Command	Icon	Arithmetic operation
Add (+)		By entering + you add two scalars, two strings, two Vib data sets or one Vib data set and one scalar or join two sets.
Sub (–)		By entering – you subtract two scalars, two Vib data sets or one Vib data set and a scalar or two sets.
Mul (*)		By entering * you multiply two scalars, two Vib data sets or one Vib data set with a scalar or intersect two sets.
Div (/)		By entering / you divide two scalars or two Vib data sets by each other, one Vib data set by a scalar or one scalar by a Vib data set.
Power (^)		By entering ^ you raise the value of x to the power of y (written: x^y, i.e. x^y); x and y can be scalars or Vib data sets.
RelativePath (TargetPath, SourcePath, UseRelative)		Entering RelativePath ensures that the link to the original file still works if you copy or move the signal processor file with the original file. But while doing so, you need to retain the relative path names in the file system. If UseRelative = 1, then the path is formed from TargetPath and the relative SourcePath given for it. Otherwise the path is only formed from the SourcePath.
CopyVibData (FilePath, Domain, Channel, Signal, DataType, 3D)		The CopyVibData (RelativePath) operation is used to copy specified measurement data from a measurement file into the signal processor (Paste command in the context menu).
LinkVibData (FilePath, Domain, Channel, Signal, DataType, 3D)		The LinkVibData operation is used to link specified measurement data from a measurement file into the signal processor (Paste Link command in the context menu). After making changes to the display in the measurement file the data will be updated.
LinkAcqData (Domain, Channel, Signal)		The LinkAcqData operation is used to display the specified signal from the measurement running in acquisition mode (Paste command in the context menu). The data is automatically updated after each single shot or after each measured scan point.

Several arithmetic operations apply to one scalar (x) or Vib data set (VibData). Optional parameters are marked [ ].

Command	Icon	Arithmetic operation
Re(x), Re(VibData)		By entering Re you calculate the real part.
Im(x), Im(VibData)		By entering Im you calculate the imaginary part.
Abs(x), Abs(VibData)		By entering Abs you calculate the absolute value.
Arg(x), Arg(VibData)		By entering Arg you calculate the phase.
Exp(x), Exp(VibData)		By entering Exp you calculate the exponential function.
Log(x), Log(VibData)		By entering Log you calculate the logarithm (base 10).
Ln(x), Ln(VibData)		By entering Ln you calculate the natural logarithm (base e).
Sqrt(x), Sqrt(VibData)		By entering Sqrt you calculate the square root.
Sin(x), Sin(VibData)		By entering Sin you calculate the sine (radian).
Cos(x), Cos(VibData)		By entering Cos you calculate the cosine (radian).
Tan(x), Tan(VibData)		By entering Tan you calculate the tangent (radian).
ASin(x), ASin(VibData)		By entering ASin you calculate the arc sine (radian).
ACos(x), ACos(VibData)		By entering ACos you calculate the arc cosine (radian).
ATan(x), ATan(VibData)		By entering ATan you calculate the arc tangent (radian).
Sign(x), Sign(VibData)		By entering Sign you calculate the sign of the value x. The result is –1 for x = negative, +1 for x = positive and 0 for x = 0.
Deg(x), Deg(VibData)		By entering Deg you convert radians into degrees.
Rad(x), Rad(VibData)		By entering Rad you convert degrees into radians.
Round(x)		By entering Round you round real numbers to the next integer e.g. Round(1.1) = 1 or Round(1.9) = 2.
Ceil(x)		By entering Ceil you round up real numbers to the next integer e.g. Ceil(1.1) = 2 or Ceil(1.9) = 2.
Floor(x)		By entering Floor you round down real numbers to the next integer e.g. Floor(1.1) = 1 or Floor(1.9) = 1.
Card(x)		By entering Card you calculate the cardinality (number of elements) in a set.
Pi()		By entering Pi you add the constant π to the formula.
Frame(VibData, IndexOrIndexSet)		By entering Frame you extract the specified frame or frames from the Vib data set. If the second parameter is a scalar, its real part is rounded up or down to the next integer and its imaginary part is ignored.
Point(VibData, IndexOrIndexSet)		By entering Point you extract the specified sample point or points from the Vib data set. If the second parameter is a scalar, its real part is rounded up or down to the next integer and its imaginary part is ignored.
Band(VibData, IndexOrIndexSet)		By entering Band you extract the specified band or bands from the Vib data set. If the second parameter is a scalar, its real part is rounded up or down to the next integer and its imaginary part is ignored.

Command	Icon	Arithmetic operation
ReIndex(VibData[, IndexOrIndexSet, IndexOrIndexSet])		By entering ReIndex you change the indices of the sample points of a Vib data set. Parameter: ReIndex(VibData, [SrcInd, DstInd]) Requirements: - SrcInd and DstInd can each be an IndexSet or an individual index. - SrcInd is optional with Default {*} and DstInd is optional with Default 0. - From SrcInd and the IndexSet of VibData an intersection is formed. This intersection must have the same number of elements as DstInd otherwise an error message appears. - The intersection of DstInd and the IndexSet of VibData must be empty. The sample points from SrcInd get then the indices of DstInd whereby their order remains conserved. Example 1: ReIndex(VibData, {*}, 0) = ReIndex(VibData) This changes the index to 0 for a Vib data set which contains an individual sample point. Example 2: VibData has the sample points 1…100; ReIndex(VibData, {21..30}, {101..110}) The result has the indices 1..20, 31..110.
FrameUnion (VibData, …)		By entering FrameUnion you combine the frames of up to 10 Vib data sets. The maximum total number of combined frames is 12. You can use this to compare a sample point with an identical index from different measurements.
PointUnion (VibData, …)		By entering PointUnion you combine the sample points of up to 10 Vib data sets. If sample points occur in more than one data set, those found first are used.
BandUnion (VibData, …)		By entering BandUnion you combine the frequency bands of up to 10 Vib data sets.
DisplayUnion (VibData, …)		By entering DisplayUnion you combine the frames of up to 10 Vib data sets. The maximum total number of combined frames is 12. You can use this to compare sample points with a different index from different measurements if each of the measurements only contains one sample point respectively.
Extract (VibData, xMin, xMax)		By entering Extract you extract the specified frequency line numbers or samples. The default starting value is 0 and the full scale value is the maximum number –1. The full scale value must be larger than the starting value.
ExtractX (VibData, xMin, xMax)		ExtractX works in the same way as Extract. Here, you only enter start and end of the range to be extracted in the physical quantity of the x-axis (Hz or s).

Command	Icon	Arithmetic operation
FFT(VibData[, Lines, Window, WindowParameter, WindowParameter])		By entering FFT you calculate the Fourier Transformation of the Vib data set. Parameter: FFT(VibData1, [Lines, Window, WndPar1, WndPar2]) Requirements: - VibData1 must be a real signal. - The lines must be in the range of 25…1048576. If you do not specify the lines, the maximum possible number will be assumed. - The length of the vector for the Vib data set must be at least twice the number of lines, rounded up to the next power of 2. - The window function (Window) has to be on of the following character strings: "Rectangle", "Bartlett", "Blackman Harris", "Exponential", "Flat Top", "Force", "Hamming", "Hanning", "Tapered Hanning" - If you do not specify any type of window, "Rectangle" will be assumed. - WndPar1 and WndPar2 are parameters of the selected window function if applicable. Please note that WndPar2 is not currently used.
IFFT(VibData)		By entering IFFT you calculate the inverse Fourier Transformation. Note: With a measurement which is not triggered, the phase in the Vib channel is replaced by the differential phase to the Ref channel. The calculation of the inverse FFT can provide unexpected results which are hard to interpret.
Integrate(VibData)		By entering Integrate you integrate the Vib data set. Requirement: The input data must be real.
Differentiate(VibData)		By entering Differentiate you differentiate the Vib data set. Requirement: The input data must be real.
Resample (VibData, Fnew)		By entering Resample you have the Vib data set sampled again at the new specified sample rate given in Hz (Fnew). Requirement: The input data must be real.
Filter (VibData, Type, Quality, Flow[, Fhigh])		By entering Filter you filter a Vib data set. Requirement: The input data must be real. For Type select one of the character strings "Low Pass", "High Pass", "Band Pass" or "Notch". For Quality select one of the character strings "Very Low", "Low", "Middle", "High" or "Very High". For Flow select the cutoff frequency in Hz for Low Pass and High Pass or the lower cutoff frequency for Band Pass and Notch. For Fhigh select the upper cutoff frequency in Hz for Band Pass and Notch.
FilterCoeff (Fsamp, Type, Quality, Flow[, Fhigh])		By entering FilterCoeff you calculate a Vib data set which contains the filter core of the selected filter. For Fsamp enter the sample frequency in Hz. For Type select one of the character strings "Low Pass", "High Pass", "Band Pass" or "Notch". For Quality select one of the character strings "Very Low", "Low", "Middle", "High" or "Very High". For Flow select the cutoff frequency in Hz for Low Pass and High Pass or the lower cutoff frequency for Band Pass and Notch. For Fhigh select the upper cutoff frequency in Hz for Band Pass and Notch.

Command	Icon	Arithmetic operation
Transmission (Fsamp, Type, Quality, Flow[, Fhigh], Count)		By entering Transmission you calculate a Vib data set which contains the frequency response function of the selected filter. For Fsamp enter the sample frequency in Hz. For Type select one of the character strings "Low Pass", "High Pass", "Band Pass" or "Notch". For Quality select one of the character strings "Very Low", "Low", "Middle", "High" or "Very High". For Flow select the cutoff frequency in Hz for Low Pass and High Pass or the lower cutoff frequency for Band Pass and Notch. For Fhigh select the upper cutoff frequency in Hz for Band Pass and Notch. For Count enter the number of frequency lines for which the transmission function is to be calculated.
Min(VibData) Max(VibData) Mean(VibData) StdDev(VibData)		By entering Min, Max, Mean or StdDev, you respectively calculate the minimum, maximum, average or standard deviation of a Vib data set which contains an individual sample point. The return value is a scalar. You can calculate Mean for real or complex data. You can only apply all the other functions to real data.
ArgMin(VibData) ArgMax(VibData)		By entering ArgMin or ArgMax the abscissa (x-coordinate) of the minimum or maximum in a real Vib data set which contains an individual sample point is given.
StatMin(VibData) StatMax(VibData) StatMean(VibData) StatStdDev(VibData)		By entering StatMin, StatMean, StatMax or StatStdDev, you respectively calculate the minimum, maximum, average or standard deviation of a Vib data set. The return value is a Vib data set with an individual sample point with the index 0. You can calculate StatMean for real or complex data. You can only apply all the other functions to real data. It can be meaningful to apply StatMean to a data set which only contains an individual sample point. This does not change the data set but means that the point index is set to 0. Points with the index 0 can be compared to all other indexed points.
Scale(VibData[, xMin, xMax, xName, xUnit, yMin, yMax, yName, yUnit, yScale, Name, 0dBReference, Power])		By entering Scale you modify scaling information of a Vib data set. This function is usually used to adapt the scaling after operations such as FFT or Integrate, as these operations change the dimensions of the axes. The changes you can make using the Scale command can be found in the following list. - VibData: Data set - xMin, xMax: Limits of the x-axis - xName: Name of the x-axis - xUnit: Unit of the x-axis - yMin, yMax: Limits of the y-axis - yName: Name of the y-axis - yUnit: Unit of the y-axis - yScale: The y-values are multiplied with yScale - Name: Name of the data set - 0dBReference: Reference value of dB display - Power: 1 or 0 (Power = 1 shows that this is a power signal. 10 dB then corresponds to a multiplication by ten. For other signals, set Power = 0. 20 dB then corresponds to a multiplication by ten.)

Command	Icon	Arithmetic operation
Resolution(VibData)		By entering Resolution the distance between the frequency lines or samples of a Vib data set is given.
NumValues(VibData)		By entering NumValues the number of samples or frequency lines of a Vib data set is given.
Points(VibData)		By entering Points the set of sample points in a Vib data set is given.
Abscissa(VibData)		By entering Abscissa a Vib data set of a sample point with the index 0 is given for which the ordinate (y-axis) shows the same values as the abscissa (x-axis).
Direction(VibData, dir)		By entering Direction you extract the data in vibrational direction of a 3D Vib data set. The direction is set with the variable dir (0 for x-direction, 1 for y-direction, 2 for z-direction)
Make3D(VibData)		By entering Make3D you combine 1D Vib data sets to a 3D Vib data set. If you combine less than three Vib data sets, the missing ones are replaced with 0.

ⓘ In the table cells, you can display either the formula or the result. To toggle, select View > Formulae in the menu bar (key combination: Ctrl + #).

Conventions

You can also enter sets and complex numbers. You place sets in curly brackets and complex numbers as an option in round brackets. In TABLE 9.4 and TABLE 9.5 the input conventions are explained using examples.

Table 9.4: Examples for entering sets

Command	Meaning
{1,2}	1 and 2
{1:5} or {1..5}	1 to 5
{1,4,7..9}	1, 4 and 7 to 9
{}	Empty set
{*}	General set

Table 9.5: Examples for entering complex numbers

Command	Meaning
0;1 or (0;1)	i
–1.23E4;–5.67E8 or (–1.23E4;–5.67E8)	–1.23E4–5.67E8 · i

Cell links

You will find a description what kind of cell links you can generate in the signal processor in SECTION 2.6.5 under Calculation table. A distinction is made between relative and absolute cell links:

Relative Cell Links: If you create a formula, the references to cells are shown relative to the position of the cell which contains the formula.

For example the cell B6 contains the formula =A5. The signal processor searches the value in the cell which is located one cell above and one cell on the left of B6. References of this type are called relative cell links. If you copy a formula in which relative cell links are used, then the links were automatically adapted after inserting the formula to refer to other cells in relative to the position of the formula.

Absolute Cell Links: If you want to prevent that the links will be adapted when copying a formula in another cell, use an absolute cell link.

If, for example, a formula multiply the cell A5 by the cell C1 (=A5*C1) and is copied into another cell, both references will be adapted. You can generate an absolute cell link to the cell C1 by entering a dollar symbol ($) in front of the part of the link that should not be changed. To generate, for example, an absolute cell link to the cell C1 enter (=A5*C1) in the formula.

9.4.3 Calculation Examples for Arithmetic Operations

Differentiation in the frequency range

In the following table you will find an example for differentiation in the frequency range.

Cell	Function	Description
A1	= Complex spectrum	From measurement
B1	=IFFT(A1)	Transformation in the time domain
B2	=Differentiate(B1)	Differentiation in the time domain
A2	=2*Pi()*(0;1)*Abscissa(A1)	Multiplication with $2\pi \cdot i \cdot f$ in the frequency range corresponds to differentiation in the time domain.
A3	=A1*A2	Multiplication
B3	=IFFT(A3)	Transformation in the time domain

To calculate B2, first of all transform into the time domain and then differentiate there. To calculate B3, multiply in the frequency range with a function which corresponds to differentiation in the time domain. The result is the same apart from fringe effects and shifts in the order of magnitude of the sampling. Multiplication with $2\pi \cdot f$ corresponds to a high pass, multiplication with i to a phase shift by 90°.

Integration in the frequency range

In the following table you will find an example for the integration in the frequency range.

Cell	Function	Description
A1	= Complex spectrum	From measurement
B1	=IFFT(A1)	Transformation in the time domain
B2	=Integrate(B1)	Integration in the time domain
A2	=2*Pi()*(0;1)*Abscissa(A1)	Division by $2\pi \cdot i \cdot f$ in the frequency range corresponds to integration in the time domain.
A3	=A1/A2	Division
B3	=IFFT(A3)	Transformation in the time domain

To calculate B2 first of all transform into the time domain and then integrate there. To calculate B3, multiply in the frequency range with a function which corresponds to integration in the time domain. The result is the same as with differentiation apart from an additive constant.

Average over partition

In the following table you will find an example for creating the average over a partition.

Cell	Function	Description
A1	= Single sample point	From measurement
A2	=(sign(Abscissa(A1)–4000)–sign(Abscissa(A1)–6000))/2	Function which has the value 1 in the interval [4 kHz…6 kHz] and has the value 0 outside
A3	=A1*A2	Values outside the interval [4 kHz…6 kHz] are set to 0.
A4	=Mean(A3)/Mean(A2)	Average in the range [4 kHz…6 kHz]

Power spectral density

In the following table you will find an example for calculating the power spectral density.

Cell	Function	Description
A1	= Complex spectrum	From measurement
A2	=abs(A1)^2/Resolution(A1)	Power spectral density

Calculation of MAC values

MAC values (Modal Assurance Criterion) are used to compare band data (Operational Deflection Shapes) or modal deflection shapes (Mode Shapes) and have values between 0 and 1. Here, a MAC value of < 0.9 means that the shapes are different, a MAC value of > 0.9 means that the shapes are similar and a MAC value of 1 means that the shapes are the same. In the following table you will find an example of calculating a MAC value from two bands.

Cell	Function	Description
A1	= Band data	From measurement or imported calculation. Display a band in a measurement file and select Copy Selected Band in the context menu. Insert the band in the cell A1.
B1	= Band data	The band to be compared
B2	=Re(B1)–(0;1)*Im(B1)	Calculates the conjugated complexes of cell B1
B3	=StatMean(A1)*StatMean(B2)	Calculated the cross correlation between the bands. StatMean is used to give both bands the band index 0 and can then be multiplied with each other.
C1	=Abs(Mean(B3))^2/(Mean(Abs(A1)^2)*Mean(Abs(B1)^2))	Calculating the MAC value

9.4.4 Setting up Signal Description

You can adapt the presentation of the recalculated measurement data in the analyzer. To do so, proceed as follows:

1. Use the right mouse button to click the cell which contains the recalculated measurement data and select Signal Description in the context menu.

   The Signal Description dialog appears.

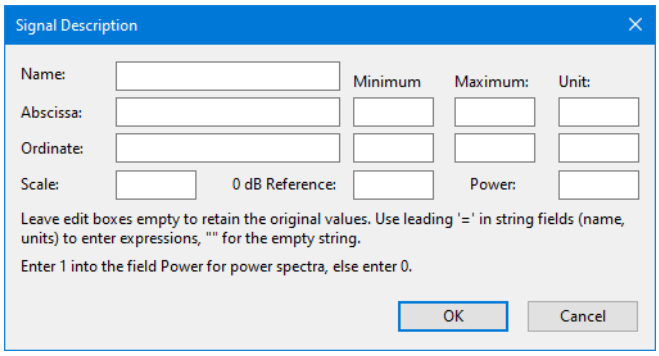


Figure 9.17: Signal Description dialog in the signal processor

2. Fill in the fields as described in the following and click OK.

ⓘ Entries in this dialog are saved and shown in any case, even if the input data for the signal calculation changes! If you do not enter anything in a field, the original value from the input data is retained or adapted.

Dialog description

Name: Enter the description of the recalculated measurement data. This name is shown in the respective cell. After saving the measurement data it will also appear on the list of available signals in the analyzer. See SECTION 9.4.5 on this as well.

Abscissa: Enter the description for the x-axis.

Ordinate: Enter the description for the y-axis.

Minimum: Enter the lower value for the scale of the x- or y-axis.

Maximum: Enter the upper value for the scale of the x- or y-axis.

Unit: Enter the abbreviation for the unit of the x- or y-axis.

Scale: Enter the factor by which the presentation of the measurement values is to be magnified or reduced.

0 dB Reference: If you have selected a view with [dB], here you enter the value which corresponds to 0 dB (reference value).

Power: If for example the signal includes a cross power spectrum, enter the value 1. If the signal is not a power signal, enter the value 0 (default).

9.4.5 Saving Calculated Data (User Defined Data Sets or Bands)

You can add the measurement data which you calculated in the signal processor to the original file as user defined data sets or bands.

ⓘ The indices of the scan points in the original file have to match the indices of the data set. If not, only the data with corresponding indices is applied. The geometry of the objects, i.e. the number of scan points should be the same. If not, a message will appear and you will be asked whether you want to apply the data sets.

Add measurement data

You have the following options for adding the calculated measurement data:

1. Mark the cell which contains the recalculated measurement data in the signal processor.
2. Click the cell with the right mouse button and select Copy in the context menu.
3. Then use the right mouse button to click the name of the original file in the project browser and select Paste in the context menu.

Or:

4. Mark the cell which contains the recalculated measurement data in the signal processor.
5. Hold the left mouse button pressed and drag the cell to the analyzer.

Or:

6. Mark the cell which contains the recalculated measurement data in the signal processor.
7. Hold the left mouse button pressed and drag the cell to the name of the original file in the project browser.

In the Object view, you will now immediately have another channel available with the name Usr, Usr X, Usr Y, Usr Z or Usr 3D. In the list of signals you will find the recalculated measurement data listed with the name which you entered in the Signal Description dialog.

ⓘ You can also copy original signals or original bands respectively and can insert them as user defined signals or bands respectively.

Display properties

You can display the properties of user defined signals and change or delete them. To do so, proceed as follows:

1. Click  and select the Usr, Usr X, Usr Y, Usr Z or Usr 3D channel in the pull-down menu.
2. Click  and select Properties in the pull-down menu.

   The Properties of User Defined Signals dialog appears.

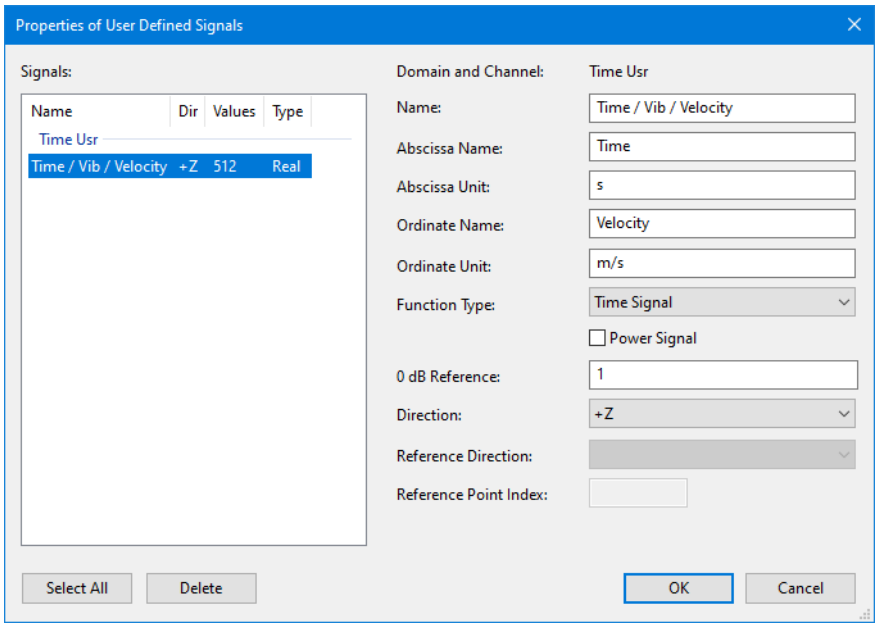


Figure 9.18: Properties of User Defined Signals dialog

Dialog description

Signals or Band Signals: On the left you will see the user defined signals or band signals with the name you entered in the Signal Description dialog.

Domain and Channel: The domain, the channel as well as the signal name and other properties are shown. You can change these properties for every signal. If you have inserted user defined bands, you will see Band Definitions right at the bottom.

Delete: You can delete user defined signals or bands. To do so, mark the signal or band and click Delete. The user defined data set is deleted from the file.

Equalize properties

To set the properties of several user defined signals to the same value, proceed as follows:

1. Mark the respective signals in the list. To do so, hold the Control key pressed and click the signals.
2. Enter the required common property on the right.
3. As soon as you now change the signal selection or click OK, the changes are applied to the signals.

9.4.6 Displaying Measurement Results

Analyzer in the signal processor

At the same time as recalculating the data, the updated graph is shown in the analyzer. The analyzer of the signal processor has the functions described in SECTION 2.6.3. If you mark several table cells, then you will be shown the respective frames for selection. If you select several frames, you will be shown the respective graphs in parallel.

Example

ⓘ Band data is plotted in the analyzer using the scan point number. The scan point number does not necessarily correspond to the scan point index shown in the presentation window.

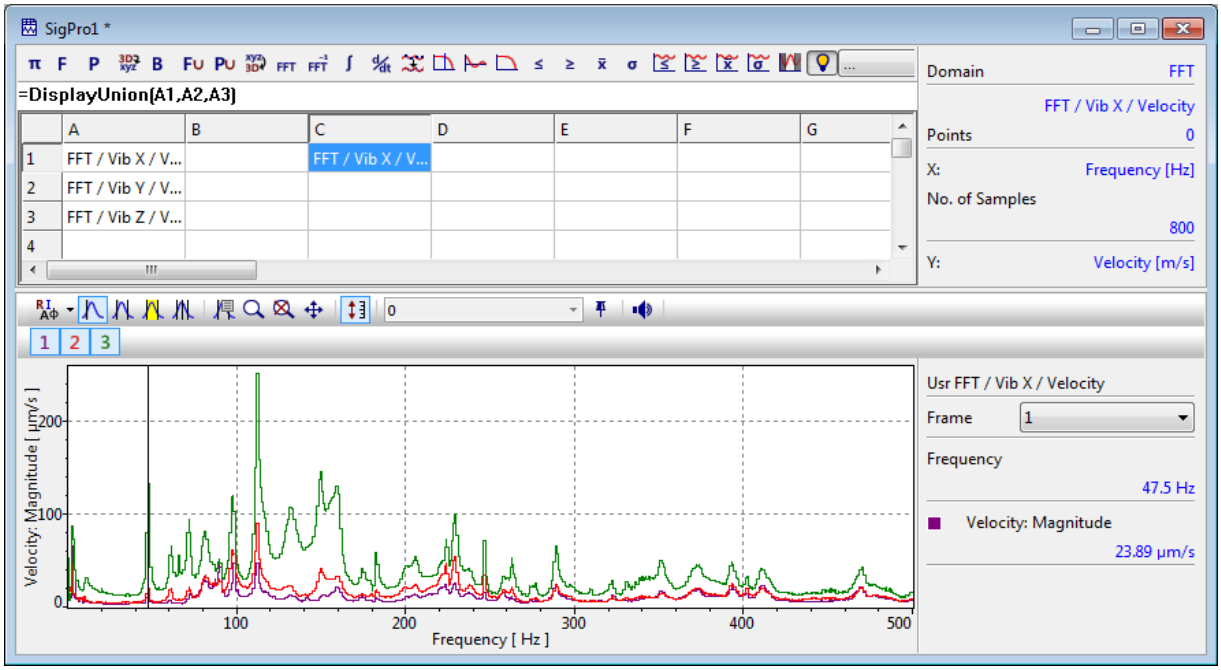


Figure 9.19: Displaying several graphs in the analyzer of the signal processor

Fix the view style

If you would like to stop updating the graph at a certain point in the arithmetic chain, click  (Fix). The presentation then corresponds to the contents of the table cell selected at that moment in time. All subsequent arithmetic steps are carried out anyway, but do not have any effect on the presentation of the graph. However, changes in the formulae before the position marked with  still have an effect on the adaptation of the presentation up to this point.

9.4.7 Saving Formulae

To save formulae after you have recalculated the measurement data, proceed as follows:

1. In the menu bar, select File > Save As (toolbar:).

   The Save As dialog appears.

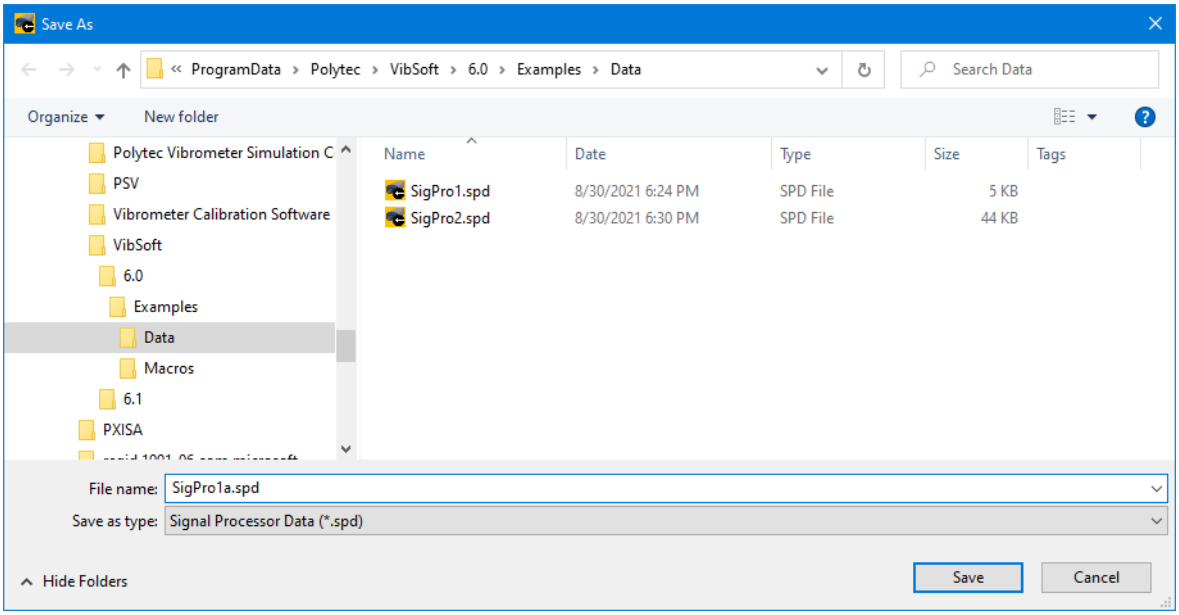


Figure 9.20: Save As dialog for signal processor files

2. Navigate to the saving location and enter the file name.
3. Click Save.

   The file saved in this way does not contain any measurement data itself, but only contains references to data. The reference among other things includes the full path name of the original file. If you move or rename the measurement, then the references to the measurement data become invalid. To permanently save your results, you have to add them to the original file.

ⓘ A file in Signal Processor Data (*.spd) format does not contain any data, only calculation specifications!

9.4.8 Finding and Replacing in Formulae

To search for arithmetic operations or expressions in formulae and replace them accordingly, proceed as follows:

1. Click the desired cell with the right mouse button and select Find and Replace in Formulae in the context menu.

   The Find and Replace in Formulae dialog appears.

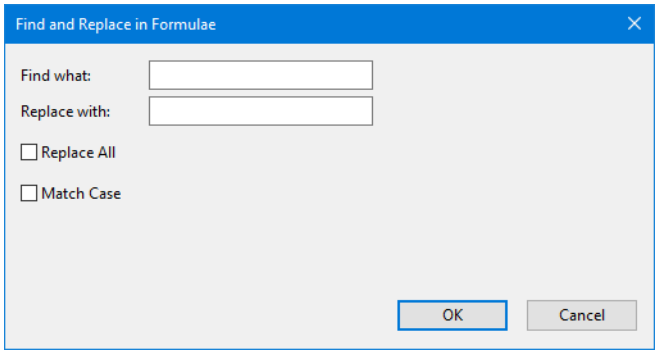


Figure 9.21: Find and Replace in Formulae dialog in the signal processor

2. Enter the expression you want to replace in the Find what field.

ⓘ If you want to specify the search, tick the Match Case check box.

3. Enter the new expression in the Replace with field.
4. If you want to replace all expressions found in this way at once, additionally tick the Replace All check box.
5. Click OK.

9.4.9 Example of the Function Wizard

You can also switch on a function wizard for the arithmetic operations which are offered as icons in the signal processor. To do so, click  in the toolbar of the signal processor.

As an example, replace steps 20 and 21 from the third example in presentation mode with the following instruction:

1. In the toolbar of the signal processor, click  .

   The Function Wizard dialog appears.

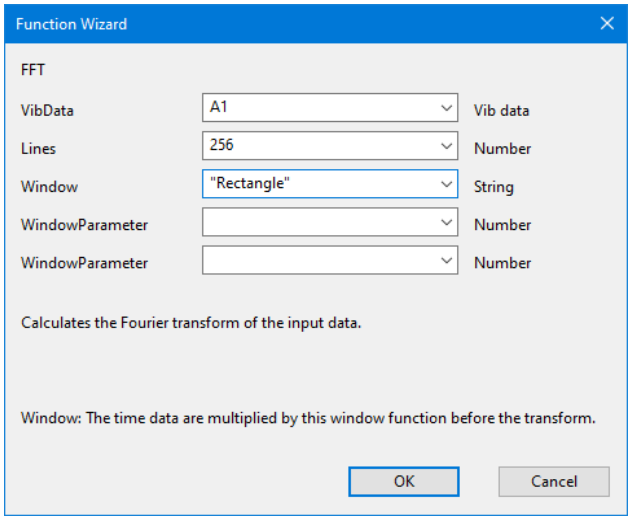


Figure 9.22: Function Wizard dialog for the signal processor

2. For VibData enter the value B2, for Lines the value 256 and for Window select the entry "Rectangle" from the list.
3. Click OK.
4. Then proceed with step 22.

10 Managing Data

The following functions are available for managing your measurement data:

You will find information on opening, loading and importing files, settings, and geometries in SECTION 10.1.

You will find information on saving and exporting files, settings, and geometries in SECTION 10.2.

10.1 Opening, Loading, Importing

You can display file information in acquisition mode. See SECTION 10.1.1 on this.

The software can open, load or import the following files in acquisition mode:

- Signal processor files in proprietary binary format Signal Processor Files (*.spd) - See SECTION 10.1.2 on this.
- Settings in proprietary binary format Settings Files (*.set) - See SECTION 10.1.3 on this.
- Importing geometries - See SECTION 6.3.7 on this.

You can open a path length alignment in *.pla format for viewing in Presentation mode. See the Presentation manual as well.

10.1.1 Displaying File Information

As soon as you have started PSV Acquisition, you can display information on the measurement system and on parameters and settings. This information is saved with the scan.

To display it, select File > Info in the menu bar (toolbar: ). The File Information dialog appears.

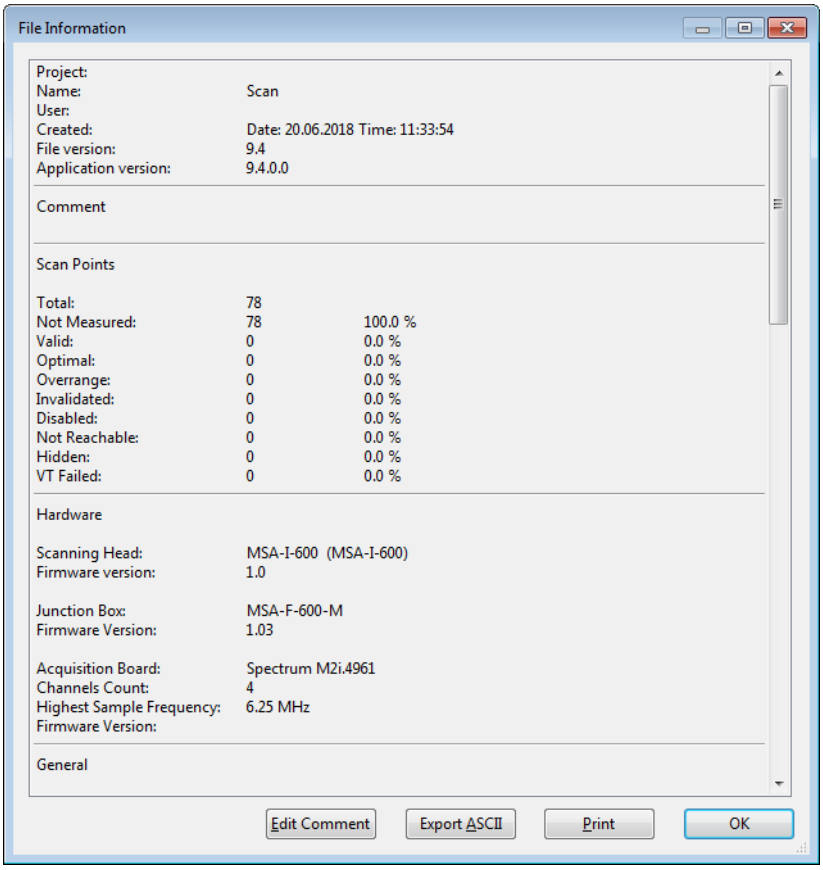


Figure 10.1: File Information dialog

ⓘ You can freely adjust the size of the dialog (width/height) to be able to read all the information.

Dialog description

File Information: The name of the scan, the current date, the file and program version are displayed.

Comment: You can enter comments on the measurement which are displayed. To do so, click Edit Comment at the bottom of the dialog. The Comment dialog appears. You can enter text and save it, by clicking OK.

Scan Points: A summary of the defined scan points is shown.

Hardware: The system equipment is shown (sensor head, junction box, data acquisition board, etc.).

General, Frequency, Sampling, Trigger, Channels, SE etc.: The parameters for data acquisition are shown.

Vibrometer 1: The front-end or controller equipment is shown respectively.

Function Generator 1: If you are using an internal function generator, then the type of function generator and its settings are shown.

Export ASCII: You can export the information in ASCII Files (*.txt) format. To do so, click Export ASCII at the bottom of the dialog. The Save As dialog appears. Navigate to the saving location, enter the file name (preset ScanInfo.txt) and click Save.

Print: You can print out the file information. To do so, click Print at the bottom of the dialog. The Print dialog appears. Select the printer and set additional print properties. Click OK to start the printing process.

10.1.2 Opening a Signal Processor Document

In acquisition mode, you can open signal processor documents in proprietary binary format Signal Processor Files (*.spd).

Open file via dialog

To open a signal processor document, proceed as follows:

1. In the menu bar, select File > Open (toolbar: ).

   The Open dialog appears.

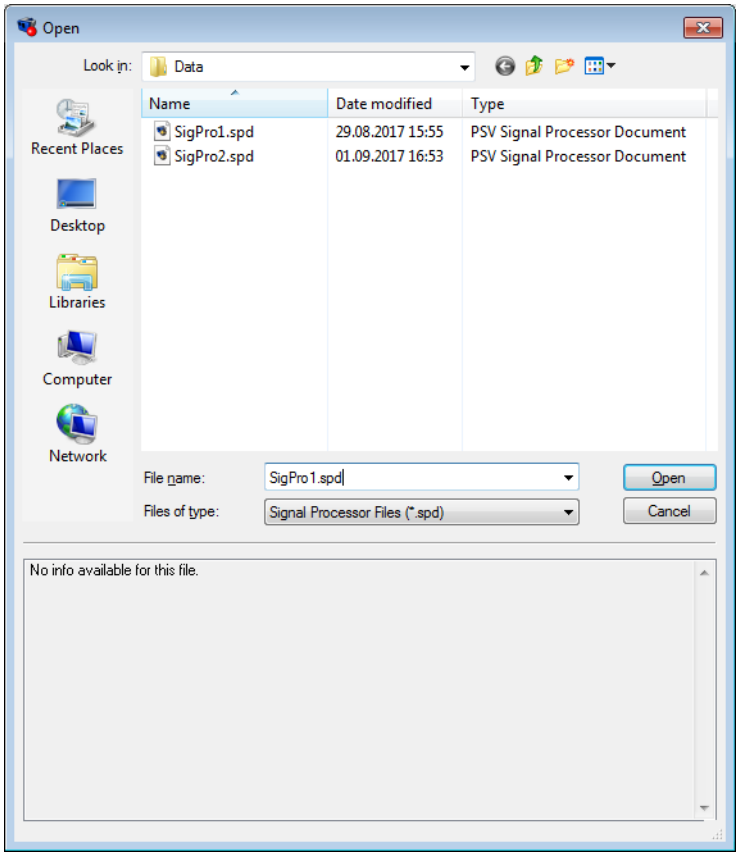


Figure 10.2: Open dialog for a signal processor document

2. Navigate to the saving location and select the file (File name)
3. Click Open.

   A signal processor appears. Working with the signal processor is described in SECTION 9.4.

10.1.3 Loading and Deleting Settings

ⓘ You will find information on saving the current settings in SECTION 10.2.3.

In acquisition mode, you can load settings from files in proprietary binary format Settings Files (*.set) and from scans or single point measurements.

ⓘ You can only delete settings if they are saved in a setting file in the Settings Files (*.set) format. You can not delete the settings for a scan or a single point measurement.

You can load the following information from setting files and scans or single point measurements:

 Frequency band definition (not possible from setting files and single point measurements)



Settings for data acquisition in the Acquisition Settings dialog (only possible for identical data acquisition boards and junction boxes)

 Scan points (scan point definition)

 2D alignment

 3D alignment

 Optics settings (zoom and focus of the camera)

 Window layout (not possible from scan files)

ⓘ If problems occur when loading settings, a message appears which in detail shows what kind of settings could not be loaded. All other settings are loaded.

To load the settings, you have the following options:

- Open dialog
- Project browser

Load settings via dialog

You can load the settings via the dialog.

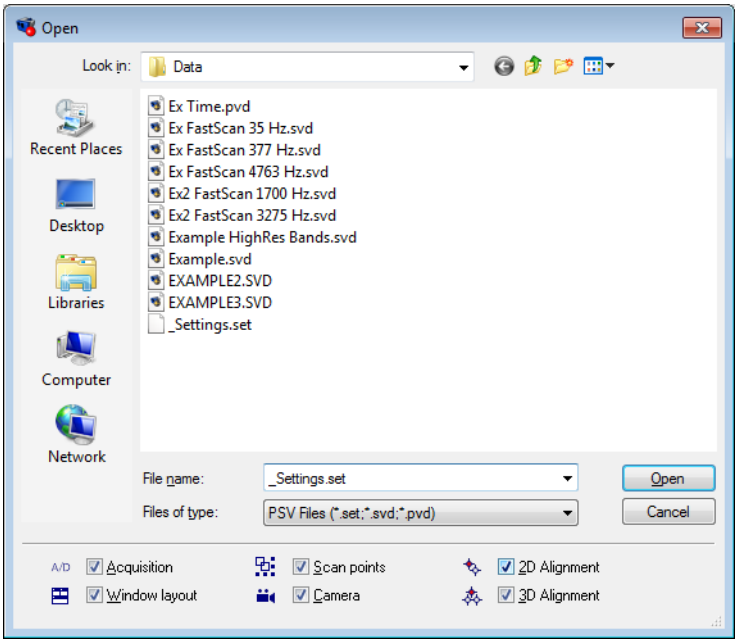


Figure 10.3: Open dialog for loading settings

Approach

To save the settings via the dialog, proceed as follows:

1. In the menu bar, select File > Load Settings (toolbar:).

   The Open dialog appears (refer to figure above).
2. Navigate to the saving location and select Files of type and File name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

3. At the bottom of the dialog, you can select which settings you want to load. To do so, click the corresponding check box.
4. Click Open.

   The selected settings will be loaded.

Load settings via project browser

You can load the settings via the project browser.

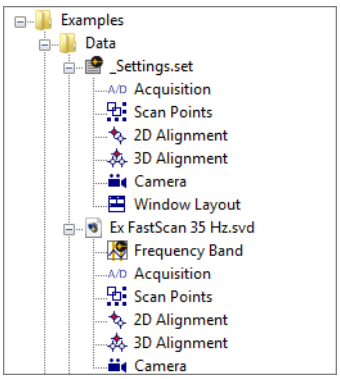


Figure 10.4: Settings in the setting file (top) and in the scan (bottom)

Approach

To load the settings via the project browser, proceed as follows:

1. Open the project browser. See SECTION 2.6.4 on this.
2. Open a directory which contains setting files or measurements.
3. Click the plus sign (+) in front of the (*.set) settings file or the measurement whose settings you want to load.

   You will see a list of settings which are saved in the file (refer to figure above).
4. If you want to load certain settings from a file, double click the respective setting.

ⓘ To mark several settings at once, hold the Control key pressed and click the respective settings one after another. Click the marked settings with the right mouse button and select Load in the context menu.

ⓘ If you want to load all settings from a setting file, double-click the file name or click the file name with the right mouse button and select Load in the context menu.

A message box appears asking whether you want to overwrite the settings you have used so far.

5. Click Yes.

   The selected settings will be loaded.

Delete settings

You can only delete settings if they are saved in a setting file in the Settings Files (*.set) format. You can not delete the settings of a measurement. To delete settings, proceed as follows:

1. Open the project browser. See SECTION 2.6.4 on this.
2. Click the plus sign (+) in front of the setting file (*.set) whose settings you want to delete.

   You will see a list of settings which are saved in the file (refer to figure above).

3. If you want to delete certain settings from the setting file, mark the respective setting.

ⓘ To mark several settings at once, hold the Control key pressed and click the respective settings one after another.

4. Click the marked settings with the right mouse button and select Delete in the context menu.

ⓘ If you want to delete the setting file, click the file name with the right mouse button and select Delete in the context menu.

10.2 Saving and Exporting

The software offers the following options for saving and exchanging data:

- Saving scans in proprietary binary format Scan Data (*.svd) - See SECTION 10.2.1 on this.
- Saving single point measurements and average spectrum in proprietary binary format Single Point Data (*.pvd) - See SECTION 10.2.2 on this.
- Saving settings in proprietary binary format Settings Files (*.set) - See SECTION 10.2.3 on this.
- Saving windows in various graphics formats - See SECTION 10.2.4 on this.
- Saving a path length alignment in Path Length Data (*.pla) format - See SECTION 10.2.5 on this.
- Exporting a path length alignment in ASCII Files (*.txt) format - See SECTION 10.2.6 on this.
- Saving recalculated scans as a signal processor document in Signal Processor Data (*.spd) format - See SECTION 9.4.5 on this.
- Copying measurement data in ASCII format, signal processor data, graphics, and windows settings onto the clipboard - See SECTION 10.2.7 on this.
- Exporting geometries in ASCII Files (*.txt) format - See SECTION 10.2.8 on this.
- Exporting geometries in Universal Files (*.unv, *.uff, *.asc) format - See SECTION 10.2.9 on this.
- With ExpME option: Exporting geometries in ME'Scope Files (*.str, *.blk, *.shp) format - See SECTION 10.2.10 on this.
- With ExpASAM option: Exporting geometries in ASAM ODS ATFX Files (*.atfx) format - See SECTION 10.2.11 on this.
- With GeoWorks option: Exporting geometries in STL Files (*.stl) stereo lithography data format - See SECTION 10.2.12 on this.

10.2.1 Saving a Scan

Complete scan

When you start a scan, the software automatically sets up a file for the scan and asks you to save.

During scanning, the software saves the following data in proprietary binary format Scan Data (*.svd):

- Settings
- Spectra of all scan points which have been measured completely (single point spectra)
- Average spectrum of all scan points
- RMS signal of amplitudes over the entire bandwidth

Even if you cancel a scan, the software has saved all scan points already measured.

10.2.2 Saving Single Point Measurement

In acquisition mode, you can save the measurement in the active analyzer in proprietary binary format Single Point Data (*.pvd).

At the same time, the software saves the following data:

- Time signals
- Spectra
- Parameters for data acquisition

To save single point measurements or spectra, proceed as follows:

1. In the menu bar, select File > Save Single Point Data As.

   The Save As dialog appears.

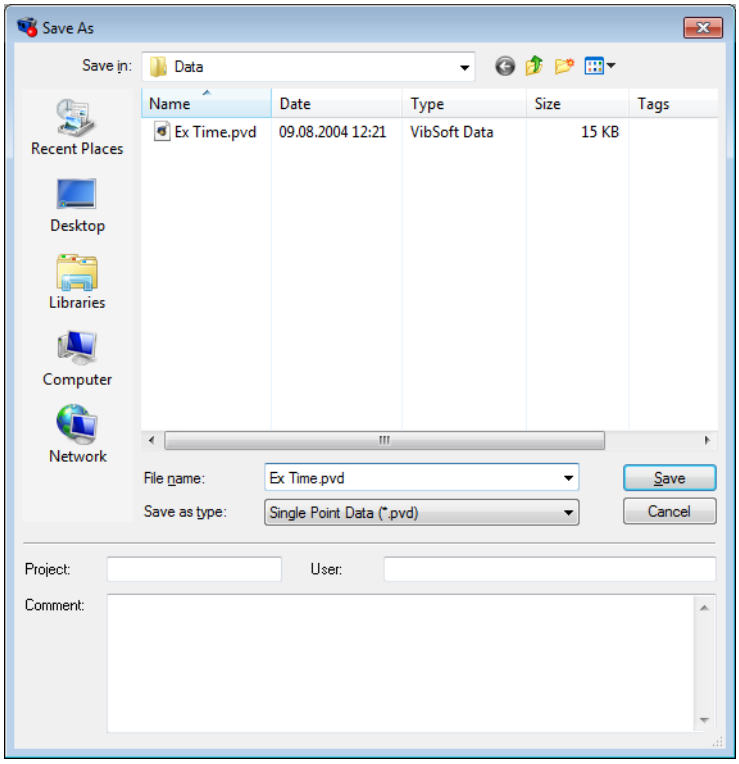


Figure 10.5: Save As dialog for saving single point data

2. Navigate to the saving location and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

3. At the bottom in the dialog you can enter additional information to characterize the file more accurately.

ⓘ This information is part of the file properties which you can also view and edit in the Explorer. It is displayed when you open a file in presentation mode and also in the File Information dialog. See SECTION 10.1.1 on this as well.

4. Click Save.

10.2.3 Saving Settings

In acquisition mode, you can save the current settings in the proprietary binary format Settings Files (*.set).

The setting files contain the following information:

- Settings for data acquisition in the Acquisition Settings dialog
- Scan points (scan point definition)
- 2D alignment
- 3D alignment
- Optics settings (zoom and focus of the camera)
- Window layout

ⓘ Frequency bands are not saved.

To save the settings, proceed as follows:

1. In the menu bar, select File > Save Current Settings As (toolbar:).
   The Save As dialog appears.

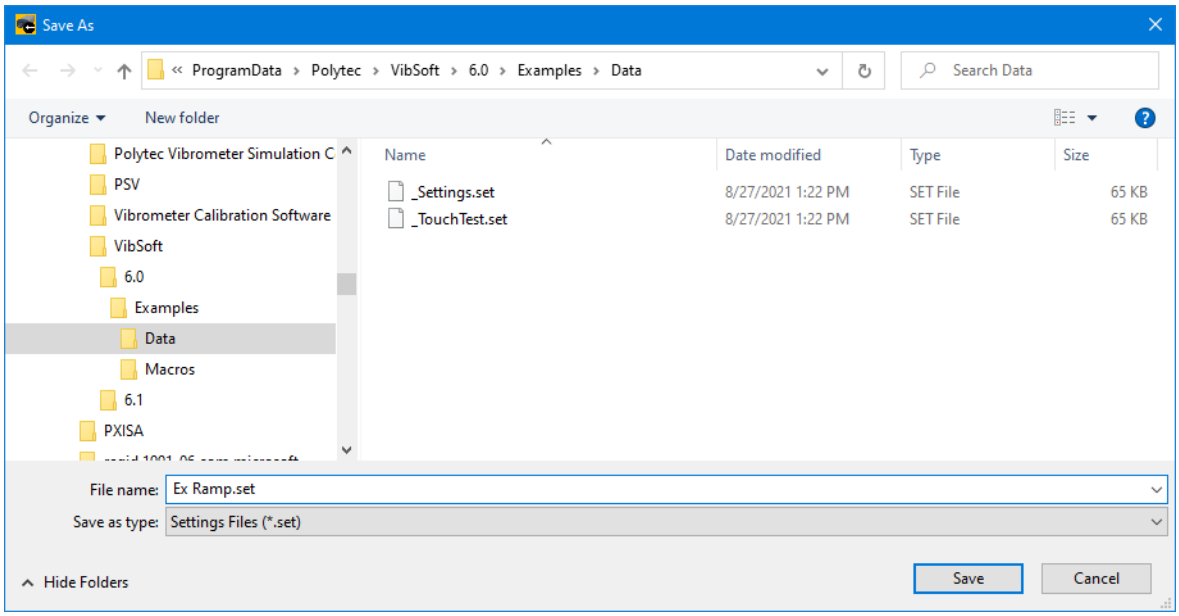


Figure 10.6: Save As dialog for settings

2. Navigate to the saving location and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

3. Click Save.
   All current settings are saved in the file.

10.2.4 Saving Graphics

You can save the active window in different graphics formats. The title bar and the frame are not saved. The live video image is saved without scan points.

The following graphics formats are available:

- Windows Bitmap Files (*.bmp)
- Portable Network Graphics Files (*.png)
- Jpeg File Interchange Format Files (*.jpg)
- Tagged Image File Format Files (*.tif)
- Graphics Interchange Format Files (*.gif)

To save graphics, proceed as follows:

1. Activate the window which you want to save as graphics. To do so, click it.
2. In the menu bar, select File > Save Graphics.

   The Save As dialog appears.

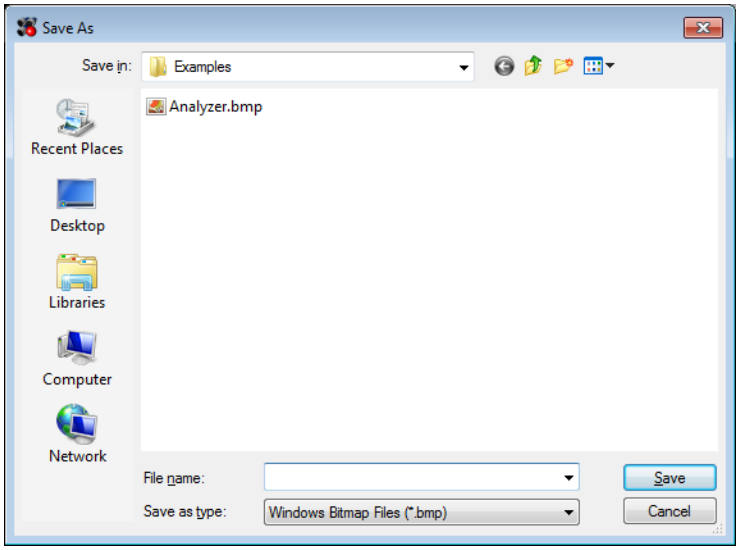


Figure 10.7: Save As dialog for graphics

3. Navigate to the saving location, select Save as type and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

4. Click Save.

   The graphics is saved in the selected format.

10.2.5 Saving a Path Length Alignment (only MSA-650)

You can save each path length alignment in proprietary binary format Path Length Data (*.pla).

1. Carry out a path length alignment as described in SECTION 5.4.
2. In the menu bar, select File > Save As.

   The Save As dialog appears.
3. Navigate to the saving location and enter the file name.
4. Click Save.

   The path length alignment will be saved in *.pla format.

10.2.6 Exporting a Path Length Alignment (only MSA-650)

You can export the data in Presentation mode as an ASCII file in ASCII Files (*.txt) format. In doing so, the individual values for time signals and path lengths are stored.

Export ASCII

To export the measurement data in ASCII format, proceed as follows:

1. Activate the window which you want to export as an ASCII file. To do so, click it.
2. In the menu bar, select File > Export > ASCII.

   The Save As dialog appears.

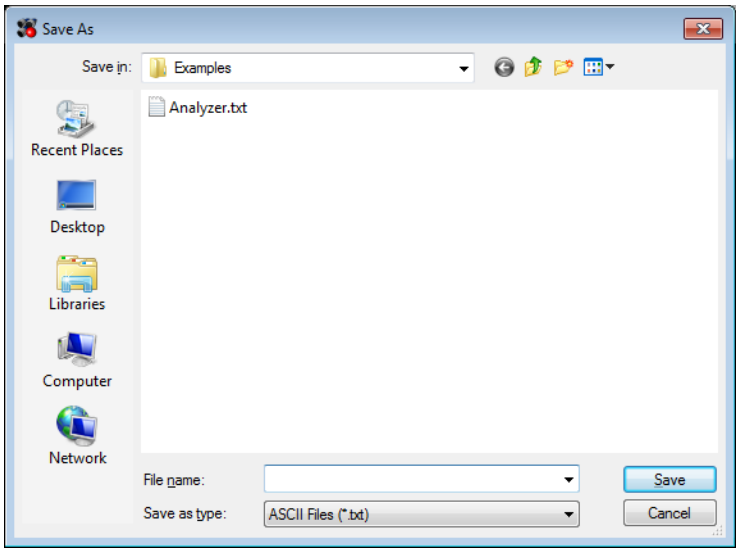


Figure 10.8: Save As dialog for ASCII export

3. Navigate to the saving location, select Save as type and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

4. Click Save.

   The data is saved in the selected format.

10.2.7 Copying Data onto Clipboard

You can copy data onto the clipboard from the active window as graphics, as signal processor data or in ASCII format. You can also copy the window settings onto the clipboard. If you select the graphics, then the title bar and the frame are not copied. The live video image is copied as graphics without scan points.

ⓘ Depending on the window selection or the view, you are offered a more or less comprehensive dialog.

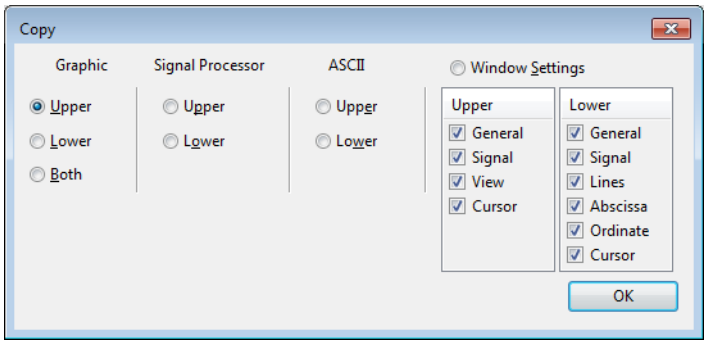


Figure 10.9: Copy dialog for clipboard

ⓘ When copying in the profile view, pay attention that you do not have selected profile lines. Otherwise these profile lines will be copied onto the clipboard.

Copy onto clipboard

To copy onto the clipboard, proceed as follows:

1. Activate the required window or the required view. To do so, click it.
2. In the menu bar, select Edit > Copy (toolbar: ).

   The Copy dialog appears (refer to figure above).
3. Select the data you want to copy onto the clipboard.
4. Click OK.

   You can now paste the data from the clipboard into other applications.

ⓘ You can also press the Ctrl + C key combination. The Copy dialog appears as long as you hold the Control key pressed. Select which data you want to copy and release the Control key. The data is copied.

Copy view onto clipboard

You can also selectively copy a certain part of the window onto the clipboard. To do so, proceed as follows:

1. Activate the corresponding window (video window or analyzer). To do so, click it.
2. In the menu bar, select Edit > Copy (toolbar: ).

   The Copy dialog appears.

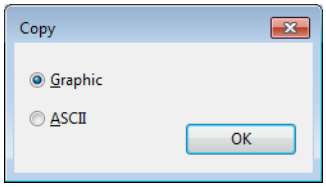


Figure 10.10: Copy dialog in the video window

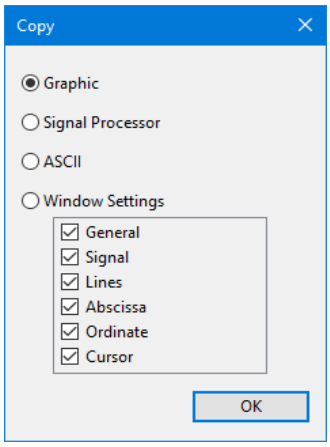


Figure 10.11: Copy dialog in the analyzer

3. Select the data you want to copy onto the clipboard.
4. Click OK.

   You can now paste the data from the clipboard into other applications.

10.2.8 Exporting Geometries in ASCII Format

You can export the data in the active analyzer or video window as an ASCII file in ASCII Files (*.txt) format. In doing so, the coordinates shown in the readout are exported.

To calculate the angles of the scanner mirrors from the coordinates, you will need the information from the 2D or 3D alignment respectively. You will get this from Polytec File Access.

Export ASCII

To export the measurement data in ASCII format, proceed as follows:

1. Activate the window which you want to export as an ASCII file. To do so, click it.
2. In the menu bar, select File > Export > ASCII.

   The Save As dialog appears.

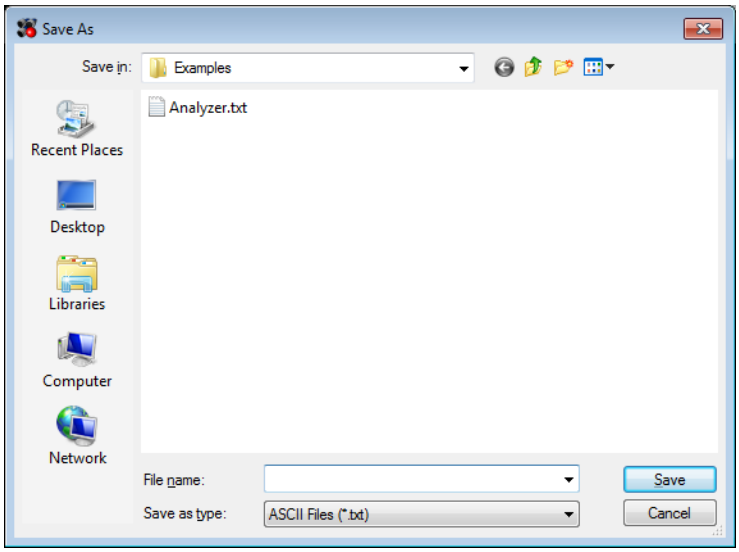


Figure 10.12: Save As dialog for ASCII export

3. Navigate to the saving location, select Save as type and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

4. Click Save.

   The data is saved in the selected format.

10.2.9 Exporting Universal File

Data sets

In both acquisition and presentation mode, you can export data in Universal File format to use it in a modal analysis software.

The following data sets are exported:

Data set	Description	Format
Header	General information, e.g. file name, date, comment	151
Units	Units of the signals and coordinates	164 Double Precision
Nodes (Geometry)	Coordinates of the scan points	2411 Double Precision
Elements (Trace Lines)	Graphic representation of the scan points and their connections	2412 + 82 Integer
Frequency Bands	A data set for every signal and every frequency band	55 Single Precision
Scan Points	A data set for every signal and every scan point	58 Single Precision

Features

You need to take the following points into consideration when exporting:

- You can export data in Universal File format in acquisition mode, but not in APS mode.
- Scan points with the Not Measured status are not exported.
- Signals which have been gained from the original signals through integration and differentiation can be exported.
- Only MSA-500 with VDD option: Measurements in I/Q measurement mode cannot be exported as Universal File.

Export UFF data

To export data in Universal Files (*.unv, *.uff, *.asc) format, proceed as follows:

1. In the menu bar, select File > Export > Universal File.

   The Save As dialog appears.

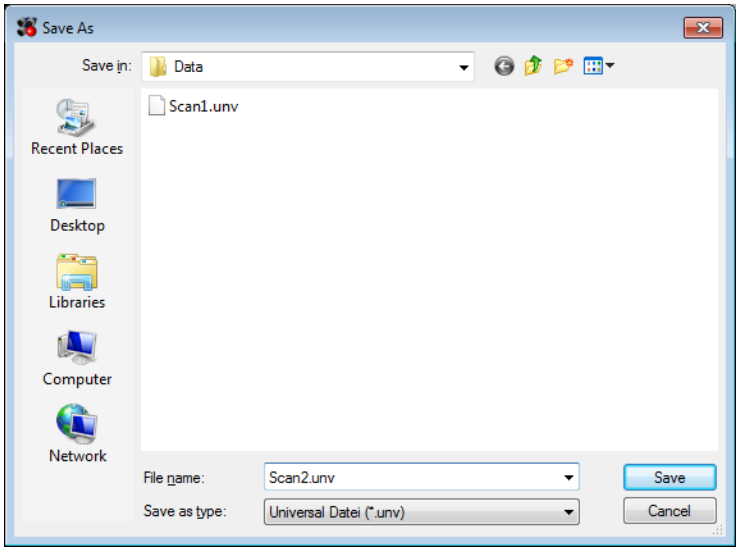


Figure 10.13: Save As dialog for Universal File export

2. Navigate to the saving location, select Save as type and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

3. Click Save.

   The Universal File Export dialog appears.

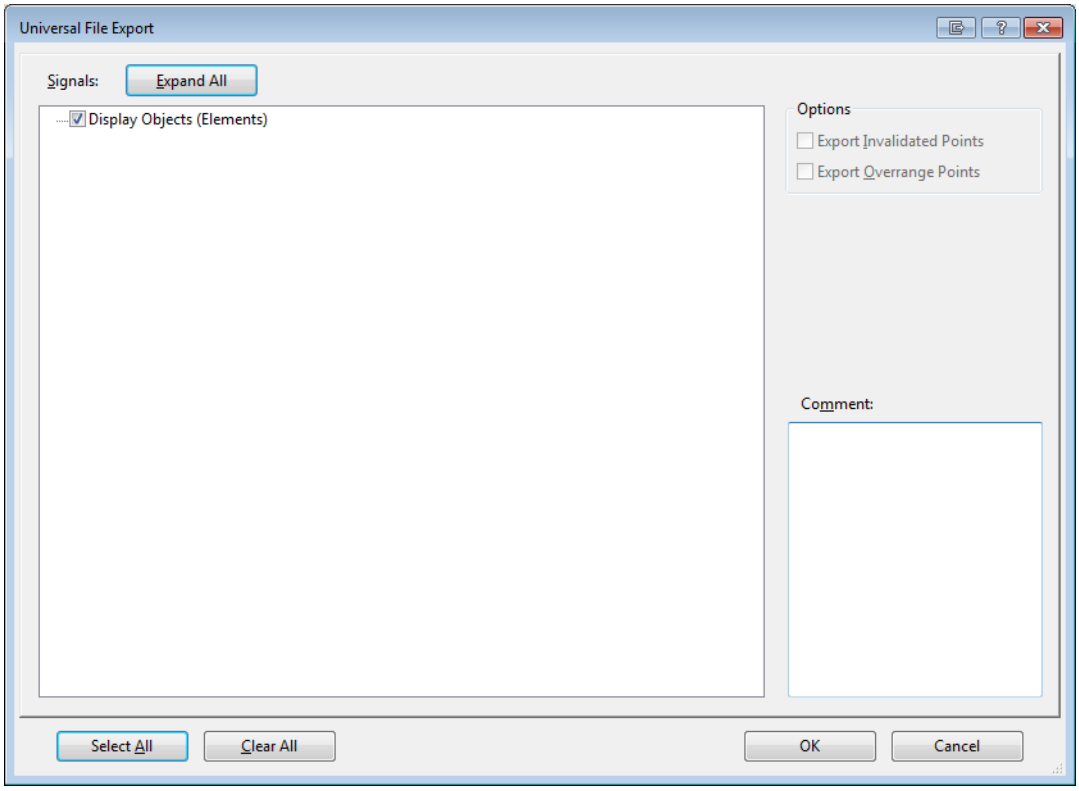


Figure 10.14: Universal File Export dialog

ⓘ You can freely adjust the size of the dialog (width/height) to be able to read all the information.

4. Mark the signals which you want to export and set the parameters for the export.
5. Click OK.

   The data is saved in the selected format.

Dialog description

On the left you mark the signals and on the right you can enter a comment for the export.

ⓘ If you have defined geometry components as described in SECTION 6.5.4, then only the component names for the data sets 82 and 58 will be taken into account for export.

Signals: Mark the signals which you want to export. If you export the data in acquisition mode, the only choice is to display objects. If Display Objects (Elements) is ticked, then a graphic representation of the scan points and their connections is exported.

Expand All: Click here to display the entire tree in detail.

Select all: Click here to select all signals.

Clear All: Click here to clear the signal selection.

Options: The options are only available in presentation mode.

Comment: You can enter a comment up to 80 characters long.

10.2.10 Exporting ME'Scope

Data sets

You can export files in ME'Scope format. For this, you need the ExpME option.

In acquisition mode, you can then export the object geometry, for example to make further use of it in other applications.

The export provides the geometry in ME'Scope Structure Files (*.str) format.

Features

You need to take the following points into consideration when exporting:

- Scan points with the Not Measured status are not exported.
- Only MSA-500 with VDD option: Measurements in I/Q measurement mode cannot be exported as ME'Scope.

Export ME'Scope data

To export the data in ME'Scope Structure Files (*.str) format, proceed as follows:

1. In the menu bar, select File > Export > ME'Scope.

   The Save As dialog appears.

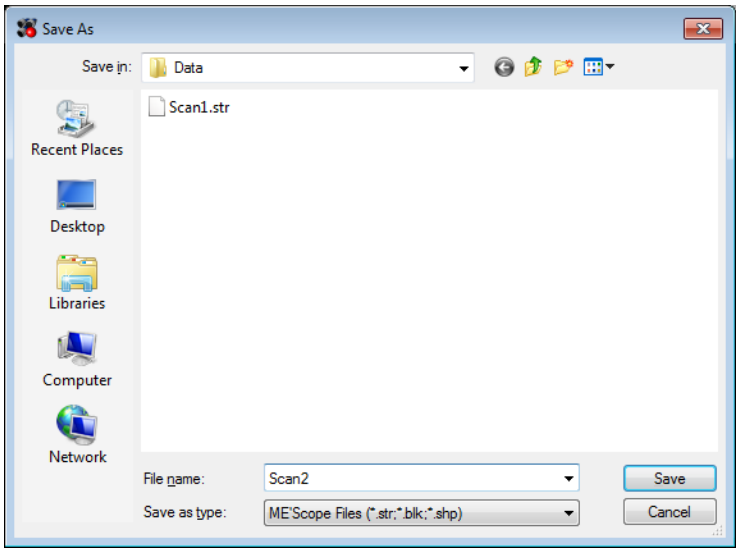


Figure 10.15: Save As dialog for ME'Scope export

2. Navigate to the saving location and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

3. Click Save.

   The ME'Scope Export dialog appears.

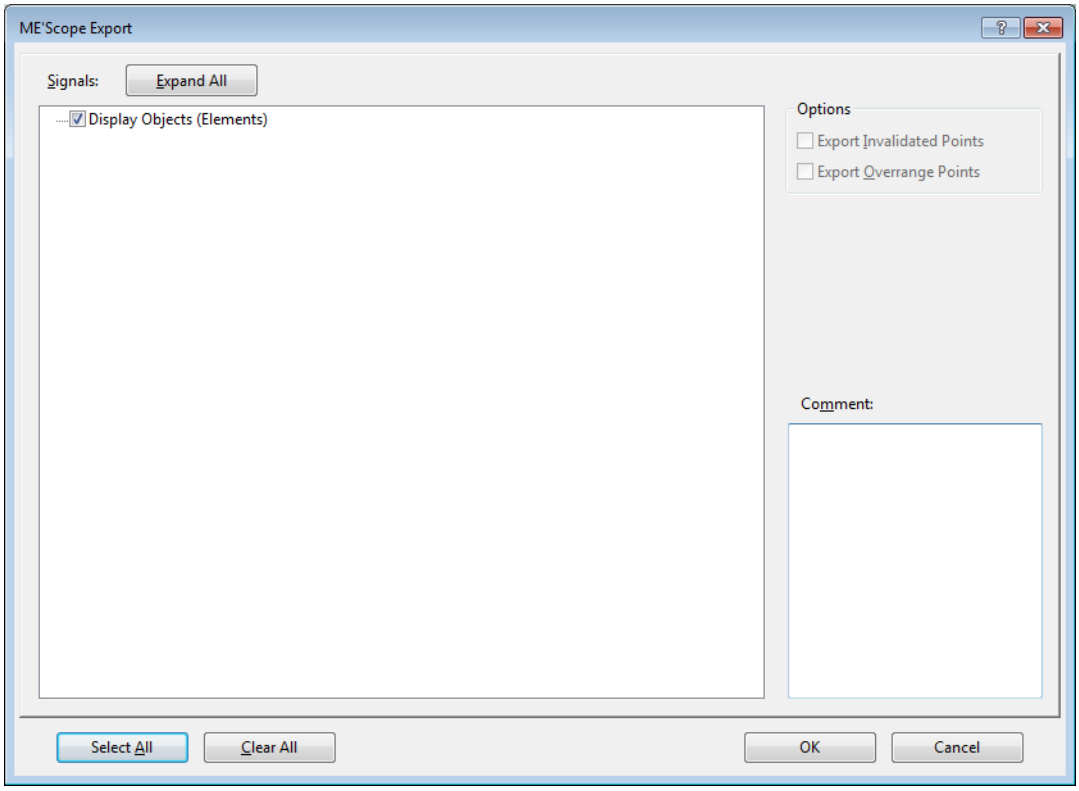


Figure 10.16: ME'Scope Export dialog

ⓘ You can freely adjust the size of the dialog (width/height) to be able to read all the information.

4. Mark the signals which you want to export and set the parameters for the export.
5. Click OK.

   The data is saved in the selected format.

Dialog description

On the left you mark the signals and on the right you can enter a comment for the export.

Signals: Mark the signals which you want to export. If you export the data in acquisition mode, the only choice you are given is to display objects. If Display Objects (Elements) is ticked, then a graphic representation of the scan points and their connections is exported.

Expand All: Click here to display the entire tree in detail.

Select all: Click here to select all signals.

Clear All: Click here to clear the signal selection.

Options: The options are only available in presentation mode.

Comment: You can enter a comment up to 80 characters long.

10.2.11 Exporting ASAM ODS ATFX

Data sets

Both in acquisition and presentation mode, you can export measurement data in the ASAM format (Standard ASAM ODS 5.3.0). For this, you need the ExpASAM option.

In acquisition mode, you can then export the object geometry, for example to make further use of it in other applications.

Export will provide two files with the following data:

- XML file in ASAM ODS ATFX Files (*.atfx) format
- Binary file with the included data in (*.btf) format

ⓘ You always have to save or move both files together. During import, only the ASAM file is selected. However, all data is required for an import.

Features

You need to take the following points into consideration when exporting:

- Scan points with the Not Measured status are not exported.
- Only MSA-500 with VDD option: Measurements in I/Q measurement mode cannot be exported as ASAM ODS ATFX.

Export ASAM data

To export the data in the ASAM ODS ATFX Files (*.atfx) format, proceed as follows:

1. In the menu bar, select File > Export > ASAM ODS ATFX.

   The Save As dialog appears.

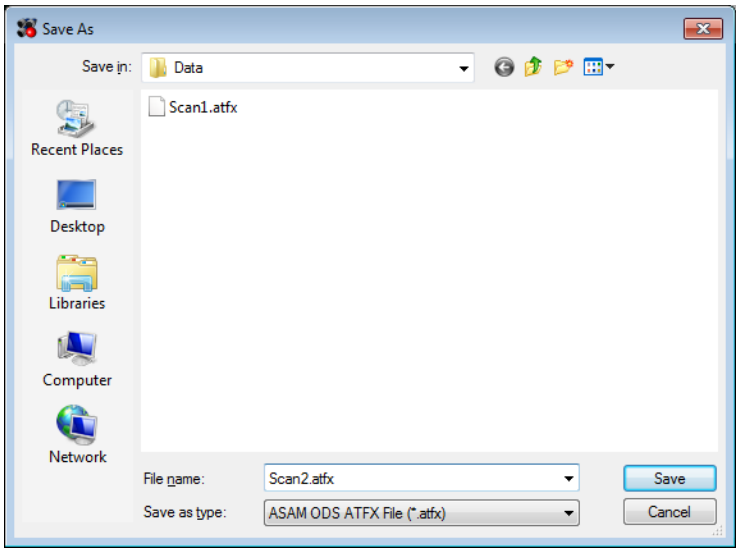


Figure 10.17: Save As dialog for ASAM ODS ATFX export

2. Navigate to the saving location and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

3. Click Save.

   The ASAM ODS ATFX Export dialog appears.

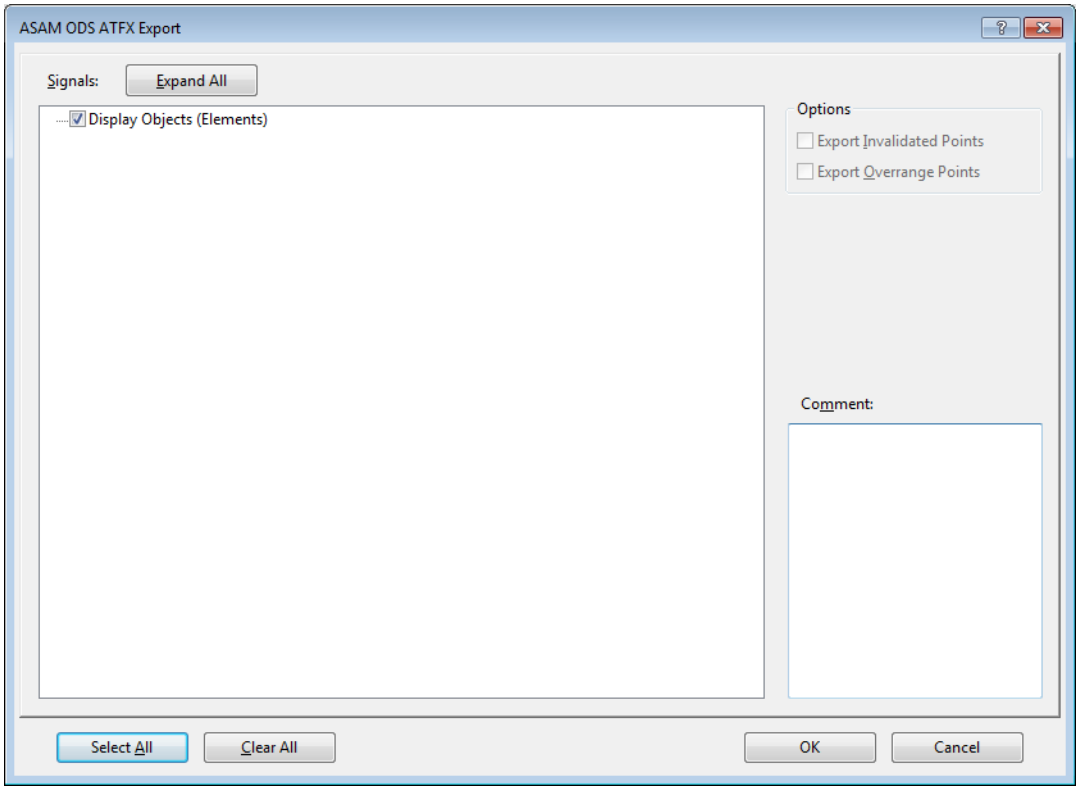


Figure 10.18: ASAM ODS ATFX Export dialog

ⓘ You can freely adjust the size of the dialog (width/height) to be able to read all the information.

4. Mark the signals which you want to export and set the parameters for the export.
5. Click OK.

   The data is saved in the selected format.

Dialog description

On the left you mark the signals and on the right you can enter a comment for the export.

Signals: Mark the signals which you want to export. If you export the data in acquisition mode, the only choice you are given is to display objects. If Display Objects (Elements) is ticked, then a graphic representation of the scan points and their connections is exported.

Expand All: Click here to display the entire tree in detail.

Select all: Click here to select all signals.

Clear All: Click here to clear the signal selection.

Options: The options are only available in presentation mode.

Comment: You can enter a comment up to 80 characters long.

10.2.12 Exporting Geometries in STL Format

You can export the geometries as an STL file in stereo lithography data format STL ASCII File (*.stl) or STL Binary File (*.stl). For this, you need the GeoWorks option.

Export STL

To export the geometries in STL format, proceed as follows:

1. In the menu bar, select File > Export > STL.

   The Save As dialog appears.

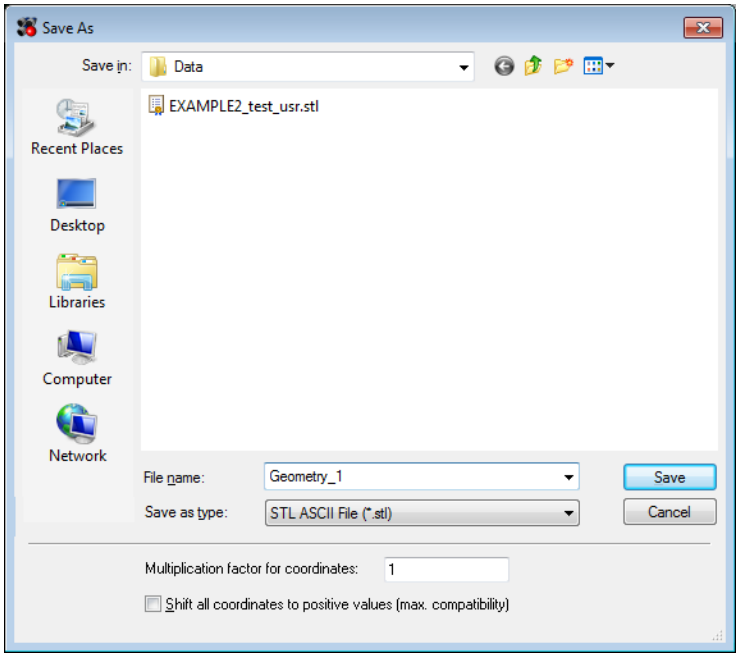


Figure 10.19: Save As dialog for STL export

2. Navigate to the saving location, select Save as type and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. See SECTION 2.6.4 on this as well.

3. In the dialog at the bottom, select which multiplication factor is to be used for the coordinates and select the maximum compatibility.
4. Click Save.

   The data is saved in the selected format.

Appendix A: Key Combinations in the Software

General

Key combination	Definition
Ctrl + O	Open Document
Ctrl + I	Display file information
Ctrl + S	Save current settings
Ctrl + G	Save active view as graphic
Ctrl + Z	Undo last action
Ctrl + Y	Redo previously undone action
Ctrl + C	Copy selection onto the clipboard
Ctrl + V	Insert selection from the clipboard
Ctrl + A	Select all
Ctrl + D	Start presentation mode
Ctrl + B	Show/hide scanning head control
Ctrl + M	Show/hide message window
Ctrl + N	Open an new analyzer
Ctrl + Pause	Stop running macro
Ctrl + 1	Switch laser beam on/off
Ctrl + F	Automatically focus laser beam
Ctrl + E	Automatically focus laser beam quickly (less precise)
Ctrl + Shift + ↓	Move laser beam down
Ctrl + Shift + ←	Move laser beam to the left
Ctrl + Shift + →	Move laser beam to the right
Ctrl + Shift + ↑	Move laser beam up
Ctrl + ↓	MSA-650: move downwards to detected levels after autofocus travel
Ctrl + ↑	MSA-650: move upwards to detected levels after autofocus travel
F5	Define acquisition settings
F6	Switch generator on/off
F8	Start a continuous measurement
Shift + F8	Start a single shot
F9	Start a scan
F10	Cancel measurement
F11	Continue scan at last measured scan point
F12	Start remeasuring invalid scan points
Shift + F12	Start remeasuring invalid scan points using the AutoRange function

In the Video Window

Key combination	Definition
Alt + Enter	Change display properties
Ctrl + Space	Disable selected scan points
Shift + Space	Enable selected scan points
Shift + Y	3D view: Sight of camera
Esc	Abort initialization of the high-contrast laser display
F4	Only for systems using XY positioning stage for scanning: Pin scan points
Alt + Shift + Enter	Two monitors connected: Video image is displayed as full screen
Ctrl + ←	Modify 3D Coordinates dialog open: Decrease x-coordinate
Ctrl + →	Modify 3D Coordinates dialog open: Increase x-coordinate
Ctrl + ↓	Modify 3D Coordinates dialog open: Decrease y-coordinate
Ctrl + ↑	Modify 3D Coordinates dialog open: Increase y-coordinate
Ctrl + Page Dn↓	Modify 3D Coordinates dialog open: Decrease z-coordinate
Ctrl + Page Up↑	Modify 3D Coordinates dialog open: Increase z-coordinate
Ctrl + –	Modify 3D Coordinates dialog open: Decrease geometry scan distance
Ctrl + +	Modify 3D Coordinates dialog open: Increase geometry scan distance

In the Video Window and APS Mode

Key combination	Definition
D	Standard object and object marked: Duplicate object
Ctrl + L	Point object: Set new scan point at current laser position
Delete	Standard or point object: Delete marked objects/scan points Point object and Connect Points active: Delete connection
C	Point object and Coarsen Grid dialog open: Coarsen grid
Pause	Point object and Coarsen Grid dialog open: Cancel action
Esc	Point object and Coarsen Grid dialog open: Close dialog
T	Point object, 3D view and Connect Points dialog open: Generate triangles for selected scan points

In the Analyzer

Key combination	Definition
Alt + Enter	Change analyzer properties
Ctrl + K	Scale ordinate to min/max values of current data
Ctrl + P	Print out analyzer

In the Project Browser

Key combination	Definition
Shift + Delete	Irrevocably delete selected file without security popup
Delete	Irrevocably delete selected file with security popup
F2	Rename selected file or folder
Enter	Execute default action in the context menu

In the Signal Processor

Key combination	Definition
Ctrl + C / Ctrl + Insert	Copy selected cell
Ctrl + X / Shift + Delete	Cut selected cell
Ctrl + V / Shift + Insert	Insert selected cell
Ctrl + Z / Alt + Backspace	Undo last action
Ctrl + Y	Redo previously undone action
Ctrl + #	For all cells switch display to show the formulae

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